

Exhibit CC-32

JEDEC STANDARD

**DDR5 Registering Clock Driver
Definition (DDR5RCD02)**

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DDR5 REGISTERING CLOCK DRIVER DEFINITION (DDR5RCD02)

(From JEDEC Board Ballot JCB-22-43, item 339.98D and JCB-23-02, item 339.98E, formulated under the cognizance of JC-40, Digital Logic).

1 Scope

This standard defines specifications of DC interface parameters, switching parameters, and test loading for definition of the DDR5 Registering Clock Driver (RCD) with parity for driving address and control nets on DDR5 RDIMM and LRDIMM applications. The DDR5RCD02 Device ID is DID = 0x0052.

The terms ‘Registering Clock Driver’, ‘RCD’, ‘register’, or ‘device’ are used interchangeably to refer to this device in the remainder of this standard.

The purpose is to provide a standard for the DDR5RCD02 (see Note) logic device, for uniformity, multiplicity of sources, elimination of confusion, ease of device specification, and ease of use.

NOTE The designation DDR5RCD02 refers to the part designation of a series of commercial logic parts common in the industry. This designation is normally preceded by a series of manufacturer specific characters to make up a complete part designation.

2 Mechanical Outline

Package options include a 240-ball Flip-Chip Fine-Pitch BGA (FCBGA) with 0.60 mm/0.70 mm ball pitch, 14 x 19 grid. The package has a number of depopulated balls. Package size is 8.70 mm x 13.50 mm as defined in MO-330 Issue A, Variation 10013.50x8.70-70x6030-240A¹. The device pinout supports outputs on the top and outer left and right columns to support easy DIMM signal routing. Corresponding inputs are placed in a way to match the corresponding pin location on the connector.

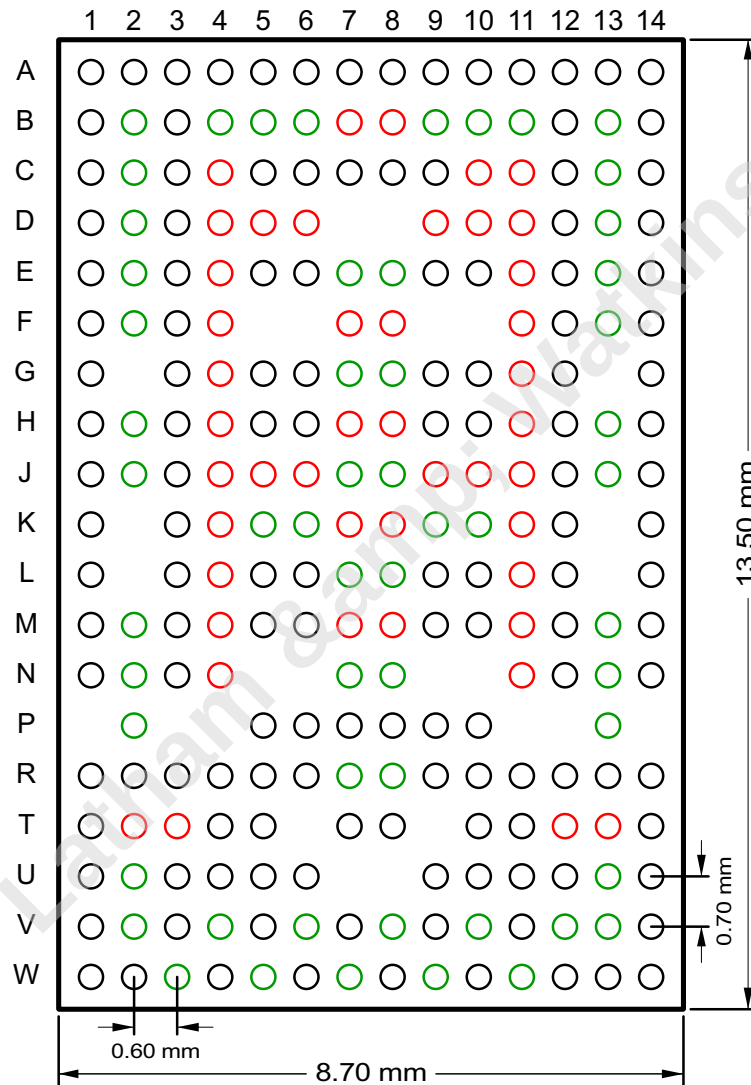


Figure 1 — Ball Configuration, TOP VIEW

Ball pitch: 0.60 mm x 0.70 mm, Size (ø 0.3 mm), SMD Pad SRO (ø 0.275 mm)
X-ray view from topside

1. This variation defines a maximum package thickness of 1.00 mm. The DDR5RCD02 must comply with a minimum thickness of 0.80 mm.

2.1 Pinout

Table 1 specifies the pinout for the DDR5RCD02.

Table 1 — Ball Assignment -240 ball FCBGA, 14 x 19 Grid, TOP VIEW

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
A	NU	QB CA7_A	QB CA3_A	QB CA13_A	QB CA11_A	QB CA12_A	QB CA10_A	QB CA10_B	QB CA12_B	QB CA11_B	QB CA13_B	QB CA3_B	QB CA7_B	NU	A
B	QB CA1_A	V _{SS}	QB CA9_A	V _{SS}	V _{SS}	V _{SS}	V _{DD}	V _{DD}	V _{SS}	V _{SS}	V _{SS}	QB CA9_B	V _{SS}	QB CA1_B	B
C	QB CA6_A	V _{SS}	QB CA8_A	V _{DD}	ZQ CAL	SCL	DERROR _IN_A_n	DERROR _IN_B_n	SDA	V _{DDIO}	V _{DD}	QB CA8_B	V _{SS}	QB CA6_B	C
D	QB CA5_A	V _{SS}	QB CA2_A	V _{DD}	V _{DD}	V _{DD}			V _{DD}	V _{DD}	V _{DD}	QB CA2_B	V _{SS}	QB CA5_B	D
E	QB CA0_A	V _{SS}	QB CA4_A	V _{DD}	QBCK _A_c	QBCK _A_t	V _{SS}	V _{SS}	QBCK _B_t	QBCK _B_c	V _{DD}	QB CA4_B	V _{SS}	QB CA0_B	E
F	QBCS0 _A_n	V _{SS}	QBCS1 _A_n	V _{DD}			V _{DD}	V _{DD}			V _{DD}	QBCS1 _B_n	V _{SS}	QBCS0 _B_n	F
G	QA CA11_A		QA CA13_A	V _{DD}	QDCK _A_c	QDCK _A_t	V _{SS}	V _{SS}	QDCK _B_t	QDCK _B_c	V _{DD}	QA CA13_B		QA CA11_B	G
H	QA CA9_A	V _{SS}	QA CA12_A	V _{DD}	RFU	RFU	V _{DD}	V _{DD}	RFU	RFU	V _{DD}	QA CA12_B	V _{SS}	QA CA9_B	H
J	QA CA10_A	V _{SS}	QA CA3_A	V _{DD}	V _{DD}	V _{DD}	V _{SS}	V _{SS}	V _{DD}	V _{DD}	V _{DD}	QA CA3_B	V _{SS}	QA CA10_B	J
K	QA CA6_A		QA CA7_A	V _{DD}	V _{SS}	V _{SS}	V _{DD}	V _{DD}	V _{SS}	V _{SS}	V _{DD}	QA CA7_B		QA CA6_B	K
L	QA CA1_A		QA CA4_A	V _{DD}	QA CA2_A	QA CA8_A	V _{SS}	V _{SS}	QA CA8_B	QA CA2_B	V _{DD}	QA CA4_B		QA CA1_B	L
M	QA CA5_A	V _{SS}	QA CA0_A	V _{DD}	QACK _A_c	QACK _A_t	V _{DD}	V _{DD}	QACK _B_t	QACK _B_c	V _{DD}	QA CA0_B	V _{SS}	QA CA5_B	M
N	QACS0 _A_n	V _{SS}	QACS1 _A_n	V _{DD}			V _{SS}	V _{SS}			V _{DD}	QACS1 _B_n	V _{SS}	QACS0 _B_n	N
P		V _{SS}			QCCK _A_c	QCCK _A_t	QLBD	QLBS	QCCK _B_t	QCCK _B_c			V _{SS}		P
R	BCS _A_n	BCOM1 _A	BCOM2 _A	BRST _A_n	BCK _A_c	BCK _A_t	V _{SS}	V _{SS}	BCK _B_t	BCK _B_c	BRST _B_n	BCOM2 _B	BCOM1 _B	BCS _B_n	R
T	BCOM0 _A	V _{DD}	V _{DD}	DCS1 _A_n	QRST _A_n		DCK_t	DCK_c		QRST _B_n	DCS0 _B_n	V _{DD}	V _{DD}	BCOM0 _B	T
U	DCA0 _A	V _{SS}	DCS0 _A_n	DLBD_A	DLBS_A	ALERT _n			DRST_n	DLBS_B	DLBD_B	DCS1 _B_n	V _{SS}	DPAR _B	U
V	DCA1 _A	V _{SS}	DCA3 _A	V _{SS}	DPAR _A	V _{SS}	DCA6 _A	V _{SS}	DCA1 _B	V _{SS}	DCA3 _B	V _{SS}	V _{SS}	DCA6 _B	V
W	NU	DCA2 _A	V _{SS}	DCA4 _A	V _{SS}	DCA5 _A	V _{SS}	DCA0 _B	V _{SS}	DCA2 _B	V _{SS}	DCA4 _B	DCA5 _B	NU	W
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	

2.2 Terminal Functions

Table 2 — Terminal Functions

Signal Group	Signal Name	Type	Description
Input Control bus	DCS[1:0]_[B:A]_n	POD V_{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.
Input Address and Command bus	DCA[6:0]_[B:A]	POD V_{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.
Parity input	DPAR_[B:A]	POD V_{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.
Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.
Loopback	DLBD_[B:A] DLBS_[B:A]	POD V_{REF} based	Loopback Inputs to RCD DLBD - loopback Data DLBS - loopback Strobe
Reset input	DRST_n	CMOS input	Active LOW asynchronous reset input. When LOW, it causes a reset of the internal latches and disables the outputs, thereby forcing the outputs to float.
Error input	DERROR_IN_[B:A]_n	Low voltage swing POD input	DRAM CRC ALERT_n output is connected to this input pin, which in turn is buffered and redriven to the ALERT_n output of the register. There is a separate signal per channel.
Data buffer control and communication outputs	BCS_[B:A]_n	POD	Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.
	BCOM[2:0]_[B:A]	POD	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.
	BRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.
	BCK_[B:A]_t BCK_[B:A]_c	POD differential	Differential clock output pair to the data buffer. Separate per DRAM channel.
Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.
Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).
Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.
Loopback	QLBD QLBS	POD	Loopback outputs to Host QLBD - loopback Data QLBS - loopback Strobe
Reset output	QRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD QRST_n to reset the DDR5 SDRAM on all DIMM topologies. The QRST outputs are asserted at power up. RCD requires MRW to independently de-assert to allow staggering of sub-channels.
Error out	ALERT_n	POD	When LOW, this output indicates that a parity error was identified associated with the CA inputs when parity checking is enabled or that the DERROR_IN_n input was asserted, regardless of whether parity checking is enabled or not. One signal for the two channels.
SidebandBus pins ¹	SDA	Open drain or push-pull I/O ²	SidebandBus Data
	SCL	CMOS input ³	SidebandBus Clock
	VDDIO	Power input	SidebandBus power input

Table 2 — Terminal Functions (cont'd)

Signal Group	Signal Name	Type	Description
Miscellaneous pins	V _{DD}	Power Input	Power supply voltage
	V _{SS}	Ground Input	Ground
	ZQCAL	Reference	Reference pin for driver calibration
	NU	Mechanical ball	Do not connect on PCB
	RFU[3:0]	I/O	Reserved for future use pins, must be left floating on DIMM and in RCD
NOTE 1 SA pins are not required for DDR5RCD02 as the address will be hard-coded. Refer to Section 7.5.1, “Target Address,” on page 88 for details.			
NOTE 2 SDA driver operation is dynamic. Depending on the SidebandBus mode of operation (I ² C RW25[5] = ‘0’ or I3C Basic RW25[5] = ‘1’), and even on the specific step (byte or bit) of a SidebandBus transaction packet, the SDA output driver can operate either in open-drain mode or push-pull mode.			
NOTE 3 These inputs are 1.0 V inputs.			

Naming Convention:

Input Example:

DCAy_N - where ‘y’ is the signal number and ‘N’ is the sub-channel.

Output Example:

QxCAY_N - where ‘y’ is the signal number, ‘x’ is the output copy (A or B) and ‘N’ is the sub-channel.

3 Device Standard

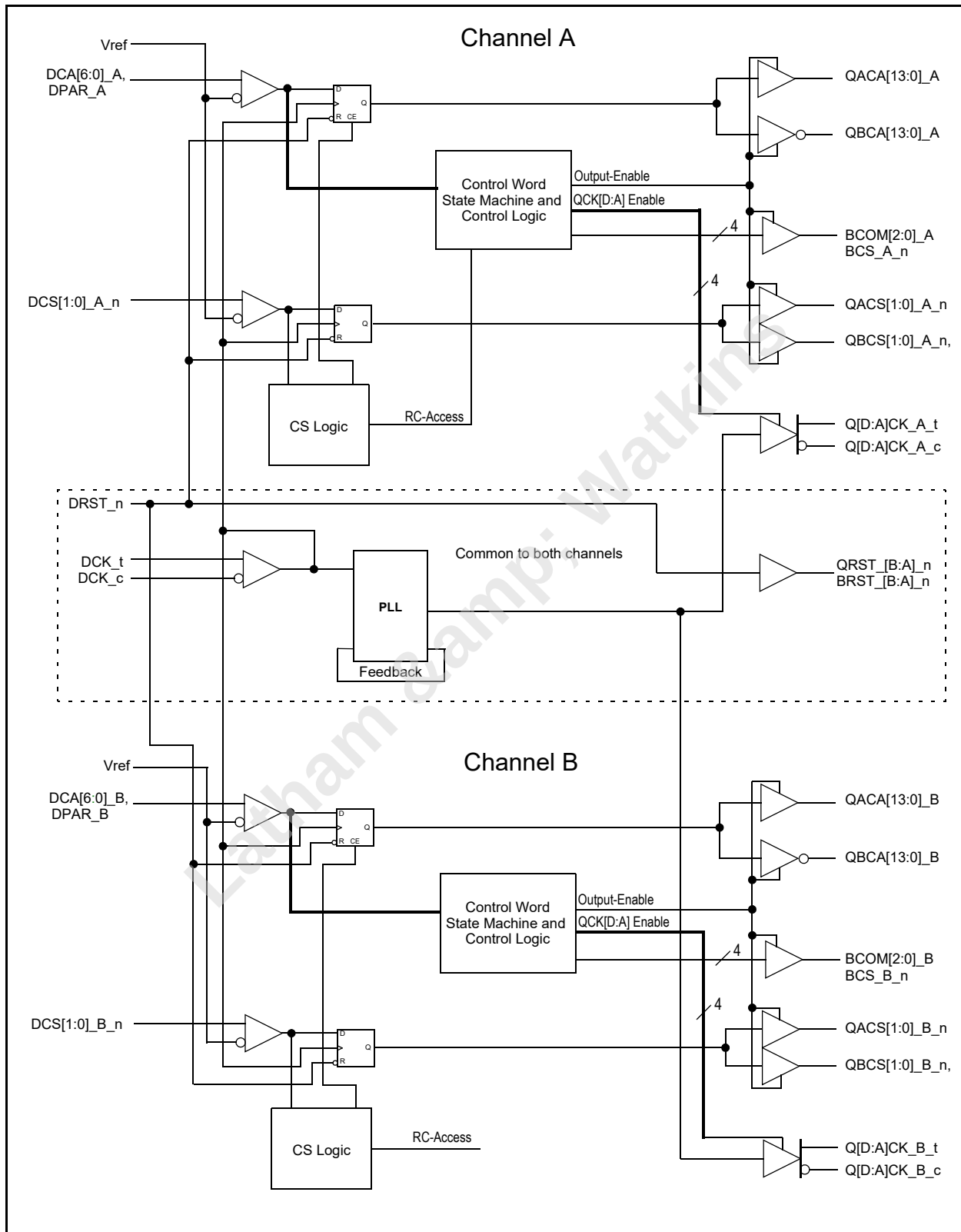
3.1 Description

The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs.

It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels.

Latham & Watkins

3.2 Logic Diagram Positive Logic



3.3 Register Operation

3.3.1 Chip Select Operation

The component has two chip select inputs per channel, DCS0_n and DCS1_n, and two copies of each chip select output, QCS0_A_n, QCS1_A_n, QCS0_B_n and QCS1_B_n. Commands are sent to a single rank or multiple ranks, as determined by the DCS[1:0]_n inputs. The number of input chip selects matches the number of output chip selects in each of the two sets (A-outputs and B-outputs).

Table 3 — DCS - QxCS Mapping

Input CS	Output CS
DCS0_N_n	QxCS0_N_n
DCS1_N_n	QxCS1_N_n

3.3.2 ODT Operation

DDR5 does not have a separate ODT (On Die Termination) signal going to the DIMM or the DRAMs. The ODT control is encoded into the chip selects. All commands that utilize the data bus require 2UI on the DCA bus in order to transfer the appropriate number of command and addressing bits. This requirement is used to signal ODT control by how the chip selects are asserted for the two UI.

Table 4 — ODT Encoding on the Chip Select

DCS_n on 1st UI	DCS_n on 2nd UI	QxCS[1:0]_n	DRAM DQ Terminations
Active	Inactive	Target of command	Read: none (device drives data) Write: RTT_WR.
Active	Active	Non-Target Termination should be applied.	Read: RTT_NOM_RD Write: RTT_NOM_WR
Inactive	Inactive	DRAM ignores commands and the Parked termination will be maintained	RTT_PARK
Inactive	Active	Illegal	Illegal

The host controller will determine the termination at the time of the command, by activating the chip selects appropriately.

3.3.2.1 Data Buffer DQ Bus Termination in LRDIMM Application

The RCD supports DQ bus termination in LRDIMM applications. DDR5 does not have a discrete ODT signal. The ODT function is encoded on the Chip Select for RD, WR, and MRR commands. A RD, WR, MRW or MRR command will normally have the BCS_n signal active for only the first UI of the command. The data buffers will terminate the host bus as is appropriate for the command (RTT_WR for write, and driving the bus for reads).

3.3.2.1 Data Buffer DQ Bus Termination in LRDIMM Application (cont'd)

The DIMM terminations for DRAM and Data Buffer are shown in Table 5 and Table 6.

Table 5 — DIMM Termination Write

DCS0		DCS1		BCS		DRAM Termination		Buffer Termination
1st UI	2nd UI	1st UI	2nd UI	1st UI	2nd UI	Rank 0	Rank 1	
L	H	H	H	L	H	RTT_WR	RTT_PARK	RTT_WR
L	H	L	L	L	H	RTT_WR	RTT_NOM_WR	RTT_WR
H	H	L	H	L	H	RTT_PARK	RTT_WR	RTT_WR
L	L	L	H	L	H	RTT_NOM_WR	RTT_WR	RTT_WR
L	L	H	H	L	L	RTT_NOM_WR	RTT_PARK	RTT_NOM_WR
H	H	L	L	L	L	RTT_PARK	RTT_NOM_WR	RTT_NOM_WR
L	L	L	L	L	L	RTT_NOM_WR	RTT_NOM_WR	RTT_NOM_WR

Table 6 — DIMM Termination Read

DCS0		DCS1		BCS		DRAM Termination		Buffer Termination
1st UI	2nd UI	1st UI	2nd UI	1st UI	2nd UI	Rank 0	Rank 1	Host-side DQ
L	H	H	H	L	H	Target	RTT_PARK	Target
L	H	L	L	L	H	Target	RTT_NOM_RD	Target
H	H	L	H	L	H	RTT_PARK	Target	Target
L	L	L	H	L	H	RTT_NOM_RD	Target	Target
L	L	H	H	L	L	RTT_NOM_RD	RTT_PARK	RTT_NOM_RD
H	H	L	L	L	L	RTT_PARK	RTT_NOM_RD	RTT_NOM_RD
L	L	L	L	L	L	RTT_NOM_RD	RTT_NOM_RD	RTT_NOM_RD

If the DCS_n signal is active during the 2nd UI of a command, it indicates a non-target termination for the DIMM. The command is targeting a different DIMM and the DRAMs or data buffers should terminate with either RTT_NOM_WR for writes or RTT_NOM_RD for reads.

For a non-target operation, the DRAMs or data buffers should remain in RTT_PARK, the command will not be sent on the BCOM bus at all. The RCD will not have received any DCS_n signals active in this case and will see Deselects.

3.3.3 QCA Bus Operation

Under normal operation, the RCD passes commands through from the D inputs to the Q outputs when a command is present on the bus, and drives HIGH on the QCA outputs when no command is being sent. The RCD passes the DCA bus through when either DCS input is active but there are exceptions to this:

- If the register detects a parity error and parity checking is enabled, the DCA bus is still passed through to DRAM and QCS is blocked.
- During the 2nd UI of a 2 UI command, the DCA bus must be passed through. There will only be a DCS signal active for the 2nd UI if a non-target termination is being signaled to a rank.
- When DRAM interface Block MR Mode is enabled **RW01[1] = 0**.
- When CA Pass-through mode is enabled **RW00[2] = 1**.

3.3.4 QCS Operation

Under normal operation the RCD passes the DCS signals through to the QCS outputs. There are several exceptions to this:

- If the register detects a parity error with parity checking enabled, on a one-cycle command or the 1st UI of two-cycle command, the DCS signal is not passed through, and the QCS outputs remain inactive.
- If the register detects a parity error with parity checking enabled, on the 2nd UI of a two-cycle command, the same QCS outputs that were active on the 1st UI will be active on the 2nd UI. This causes the command to be canceled in the DRAM. For read and write commands it causes the command to be converted to non-target termination.
- When DRAM interface Block MR Mode is enabled $RW01[1] = 0$.
- When CA Pass-through mode is enabled $RW00[2] = 1$.

3.3.5 Output Inversion and Mirroring

Output Inversion is enabled by default, after DRST_n is de-asserted, to reduce simultaneous output switching current.

- All QACA[13:0]_[B:A] outputs will follow the equivalent inputs. All QBCA[13:0]_[B:A] outputs will be inverted. In other words, when sending commands, the “B” copy will be the invert of the “A” copy.
- When no commands are being sent, both the “A” and “B” copies will drive HIGH.
- The DRAM devices connected to the “B” outputs will un-invert the signals internally before using them. This eliminates the need for the host controller to send separate MRW commands for the two CA output sets, and allows every CA output to be inverted on the “B” set.

Address mirroring is also handled by the DRAM device, so there is no RCD requirements for mirroring.

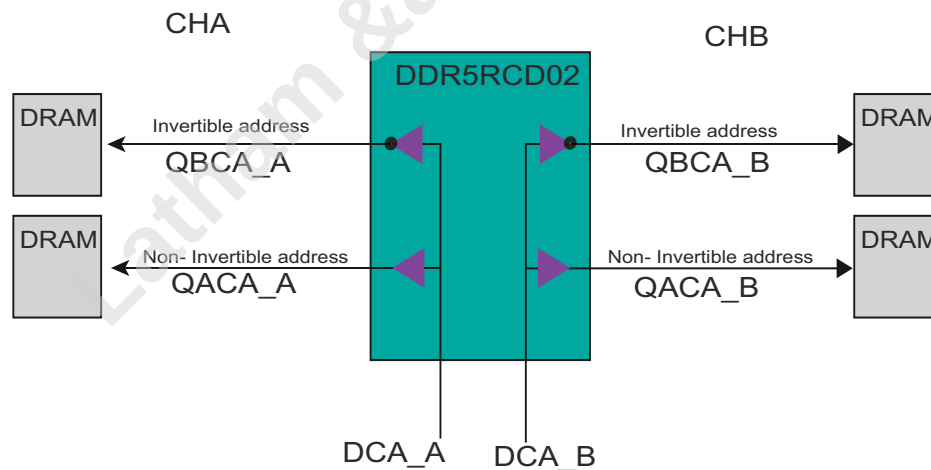


Figure 2 — RCD Output Inversion

3.3.6 ZQ Calibration

The RCD snoops the MPC command to initiate ZQ Calibration, which calibrates the output driver impedance across process, temperature, and voltage. ZQ Calibration occurs in the background of device operation.

There are two ZQ Calibration commands initiated with the MPC command: ZQCal Start, and ZQCal Latch. ZQCal Start initiates the RCD's calibration procedure, and ZQCal Latch captures the result and loads it into the RCD's drivers.

A ZQCal Start command may be issued anytime the DDR5 SDRAM is not in a power-down state. There are two timing parameters associated with ZQ Calibration. t_{ZQCAL} is the time from when the ZQCal Start MPC command is sent to when the host can send the ZQCal Latch MPC command. t_{ZQLAT} is the time from when the ZQCal Latch MPC command is sent by the host to when the CA bus (and subsequently the DQ bus) can be used for normal operation. A ZQCal Latch Command may be issued anytime outside of power-down after t_{ZQCAL} has expired and all CA bus operations have completed. The CA Bus must maintain a Deselect state during t_{ZQLAT} to allow CA ODT calibration settings to be updated.

ZQCal commands are issued independently on a sub-channel level. After a ZQCal Start and until t_{ZQCAL} finishes, neither another ZQCal Start nor a ZQCal Latch is allowed on this same sub-channel.

To use the ZQ calibration function, a $240\ \Omega \pm 1\%$ tolerance external resistor must be connected between the ZQ pin and VSS.

3.3.7 Latency Equalization Support

3D Stacked SDRAMs have a higher CAS latency than monolithic devices. For greater platform flexibility, it is highly desirable to mix DIMMs with both 3DS and mono SDRAMs on the same DDR5 channel. Since memory controllers typically can only handle devices with equal CAS latencies in the same channel, it is required that the DDR5 register has a mechanism to increase the CAS latency of mono SDRAMs to match the CAS latency of 3DS SDRAMs. However, this mechanism only equalizes the DRAM latencies, not the total DIMM latencies. Hence, this DDR5 register mechanism does not by itself allow mixing DDR5 RDIMMs and LRDIMMs in the same channel.

To equalize different SDRAM latencies, the DDR5 register supports a programmable latency adder of 0 nCK, 1 nCK, 2 nCK, 3 nCK or 4 nCK for all DRAM commands—see conceptual diagram in Figure 3. The power-up default is 0 nCK latency adder.

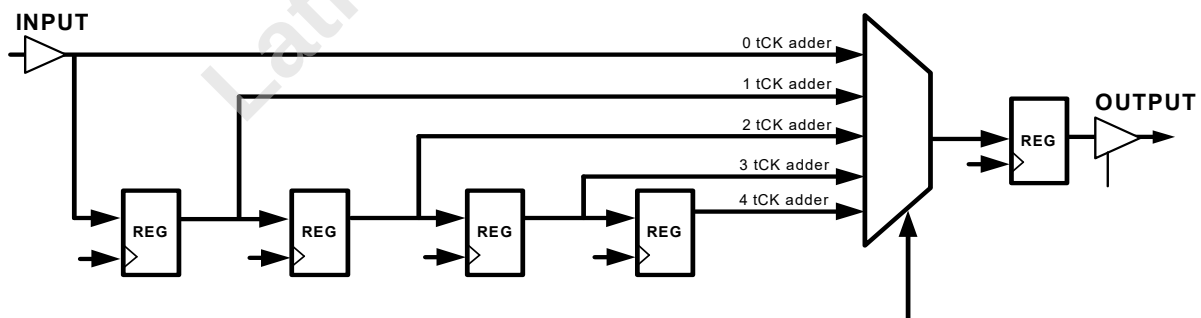


Figure 3 — Latency Equalizer Delays

With a latency adder enabled, the DDR5 register will delay assertion of the QxCSy_N_n and QxCAy_N outputs by the corresponding number of clock cycles. The output timing is determined by the Command Latency Adder control word [RW11](#) regardless of whether the device has parity checking enabled or disabled.

3.3.8 Per-group Output Delay Control

The RCD supports output delay control for output signal groups defined and located in [RW12](#) to [RW1E](#). The output delay settings are incremental CK fractions of $1/64$ tCK. Each signal group can be enabled or disabled independently through RCD Control Words. The Host must wait t_{ODU} after the Output Delay Control Word has been written to a signal group before the output becomes stable. The output delay feature controlled by [RW12](#) to [RW1E](#) is not supported when the RCD is configured in Test Frequency mode (Band 2, [RW05\[7\] = 1](#)) or in PLL bypass mode ([RW05\[3:0\] = 1111](#)).

3.3.9 Per-bit QCA Output Delay Control

In DDR5RCD02, fine adjustment of the phase of individual QCA bit lanes relative to QCK is required. For this purpose, the host controller can utilize the per-bit QCA output delay control words in [PG\[5\]RW\[7B:60\]](#) to control the phase of A-copy and B-copy QCA bits respectively. Since all bits within a copy are generally aligned by routing, only a small range of [0, 40ps] is provided for fine-grained adjustment of individual bit lane delay differences. A positive delay in these control words means that the particular lane requires slightly more delay than the previously established QCA delay in [RW1B](#) and [RW1C](#) for the entire copy.

3.3.10 Power Down Operation

DDR5 does not have a separate CKE signal going to the DIMM or the DRAMs. Power Down Entry, including Self Refresh, is signaled via a single UI command. Following the Power Down Entry command, the CS signal effectively becomes the equivalent of the CKE signal. An inactive CS keeps the DRAM devices in the Power Down state. CS going active will take the DRAMs out of the Power Down state. A NOP command is placed on the CA bus along with CS going active so that no commands are executed by the DRAM devices. DRAMs can be put in Power Down mode with or without non-target ODT control, see Chapter 4.

3.3.11 VHost Mode

The Virtual Host mode or “VHost mode” allows a sequence of commands to be sent on the DRAM interface without receiving commands via the host interface. Commands to be sent on the DRAM interface are programmed by writing RCD control words. The intention is to operate this RCD feature via SidebandBus in a test environment. VHost mode is not intended for normal operation. A sequence of commands is programmed into the RCD, then executed by writing a START MRW bit.

Virtual host mode has the following features:

- Up to 4 commands may be programmed.
- Each command may be a 1 or 2 UI command.
- Control words specify the complete CA[13:0] bus and the two chip selects for each UI of the commands.
- There is a programmable number of clocks between commands, programmed separately for the spacing between each command.
- Command spacing can be 2, 4, 8, 16, 64, 256, or 1024 clock.
- Following the last command, the sequence may then optionally repeat starting at command 0, 1, 2, or 3.
- Shorter sequences can be supported by programming Deselects (neither chip select active) for some of the commands.
- Non-target termination can be signaled by programming the chip select to be active for both UI of a command that supports non-target termination.

Each command is programmed using four 8-bit control words, to specify the state of CA[13:0], CS0_n and CS1_n for each of the 2 UI. For a 1UI command, the 2nd UI is programmed as a Deselect, with both chip selects inactive. Note that two back to back one UI commands can be programmed instead of a 2UI command, but these will always be back to back, with no programmable spacing.

3.3.11 VHost Mode (cont'd)

There are four three-bit delay fields to specify the delay between commands 0 and 1, commands 1 and 2, commands 2 and 3, and command 3 to start of the repeat sequence.

Following the execution of command 3, the sequence will optionally repeat. The sequence can repeat starting at any command (0 through 3) and will continue again through command 3.

16 control words specify the four commands (4 registers each). Two registers specify the gap between commands. One register specifies the command to repeat from and enables repeating. One register contains a start bit for each channel. Writing a 1 to a start bit begins the sequence on that channel. Writing a 0 stops the sequence. The sequencer is common to the two channels, so the sequence cannot be started on the two channels at different times. The commands operate in either 1N or 2N mode as specified by the existing 1N/2N mode bit. VHost Commands are sent across the BCOM bus to the data buffers. VHost is not supported during SRE/PDE/Initialization or when any training mode is enabled. The commands processed by the VHost logic are not interpreted by the RCD. For example, if the programmed commands include PDE or SRE, those commands will be sent to the DRAMs but they will not cause the RCD to power down. VHost settings can-not be modified while the VHost state machine is running. The RCD will auto stop after cmd3 executes if repeat is not enabled. RST, DCK signals must be applied on the host interface during VHost mode.

3.3.11.1 VHost Lockout mode

When the RCD receives an In-band MRW to [RW01](#), the RCD will Lockout the VHost mode until the next DRST_n assertion. SMBus sideband write to RW01 will not trigger the lockout mechanism.

Suppliers must guarantee a permanent lock out with a fuse option.

3.3.12 Transparent Mode

The DDR5RCD02 supports a transparent mode feature of the data buffer that enables testing of LRDIMMs. The RCD will disable parity checking automatically when this mode is entered.

To enter transparent mode, the host first sends an MRW command with [RW82\[0\]](#) = 1 to enable transparent mode in the DDR5DB02. Then it sends an MRW command with [RW00\[7\]](#) = 1 to enable transparent mode in the RCD.

While in transparent mode, the DDR5DB02 does not interpret BCS_n and BCOM commands. For BCS_n input value of “1” and BCOM[2:0] input values of “111”, it directs the data flow from host interface to DRAM interface. For BCS_n input value of “1” and BCOM[2:0] input values of “000”, it directs the data flow from DRAM interface to host interface. Any other input values of BCS_n and BCOM[2:0] are invalid.

While in transparent mode, the RCD drives BCS_n statically to “1” and BCOM[2:0] statically to “111” (DES) when the DPAR input is LOW, and BCS_n to “1” and BCOM[2:0] to “000” when the DPAR input is HIGH. All output signals in the RCD, other than BCS_n and BCOM[2:0], continue working as they do in the normal mode of operation. In other words, all the RCD features and functions not directly related to DPAR, BCS_n and BCOM[2:0] remain operational in transparent mode. In addition, all the clock frequencies and PLL operational modes available in low-speed test mode and normal operation mode will be supported in transparent mode. Input DPAR_A controls outputs BCOM[2:0]_A and input DPAR_B controls outputs BCOM[2:0]_B, correspondingly.

To exit transparent mode, the host sends an MRW command with [RW00\[7\]](#) = 0 to exit RCD transparent mode following by issuing a Data Buffer reset command via [RW04](#) to exit DDR5DB02 transparent mode.

3.3.13 CA Validation Pass-Through Mode

To facilitate DRAM or RCD validation the RCD will be required to support a validation pass-through mode. In this mode, the DCA signals are passed through to QCA, regardless of DCS assertion. In this mode, the DCS0_n and DCS1_n signals are passed through to QCS0_n and QCS1_n respectively, regardless of DCA signal state. Parity checking must be disabled prior to and during validation pass-through mode. Validation pass-through mode is entered when [RW07\[4\]](#)=1. When [RW07\[5\]](#)=1 in CA Validation Pass-through mode, all MRW which write to the RCD are blocked and further configuration is expected via the SMBus. This SMBus is not required to be driven by the controller.

3.3.13 CA Validation Pass-Through Mode (cont'd)

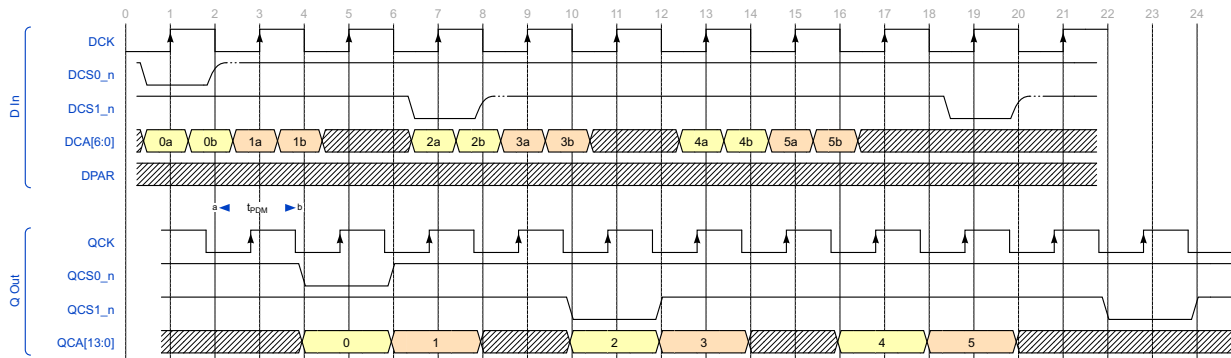


Figure 4 — Validation Pass-Through Mode - DDR

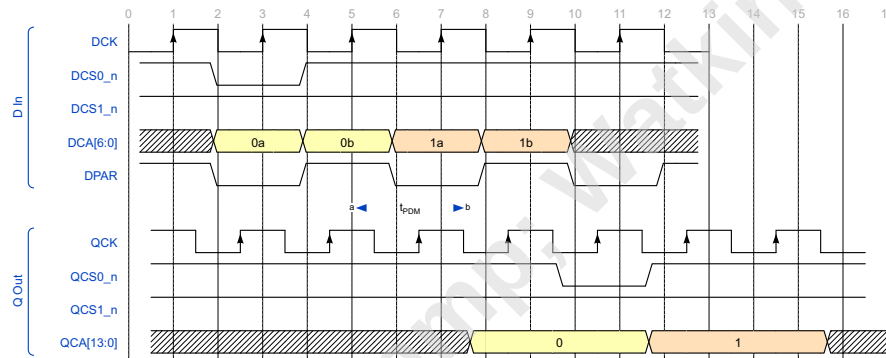


Figure 5 — Validation Pass-Through Mode - SDR1

Similar to CA Pass-Through Mode described in Section 5.2.3, when Validation Pass-Through Mode is enabled in SDR Mode, the DPAR signal will be used to indicate 1st UI QCA[6:0] (DPAR = LOW) and 2nd UI QCA[13:7] (DPAR = HIGH).

3.3.14 DRAM Interface and Data Buffer Interface Training Support

In order for the Host to train the DRAM and Data Buffer interfaces, the RCD will support two Pass Through modes: CA Pass-Through mode, described in Section 5.2.3, and BCOM Pass-Through mode, described in Section 5.2.5.

3.4 Command Address Bus

The DDR5RCD02 can support two data rates on the Host Command Address signals. The first setting is Double Data Rate (DDR) on the DCA with 1N timing on the QCA. The second is Single Data Rate (SDR) on DCA with 2N timing on the QCA to the DRAM interface. Chip Select is SDR for both settings on DCA and QCA.

Table 7 — Command Address Rates

RW00[1:0]	DCA	QCA
00	SDR1	2N timing (extra setup time)
01	DDR	1N timing
10	SDR2	2N timing (extra hold time)

3.4.1 Double Data Rate DCA

The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6.

Table 8 — DDR CA Mapping DCA to QCA

UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13

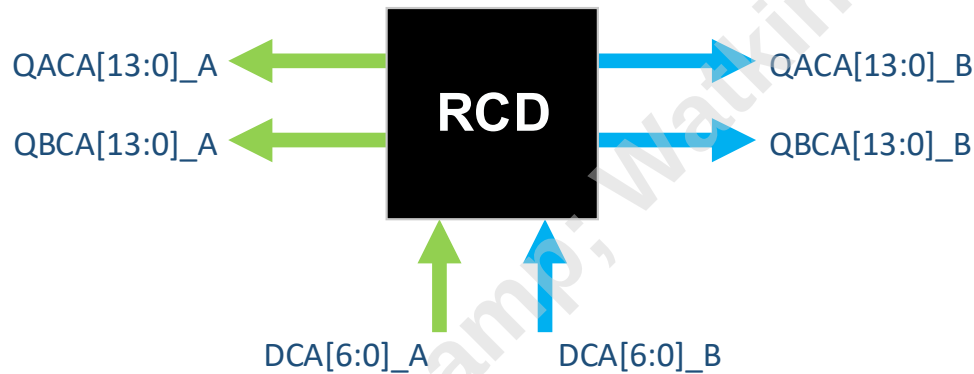
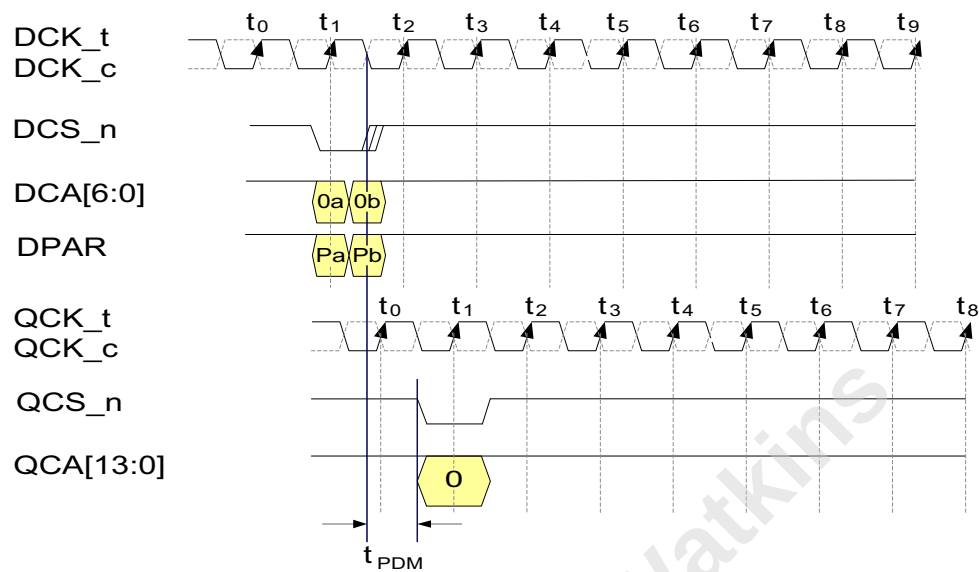


Figure 6 — RCD DCA and QCA Ports

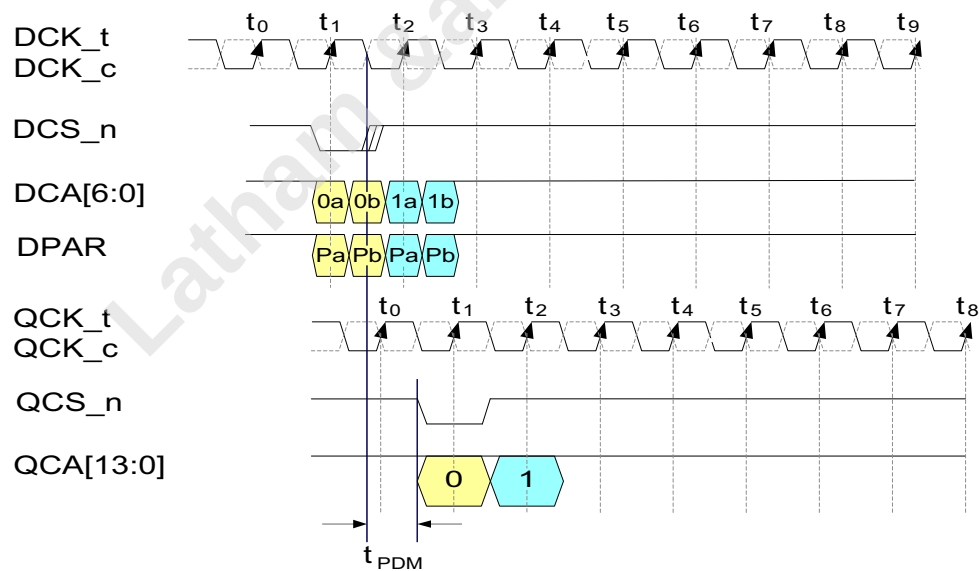
The following timing diagrams in Figures 7, 8, and 9 show the timing relationship from the Dn inputs to the Qn outputs for a 1UI command, a 2UI command, and multiple commands.

3.4.1 Double Data Rate DCA (cont'd)



NOTE Timing Diagram for 0 nCK command latency adder.

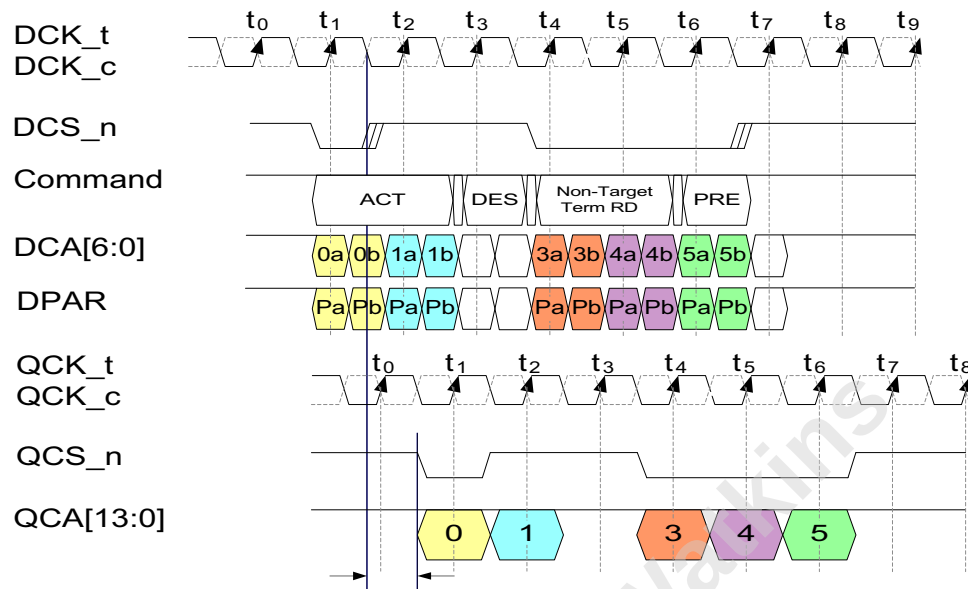
Figure 7 — One UI DRAM Command Timing Diagram



NOTE Timing Diagram for 0 nCK command latency adder.

Figure 8 — Two UI DRAM Command Timing Diagram

3.4.1 Double Data Rate DCA (cont'd)



NOTE Timing Diagram for 0 nCK command latency adder.

Figure 9 — DDR CA Multi-Command Timing Diagram Example

The DCS_n signal is SDR and is sampled by the RCD on the rising edge of DCK. See Section 13.11 on page 227 for details on DCS input receiver electrical and timing parameters. The DDR5RCD must maintain output timings to the DRAM and comply with the output tolerance skew parameter t_{QSK} as shown in Figure 10.

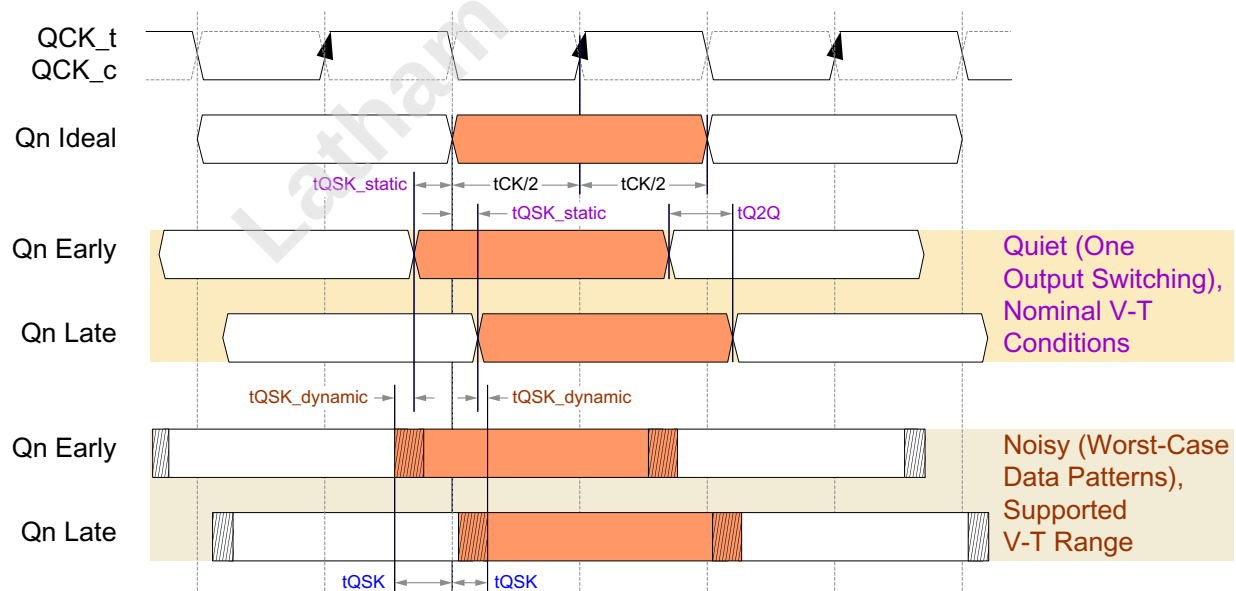


Figure 10 — Qn Output Skew

NOTE For Figure 10 outputs as specified in Table 200, “Output Timing Requirements1” Footnote 7.

3.4.2 Single Data Rate DCA

The DDR5RCD02 will support SDR (Single Data Rate) on the host-interface DCA signals. In this case, the RCD will apply 2N timing to the DRAM interface signals. This allows more setup or more hold time on the QCA bus. This mode is enabled when $RW00[0] = 0$. DDR5 has defined two-cycle commands that require the RCD to capture the commands differently between DDR and SDR modes. In both modes, the first half of the command is sampled on the rising edge of clock when the chip select is active. In SDR mode, the second half of the command is sampled on the next rising clock edge. Non-target ODT signaling (through chip select inputs) is also delayed by a clock in SDR mode. For applications that require SDR on the Host interface but do not need extra setup time on the DRAM interface, QCS pulses can be pulled earlier by one clock cycle by setting $RW00[1] = 1$.

The QCA will follow the DRAM specification for 2N mode.

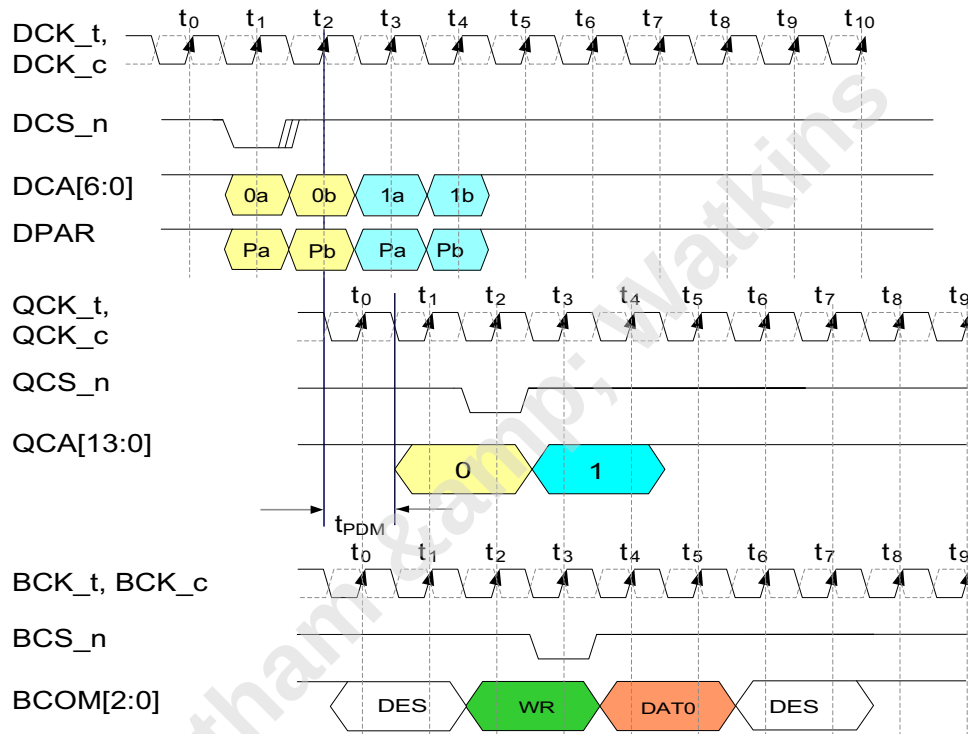


Figure 11 — SDR1 Mode Example of DRAM WR Command with $RW00[1] = 0$

3.4.2 Single Data Rate DCA (cont'd)

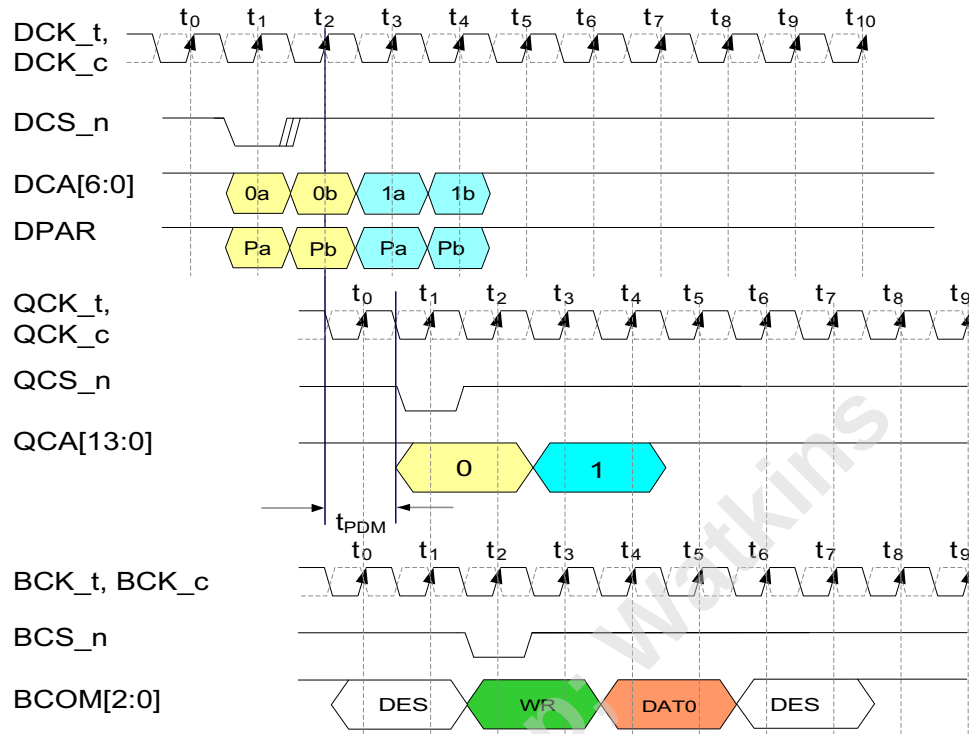


Figure 12 — SDR2 Mode Example of DRAM WR Command with $RW00[1] = 1$

3.4.3 SDR to DDR Transition

This mode will be set through SMBus during initialization between t_{STAB01} and t_{STAB02} , or after t_{STAB01} following Exit from Self Refresh Mode with Clock Stop (see Section 4.2.2 on page 47). There is no transition during normal operation. Transitions for SDR to DDR or SDR1 to SDR2 occur prior to Host Interface Training.

3.5 Parity

The Parity function in DDR5 differs from DDR4 in several ways

- The parity signal is received on the same cycle as the command/address bits rather than one cycle later.
- 2UI DRAM commands require a different protocol to block commands when the parity error is on the 2nd UI.
- No parity signal is sent on to the DRAMs.

The convention of parity is even parity across the $DCA_N[6:0]$ and $DPAR_N$ signals. The Parity is checked separately per UI on the DCA bus.

Parity is checked separately on the two sub channels. A parity error on one sub channel does not affect the operation of the other sub channel.

When DDR Command Address Rate is selected, $RW00[0] = 1$, ALERT_n will be asserted three or four input clocks after rising edge of first UI of DCA in which the erroneous command is registered.

3.5 Parity (cont'd)

In SDR mode, $RW00[0] = 0$, $ALERT_n$ will be asserted three or four input clocks after the rising edge corresponding to the second half of the erroneous command. This is the case regardless of the error being in the first, second, or both halves.

If parity checking is disabled, the register forwards sampled commands to the outputs unconditionally, the command appears on the outputs regardless of whether a parity error occurred or not.

For single and 2 UI DRAM commands, if parity checking is enabled and there is no parity error, the register will forward the sampled commands to the outputs.

3.5.1 Single UI DRAM Commands with Parity Enabled

For single UI DRAM commands, if parity checking is enabled and a parity error is detected on either UI of the DCA bus, the register will forward the command on QCA bus but will not assert any chip select to output.

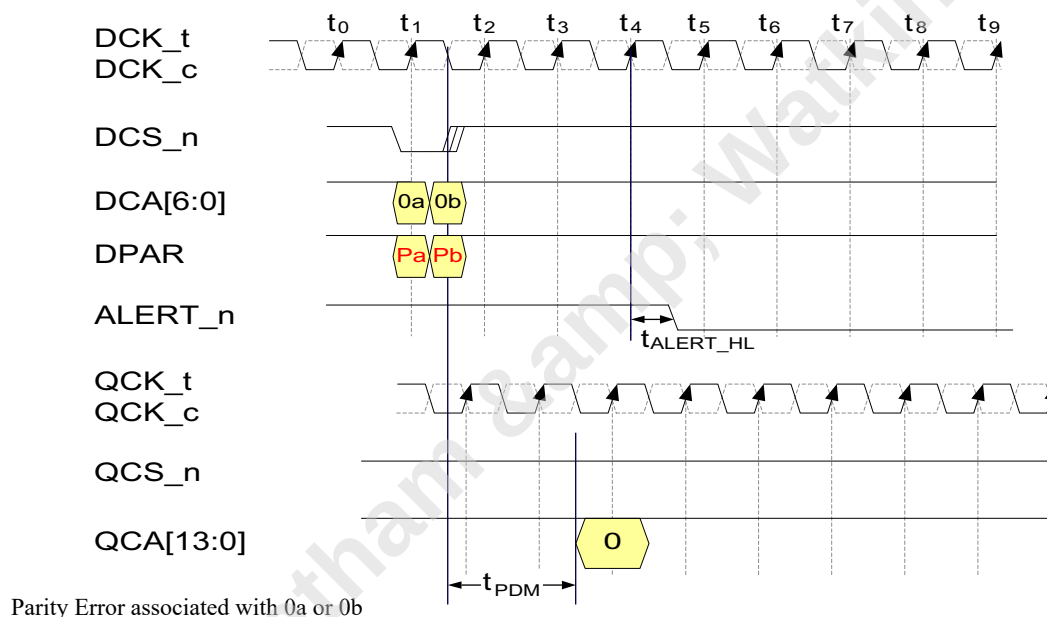


Figure 13 — 1 UI DRAM Command with Parity Checking Enabled and $CLA = 0$

3.5.1 Single UI DRAM Commands with Parity Enabled (cont'd)

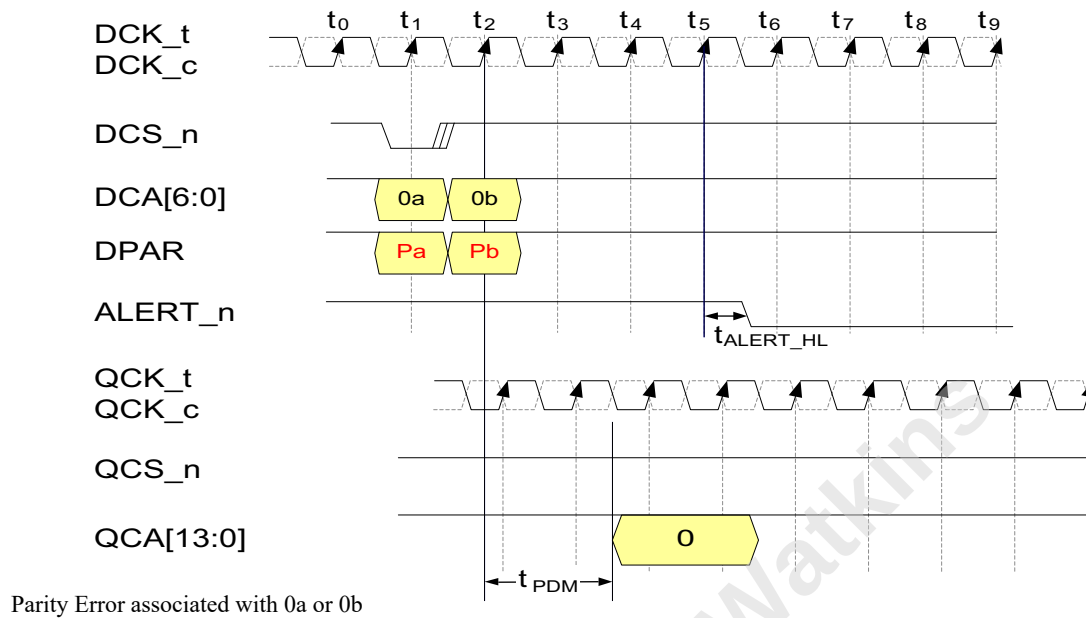
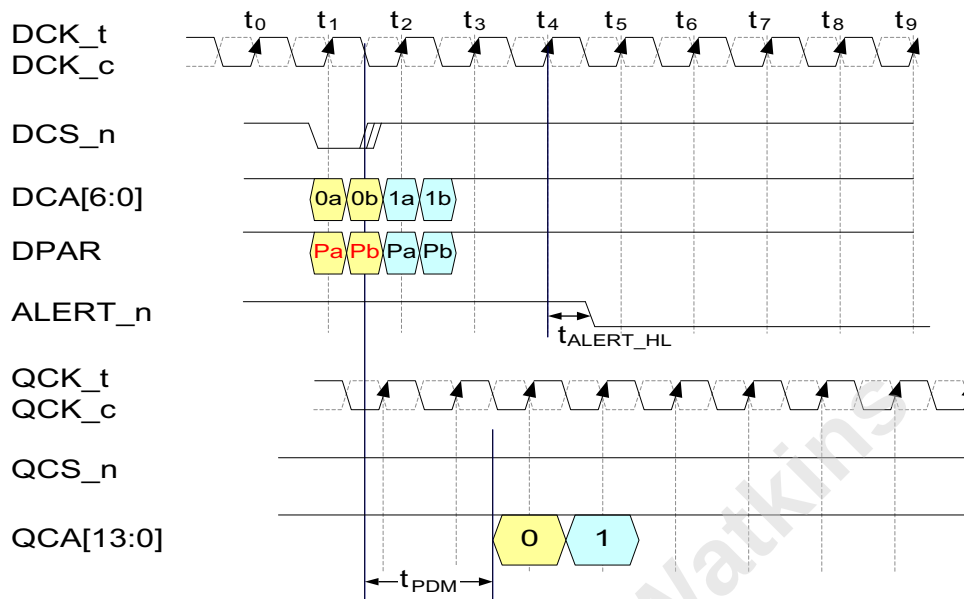


Figure 14 — SDR Mode 1 UI DRAM Command with Parity Checking Enabled and CLA = 0

3.5.2 2UI DRAM Commands with Parity Enabled

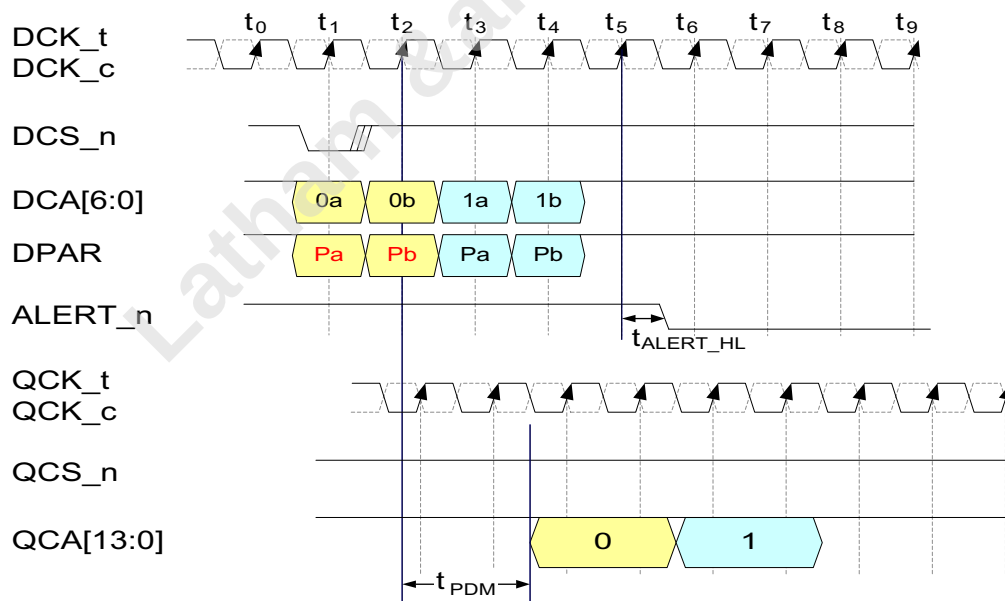
For 2UI DRAM commands, if parity checking is enabled and a parity error is detected on either UI of the DCA bus for the 1st UI of the DRAM command the register will forward the command on the QCA but will not assert any chip select output as shown in Figure 15.

3.5.2 2UI DRAM Commands with Parity Enabled (cont'd)



Parity Error associated with 0a or 0b

Figure 15 — Error in 1st UI of 2 UI DRAM Command with Parity Checking Enabled



Parity Error associated with 0a or 0b

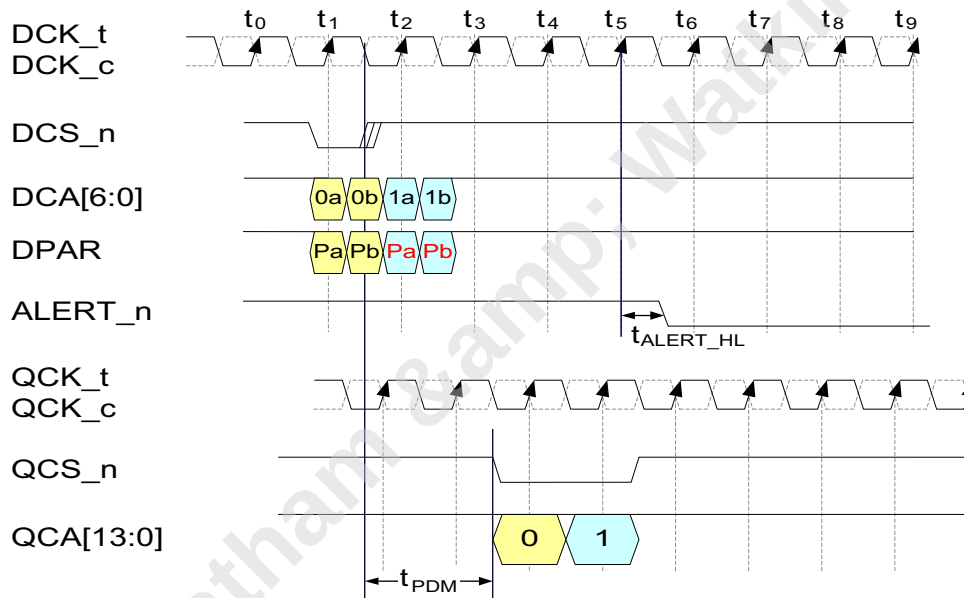
Figure 16 — SDR Mode Error in 1st UI of 2 UI DRAM Command with Parity Checking Enabled

3.5.2 2UI DRAM Commands with Parity Enabled (cont'd)

For 2UI DRAM commands, if the parity error occurs in either UI of the DCA bus for the 2nd UI of the DRAM command and the 1st had correct parity, the command will have already been sent to the DRAMs. In this case, the command itself will be known to be correct, but the additional address information contained in the 2nd UI will be incorrect. In order to prevent the command from being executed by the DRAM, the RCD will assert the chip select in the 2nd UI also for all ranks which received a chip select in the 1st UI as shown in Figure 17.

For RD, WR, and MRR commands, this will convert the command to a non-target termination encoding. All ranks receiving the chip select will terminate for this command, but no rank will execute the command.

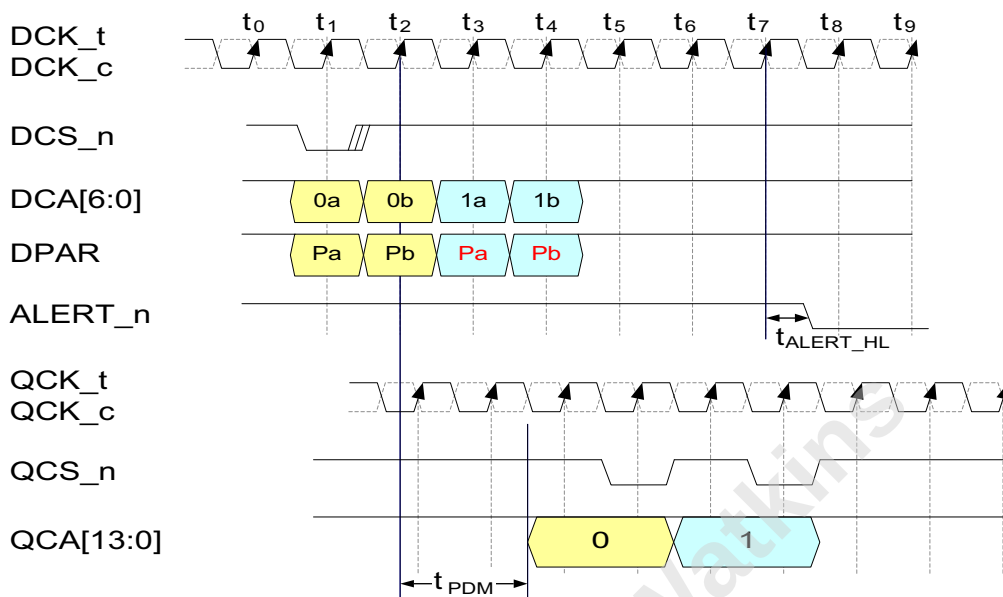
The only other 2 UI DRAM commands are the Activate, MRW, and Write Pattern commands. A DRAM seeing the chip select active for both UI of an Activate or Write Pattern command will ignore the command. Note that an MRW command will not use the data bus. The DRAM will not execute the MRW command if the chip select is asserted in both UI.



Parity Error associated with 1a or 1b

Figure 17 — Error in 2nd UI of 2 UI DRAM Command with Parity Checking Enabled

3.5.2 2UI DRAM Commands with Parity Enabled (cont'd)



Parity Error associated with 1a or 1b

Figure 18 — SDR1 Mode Error in 2nd UI of 2 UI DRAM Command with Parity Checking Enabled

If a parity error occurs on the 1st UI of a DRAM command, regardless of the command length, the command encoding itself may be corrupt. The register will be required to block all future commands until command forwarding is re-enabled. When commands are re-enabled, the register must not enable commands until it sees a UI with all CS inactive in order to avoid re-enabling in the middle of a command.

Table 9 — Blocking Commands on Parity Error

Parity error on 1st DRAM UI	Parity error on 2nd DRAM UI	Outputs
No	No	Command is forwarded to DRAM
No	Yes	First UI of command forward to DRAM. 2 nd UI of command forward to DRAM, but with QCS asserted on ALL ranks which received a DCS on the 1st UI. Future commands not sent until parity checking is re-enabled.
Yes	Yes or No	First UI and Second UI of the command forward to DRAM but QCS is not asserted. Future commands not sent until parity checking is re-enabled.

3.5.3 ALERT_n Assertion

The behavior of ALERT_n with parity checking disabled is the same as when parity checking is enabled with the exception that ALERT_n is never asserted as result of a CA parity error. However, when parity checking is disabled, ALERT_n can still be asserted as a result of an assertion of the DERROR_IN_n input.

After the RCD receives DPAR_N from the memory controller, it compares it with the data received on the CA inputs and indicates on its open-drain ALERT_n pin (active LOW) whether a parity error has occurred. The computation only takes place for data which is qualified by at least one of the DCS[1:0]_n signals being Active, or the 2nd UI of a 2UI DRAM Command.

3.5.3 ALERT_n Assertion (cont'd)

The convention of parity is even parity across the DCA and DPAR signals. Parity is checked independently for the two channels.

If a parity error occurs and parity checking is enabled in RW01, the RCD sets the 'CA Parity Error Status' bit in RW24 to '1' and disables parity checking. ALERT_n is asserted three or four input clocks after the erroneous command is registered.

If the 'CA Parity Error Status' bit is '0', the RCD logs the error by storing the erroneous command and address bits in the Error Log Register. ALERT_n stays asserted LOW until a 'Clear CA Parity Error Status' command is sent if the 'ALERT_n Assertion' bit in the Parity Control Word RW01 is '0'. In this case the erroneous command and all subsequent commands are not forwarded (i.e., QxCsy_N_n are not asserted) to the DRAM until the memory controller issues a 'Clear CA Parity Error' command, which will also clear the 'CA Parity Error Status' bit in the Error Log Register.

If the 'ALERT_n Assertion' bit is '1', the ALERT_n pulse width is as defined in Table 102 on page 136 with start of the ALERT_n pulse width counted from the third or fourth input clock edge after the 1st parity error. The erroneous command and all subsequent commands are not forwarded (i.e., QxCsy_N_n are not asserted) to the DRAM until the end of the ALERT_n pulse width. After the ALERT_n pulse, the device will automatically resume forwarding commands to the DRAM.

- If the 'ALERT_n Re-enable' bit is '0', parity checking remains disabled until a 'Clear CA Parity Error' command is sent, which will also clear the 'CA Parity Error Status' bit in the Error Log Register.
- If the 'ALERT_n Re-enable' bit is '1', the device will re-enable parity checking after the ALERT_n pulse. The 'CA Parity Error Status' bit will remain set. If a subsequent parity error is detected, the device will re-enter the parity error state and set the '> 1 Error' bit. The other bits in the Error Log Register are not updated on this subsequent error. Both the 'CA Parity Error Status' bit as well as the '> 1 Error' bit will be cleared by sending a 'Clear CA Parity Error' command.

3.6 DCA Decision Feedback Equalization

Since the DCA will be Double Data Rate, the RCD will support a 4 Tap DFE on the DCA host-interface receivers. The 4-tap DFE subsystem consists of a gain amplifier, a DFE summer, 4 CA slicers (also called Taps) with outputs that loop back to the DFE summer, and a coefficient multiplier for each Tap. The gain control of the front end is used to ensure that the cursor or the current bit is in a congruent relationship with the ISI correction required for the channel. The taps C1, C2, C3, C4 coefficients provide the corrections needed to the current bit by adding or subtracting the effects of ISI of the previous bits. The host can enable 1, 2, 3, or 4 taps in a consecutive order always starting with Tap 1 as shown in Table 10, "DFE Tap Configurations". DFE input gain adjustment and DFE Tap Coefficients 1 through 4 are set by control words located in PG[1:0]. These DFE control words allow programming of each parameter on a DCA I/O-specific basis, and they are intended to be trained by the host. The DDR5RCD02 device does not support adaptive DFE, and it only supports DFE when its DCA host interface is configured in DDR mode (i.e., RW00[0] = 1).

3.6.1 DFE Tap Configurations

Table 10 — DFE Tap Configurations

RW31 OP4	RW31 OP5	RW31 OP6	RW31 OP7	Tap 1	Tap 2	Tap 3	Tap 4
0	x	x	x	Disabled	Disabled ¹	Disabled ¹	Disabled ¹
1	0	x	x	Enabled	Disabled	Disabled ²	Disabled ²
1	1	0	x	Enabled	Enabled	Disabled	Disabled ³
1	1	1	0	Enabled	Enabled	Enabled	Disabled
1	1	1	1	Enabled	Enabled	Enabled	Enabled

NOTE 1 DDR5RCD02 hardware disabled when RW31[4] = 0 and RW31[7:5] are don't care and ignored by the RCD.

NOTE 2 DDR5RCD02 hardware disabled when RW31[5] = 0 and RW31[7:6] are don't care and ignored by the RCD.

NOTE 3 DDR5RCD02 Hardware disabled when RW31[6] = 0, and RW31[7] is a don't care and ignored by the RCD.

3.6.2 DFE Gain and Tap Range

Table 11 — DFE Gain and Tap Total Control Word Range

DFE Parameter	Min	Max	Unit
Gain	-6	6	dB
Tap 1	- 150	150	mV
Tap 2	- 60	60	mV
Tap 3	- 45	45	mV
Tap 4	- 45	45	mV

Table 12 — DFE Gain and Tap Coefficient Step Parameters Required per Speed Bin

DFE Parameter	DDR5-3200 - 5600			Unit
	Min	Typ	Max	
DFE Gain Bias Step Size ^{1,2}	1.7	2	2.3	dB
DFE Gain Bias Tolerance (INL) ^{1,2}	- 0.5	-	0.5	dB
DFE Gain Bias Step Size Tolerance (DNL) ^{1,2}	- 0.3	-	0.3	dB
DFE Gain Bias Step Time ¹	-	-	300	ns
DFE Tap Bias Step Size ^{1,2}	1	3	5	mV
DFE Tap Bias Tolerance (INL) (-150 mV to 0 mV) ^{1,2}	Min (V _{ideal} x 115%, V _{ideal} - 3 * LSB) ^{3,4}	-	Max (V _{ideal} x 85%, V _{ideal} + 3 * LSB) ^{3,4}	mV
DFE Tap Bias Tolerance (INL) (0 mV to 45 mV) ^{1,2}	V _{ideal} - 3 * LSB ^{3,4}	-	V _{ideal} + 3 * LSB ^{3,4}	mV
DFE Tap Bias Step Size Tolerance (DNL) ²	- 66.67	-	66.67	% of Step Size
DFE Tap Bias Step Time	-	-	64	tCK

NOTE 1 Host Vref fixed at 0.75 x V_{DD}, typical Voltage-Temperature condition (i.e., V_{DD} = 1.1 V, 25 °C), and default gain setting.

NOTE 2 These parameters are neither subject to silicon validation nor production testing.

NOTE 3 V_{ideal} refers to the ideal DFE Tap value based on the setting.

NOTE 4 LSB = 3 mV.

3.6.3 DCA DFE Tap Measurement

When DFE taps are enabled, they add offset to the DFE summer, which affects the input to the slicer. The feedback to the taps may be positive or negative, based on the sign of the coefficient and the level of the input, resulting in four cases, all of which must be tested. Each tap is measured individually with other tap coefficients set to 0. Other combinations can be extrapolated from the results.

3.6.3.1 HIGH-to-LOW Transition Test Method

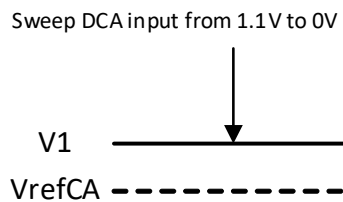
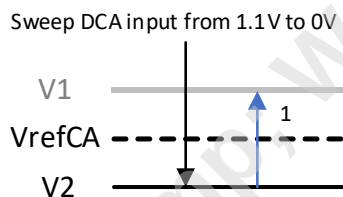
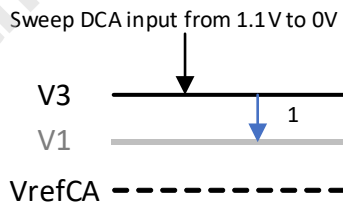


Figure 19 — HIGH-to-LOW Sweep: No DFE Applied



Note 1: Positive feedback results in an increase in voltage as seen by the slicer

Figure 20 — HIGH-to-LOW Sweep: DFE Positive Bias Applied



Note 1: Negative feedback results in an decrease in voltage as seen by the slicer

Figure 21 — HIGH-to-LOW Sweep: DFE Negative Bias Applied

1. Make sure VrefCA (RW[47:40]) is set to the default, DFE_Vref (PG[2]RW[62, 66, 6A, 6E, 72, 76, 7A, 7E]) is set to 0, power supply is 1.1V, and DFE is disabled.
2. Sweep the input signal from 1.1V to 0V to find the lowest level that produces a stable HIGH output and record this input level as V1. No overshoot is allowed on the input. See [Figure 19](#).
3. Enable DFE (RW31[0]). Enable the tap under test and all previous taps in RW31[7:4]. Set the coefficients for the previous taps to 0. All configuration in this procedure is via Sideband Bus access.
4. Configure the sign bit to 0 and the coefficient to n steps for the tap under test, where n is a loop variable.
5. Sweep the input signal from 1.1V to 0V to find the lowest level that produces a stable HIGH output and record this input level as V2. No overshoot is allowed on the input. See [Figure 20](#).

3.6.3.1 HIGH-to-LOW Transition Test Method (cont'd)

6. The DFE Tap Bias $V(+)$ = $(V1 - V2)$ for positive feedback. Validate $V(+)$ is within spec.
7. Configure the sign bit to 1 and the coefficient to n steps for the tap under test, where n is a loop variable.
8. Sweep the input signal from 1.1V to 0V to find the lowest level that produces a stable HIGH output and record this input level as V3. No overshoot is allowed on the input. See Figure 21.
9. The DFE Tap Bias $V(-)$ = $(V1 - V3)$ for negative feedback. Validate $V(-)$ is within spec.
10. Repeat steps 4-9 until all the Tap steps for Taps 1-4 are measured.

3.6.3.2 LOW-to-HIGH Test Transition Method

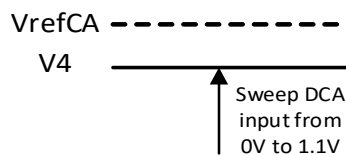
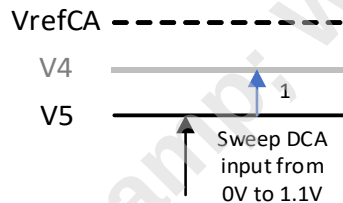
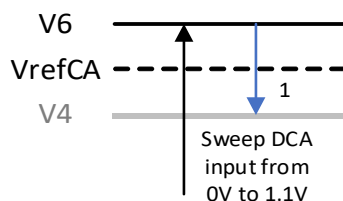


Figure 22 — LOW-to-HIGH Sweep: No DFE Applied



Note1: Positive feedback results in an increase in voltage as seen by the slicer

Figure 23 — LOW-to-HIGH Sweep: DFE Positive Bias Applied



Note 1: Negative feedback results in an decrease in voltage as seen by the slicer

Figure 24 — LOW-to-HIGH Sweep: DFE Negative Bias Applied

1. Make sure VrefCA (RW[47:40]) is set to the default, DFE_Vref (PG[2]RW[62, 66, 6A, 6E, 72, 76, 7A, 7E]) is set to 0, power supply is 1.1V, and DFE is disabled.
2. Sweep the input signal from 0V to 1.1V to find the highest level that produces a stable LOW output and record this input level as V4. No overshoot is allowed on the input. See Figure 22.
3. Enable DFE (RW31[0]). Enable the tap under test and all previous taps in RW31[7:4]. Set the coefficients for the previous taps to 0. All configuration in this procedure is via Sideband Bus access.
4. Configure the sign bit to 1 and the coefficient to n steps for the tap under test, where n is a loop variable.

3.6.3.2 LOW-to-HIGH Test Transition Method (cont'd)

5. Sweep the input signal from 0V to 1.1V to find the highest level that produces a stable LOW output and record this input level as V5. No overshoot is allowed on the input. See [Figure 23](#).
6. The DFE Tap Bias $V(+) = (V4 - V5)$ for positive feedback. Validate $V(+)$ is within spec.
7. Configure the sign bit to 0 and the coefficient to n steps for the tap under test, where n is a loop variable.
8. Sweep the input signal from 0V to 1.1V to find the highest level that produces a stable LOW output and record this input level as V6. No overshoot is allowed on the input. See [Figure 24](#).
9. The DFE Tap Bias $V(-) = (V4 - V6)$ for negative feedback. Validate $V(-)$ is within spec.
10. Repeat steps 4-9 until all the Tap steps for Taps 1-4 are measured.

3.7 Continuous Time Linear Equalization (CTLE)

3.7.1 Background

Continuous time linear equalization can be an effective method to improve signal margins at input receivers of the DDR5RCD02 device.

3.7.2 CTLE Transfer Function

The equation below represents the CTLE transfer function implemented in the DDR5RCD02 device.

$$\frac{A_{dc} \left(1 + \frac{s}{Z} \right)}{\left(1 + \frac{s}{P_1} \right) \left(1 + \frac{s}{P_2} \right)}$$

Where

- A_{dc} is the gain of the CTLE at DC,
- Z is the location of the zero,
- P_1 is the location of the first pole, and
- P_2 is the location of the second pole.

The DDR5RCD02 receivers support 16 possible combinations of CTLE parameter values as shown in Table 13 below. The device revision can only support one of the CTLE Configuration Ranges, A, B, C, or D. The host can read out the supported range from the CTLE Configuration Control Word, [RW50\[7:5\]](#). Each combination can be selected through 4-bit control word settings [RW52\[3:0\]](#) (CTLE Parameter Set A) and [RW53\[3:0\]](#) (CTLE Parameter Set B). There is a CTLE global feature enable control bit in [RW50\[0\]](#), and enable/disable controls per-pin in [RW51\[7:0\]](#). An 8-bit control word, [RW54\[7:0\]](#), provides per-pin control to determine whether a receiver uses CTLE Parameter Set A controlled by [RW52\[3:0\]](#), or Parameter Set B controlled by [RW53\[3:0\]](#).

3.7.2 CTLE Transfer Function (cont'd)

Table 13 — Supported CTLE Parameter Value Combinations

Range	CW Bits OP[3:0] in RW52 or RW53	Gain (A_{dc}) ¹	Zero (Z)	Pole 1 (P_1)	Pole 2 (P_2)
A	0000 _B	TBD	TBD	TBD	TBD
	0001 _B	TBD	TBD	TBD	TBD
	0010 _B	TBD	TBD	TBD	TBD
	0011 _B	TBD	TBD	TBD	TBD
	0100 _B	TBD	TBD	TBD	TBD
	0101 _B	TBD	TBD	TBD	TBD
	0110 _B	TBD	TBD	TBD	TBD
	0111 _B	TBD	TBD	TBD	TBD
	1000 _B	TBD	TBD	TBD	TBD
	1001 _B	TBD	TBD	TBD	TBD
	1010 _B	TBD	TBD	TBD	TBD
	1011 _B	TBD	TBD	TBD	TBD
	1100 _B	TBD	TBD	TBD	TBD
	1101 _B	TBD	TBD	TBD	TBD
	1110 _B	TBD	TBD	TBD	TBD
	1111 _B	TBD	TBD	TBD	TBD
B	0000 _B	TBD	TBD	TBD	TBD
	0001 _B	TBD	TBD	TBD	TBD
	0010 _B	TBD	TBD	TBD	TBD
	0011 _B	TBD	TBD	TBD	TBD
	0100 _B	TBD	TBD	TBD	TBD
	0101 _B	TBD	TBD	TBD	TBD
	0110 _B	TBD	TBD	TBD	TBD
	0111 _B	TBD	TBD	TBD	TBD
	1000 _B	TBD	TBD	TBD	TBD
	1001 _B	TBD	TBD	TBD	TBD
	1010 _B	TBD	TBD	TBD	TBD
	1011 _B	TBD	TBD	TBD	TBD
	1100 _B	TBD	TBD	TBD	TBD
	1101 _B	TBD	TBD	TBD	TBD
	1110 _B	TBD	TBD	TBD	TBD
	1111 _B	TBD	TBD	TBD	TBD

Table 13 — Supported CTLE Parameter Value Combinations (cont'd)

C	0000 _B	TBD	TBD	TBD	TBD
	0001 _B	TBD	TBD	TBD	TBD
	0010 _B	TBD	TBD	TBD	TBD
	0011 _B	TBD	TBD	TBD	TBD
	0100 _B	TBD	TBD	TBD	TBD
	0101 _B	TBD	TBD	TBD	TBD
	0110 _B	TBD	TBD	TBD	TBD
	0111 _B	TBD	TBD	TBD	TBD
	1000 _B	TBD	TBD	TBD	TBD
	1001 _B	TBD	TBD	TBD	TBD
	1010 _B	TBD	TBD	TBD	TBD
	1011 _B	TBD	TBD	TBD	TBD
	1100 _B	TBD	TBD	TBD	TBD
	1101 _B	TBD	TBD	TBD	TBD
	1110 _B	TBD	TBD	TBD	TBD
	1111 _B	TBD	TBD	TBD	TBD
D	0000 _B	TBD	TBD	TBD	TBD
	0001 _B	TBD	TBD	TBD	TBD
	0010 _B	TBD	TBD	TBD	TBD
	0011 _B	TBD	TBD	TBD	TBD
	0100 _B	TBD	TBD	TBD	TBD
	0101 _B	TBD	TBD	TBD	TBD
	0110 _B	TBD	TBD	TBD	TBD
	0111 _B	TBD	TBD	TBD	TBD
	1000 _B	TBD	TBD	TBD	TBD
	1001 _B	TBD	TBD	TBD	TBD
	1010 _B	TBD	TBD	TBD	TBD
	1011 _B	TBD	TBD	TBD	TBD
	1100 _B	TBD	TBD	TBD	TBD
	1101 _B	TBD	TBD	TBD	TBD
	1110 _B	TBD	TBD	TBD	TBD
	1111 _B	TBD	TBD	TBD	TBD
NOTE 1 The A _{dc} gain parameter is implemented by overriding the DFE Gain coefficient settings in PG[1:0]RW[60, 68, 70, 78]. When CTLE is enabled in a given receiver through RW50[0] and RW51[7:0], the gain settings for that receiver are determined only by the values programmed in RW52[3:0], RW53[3:0], and RW54[7:0]; and the values in PG[1:0]RW[60, 68, 70, 78] are ignored for that receiver. This applies regardless of any one or more of the following features being enabled or disabled: DFE, DFE Training, or DFE Training Acceleration					

3.8 RX Loopback

3.8.1 Loopback Mode

When the loopback circuit is enabled in [RW26](#), the DDR5RCD02 can feed a received signal or internal data back to the host for debugging, testing, and/or training purposes. Loopback Mode and DFE Training Mode shall not be enabled simultaneously. Loopback Mode is not supported in Test Mode (Frequency Band 2). Loopback is necessary in order for the host memory controller or test instrument to immediately read back data that was written to the RCD, DRAM or Data Buffer without having to issue multiple DRAM WRITE/READ commands. There are also inherent limitations when characterizing the receiver using statistical analysis methods such as Bit Error Rate (BER) analysis.

For example, at $BER=10^{-16}$:

There is not enough memory depth in the DRAM to store all the 10^{16} data.

The amount of time to perform multiple WRITE/READ commands to and from the array is prohibitively long.

Since the amount of time involved performing these operations is much longer than the DRAM refresh rate interval, the host or memory controller must also manage Refreshes during testing to ensure data retention.

Limited pattern depth means limited Inter symbol Interference (ISI). Loopback is a necessity for characterizing the receiver without the limitations and complexities of other traditional methods.

3.8.2 Loopback Ports

There are five RCD loopback ports, four input ports and one output port. The input ports consist of two internal and two external port as shown in Figure . Each port contains one data signal and one strobe signal. [RW26](#) selects the internal or external signals to loopback to the Host on the output port. Transitions in the incoming data signal are aligned to the rising edges of the incoming strobe. The RCD will sample incoming data with the falling edges of the incoming strobe and it will drive the strobe and data signals out to the host controller. The RCD will apply logic inversion in the outgoing strobe signal to ensure output data signal transitions are aligned to the rising edges of the corresponding strobe signal as shown in Figure 26.

Internal RCD Loopback is only intended to send back the output of the functional path slicer. Internal RCD Loopback is independent from [RW32\[2:1\]](#) Training Source selection field.

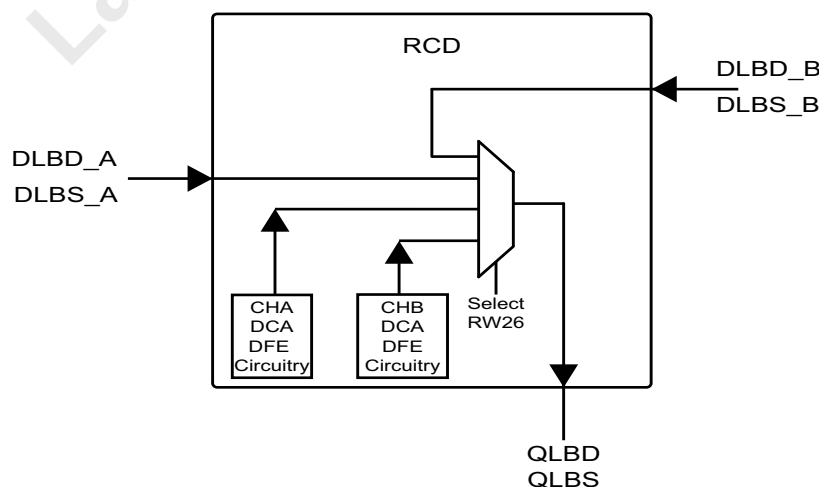


Figure 25 — RCD Loopback Ports

3.8.3 Pin Assignment for Loopback, Timing Diagrams, and Timing Parameters

Inputs (From DB or DRAM)

DLBD_A

DLBS_A

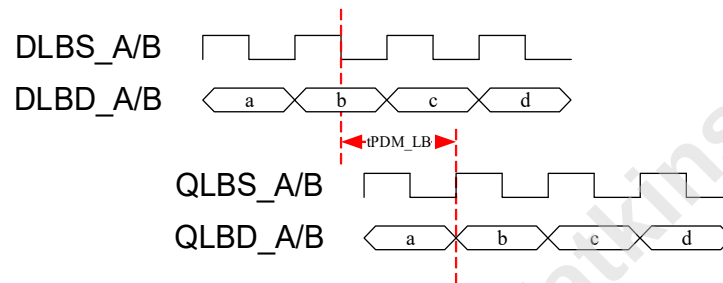
DLBD_B

DLBS_B

Outputs (To Host)

QLBD

QLBS



Note: A waiting time $t_{Ext_LB_Entry}$ applies when External Loopback is enabled in RW26[2:0].

Figure 26 — External Loopback Timing Diagram

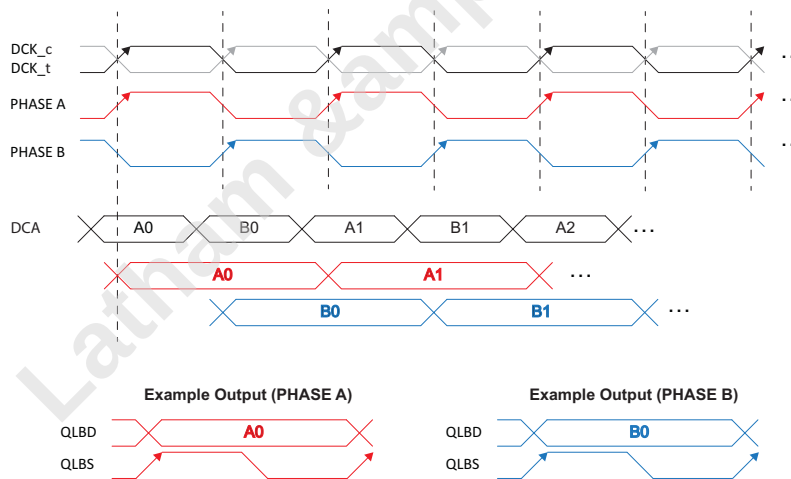


Figure 27 — Internal Loopback 2-Phase A and B Phase

3.8.3 Pin Assignment for Loopback, Timing Diagrams, and Timing Parameters (cont'd)

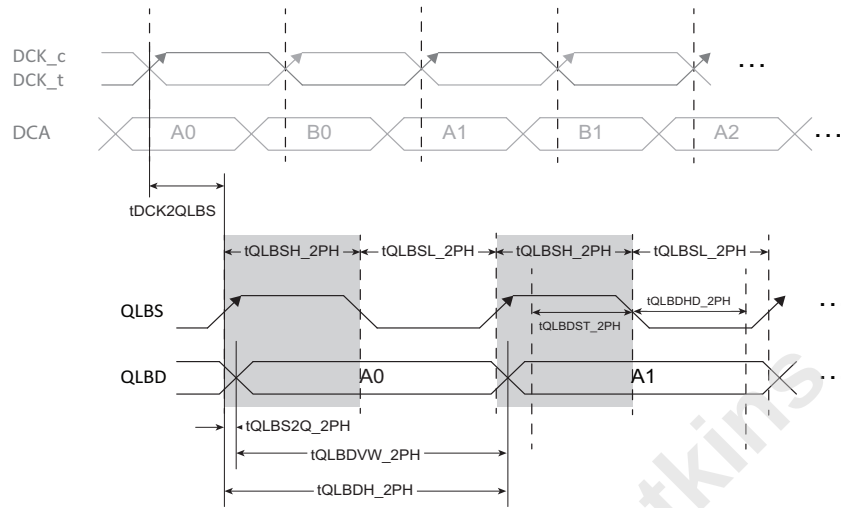


Figure 28 — Loopback Output 2-Phase Timing Parameters

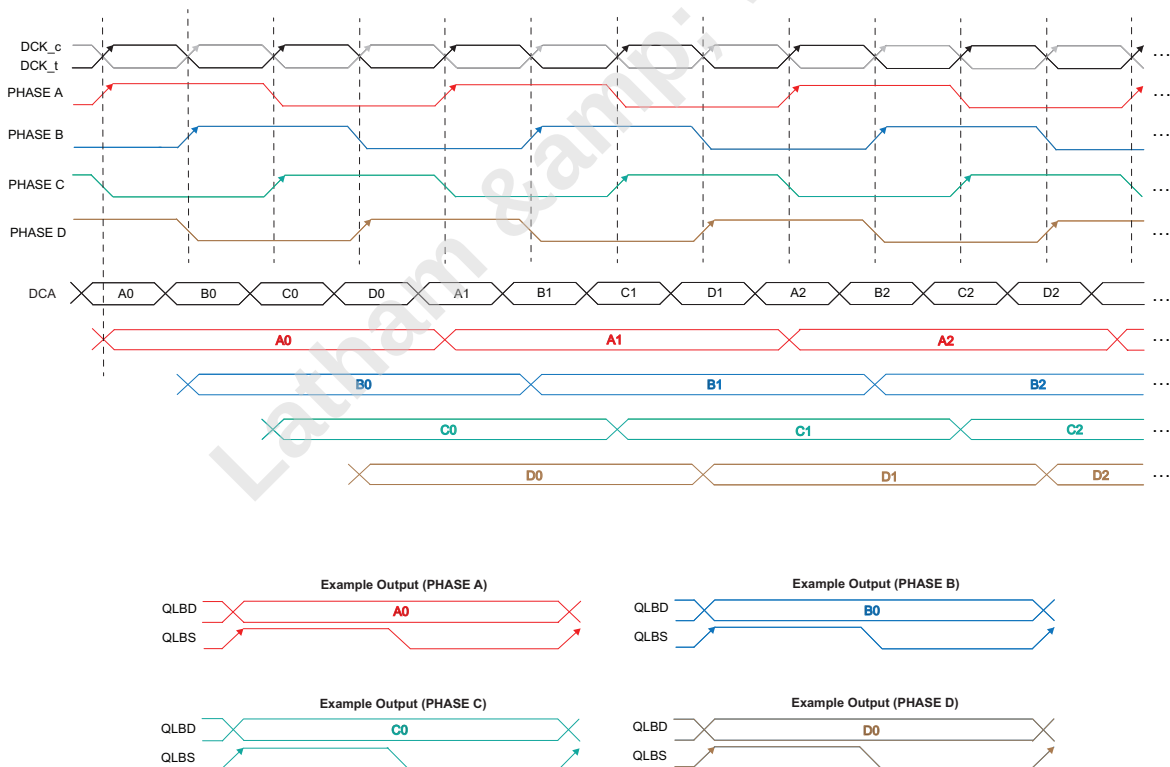


Figure 29 — Internal Loopback 4-Phase A, B, C, and D Phase

3.8.3 Pin Assignment for Loopback, Timing Diagrams, and Timing Parameters (cont'd)

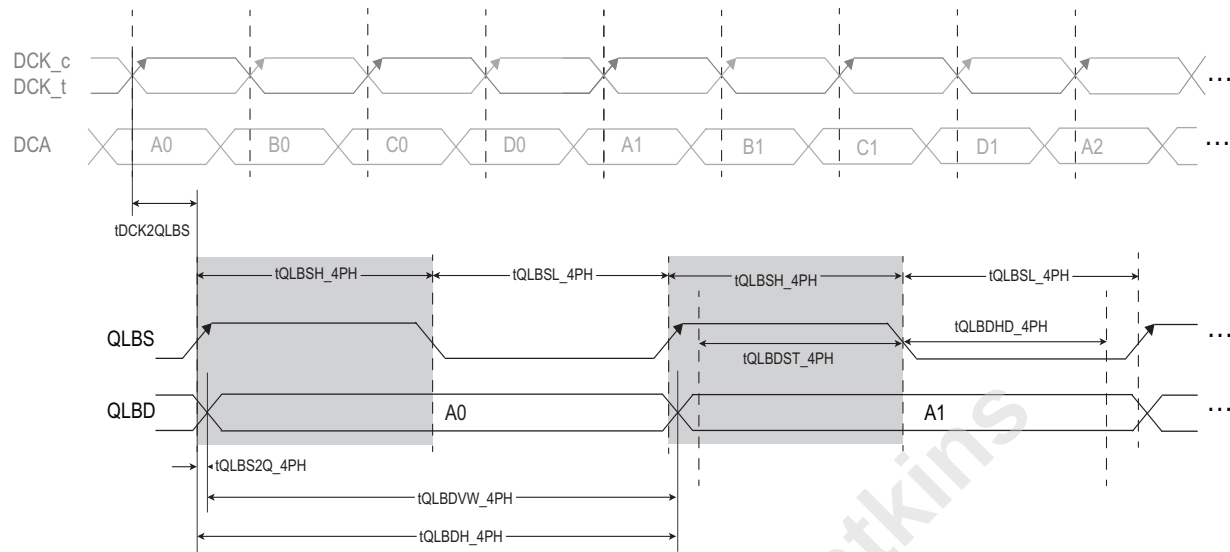


Figure 30 — Loopback Output 4-Phase Timing Parameters

3.8.4 Internal Loopback Mode Entry

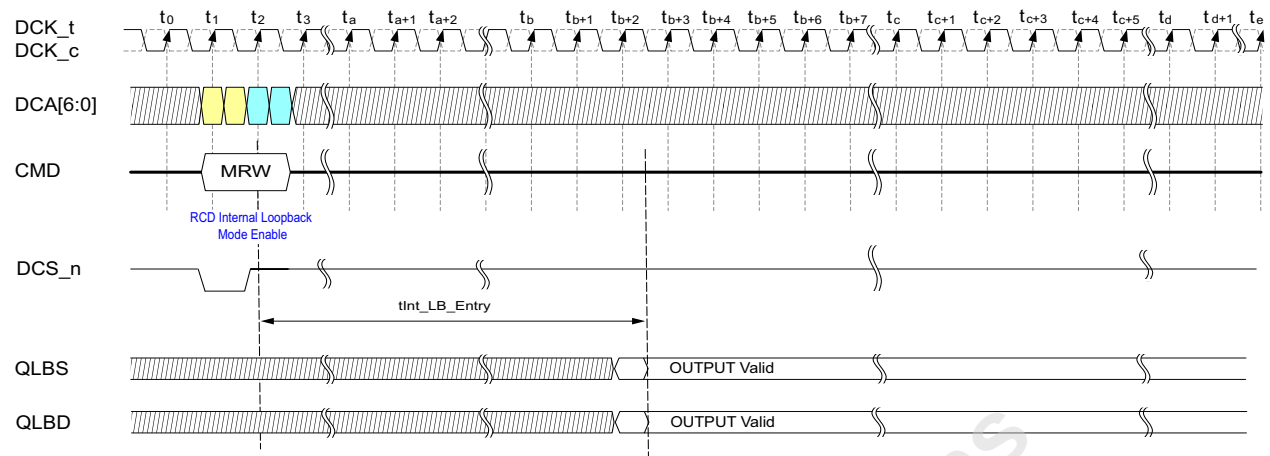


Figure 31 — RCD02 Internal Loopback Mode Entry

- Once the 2-Phase or 4-Phase Internal Loopback mode is registered through In-Band MRW to [RW26](#), the DCA sampled by the first DCK *rising* edge after tInt_LB_Entry will be the first Phase A output. The DCA sampled by the first DCK *falling* edge after tInt_LB_Entry will be the first Phase B output. Similarly, after tInt_LB_Entry, the 2nd rising edge DCK sampled DCA is the first Phase C output and the 2nd falling edge DCK sampled DCA is the first Phase D output.
- If Internal Loopback is enabled through Sideband, the entry time will be much longer and asynchronous.
- While Internal Loopback is enabled, switching Phases (such as In-Band or SideBand MRW to [RW27\[7\]](#) = 0 or 1) is supported. Once the register write is received, the QLBS and QLBD output will move to the new Phase.
- Because the 4-phase Loopback phase is non-deterministic until the first UI has been captured after a clock discontinuity, testing each of the rising edge phases or each of the falling edge phases can be achieved by sending the same stimulus in consecutive bursts, offset by an odd number of clock cycles, and with no resets, clock stops, or MRWs between them.

3.9 DLBD and DLBS Fine Vref Control and Input Delay Control

In DDR5RCD02, fine-grained DLBD and DLBS Vref control is required. The host controller can utilize the Loop-Back Vref control words, [RW4B](#), to control the Vref of DLBD and DLBS.

In DDR5RCD02, adjustment of the phase of the received DLBD and DLBS is required. The host controller can utilize the Loop-Back input delay control words, [RW4C](#), to control the phase of the received DLBD and DLBS.

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3.10 Device Initialization

To ensure defined outputs from the register before a stable clock has been supplied, the register must enter the reset state during power-up. After the voltage ramp, stable power is provided for a minimum of t_{RINIT1} μ s with $DRST_n$ asserted. When the reset input $DRST_n$ is LOW, all input receivers are disabled, and shall be left floating. In addition, when $DRST_n$ is LOW, all non-sticky control registers are restored to their default states, which is all '0's unless explicitly stated otherwise. The outputs $QRST_n$, and $BRST_n$ must drive LOW. All other Qx outputs must float with the exception of the $QCS[x]_n$ outputs, which are driven LOW. The RCD must float $QLBD$, $QLBS$ and $ALERT_n$ outputs. As long as the $DRST_n$ input is pulled LOW, the register is in low-power state and input termination is not present.

The DDR5RCD02 device will not affect $DRST_n$ signal voltage levels even when V_{DD} is not powered on. An RCD with V_{DD} powered off shall not prevent other devices connected to the same $DRST_n$ net from receiving the correct High and Low logic voltage levels driven by the host on that net. Also, the DDR5RCD02 device will not sustain permanent damage if $DRST_n$ is driven HIGH (to V_{IH} levels) when V_{DD} is powered off, and it must be able to support all the specified features and functions once power is restored.

Before the $DRST_n$ signal goes HIGH, all DCS_n inputs are driven LOW by the host controller. This puts the signals in a similar state as self refresh allowing the initialization and self refresh protocol to be similar. After $DRST_n$ goes HIGH, the DCS_n input signals will go HIGH. The host may stagger (i.e., delay) this transition between sub-channels by up to $t_{DCSLHSK1}$, but $DCS0_n$ and $DCS1_n$ within each sub-channel can only be staggered by up to $t_{DCSLHSK2}$. This transition will enable the termination on the DCS_n , DCA , and DCK inputs. The host will start the DCK clock early enough to meet t_{CKACT}^1 . After t_{STAB01} , the SidebandBus Interface is operational and the host controller will write the coarse and fine grain frequency registers and put the RCD in the proper DCA input mode (DDR or SDR). The host controller must then wait t_{STAB02} to allow the PLL to re-lock.

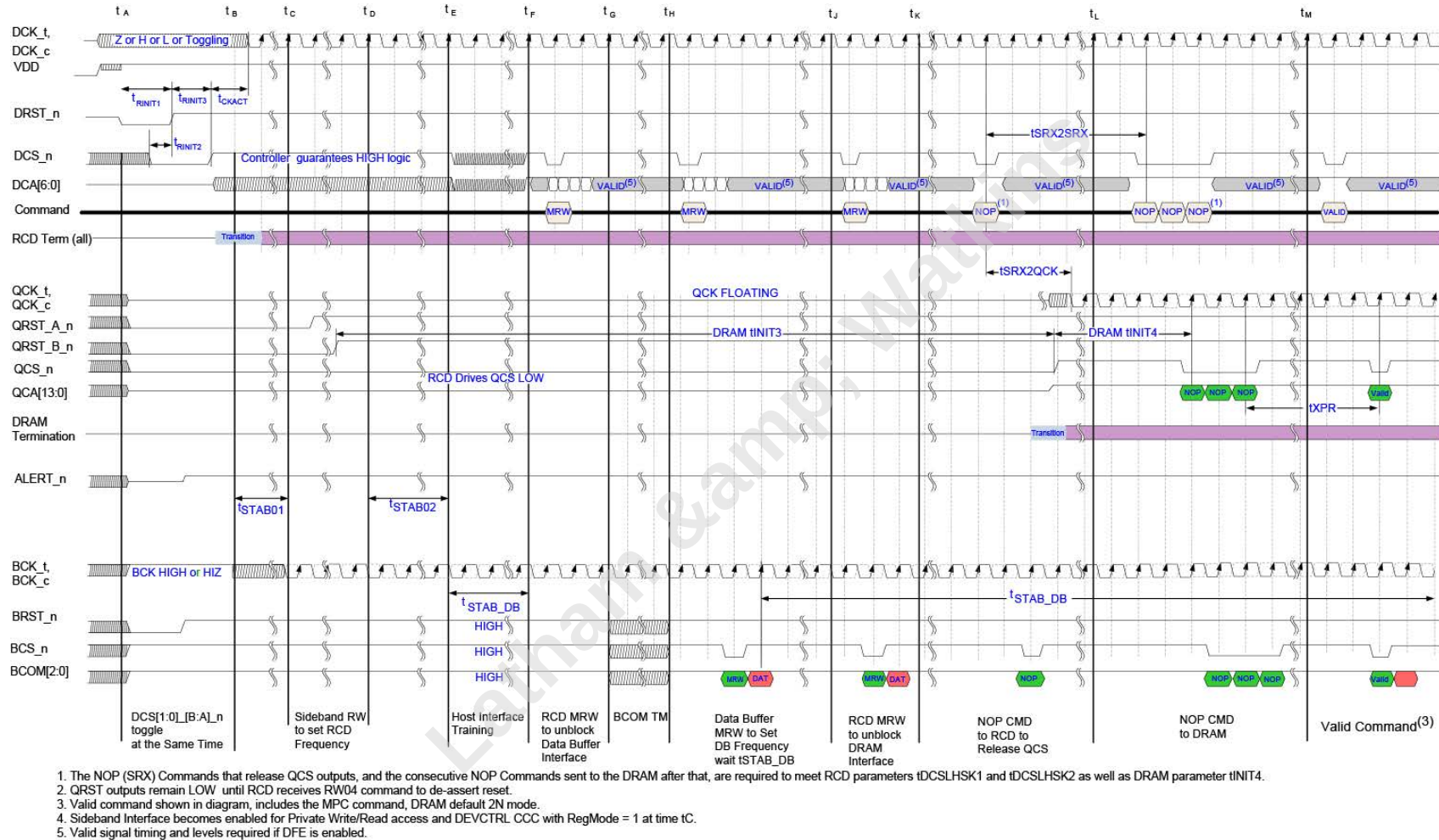
Training of the DCS and DCA signals now takes place, followed by BCS and $BCOM$ training if applicable. Commands to the DRAMs are blocked by default, and the DRAM QCS signals will remain LOW.

The $QRST_n$ outputs will remain LOW after $DRST_n$ goes HIGH. The $QRST_n$ signals do not go HIGH until the host writes the Command Control Word with the commands to do so. This may be done via the SidebandBus or in-band. The $QRST_n$ signal for each sub channel is controlled separately to allow staggering of the signals for power supply inrush control.

As part of the power-on initialization, all control words are reset to their default state. After initialization, the Host must write to those control registers whose contents are required for frequency and DIMM configuration.

1. Since DCS_n Low-to-High transitions can be staggered by up to $t_{DCSLHSK1}$ (for the entire chip) or $t_{DCSLHSK2}$ (within the same sub-channel), t_{CKACT} must be met with respect to the DCS_n signal with the earliest rising transition.

3.10 Device Initialization (cont'd)



NOTE For RDIMM, Steps t_F through t_J are not required.

Figure 32 — Timing of Clock and Data During Initialization Sequence

3.10 Device Initialization (cont'd)

From a device perspective, the initialization sequence must be as shown in Table 14.

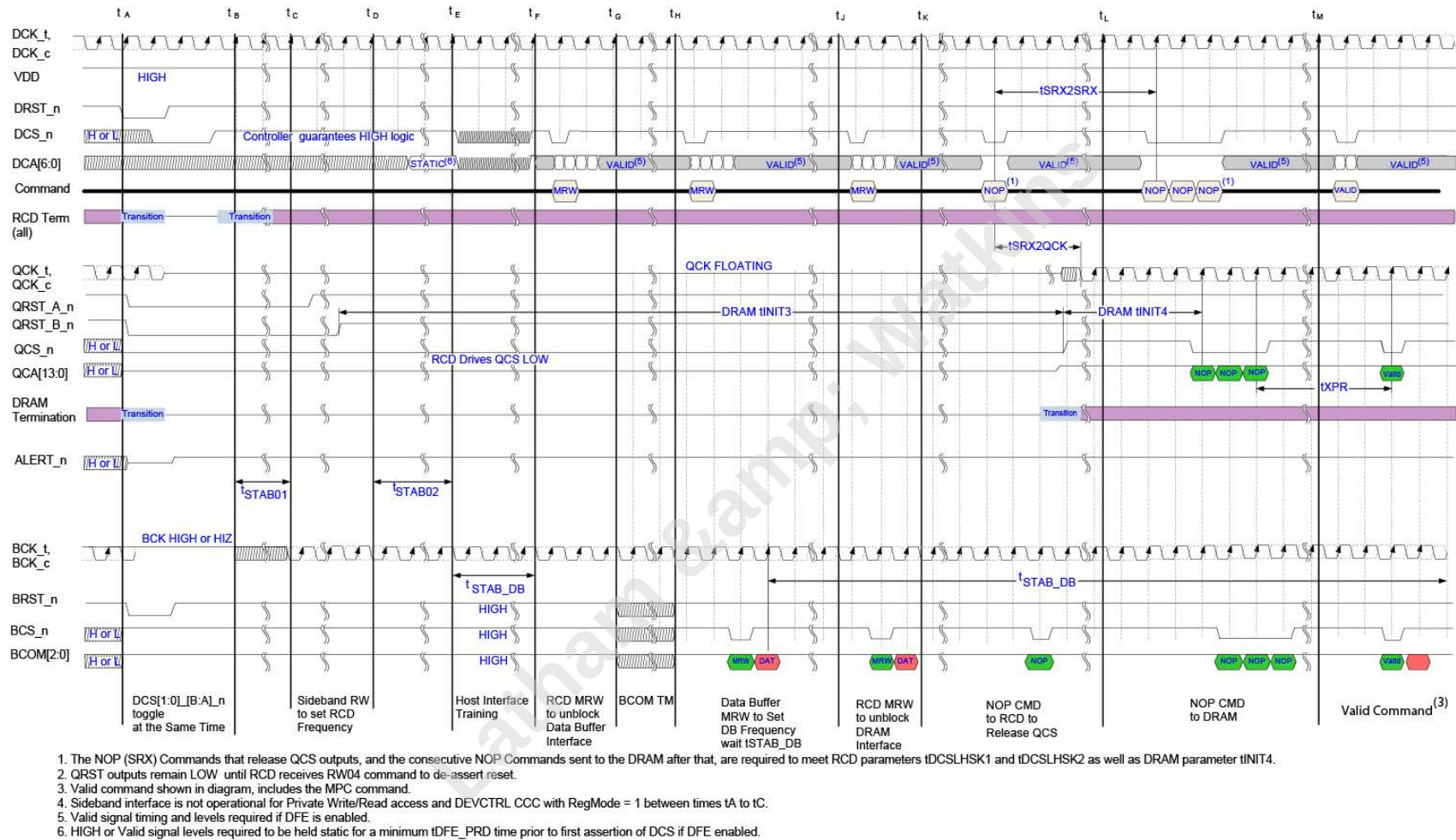
Table 14 — DDR5RCD02 Device Initialization Sequence at Power-on Ramping¹

Step	Power	Inputs: Signals Provided by the Controller					RCD Input Termination	Outputs: Signals Provided by the Device				
	V_{DD}	DRST_n	DCS[1:0]_n	DA/C	DPAR	DCK_t, DCK_c	DCS, DCA, DCK Term	QCS[1:0]_n	QRST_n	QCA	ALERT_n	QCK_t, QCK_c
0	0V	X or Z	X or Z	X or Z	X or Z	X or Z	OFF	Z	Z	Z	Z	Z
1	0→ V_{DD}	X or Z	X or Z	X or Z	X or Z	X or Z	OFF	X or Z	X or Z	X or Z	X or Z	X or Z
2	V_{DD}	L	X or Z	X or Z	X or Z	X	OFF	L	L	Z	H	Z
3	V_{DD}	L	L	X or Z	X or Z	X	OFF	L	L	Z	H	Z
4	V_{DD}	H	L	X or Z	X or Z	X	OFF	L	L	Z	H	Z
5	V_{DD}	H	H	X	X	running	ON	L	NOTE 2	Z	H	Z
NOTE 1 1X = Logic LOW or Logic HIGH. Z = Floating												
NOTE 2 QRST_n goes HIGH with the proper Command Control Word write.												

3.11 Reset Initialization with Stable Power

The timing diagram in Figure 33 depicts the initialization sequence with stable power and clock. This will apply to the situation when we have a soft reset in the system. DRST_n will be asserted for minimum 1 μ s. This DRST_n timing is based on DDR5 DRAM Reset Initialization with Stable Power requirement, and is a minimum requirement. Actual DRST_n timing can vary based on specific system requirement, but it cannot be less than 1 μ s. Some of the steps in Figure 33 could be skipped by taking advantage of control words with sticky attributes.

3.11 Reset Initialization with Stable Power (cont'd)



NOTE For RDIMM, Steps tF through tJ are not required.

Figure 33 — Reset Initialization With Stable Power

3.11 Reset Initialization with Stable Power (cont'd)**Table 15 — DDR5RCD02 Device Initialization Sequence when Power and Clock are Stable¹**

Step	Power	Inputs: Signals Provided by the Controller					RCD Input Termination	Outputs: Signals Provided by the Device				
		DRST_n	DCS[1:0]_n	DA/C	DPAR	DCK_t, DCK_c		QCS[1:0]_n	QRST_n	QCA	ALERT_n	QCK_t, QCK_c
1	V _{DD}	H	X	X	X	running	ON	X	H	X	X	running
2	V _{DD}	L	X	X	X	X	Transitioning	L	L	Z	H	Z
3	V _{DD}	L	L	X	X	X	OFF	L	L	Z	H	Z
4	V _{DD}	H	L	X	X	X	OFF	L	L	Z	H	Z
5	V _{DD}	H	H	X	X	running	ON	L	NOTE 2	Z	H	Z

NOTE 1 X = Logic LOW or Logic HIGH. Z = Floating

NOTE 2 QRST_n goes HIGH with the proper Command Control Word write.

4 Power Down Operation

DDR5 does not have a separate CKE signal going to the DIMM or the DRAMs. Power Down Entry, including self refresh, is signaled via a single UI command. Following the Power Down Entry command, the CS signal effectively becomes the equivalent of the CKE signal. An inactive CS keeps the DRAM devices in the power down state. CS going active will take the DRAMs out of the power down state. A NOP command is placed on the CA bus along with CS going active so that no commands are executed by the DRAM devices. DRAMs can be put in power down mode with or without non-target ODT control.

4.1 Power Savings Modes

The DDR5RCD02 device supports the following power saving mechanisms:

- PDE Power Down Mode. The RCD goes into a low-power mode when all DRAM ranks are put into PDE Power Down.
- PDE Power Down with ODT control. In this mode ranks can be in power down mode but will still require ODT commands for termination, therefore the RCD and Data buffer will remain in normal operating mode.
- Self Refresh Modes. With and without Clock Stop.

The DRST_n input has priority over all other power saving mechanisms. When DRST_n is driven LOW, the RCD will force the Qn outputs to float (except QCS_n, QRST_n and BRST_n, which are driven LOW), de-asserts the ALERT_n output, disables Input Bus Termination (IBT) and forces BCK_t/BCK_c, BCS_n, and BCOM[2:0] HIGH or HI-Z.

4.1.1 PDE Power Down Mode

The RCD Power Down mode is handled independently between the sub-channels.

The RCD device enters power down mode when all DRAM ranks on the sub-channel have entered power down. The host controller puts DRAM ranks into PDE Power Down Mode by sending a PDE command to the rank. Due to the V_{DD} termination, all input termination remains enabled during PDE power down mode and all outputs may remain enabled. The QCA bus outputs will be all HIGH during PDE Power Down mode as they are for Deselect commands. The RCD may reduce power internally as it is able. No command may be sent on the clock cycle following the PDE command to allow the RCD to safely update its configuration for command decoding, because the DCA input receivers will be disabled and require a clock cycle to be ready to receive the NOP (PDX) command without checking parity.

Power down exit is signaled by taking one or both DCS inputs active with a PDX on the bus. The RCD will not check CA Parity when the PDX command is received while both ranks are in power down as the DCA input receivers would be disabled.

The PDE command is sent to the Data Buffers on the BCOM bus only when the last rank has entered PDE mode. The PDX command is sent to the Data Buffers on the BCOM bus when the first rank receives the PDX command. Therefore, the Data Buffers are in power down mode only when ALL ranks on the sub channel are in power down mode.

4.1.2 PDE Power Down Mode with ODT Control

In PDE Power Down Mode with ODT control, the powered down rank(s) and the Data Buffers may still receive non-target termination commands. The RCD itself does not power down if either rank is in PDE with ODT control, as it must pass on the non-target termination commands. QCA11 determines whether the DRAMs are put into PDE with or without ODT control. QCA11 = 1 for no ODT control, and QCA11 = 0 for ODT control.

While in PDE with ODT control, RD, WR, and MRR commands may be sent to the DRAM rank with non-target termination signaled (CS active for both UI of the command). It is the responsibility of the host controller to not send commands to the DRAMs which are “Not Allowed”, as defined in Table 16.

In case of Parity Error state in ‘*Static ALERT_n Assertion Mode*’ (RW01[6] = ‘0’), the host needs to send MRW to the exit the Error state before the RCD can forward a PDX command on any sub-channel that had Parity Error. To avoid violating the restrictions defined in Table 16 below, MRW must not be sent to DRAMs in PDE state with ODT Control. Therefore, it is necessary to first send PDX commands so that the sub-channels that are not in parity error state can take their DRAMs out from Power-Down state with ODT control. After sending MRW to RW04, it will be necessary to send PDX for the sub-channels that were in Parity Error state initially. Based on these requirements, an example of a recovery sequence for this case could be the following.

If parity error is detected during PDE with ODT control:

1. Send PDX (NOP) to CH_A and CH_B.
2. Send MRW for RW04 CMD10 ‘CH_B Clear Parity Error’.
3. Send MRW for RW04 CMD9 ‘CH_A Clear Parity Error’.
4. Send PDX (NOP) to CH_A and CH_B.

The above example sequence in is intended to work for any combination of parity error state and PDE with ODT control per sub-channel (error or PDE in CH_A only, CH_B only, or CH_A and CH_B at the same time).

In case of Parity Error state in ‘*Pulsed ALERT_n Assertion Mode*’ (RW01[6] = ‘1’), the recovery procedure can be reduced simply to the first step in the list because the DDR5RCD02 device resumes execution of DCA commands automatically after the end of the ALERT_n pulse.

The non-target termination commands must be sent to the Data Buffers if either rank is in PDE with ODT control.

Table 16 — ODT Control During Power Down Mode¹

Mode	NOP/PDX command	RD w/ non-target term	WR w/ non-target term	MRR w/ non-target term	All Others
PDE w/o ODT control	Allowed. Exits PDE mode	Not Allowed	Not Allowed	Not Allowed	Not Allowed
PDE w/ODT control	Allowed. Exits PDE Mode	Allowed	Allowed	Allowed	Not Allowed
NOTE 1 This table applies to the commands sent from the RCD to the DRAMs when the DRAMs are in PDE state with ODT Control.					

4.1.2 PDE Power Down Mode with ODT Control (cont'd)

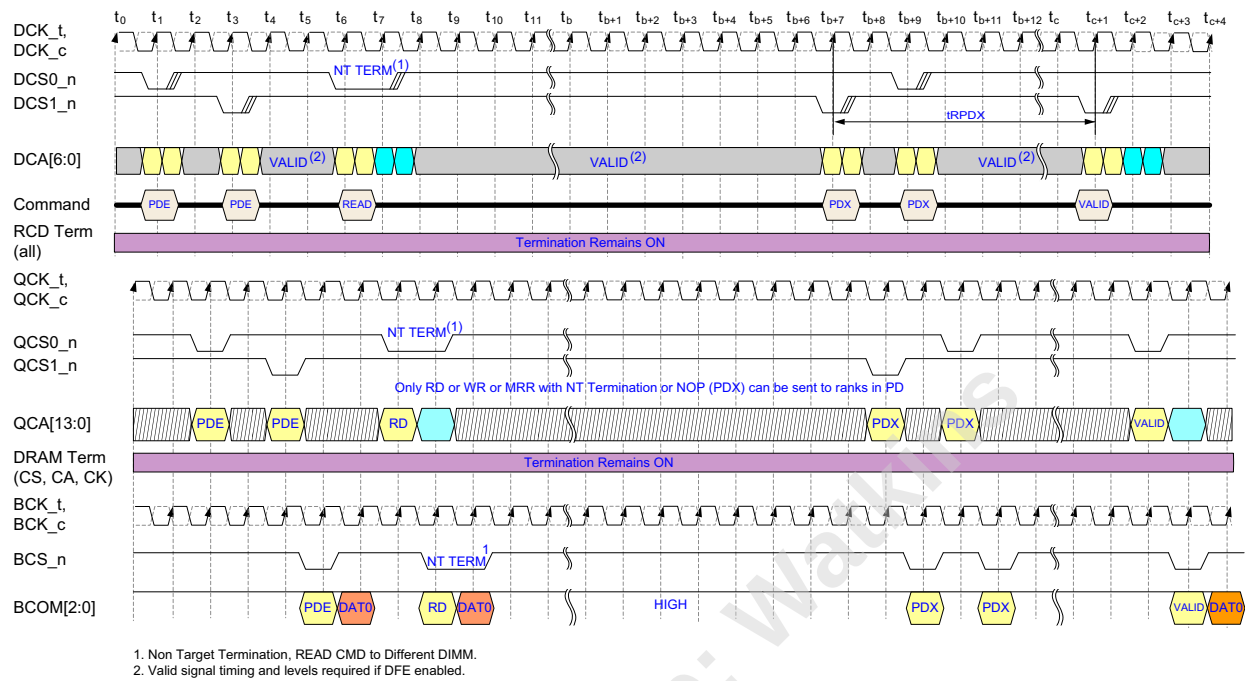


Figure 34 — Power Down Mode Entry and Exit

4.2 Self Refresh Modes

Following the sending of SRE command, the host will EITHER assert the DCS[x]_n signals for a single clock pulse (indicating Self Refresh WITHOUT clock stop), or for the t_{CSSR} minimum (indicating Self Refresh WITH clock stop). No pulse width in between is permitted. All DCS[x]_n signals must be asserted in the same manner.

4.2.1 Self Refresh Mode with Clock Stop Entry

Self Refresh Mode is a very low power state in which the input clocks will stop, and both the RCD and DRAM inputs terminations are disabled.

Self-refresh entry is signaled when the host sends SRE commands to both ranks within a sub-channel at the same time.

After all ranks on both sub-channels have been put into self refresh with no parity error detected, the host controller will pulse the DCS[1:0]_[B:A]_n inputs for t_{CSSR} to maintain self refresh. No parity checking is done on this operation, as the DCA bus is not used.

Unlike the DDR4 RCD, there is no detection done on the clock inputs, instead the RCD will detect the last DCS_n transitions to LOW of the CS pulse to disable DCK, DCA and DCS_n terminations. The DCS transition to LOW is required to apply valid DCS input timings with respect to DCK. The host controller will stop the clock. DCK_t and DCK_c can be driven LOW or can be floating. The PLL is disabled and DCS[1:0]_[B:A]_n inputs are kept LOW to maintain self refresh.

The RCD outputs will be disabled with the exception of the QRST_n, BRST_n and QCS[x]_n signals to keep the DRAMs in self refresh. QCS_n signals are driven LOW. BCK_t/BCK_c, BCS_n, and BCOM[2:0] signals will be driven High or HI-Z by RCD during the clock-stop event. Note that the BCS_n is NOT driven LOW when in self refresh.

If a parity error is detected on any of the self refresh entry commands the host will take any DIMMs or sub-channels already in self refresh out of self refresh and retry the entry operation.

4.2.2 Exit from Self Refresh Mode with Clock Stop

Exiting Self Refresh Mode is accomplished when the first DCS_n input transitions HIGH. The host may stagger (i.e., delay) this transition between sub-channels by up to $t_{DCSLHSK1}$, but DCS_{0_n} and DCS_{1_n} within each sub-channel can only be staggered by up to $t_{DCSLHSK2}$. This transition will re-enable the termination on the DCK, DCS, and DCA inputs and enable the PLL. The host will start the DCK clock early enough to meet t_{CKACT}^1 , and wait for the PLL(s) to lock. After t_{STAB01} , the SidebandBus Interface is operational. Note that the DCS[x]_n inputs going HIGH does NOT pass through to the DRAMs at this point.

Once the PLL is locked, using the SidebandBus Interface, the host is allowed to change the Command/Address bus rate settings in RW00[1:0] (requires meeting Footnote 7 in Table 199). The host is also allowed to use DCS, DCA, and DFE Training Modes, including DFE Sweep Accelerator, through SidebandBus to train the DCS and DCA signals if needed due to a modification of the DCK frequency. In case of LRDIMM, if there was a change in DCK frequency, it may also be necessary for the host to train the BCOM signal timings from RCD to Data Buffers using BCOM Pass-Through Mode in the RCD and BCOM Training Mode in the DB. The host controller will send NOP (SRX) commands to each rank, which will take the QCS[x]_n outputs HIGH. This will enable the termination on the DRAMs. The RCD starts the QCLK outputs at this time and begins passing commands through to the DRAMs. The host controller will then send three back-to-back NOP commands to each rank, which completes the exit of self refresh in the DRAMs.

4.2.3 Self Refresh Mode without Clock Stop Entry

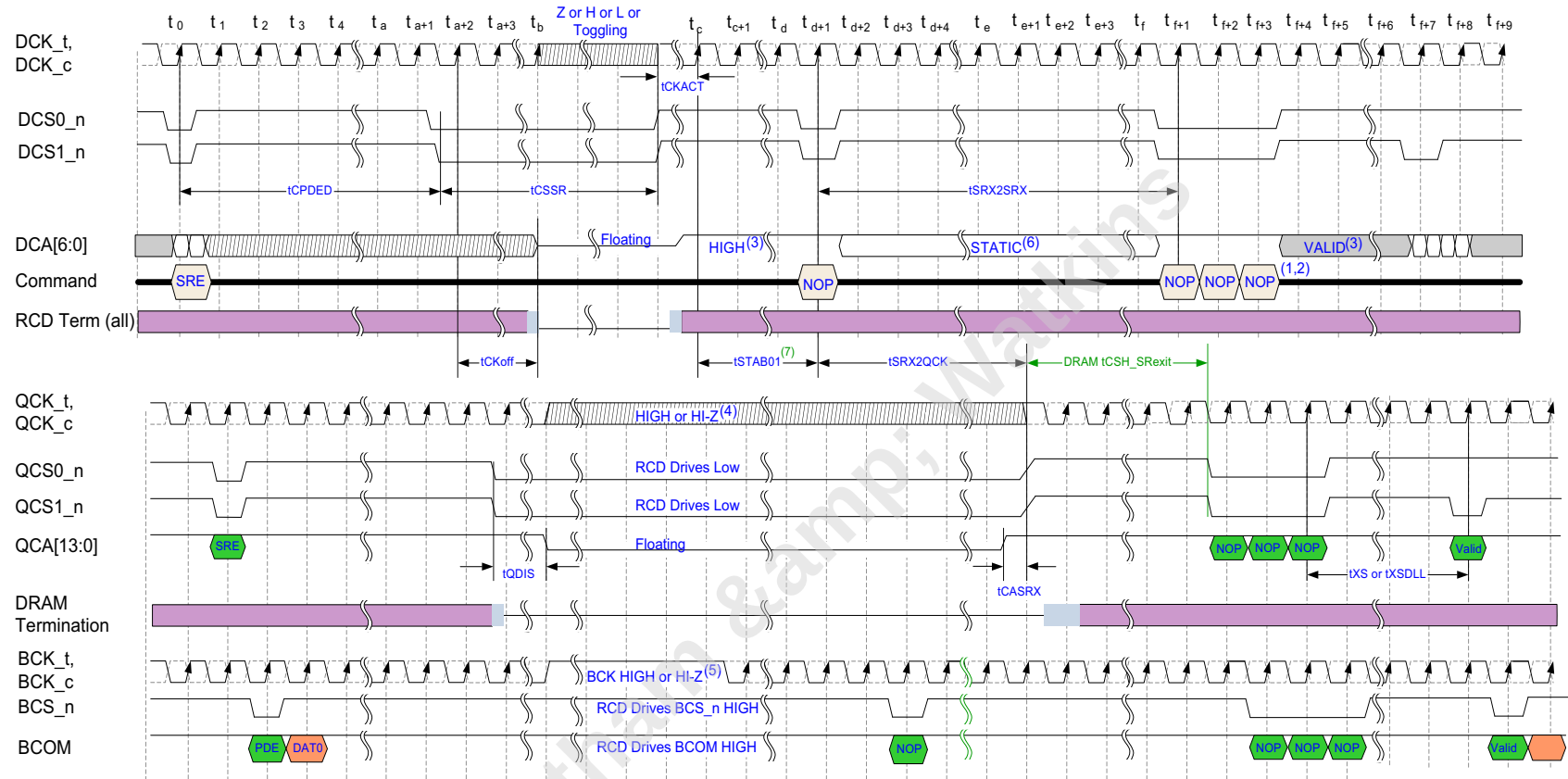
Self Refresh without Clock Stop is similar to Self Refresh with Clock Stop mode except that the host controller will not stop the DCK input. After both sub-channels have received the SRE command, the host signals the RCD to enter this mode by sending a one clock wide pulse on DCS[1:0]_[B:A]_n instead of holding it LOW for multiple cycles at t_7 in Figure 39. The RCD will not disable any of its terminations and will not disable the PLL during Self Refresh without clock stop. The RCD will also not stop the QCK or BCK clock outputs in this mode.

4.2.4 Self Refresh Mode without Clock Stop Exit

Exiting this mode is similar to Self Refresh with Clock Stop, except that the host will not wait t_{STAB} upon exit, as the PLL is already locked. Since the DCS[x]_n signals are already HIGH, the first indication of the exit is the NOPs that release the QCS[x]_n signals.

1. Since DCS_n Low-to-High transitions can be staggered by up to $t_{DCSLHSK1}$ (for the entire chip) or $t_{DCSLHSK2}$ (within the same sub-channel), t_{CKACT} must be met with respect to the DCS_n signal with the earliest rising transition.

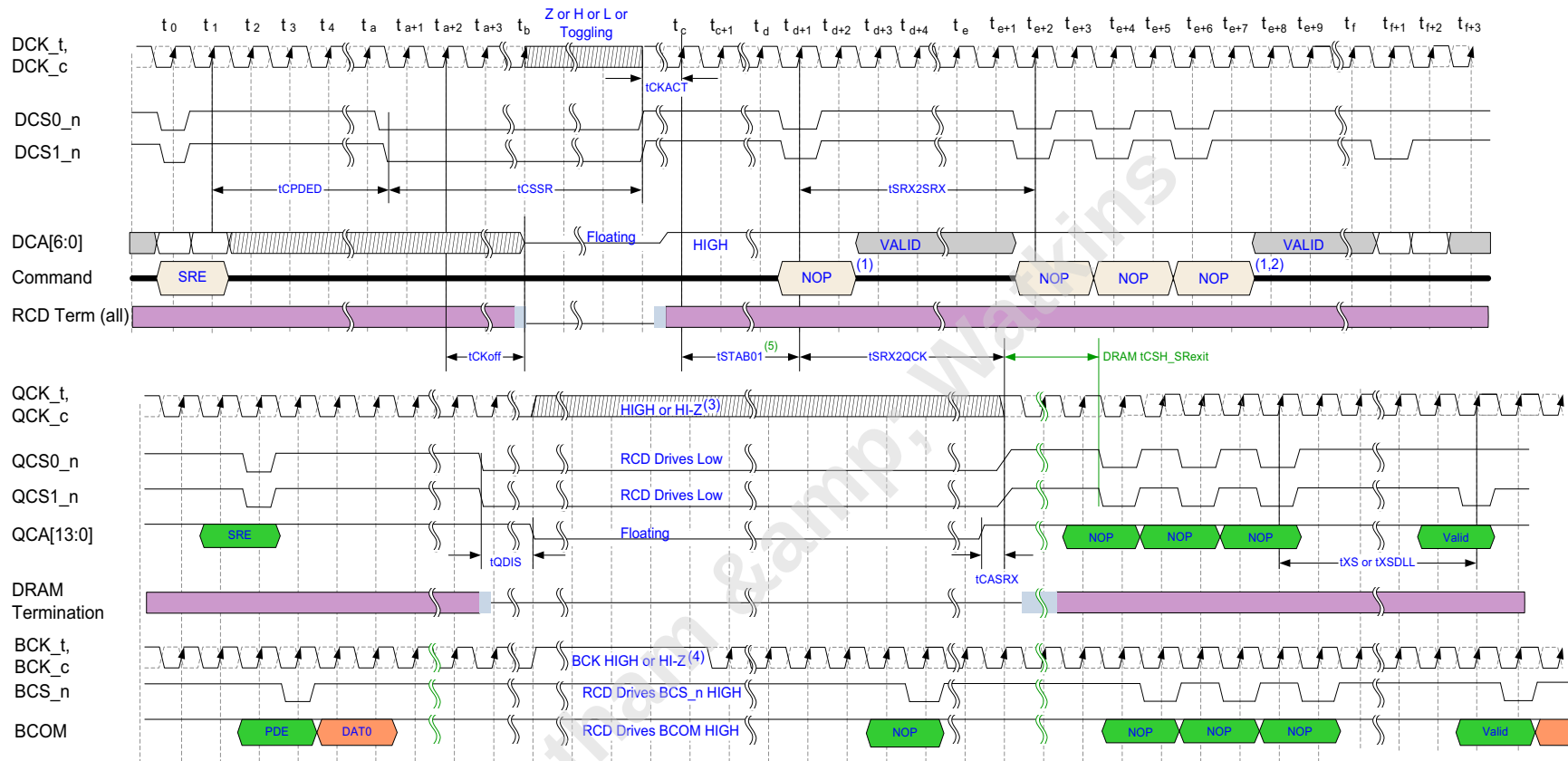
4.2.4 Self Refresh Mode without Clock Stop Exit (cont'd)



1. The NOP (SRX) Commands that release QCS outputs, and the consecutive NOP Commands sent to the DRAM after that, are required to meet RCD parameters tDCSLHSK1 and tDCSLHSK2 as well as DRAM parameter tCSH_SRExit.
2. NOP (SRX) Command to DRAM Rank 0 & 1 to take the DRAM out of Self Refresh.
3. HIGH or Valid signal timing and levels required if DFE enabled.
4. The RCD is allowed to drive HIGH or HI-Z.
5. The RCD is allowed to drive HIGH or HI-Z. The clock nets will be pulled HIGH by either the RCD output drivers or by the external termination to VDD which is required on the module PCB.
6. HIGH or Valid signal levels required to be held static for a minimum tDFE_PRD time prior to first assertion of DCS if DFE enabled.
7. In LRDIMM, tSTAB01 + tSTAB_DB must be met from start of RCD DCK to the first DCA command that gets forwarded to the LRDIMM BCOM bus interface but does not use the DQ data path (e.g., SRX NOP).

Figure 35 — Self Refresh Entry Exit with Clock Stop with DCK Frequency Change

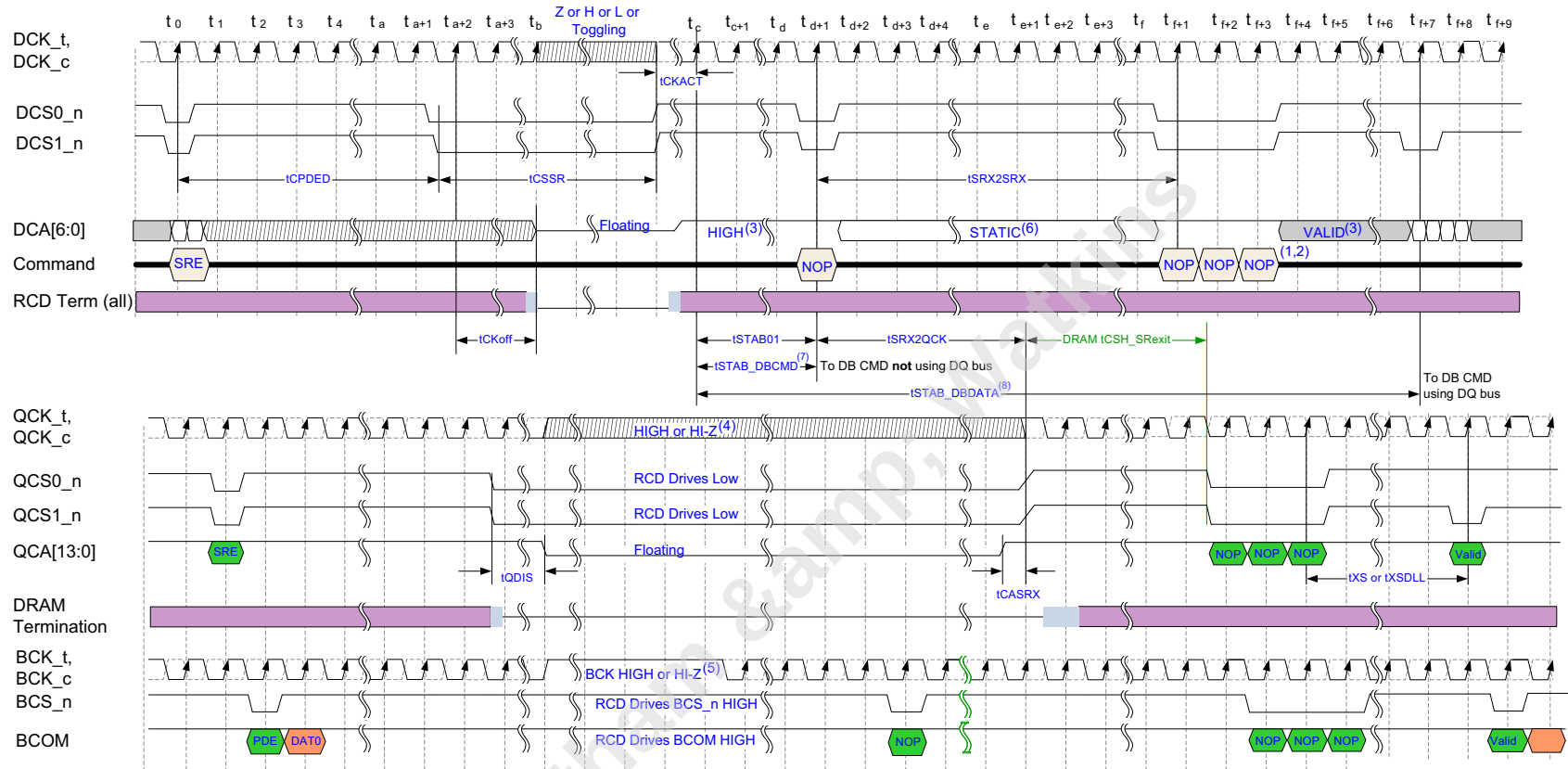
4.2.4 Self Refresh Mode without Clock Stop Exit (cont'd)



1. The NOP (SRX) Commands that release QCS outputs, and the consecutive NOP Commands sent to the DRAM after that, are required to meet RCD parameters tDCSLHSK1 and tDCSLHSK2 as well as DRAM parameter tCSH_SRexit.
2. NOP (SRX) Command to DRAM Rank 0 & 1 to take the DRAM out of Self Refresh.
3. The RCD is allowed to drive HIGH or HI-Z.
4. The RCD is allowed to drive HIGH or HI-Z. The clock nets will be pulled HIGH by either the RCD output drivers or by the external termination to VDD which is required on the module PCB.
5. In LRDIMM, tSTAB01 + tSTAB_DB must be met from start of RCD DCK to the first DCA command that gets forwarded to the LRDIMM BCOM bus interface but does not use the DQ data path (e.g., SRX NOP).

Figure 36 — Self Refresh Entry Exit with Clock Stop in SDR1 Mode with DCK Frequency Change

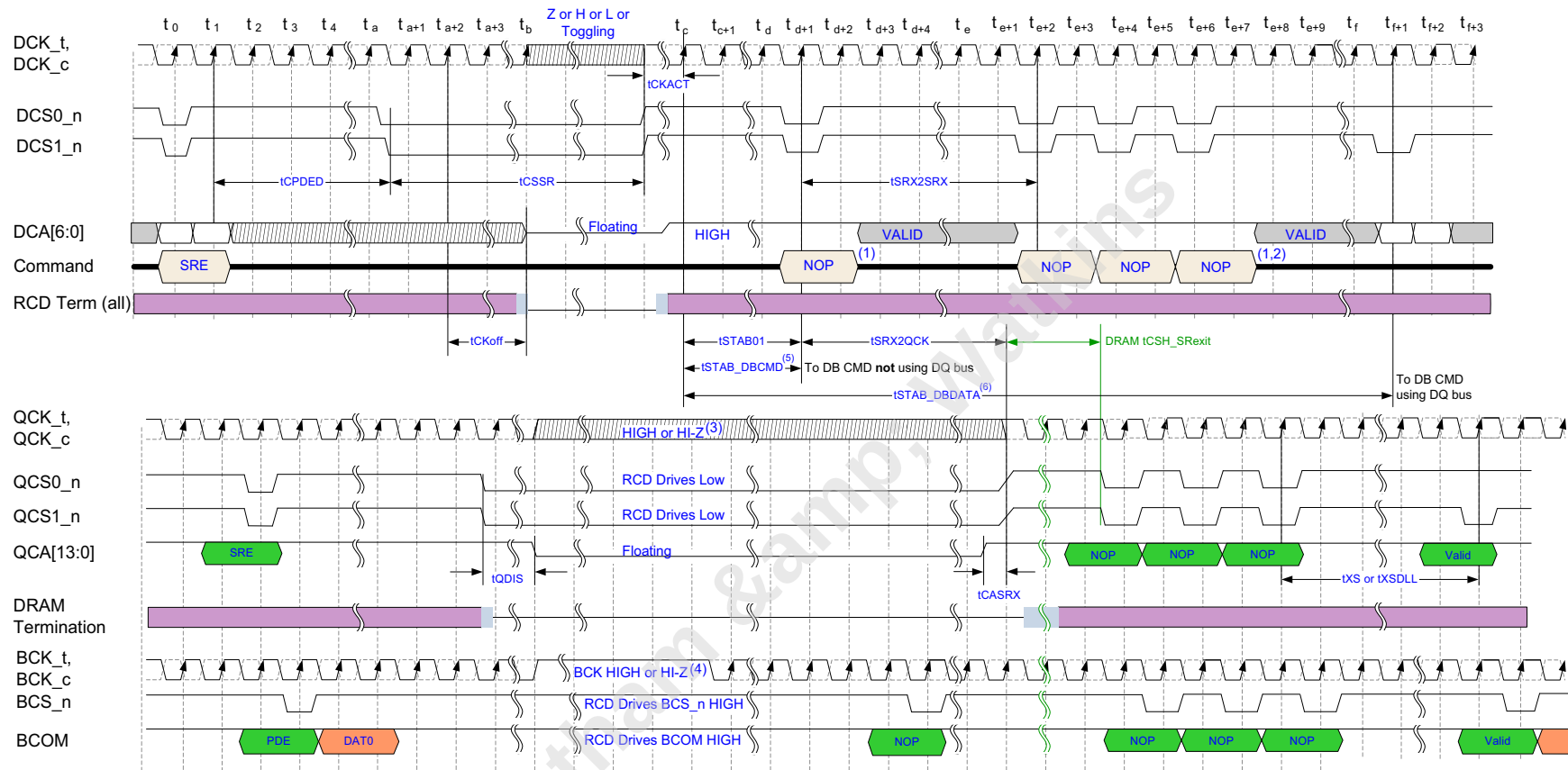
4.2.4 Self Refresh Mode without Clock Stop Exit (cont'd)



1. The NOP (SRX) Commands that release QCS outputs, and the consecutive NOP Commands sent to the DRAM after that, are required to meet RCD parameters $t_{DCSLHSK1}$ and $t_{DCSLHSK2}$ as well as DRAM parameter t_{CSH_SRExit} .
2. NOP (SRX) Command to DRAM Rank 0 & 1 to take the DRAM out of Self Refresh.
3. HIGH or Valid signal timing and levels required if DFE enabled.
4. The RCD is allowed to drive HIGH or HI-Z.
5. The RCD is allowed to drive HIGH or HI-Z. The clock nets will be pulled HIGH by either the RCD output drivers or by the external termination to VDD which is required on the module PCB.
6. HIGH or Valid signal levels required to be held static for a minimum t_{DFE_PRD} time prior to first assertion of DCS if DFE enabled.
7. In LRDIMM, t_{STAB_DBCMD} must be met from start of RCD DCK to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface but does **not** use the DQ data path (e.g., SRX, NOP, etc.).
8. In LRDIMM, t_{STAB_DBDATA} must be met from start of RCD DCK to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface and uses the DQ data path (e.g., WR, RD).

Figure 37 — Self Refresh Entry Exit with Clock Stop without DCK Frequency Change

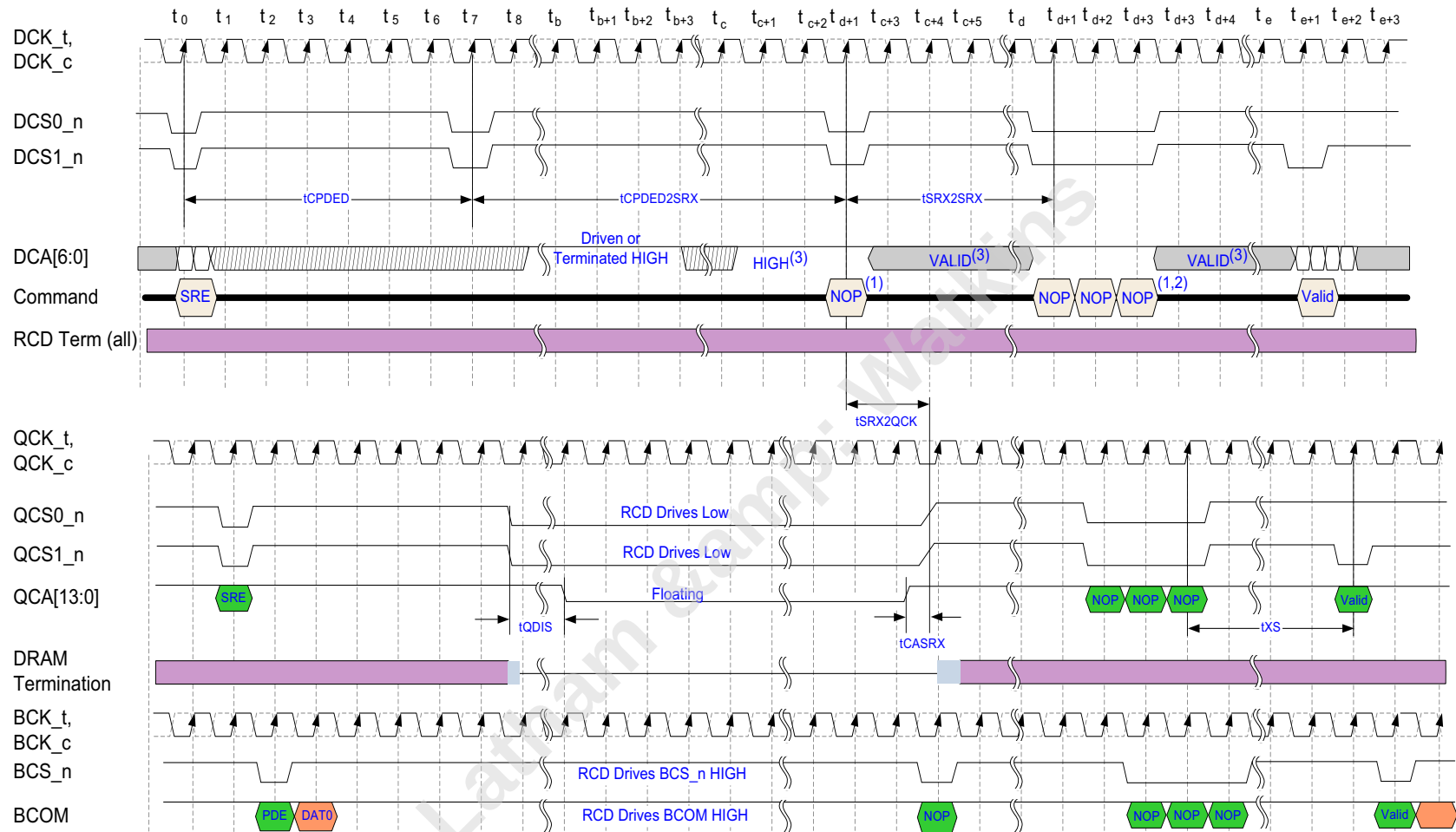
4.2.4 Self Refresh Mode without Clock Stop Exit (cont'd)



1. The NOP (SRX) Commands that release QCS outputs, and the consecutive NOP Commands sent to the DRAM after that, are required to meet RCD parameters tDCSLHSK1 and tDCSLHSK2 as well as DRAM parameter tCSH_SReXit.
2. NOP (SRX) Command to DRAM Rank 0 & 1 to take the DRAM out of Self Refresh.
3. The RCD is allowed to drive HIGH or HI-Z.
4. The RCD is allowed to drive HIGH or HI-Z. The clock nets will be pulled HIGH by either the RCD output drivers or by the external termination to VDD which is required on the module PCB.
5. In LRDIMM, iSTAB_DBCMD must be met from start of RCD DCK to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface but does **not** use the DQ data path (e.g., SRX, NOP, etc.).
6. In LRDIMM, iSTAB_DBDATA must be met from start of RCD DCK to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface and uses the DQ data path (e.g., WR, RD).

Figure 38 — Self Refresh Entry Exit with Clock Stop in SDR1 Mode without DCK Frequency Change

4.2.4 Self Refresh Mode without Clock Stop Exit (cont'd)



1. The NOP (SRX) Commands that release QCS outputs, and the consecutive NOP Commands sent to the DRAM after that, are required to meet RCD parameters tDCSLHSK1 and tDCSLHSK2 as well as DRAM parameter tCSH_SRExit.
2. NOP (SRX) Command to DRAM Rank1 & 2 out of Self Refresh.
3. HIGH or Valid signal timing and levels required if DFE enabled.

Figure 39 — Self Refresh Entry and Exit without Clock Stop

5 Training Modes

5.1 CS Training Modes

5.1.1 DCS Training Mode (DCSTM) - Host Interface

The DCS Training Mode is a method to facilitate the loopback of a sampled sequence of the DCS_n signal. In this mode, the DCK is running, and the DCA and DPAR signals are ignored by the RCD. Once this mode is enabled, the RCD is selected to actively sample and drive feedback. The RCD will sample the DCS_n signal on the rising edge of DCK. Every set of four DCK rising edge samples will be included in a logical computation to determine the DCSTM Output result that is sent back to the host on ALERT_n or QLBx configured in RW01[5]. Once sampling begins, the RCD must maintain the consecutive grouping of the samples every 4 tCK. When the DCS_n Sample[0] and Sample[2] results in a logic 0 and the DCS_n Sample[1] and Sample[3] results in a logic 1, the RCD will drive a 0 on ALERT_n or QLBx. The output signal can transition as often as every 4 tCK. All commands to DRAM and Data Buffer are blocked by the RCD when DCS Training mode is enabled. The host shall not enable Power Down mode in RW00 until after all Host Interface Training is complete.

5.1.1.1 Entry and Exit for DCS Training Mode

The DCS Training Mode is enabled by a SMBus write to RW02[3:0]. The signal used to provide the feedback is configured by RW01[5]. RCD can be configured to use ALERT_n (which requires training each sub-channel independently and one after the other), or use both QLBD and QLBS (which allows training both sub-channels in parallel). For the remainder of this section, we assume ALERT_n was selected for feedback. Once the RCD has DCS Training Mode enabled, the RCD begins sampling on every rising CK edge, with the 4-sample groups looping consecutively. Depending on the value of the samples, ALERT_n is driven HIGH or LOW. RCD hardware will block the signals coming from the DERROR_IN_n receivers if ALERT_n has been selected for feedback in RW01[5]. Prior to entering DCS Training Mode, the ALERT_n or QLBx shall remain High. The DCS Training Mode is disabled by a SMBus write to RW02[3:0].

5.1.1.2 DCS Training Mode Operation

In DCS Training Mode, the DCS_n values are sampled on all DCK rising edges. Each group of 4 consecutive samples is evaluated in pairs, and then the two pairs are combined with a logical OR prior to sending on ALERT_n. The samples evaluation to determine the output is as follows:

Table 17 — Sample Evaluation for Intermediate Output[0]

Output[0]	DCS_n Sample[0]	DCS_n Sample[1]
1	0	0
0	0	1
1	1	0
1	1	1

Table 18 — Sample Evaluation for Intermediate Output[1]

Output[1]	DCS_n Sample [2]	DCS_n Sample[3]
1	0	0
0	0	1
1	1	0
1	1	1

5.1.1.2 DCS Training Mode Operation (cont'd)

Table 19 — Sample Evaluation for Final DCSTM Output

DCSTM Output ^{1,2}	Output[0]	Output[1]
0	0	0
1	0	1
1	1	0
1	1	1

NOTE 1 When there is no change on the DCSTM Output from previous evaluation, ALERT_n shall continue to drive same value continuously with no switching on the bus.

NOTE 2 These values are driven asynchronously, but may switch as often as every 4 tCK.

As shown in Table 17 through Table 19, DCSTM Output will only be 0 when DCS_n Sample[0] = Sample[2] = 0 and Sample[1] = Sample[3] = 1. The DCSTM output will be 1 in all other cases.

During DCS Training Mode the DCS ODT is enabled as for functional operation. The VrefCS is set in [RW\[49:48\]](#). The delay from when the DCS signals are sampled during the fourth DCK rising edge (Sample[3]) to when the output of the sample evaluation is driven to a stable value on ALERT_n is specified as tDCSTM_Valid, as shown in the following figure. The details of the tDCSTM_Entry, tDCSTM_Exit, and tDCSTM_ALERT_Window are also illustrated.

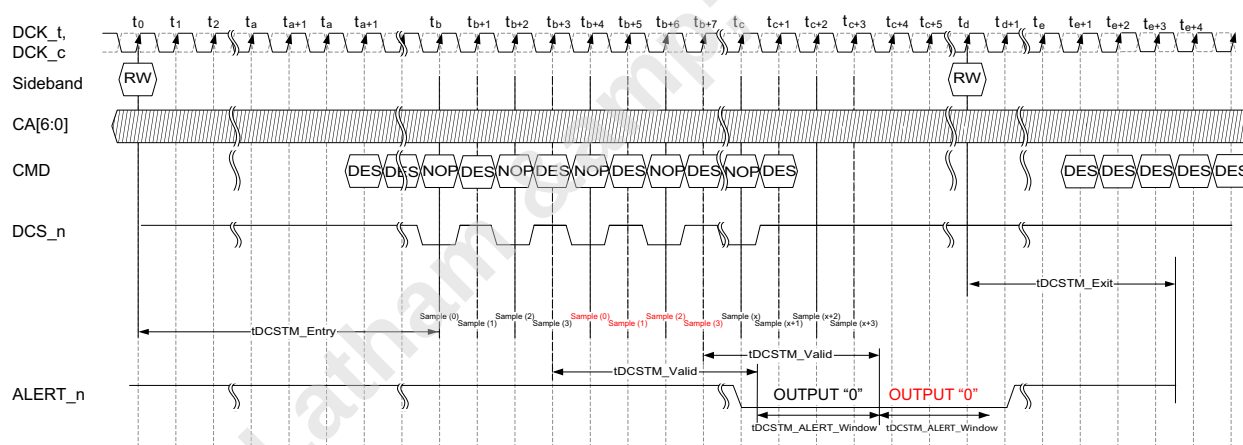


Figure 40 — Timing Diagram for DCS Training Mode with Consecutive Output Samples = 0

Figure 41 illustrates an example where the ALERT_n Output switches from a logic 0 to a logic 1 value, demonstrating the minimum tDCSTM_ALERT_Window:

5.1.1.2 DCS Training Mode Operation (cont'd)

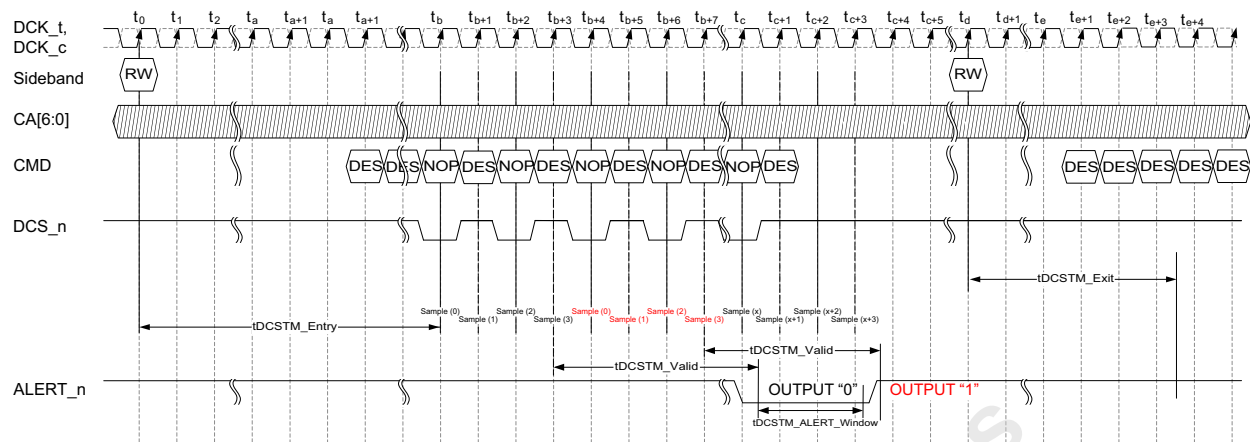


Figure 41 — Timing Diagram for DCS Training Mode with Output Sample Toggle

5.1.2 QCS Training Mode (QCSTM) - DRAM Interface

The DDR5RCD02 must support the DRAM CS training mode. Since the MPC command must be sent to the DRAM prior to completing training of the QCS and QCA signals, the RCD shall support the forwarding of DRAM MPC and VrefCA commands with extended setup and hold CA timings by using the RCD CA Pass-through mode set in the Global Features Control word. The host must disable Power Down mode in RCD prior and during QCS Training.

During QCSTM the RCD must be able to support the DDR5 DRAM CS training Mode with the following features:

- QCS_n pattern generation. When the QCS_n signal is selected in [RW03](#) the RCD will generate a continuous clock pattern to the DRAM, while sending a NOP encoding with proper inversion applied according to [RW00](#) on the associated QCA signals
- The host does not need to set the DRAM Interface blocking explicitly. The QCSTM mode will automatically block commands to the DRAM Interface.
- The RCD must adjust the QCS_n output delay when the host sends an MRW to [RW\[1A:17\]](#).
- It is recommended that the host block commands to the Data Buffer when in this mode. The host must send an MRW to [RW01](#) to set the BCOM blocking.

5.1.2 QCS Training Mode (QCSTM) - DRAM Interface (cont'd)

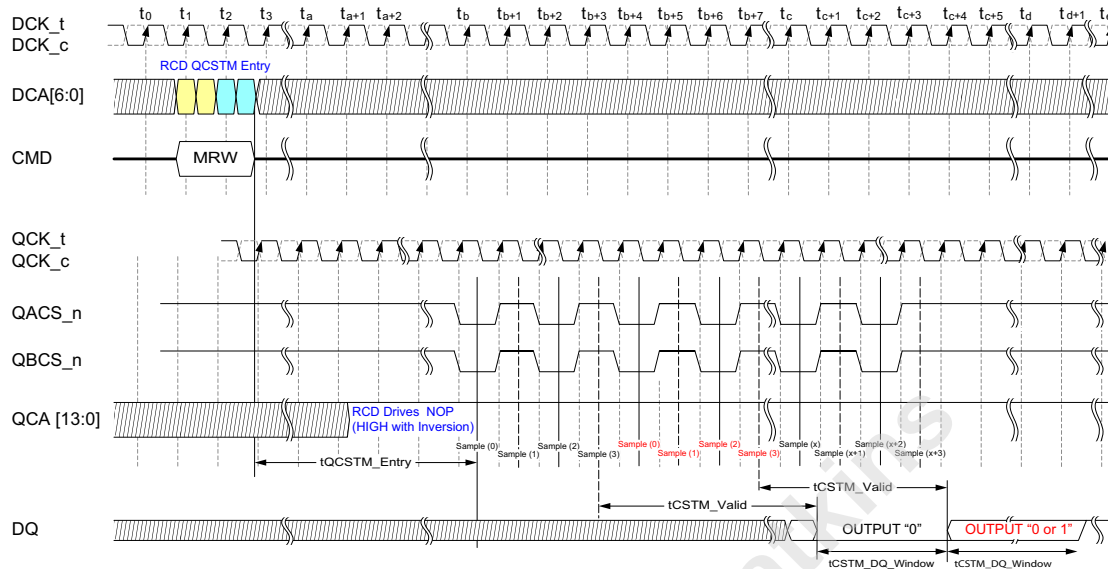


Figure 42 — Timing Diagram for QCSTM with Output Sample Toggle

5.2 CA Training Modes

5.2.1 DCA Training Mode (DCATM) - Host Interface

The DCA Training Mode is a method to facilitate the loopback of a logical combination of the sampled DCA[6:0] and DPAR signals. This mode can only be entered after DCS training has been performed. In DCATM, the DCK is running, and the DCS0_n qualifies when the DCK samples the DCA signals. A loopback equation that includes all the DCA and DPAR signals results in an output value that is sent asynchronously on the ALERT_n or QLBx signal back to the host memory controller. The host timings between DCS_n, DCK, DCA[6:0] and DPAR signals can then be optimized for proper alignment. DCK must stay valid with constant phase and frequency. When the RCD is in this mode, no functional commands are executed. The functional command interface is restored only after exiting this mode. All commands to DRAM and Data Buffer are blocked by the RCD during DCA Training mode. The host must ensure Parity Checking is disabled in RCD prior to and during DCA training. Power Down mode must be disabled in RCD prior and during DCA Training.

5.2.1.1 Entry and Exit for DCA Training Mode

The DCA Training Mode is enabled through an SMBus write to RW02[3:0]. The signal used to provide the feedback is configured by RW01[5]. RCD can be configured to use ALERT_n (which requires training each sub-channel independently and one after the other), or use both QLBD and QLBS (which allows training both sub-channels in parallel). For the remainder of this section, we assume ALERT_n was selected for feedback. Once this SMBus command has executed, no other commands will be interpreted by the RCD in the sub-channel that enters DCA Training Mode. The sub-channel that is not in Training Mode will interpret commands as in normal mode of operation. The host is responsible for avoiding undesirable signal events or commands on the RCD sub-channel that is not in Training Mode. That includes avoiding commands that may interfere with the resources needed to support DCA Training Mode such as, for example, MRW to enable the Parity Checking. For the sub-channel that is in Training Mode, only the sampling of the DCA signals, evaluation of the XOR result, and loop back to the ALERT_n will occur.

5.2.1.1 Entry and Exit for DCA Training Mode (cont'd)

While in DCA Training Mode, the DCS0_n signal will only assert for a single tCK at a time. The maximum sampling rate on the DCA signals will be every 4tCK. RCD hardware will block the signals coming from the DERROR_IN_n receivers if ALERT_n has been selected for feedback in RW01[5].

The DCA Training Mode is disabled by SMBus write to [RW02\[3:0\]](#) or by asserting DCS0_n for 2 or more cycles in a row, while sending a NOP command on the DCA bus.

5.2.1.2 DCA Training Mode (DCATM) Operation

In DCA Training Mode, the DCA and DPAR values are only sampled when DCS0_n is asserted. The sample is captured by the rising, falling or both DCK edges. The DCA XOR sampling edge is selected in RW02[5:4]. Unlike functional operation, there is no concept of multiple cycle commands in DCA Training Mode. Once the DCA and DPAR signals are sampled, the values are XORed to produce an output value. This output value is driven on the ALERT_n signal, as a static value. This output value will be held until the next sample is captured on the DCA bus, according to the DCS0_n assertion.

During DCA Training Mode the DCA ODT is enabled as for functional operation. The VrefCA is set according to RW[47:40]. The timing requirements for the Host Interface signals are the same as for functional operation.

The delay from when the DCA and DPAR signals are sampled during the DCS0_n assertion and when the output of the XOR computation is driven on the ALERT_n signal is specified as $t_{\text{DCATM_Valid}}$, as shown in the following figure. DCS_n may be asserted every 4tCK, and thus the DCA XOR output may transition every 4tCK. The following figure demonstrates an example where two DCS0_n assertions occur within 4tCK. The RCD will exit DCA Training Mode by SMBus write to RW02[3:0] or when the DCS0_n is asserted for 2 or more consecutive cycles but limited to 8 cycles.

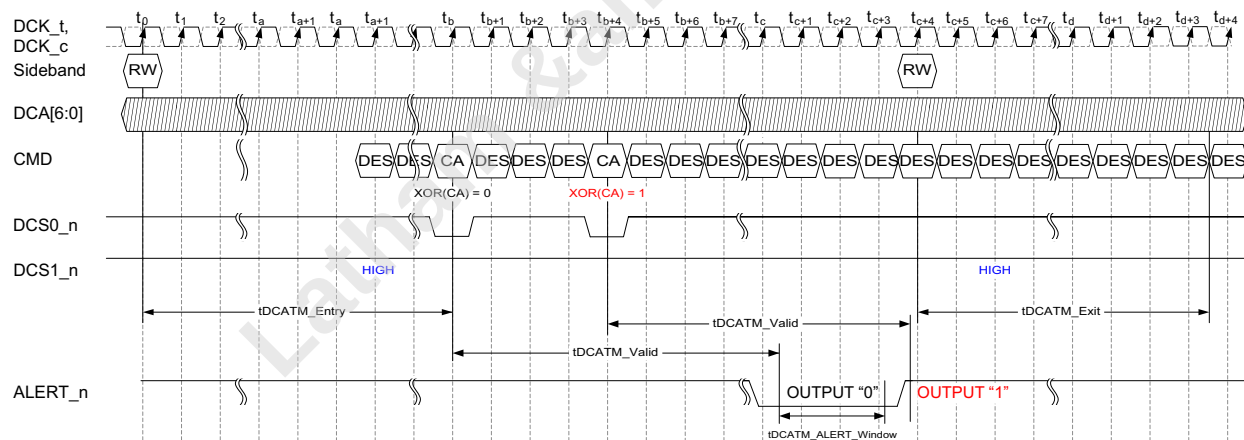


Figure 43 — Timing Diagram for DCA Training Mode SMBus Exit

[illegible]

Figure 44 — Timing Diagram for DCA Training Mode Double CS Exit

The DCATM Output is computed based on the DCS_n assertion and the values of the DCA and DPAR inputs. Table 20 clarifies the output computation.

DCS_n	DCATM Output
0	XOR (DPAR, DCA[6:0]) ^{1,2,3,4}
1	Hold Previous value

NOTE 1 For any DCA or DPAR signals not supported (i.e., disabled) in the RCD, the logical value of these signals shall be considered 0 for the XOR computation

NOTE 2 Clock sampling edge for XOR computation can be DCK rising, DCK falling or both DCK rising and falling edges and is selected in RW02[5:4].

NOTE 3 The DCK rising edge that captures DCS_n LOW must be used in cases with rising edges enabled in RW02[5:4]. The DCK falling edge immediately following the rising edge that captures DCS_n LOW must be used in cases with falling edges enabled in RW02[5:4].

NOTE 4 When both rising and falling DCK edges are enabled, XOR (rising edge DPAR, rising edge CA[6:0], falling edge DPAR, falling edge CA[6:0]).

There is no RCD QCA Training Mode. In order for the RCD to align the QCA signals to QCK, the RCD must be able to support the DDR5 DRAM CA training Mode. During RCD QCA Training the host will enable CA Pass-Through mode in [RW00\[2\]](#). The RCD will need to process MRW commands and block DRAM MRW commands while in CA Pass-Through mode. The RCD must be capable of QCA output delay adjustment when the host sends an MRW to [RW\[1C:1B\]](#) and [PG\[5\]RW\[7B:60\]](#). Power Down Mode and CA Parity Checking must be disabled in RCD prior to QCA Training, however the DPAR receivers will remain enabled to support SDR CA Pass-Through mode. CA Pass-Through mode can only be enabled when the RCD is forwarding DRAM commands, [RW01\[1\] = 1](#).

5.2.3 CA Pass-Through Mode

In order to train the DRAM interface, the RCD will be required to support a pass-through mode. In this mode the DCA signals are passed through to QCA regardless of DCS assertion. This allows the host to send multiple cycles of QCA setup/hold and multiple cycles of QCS assertion, while also controlling the duration and timing relationship of these signals at a clock cycle granularity. The Host Interface DCA must be stable before the RCD can be placed into CA Pass-through mode. Parity checking must be disabled prior to and during Pass-Through mode. CA Pass-Through is performed on a per Rank basis and entered via MRW [RW00](#). The Host is only allowed to enable one Rank at a time for CA Pass-through mode, to ensure the ability to still send MRWs on the other Rank. The RCD executes DCS_n/DCA commands on the rank that is not selected for Pass-Through mode, but it is illegal for the host to send commands other than MRW targeting the RCD. MRW to DRAM MR63 (for RCD Control Word read sequence) is not supported in CA pass-through mode. When Pass-Through mode is enabled for Rank 0 or Rank 1 all DCA signals will pass through to QCA. When [RW00\[3\]](#) = 0, Rank 0 is selected then only DCS0 will get passed through to QCS0 and the RCD will process RW commands on Rank 1 and not process any commands for Rank 0. When [RW00\[3\]](#)=1, Rank 1 is selected then only DCS1 will get passed through to QCS1 and the RCD will process RW commands on Rank 0 and not process any commands for Rank 1. The RCD keeps QCS_n static High for the rank that is not selected for Pass-Through mode. If the DRAM Interface Block all CMDs feature is enabled ([RW01\[1\]](#) = 0), the DCS_n input will not be passed through to QCS_n because all QCS_n outputs will be held statically HIGH by the RCD. The host must ensure Power Down mode is disabled in [RW00\[6\]](#) prior to and during Pass-Through mode. CA Pass-through mode is exited by disabling CA Pass-Through mode via [RW00](#).

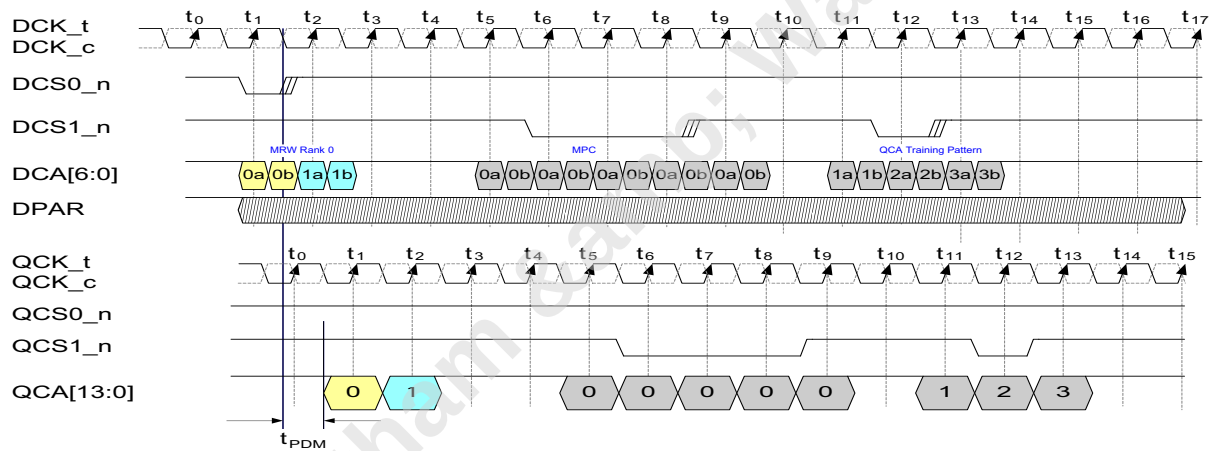


Figure 45 — MRW Example of MRW on DCS0 and CA Pass-Through on DCS1

In SDR Mode when Pass-Through mode is enabled, DPAR will be used to indicate 1st UI QCA[6:0] (DPAR = LOW) and 2nd UI QCA[13:7] (DPAR = HIGH) as shown in Figure 46. When the DPAR signal starts toggling, it may take up to two clock cycles (1 DPAR cycle) for the correct bit sequence to get established at the QCA[13:0] outputs of the RCD. In addition, the DDR5RCD02 device cannot guarantee the logic values of QCA[13:0] outputs unless the DPAR input is toggling continuously (i.e., once per DCK clock cycle) as shown in Figure 46. Therefore, the host shall keep the DCA[6:0] input signals static HIGH and not assert DCS during the first two clock cycles of toggling parity (1 DPAR cycle) and the host shall keep the DCA[6:0] input signals static HIGH and not assert DCS during the last two clock cycles of toggling parity (1 DPAR cycle, low to high transition).

Prior to entering or exiting CA Pass-Through or modifying the rank selection bit ([RW00\[3\]](#)), the host shall keep the DCA[6:0] input signals static HIGH and not assert DCS for a minimum of two clock cycles.

5.2.3 CA Pass-Through Mode (cont'd)

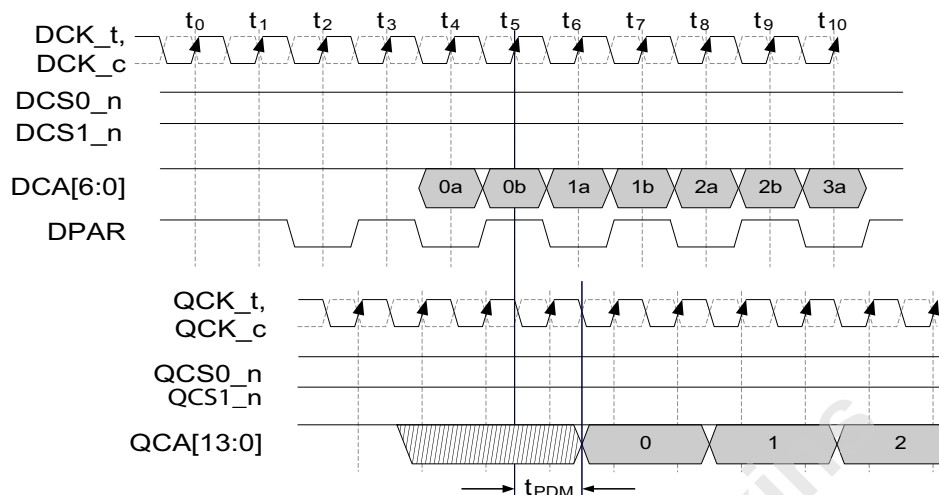


Figure 46 — SDR CA Pass-Through Mode

5.2.4 BCS and BCOM Training - Data Buffer Interface

In order for the RCD to align the BCS and BCOM signals to BCK, the RCD must be able to support the DDR5 Data Buffer BCOM training mode. During RCD BCOM Training, the host will enable BCOM pass-through mode in [RW00\[4\]](#). The RCD will need to process MRW command and block DRAM commands while in BCOM pass-through mode. Power down mode must be disabled in RCD prior to BCOM training.

5.2.5 BCOM Pass-Through Mode

In order to train the Data buffer interface, the RCD will be required to support a BCOM Pass-Through mode. In this mode the DCA4, DCA3, DCA1 signals are passed on the BCOM bus regardless of DCS assertion. DCS0_n is passed through to BCS_n but DCS1_n does not get passed through in this mode. Instead the RCD will process RCD targeted MRWs to Rank1 shown in Figure 47. This allows the host to control combinations of BCOM and BCS patterns during BCOM Training. The DRAM interface Blocking must be set by the Host during BCOM Training. The DDR5RCD02 is not required to transfer commands to the QCA outputs when BCOM pass-through mode is enabled.

The RCD BCOM Pass-through mode is enabled when [RW00\[4\]](#) = 1. The DDR5RCD02 does not support BCOM pass-through mode and CA pass-through mode enabled at the same time.

In BCOM Pass-Through mode the RCD samples the DCA4, DCA3 and DCA1 inputs on DCK rising edge in both SDR and DDR Modes.

Table 21 — BCOM Bus Mapping

DDR5RCD DCA Bit	BCOM Bit
DCA4	BCOM0
DCA3	BCOM1
DCA1	BCOM2

To achieve 2N timing on the BCOM Interface, the Host must hold the value for two tCK.

The RCD will exit BCOM Pass-Through with MRW to [RW00\[4\]](#) = 0.

5.2.5 BCOM Pass-Through Mode (cont'd)

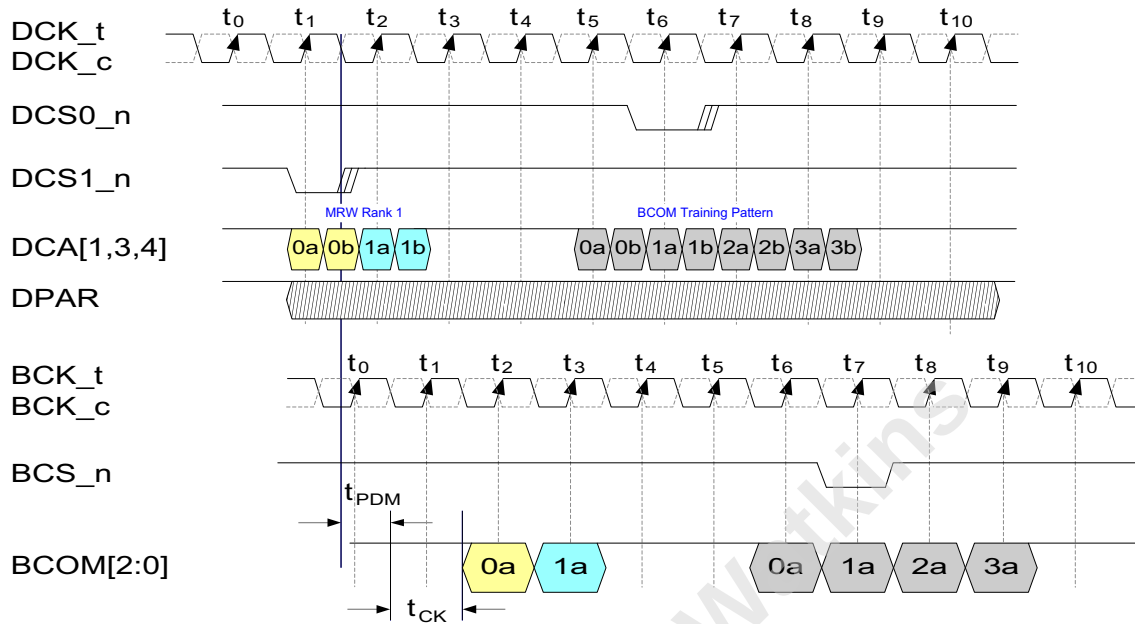


Figure 47 — BCOM Pass-Through

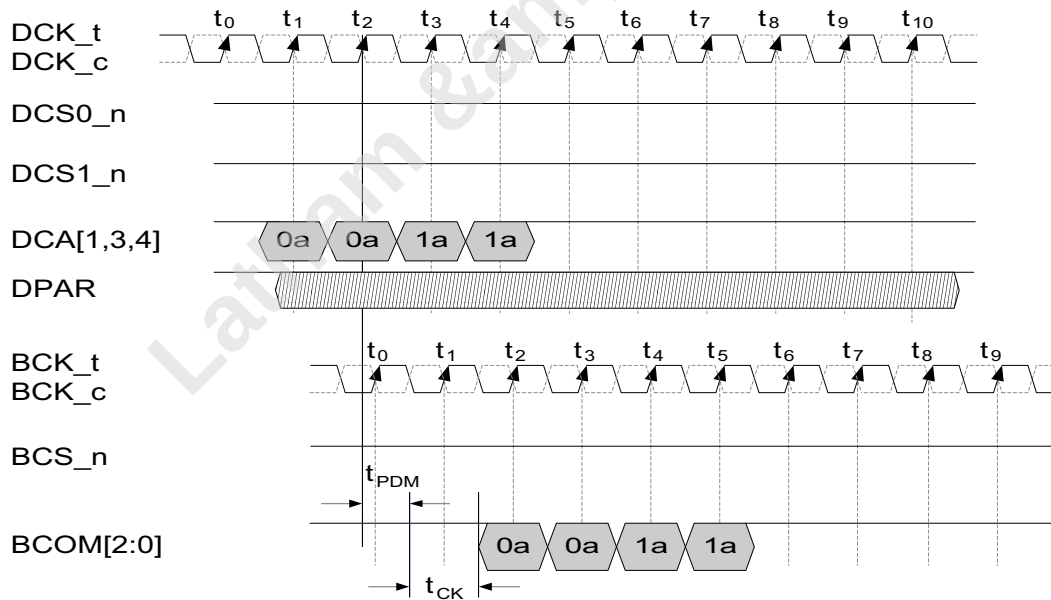


Figure 48 — BCOM Pass-Through Mode when RCD is in SDR

5.3 DFE Training Mode (DFETM)

The DDR5RCD02 will support a DCA DFE training mode. This Training Mode enables the use of pattern comparison and feedback features within the RCD during training. Figure 49 shows some of the DFE Training Circuitry associated with each individual DCA receiver. These features include a margin monitor, an XOR to compare the functional receive path to the monitor output, and LFSR pattern generator and XOR to compare the monitor output to a known expected pattern, an error counter, and a path to the ALERT_n signal, QLBD signal, and QLBS signal to provide feedback to the host controller. The host must ensure Parity Checking is disabled in RCD prior to and during DFETM. Power Down mode must be disabled in RCD prior to and during DFETM.

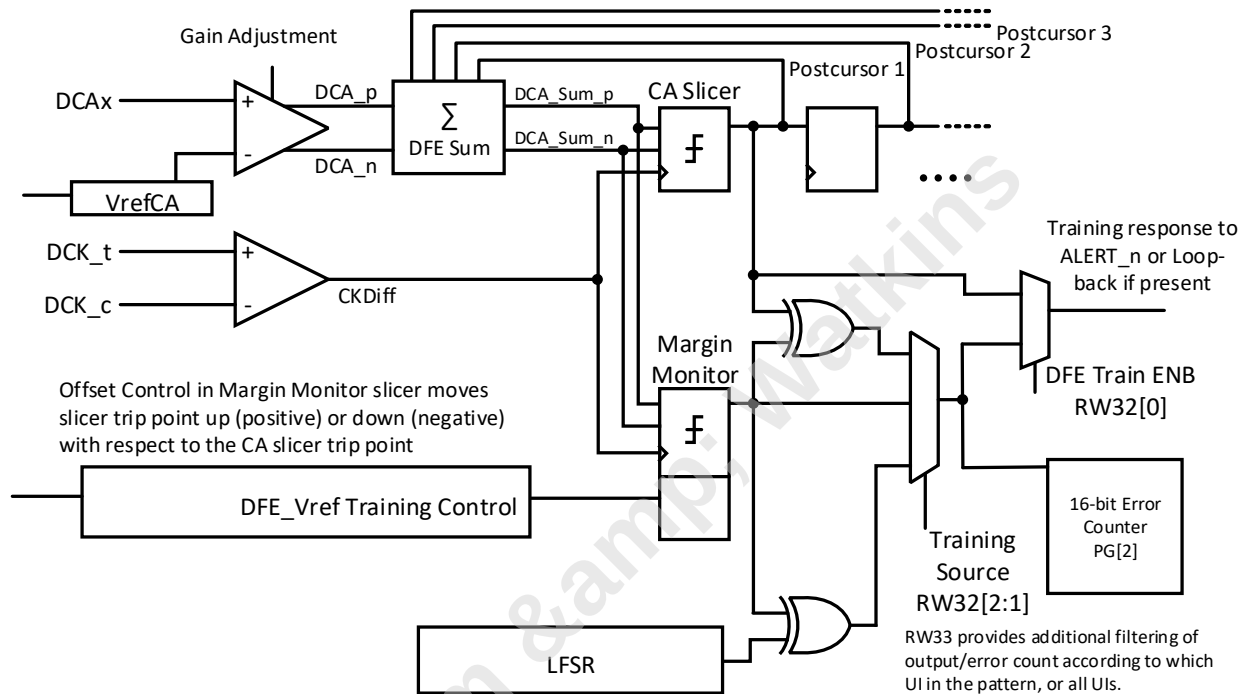


Figure 49 — DFE Training Circuitry

In addition to this fundamental set of DFE training features, the RCD supports an accelerator to sweep combinations of DFE Tap settings, Vref settings, and UI filters to enable more efficient DCA DFE Training.

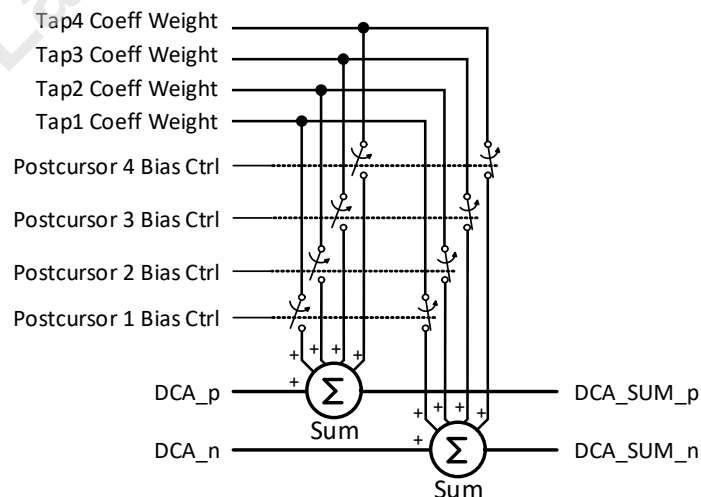


Figure 50 — DFE Summer

5.3 DFE Training Mode (DFETM) (cont'd)

Table 22 — Post-cursor Bias Ctrl Definition

Tap k Sign Bit	Post-cursor k	Switch to DCA_p Sum	Switch to DCA_n Sum
0	0	Open	Closed
	1	Closed	Open
1	0	Closed	Open
	1	Open	Closed

5.3.1 DFE Training Pattern Comparison and Feedback Features

The DFE Training Pattern Comparison features include many different configuration options to compare the output of the receiver to an expected pattern, and the ability to change the monitor DFE_Vref setting, for margining. The RCD also supports the ability to filter the comparison based on UI location in the pattern sequence. Some of these configuration settings are included in RW32 DPAR and DCA[6:0] DFE Training Mode Control Word. The UI Filtering configuration are included in RW33 Additional Filtering for DFE Training Mode Control Word. The LFSR Seed Value sets the initial state of the LFSR, and all UI's of the DCA pattern are compared to the output of the LFSR. A new seed setting is only applied when the PG[2]RW[63, 67, 6B, 6F, 73, 77, 7B, 7F] LFSR Seed value is written. This seed is never restored as the LFSR state during training unless the host writes the value to the PG[2]RW[63, 67, 6B, 6F, 73, 77, 7B, 7F] register again.

The LFSR is reset to its default state under the following conditions:

- Power-up initialization
- DRST_n assertion

Other reset conditions, like Self Refresh and Power-down Entry, are not applicable because the LFSR training pattern generator of the RCD is only supported in DFE Training Mode, where command decoding is not supported.

The LFSR is an 8-bit Galois LFSR. The polynomial for the Galois LFSR is $x^8 + x^6 + x^5 + x^4 + 1$. When a seed is programmed in the PG[2] register field, the source of the LFSR pattern must take into account the Galois LFSR implementation. The LFSR state advances once per UI in the DFE pattern, only when DCS0_n is asserted.

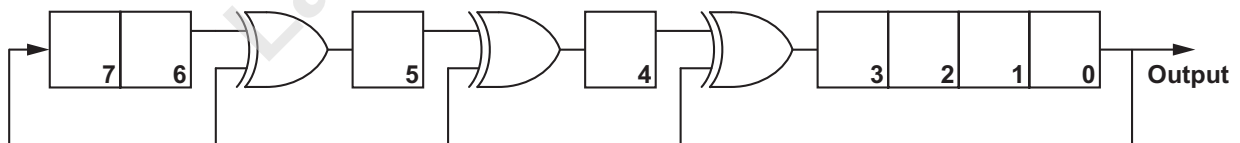


Figure 51 — Logical Representation of the Galois LFSR of Polynomial $x^8 + x^6 + x^5 + x^4 + 1$ for DCA Training

When UI Filtering is enabled, only the XOR comparison or the LFSR comparison, or the raw data (based on Training Source setting) corresponding to the UI Filter Setting is fed back to the ALERT_n (or QLBD or QLBS) signal. This is also the only UI (or group of UI's for the even/odd configuration) that triggers any events in the error counter. When UI Filtering is enabled and set to UI[n] as the UI Filter Setting, only the nth UI of the DCA pattern will be compared/recorded/looped back. When Per-UI Filtering is enabled, only patterns that are 8 UI in length can be sent to the RCD by the host. For all other patterns, the pattern length must be a multiple of 8 UI. When either the Even UI Filtering is enabled or the Odd UI Filtering is enabled, the pattern length must also be a multiple of 8 UI. The even UI's are all rising edge samples. The odd UI's are all falling edge samples.

5.3.1 DFE Training Pattern Comparison and Feedback Features (cont'd)

The following timing diagram shows an example of the LFSR pattern sent by the host to the RCD. Within each tDFETM_Interval time period (which is 4tCK), if any mismatches occur, the RCD will pull the ALERT_n (or QLBD or QLBS) signal LOW after tDFETM_Valid. If no mismatches occur during the tDFETM_Interval time period, the ALERT_n (or QLBD or QLBS) signal will remain HIGH.

Configuration of the output signal for the DFETM output is set in the RW01 Parity and Alert Global Control Word.

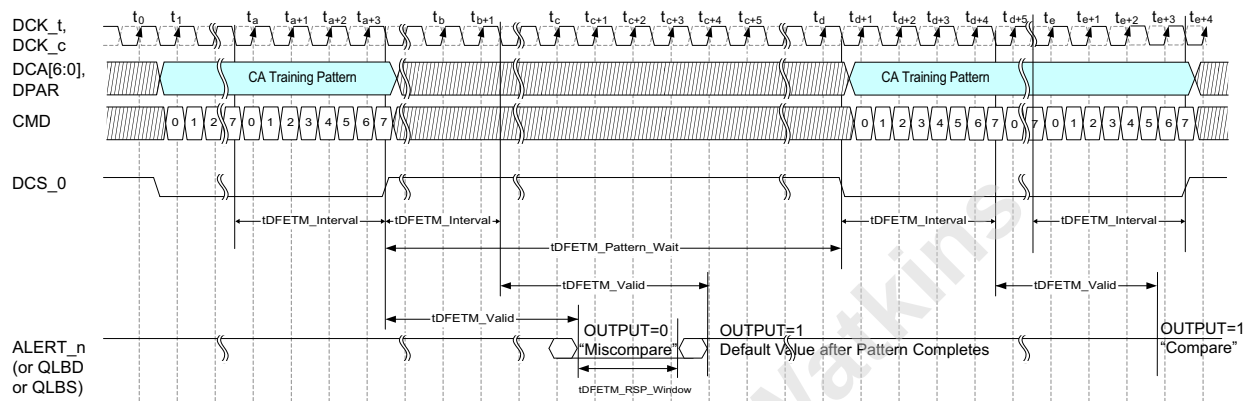


Figure 52 — LFSR CA DFE Training Pattern

The first tDFETM_Interval begins on the first UI of the pattern. Once the first tDFETM_Interval has elapsed (an additional 8 UI, or 4tCK) the first Output is generated on the ALERT_n (or QLBD or QLBS) pin, following the tDFETM_Valid time. Every tDFETM_Interval after will also generate a new Output value. For the different training configuration options, the following feedback will apply:

LFSR XOR - If any XOR result in the tDFETM_Interval = 1, the ALERT_n (or QLBD or QLBS) output will be 0.

Monitor XOR - If any XOR result in the tDFETM_Interval = 1, the ALERT_n (or QLBD or QLBS) output will be 0.

Monitor - If any Monitor sample in the tDFETM_Interval = 1, the ALERT_n (or QLBD or QLBS) output will be 0.

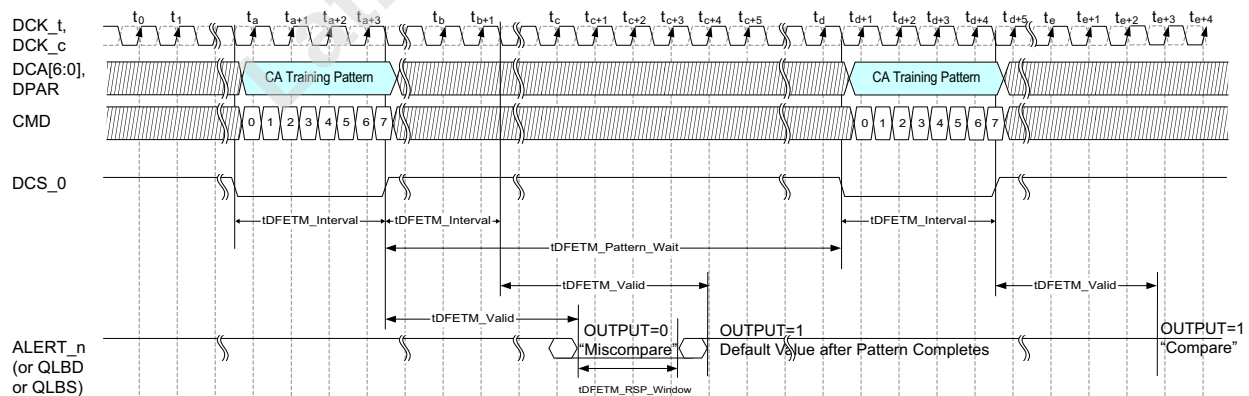
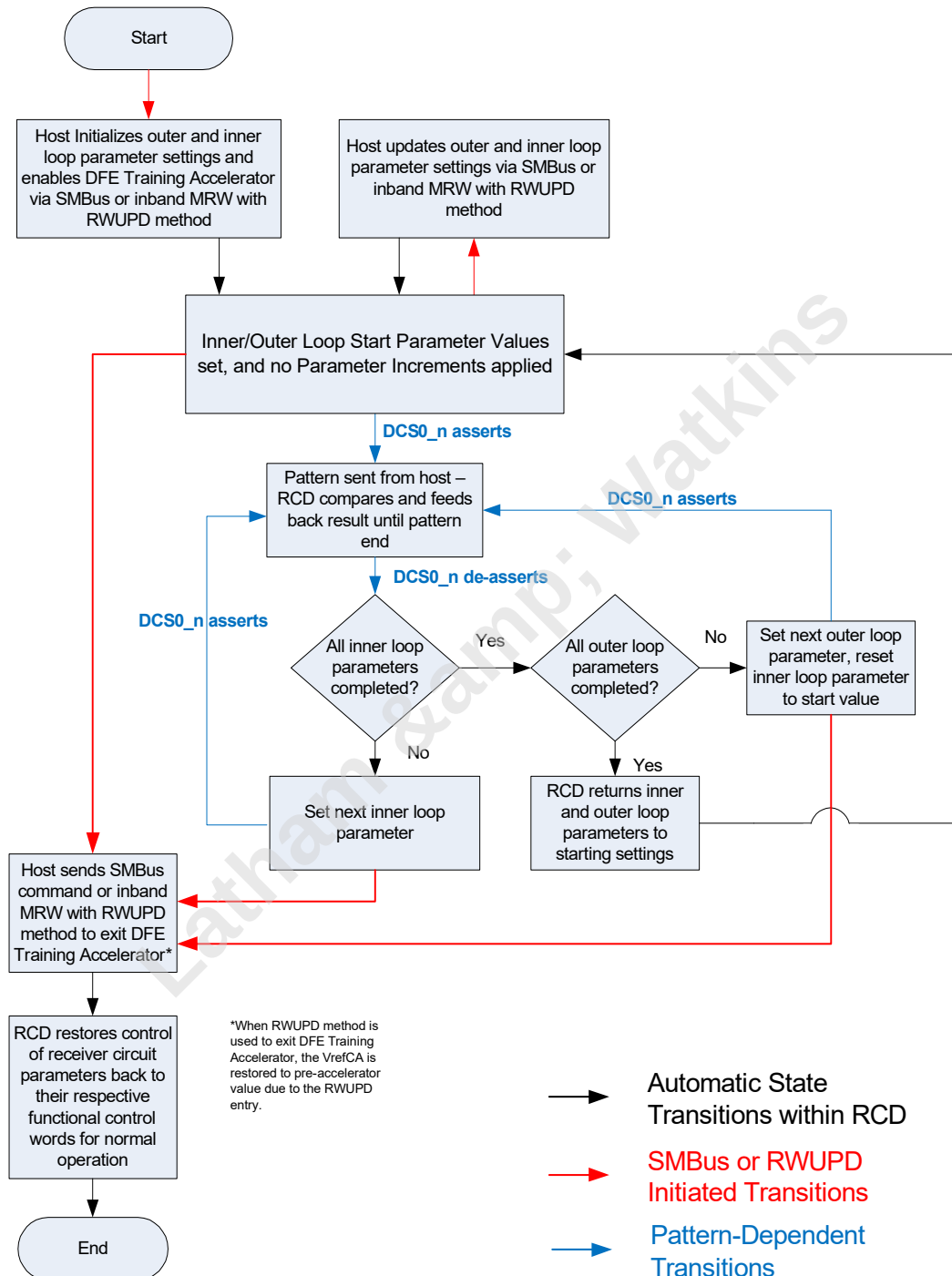


Figure 53 — CA DFE UI Filter Pattern

When UI filtering is enabled, only the (Monitor or LFSR) XOR result or the Monitor sample associated with the selected UI is fed back on the ALERT_n (or QLBD or QLBS) signal. As described above, the (Monitor or LFSR) XOR or Monitor sample result is inverted on the ALERT_n (or QLBD or QLBS) signal.

5.3.2 DFE Parameter Sweep Acceleration

In addition to the features used to compare patterns and record or feedback the results of those comparisons, the RCD shall support the ability to accelerate the training sweeps of the critical DFE training parameters.



Note: To read Loop Parameter Current Values when DFE Parameter Sweep Acceleration is enabled in the RCD, the host must use the sideband management interface and it must keep DCS0_n HIGH to stop training patterns. If these conditions are satisfied, the RCD shall provide the Loop Parameter Values intended for the next training pattern.

Figure 54 — RCD Operation During DFE Parameter Sweep Acceleration

5.3.2 DFE Parameter Sweep Acceleration (cont'd)

The parameters that are included as options for the outer loop sweep are:

- UI - (RW33[4:2])
- Tap1 - (PG[1:0]RW[61,69,71,79], depending on DCA or DPAR signal selected for training in the RW32[5:3] register)
- Tap2 - (PG[1:0]RW[62,6A,72,7A], depending on DCA or DPAR signal selected for training in the RW32[5:3] register)
- Tap3 - (PG[1:0]RW[63,6B,73,7B], depending on DCA or DPAR signal selected for training in the RW32[5:3] register)
- Tap4 - (PG[1:0]RW[64,6C,74,7C], depending on DCA or DPAR signal selected for training in the RW32[5:3] register)
- Null

The parameters that are included as options for the inner loop sweep are:

- VrefCA - (RW[47:40])
- DFE_Vref - (PG[2]RW[62,66,6A,6E,72,76,7A,7E], depending on DCA or DPAR signal selected for training in the RW32[5:3] register)
- Null

The tDFETM_Pattern_Wait time is the amount of time the host must wait between de-assertion of DCS0 for one pattern to DCS0 assertion of the next pattern. The tDFETM_Pattern_Wait_Accel restriction only applies when the DFE Training Accelerator is operating, and is defined to allow for enough time in the RCD to change settings for the inner and outer loop parameters. The actual register values associated with the inner and outer loop parameters may or may not change during the execution of the DFE training accelerator. The settings may be applied directly from the accelerator logic output, bypassing the functional operation registers.

The outer loop tap parameters are defined as signed magnitude fields. For the Outer Loop Start Value in RW37, the format is two's complement. This simplifies the logic for incrementing from negative to positive values. The RCD shall convert the final parameter value per increment to the signed magnitude values required by the actual Tap settings.

The host is responsible for avoiding incorrect combinations of the loop parameter control word settings.

When the DFE accelerator exits, the inner and outer loop parameters are returned to the values prior to enabling the accelerator.

5.3.3 Method to Update RW In-band During Training

When DDR5RCD02 is in the wait state of the DFE training mode and if DCS1_[B:A]_n is asserted then the device shall enter a mode where it allows multi cycle RW writes as specified in the following paragraphs. The wait state of the DFE training mode is defined as any state in which no iteration is active and the RCD can accept changes to the DFE training accelerator parameters.

The mode is referred to as RWUPD and it is specified as follows:

- RWUPD Entry:
DDR5RCD02 is in training mode non active states (outside of inner and outer loop iterations) and DCS1_[B:A]_n is asserted. On entry, the device shall restore DCA VrefCA trained during DCATM and DFE shall be disabled. Figure 55 and Figure 56 show how entering RWUPD and being ready to receive the first RW address requires two DCS1_[B:A]_n de-assertions.
- RWUPD Operation:
After entry in RWUPD mode the DDR5RCD02 will expect pairs of DCS1_[B:A]_n assertions. On the first assertion the device will sample the target RW address on {DCA[6:0]_[B:A], DPAR_[B:A]}. On the second assertion, it will sample the value to write to the target RW.

5.3.3 Method to Update RW In-band During Training (cont'd)

- **RWUPD Exit:**
If **RW32[7]** is written to 1 then device shall exit the RWUPD branch and return to the starting training mode state. This exit **RW32** shall also be SMBUS writable to allow the exit being also triggered out of band. The host is responsible for exiting the RWUPD before exiting the DFE training mode. {DCA[6:0]_B:A}, DPAR_B:A} must remain static High during $t_{RWUPD\ Exit}$ to ensure deterministic initialization of DFE circuits as they get re-enabled.

During the entire RWUPD duration, the host is not allowed to assert DCS0_[B:A]_n.

RW32[6] will need to be set to 1 to enable this feature.

Figure 55 shows a state machine implementing the described mode and Figure 56 shows a timing diagram demonstrating operation. Table 199 on page 191 has the relevant timing parameters.

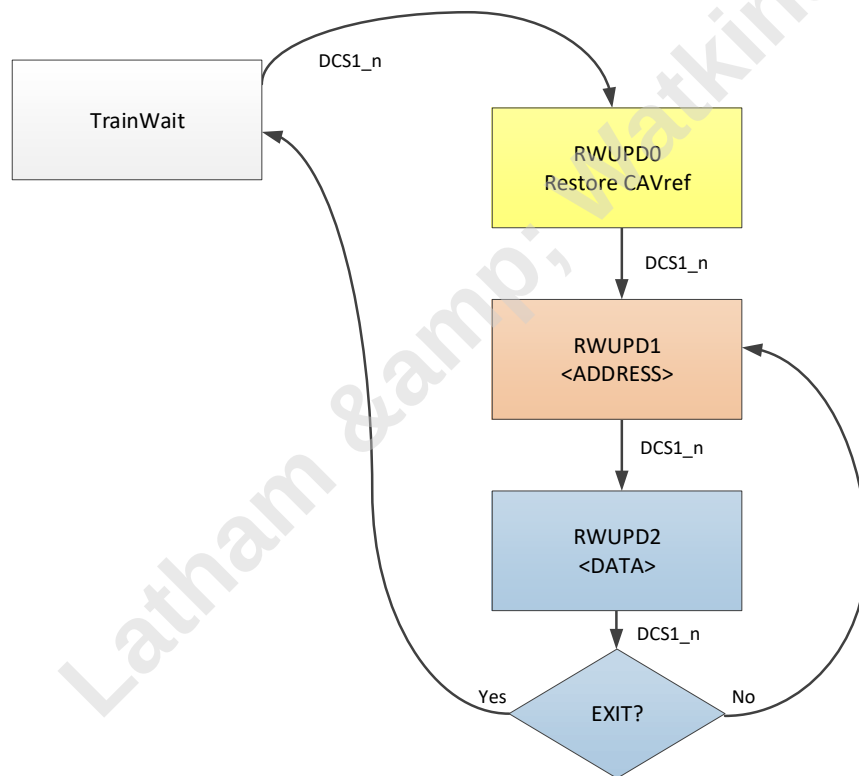


Figure 55 — RWUPD State Machine

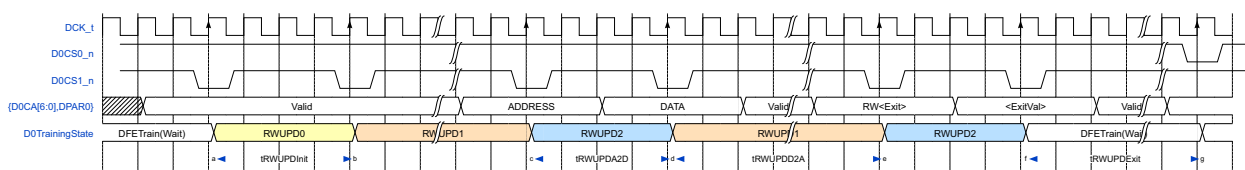


Figure 56 — RWUPD Timing Diagram for Channel A

5.4 Dual Frequency Support

The DDR5 register supports operation at a second, i.e. lower than nominal, frequency as a means to save buffer and DRAM power when the memory bandwidth demand allows.

To enable fast frequency switching without the need for retraining every time the frequency is changed, the DDR5 register can be trained at two, or more, different frequencies at boot up time. When only two frequencies need to be supported, the DDR5RCD02 can retain register settings associated with each of the two frequencies. The two sets of RCD registers are listed in Table 23. The RCD hardware will take care of updating any other frequency-dependent internal settings in each frequency context as needed. In cases where more than two frequencies of operation are necessary, RCD training information will need to be stored in memory space available to the host controller for this purpose since the DDR5RCD02 device only contains two sets of frequency context registers.

Table 23 — Control Words Duplicated for Frequency Context Support

Register Address	Description
RW0A	QCK Signals Driver Characteristics Control Word
RW0C	QxCA and QxCS_n Signals Driver Characteristics Control Word
RW0D	Data Buffer Interface Driver Characteristics Control Word
RW10	IBT Control Word
RW11[3:0]	Command Latency Adder Configuration Control Word
RW12	QACK Output Delay Control Word
RW13	QBCK Output Delay Control Word
RW14	QCKK Output Delay Control Word
RW15	QDCK Output Delay Control Word
RW17 .. RW1A	QxCS_n Output Delay Control Word
RW1B, RW1C	QxCA Output Delay Control Word
RW1D	BCS_n and BCOM[2:0] Output Delay Control Word
RW1E	BCK Output Delay Control Word
RW31[7:4, 0]	DFE Configuration Control Word
RW40 .. RW47	Internal VrefCA Control Word
RW48, RW49	Internal VrefCS Control Word
RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words
PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver
PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words

Switching between the two frequency contexts is achieved by writing to control word **RW05**. **RW05[6]** = 0 selects the default Frequency Context 1 while **RW05[6]** = 1 selects Frequency Context 2. Changing the setting in **RW05[6]** by itself only affects which copy of frequency dependent registers responds to MRW writes, MRR reads and SMBus transactions. However, when **RW05[6]** is different from the value it had in the previous Exit from Clock Stop event, the new settings any updates in **RW05[3:0]**, **RW06**, and the dual-context registers listed in Table 23 will not be applied internally in the RCD until after the RCD and DRAMs have entered and exited Self Refresh with Clock Stop mode. The RCD will use the value of **RW05[6]** at the time of Exit from Self Refresh with Clock Stop to determine which context to switch to. A context switch only occurs when the value of **RW05[6]** is different from the value it had in the previous Exit from Self Refresh with Clock Stop event.

5.4.1 Initial Frequency Setting

Since **RW05[6]** is a sticky control bit, that is, only cleared by power cycle, the internal current-context state maintained by the RCD must also be sticky. After DRST_n event with stable power, the internal current-context state and **RW05[6]** will retain the same values they had prior to the DRST_n event.

After Power-on Reset, the internal current-context state will match the Context-1 setting **RW05[6]** = '0'. Therefore, in this case, any updates in **RW05[3:0]**, **RW06**, and the dual-context registers listed in Table 23 will be applied immediately to internal circuits and logic without waiting for Exit from Self Refresh with Clock Stop.

5.4.2 Input Clock Frequency Change

Once the DDR5RCD02 is initialized, the DDR5RCD02 requires the clock to be stable during almost all states of normal operation. This means that, once the clock frequency has been set and is to be in the stable state, the clock period is not allowed to deviate except for what is allowed for by the clock jitter and SSC (spread spectrum clocking) specifications.

The input clock frequency can be changed from one stable clock rate to another stable clock rate only by going through Self Refresh with Clock Stop power down mode. The sequence must allow the host controller to update the frequency dependent registers in the background during normal operation. The sequence of steps for an input frequency change to an RDIMM is outlined below. Note that additional steps are required for an input frequency change to an LRDIMM, and they are documented in the DDR5DB02 specification.

1. Set [RW05\[6\]](#) to choose a frequency context different from the current one. At the same time, update the Frequency Information settings for the next operating speed in [RW05\[3:0\]](#). These updates will only be applied internally after the RCD exits from Self Refresh with Clock Stop.¹
2. Write Fine Granularity Frequency information for the next operating speed in [RW06](#). The new setting will only be applied internally after the RCD exits from Self Refresh with Clock Stop.¹
3. Set new values for the frequency specific registers listed in Table 23 in the background as needed during normal operation. These changes will only be applied internally after the RCD exits from Self Refresh with Clock Stop mode.
4. Go through Entry and Exit from Self Refresh with Clock Stop. The RCD will apply the new settings upon Exit from Self Refresh with Clock Stop.

With the exception of Self Refresh Entry, normal DRAM traffic and operations are supported intermingled with Steps 2 and 3 of the transition sequence defined above.

1. Due to postponed application of [RW05/RW06](#), a t_{STAB01} waiting time is always sufficient after exit from Self Refresh with Clock Stop (i.e., t_{STAB02} is not required). See timing diagrams in Figure 35 on page 48 and Figure 36 on page 49.

6 Data Buffer Control Bus

This clause describes the RCD signals used in the DDR5 LRDIMM application. The Data Buffer interface connects the DDR5 Register with each of the ten DDR5 Data Buffer (DB). This interface is enabled on power-on and can only be disabled by setting both [RW08\[5\]](#) and [RW09\[3\]](#) to ‘1’.

6.1 Control Bus Signals

Table 24 — List of Signals for Data Buffer Control Signals

Name	Description	Signal Count per Channel
B[1:0]CS_n	BCOM Chip Select	1
B[1:0]COM[2:0]	Data buffer command signals	3
B[1:0]RST_n	BCOM Reset	1
B[1:0]CK_t, B[1:0]CK_c	Clock outputs for the data buffers	2
Total		7

6.1.1 Control Bus Signal Termination

BCK_t, BCK_c, BCS_n and BCOM[2:0] signals are terminated to VDDQ on the DDR5 LRDIMM as shown in Figure 57.

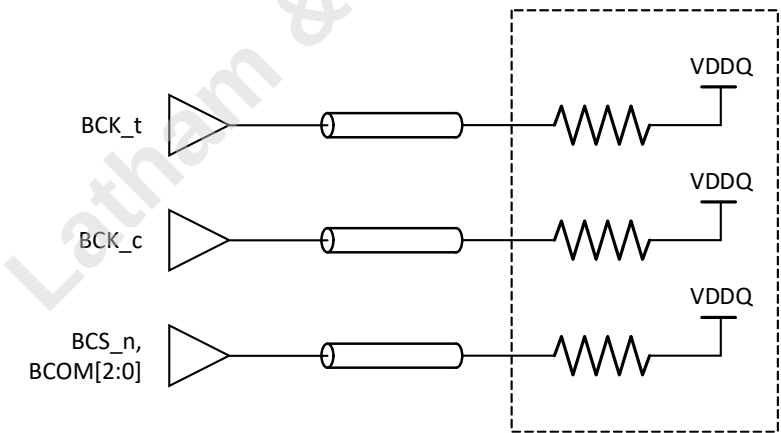


Figure 57 — Control Bus Termination

6.2 Control Bus Timing

The output BCOM signals are driven one clock later than the QxCA_N outputs as shown in Figure 58. Like the QxCA_N outputs, they are delayed by the Command Latency Adder Control Word [RW11](#).

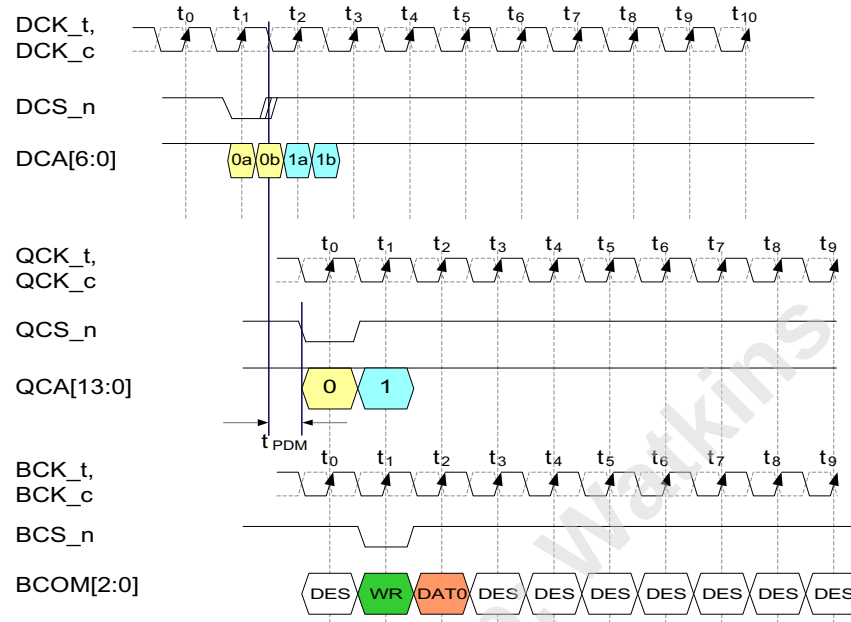


Figure 58 — Example BCOM WR Command

BCOM[2:0] are encoded such that DDR5 command bits can be used directly from the first UI of the command sequence on the RCD host interface. The mapping is as follows:

Table 25 — BCOM Bus Mapping

DDR5RCD CA Bit	BCOM Bit
CA4	BCOM0
CA3	BCOM1
CA1	BCOM2

6.3 Control Bus Commands

6.3.1 Data Buffer Control Bus Command Truth Table

Table 26 — Data Buffer Control Bus Command Truth Table

Command	Description	BCS_n	BCOM[2:0] Encoding
MRW	Mode Register Write	L	000
MRR	Mode Register Read	L	001
WR	Write	L	010
RD	Read	L	011
RFU	Reserved for Future Use	L	100
PDE	Power Down Entry	L	101
MPC	Multi-purpose CMD	L	110
NOP / PDX	Power Down exit if in power down mode.	L	111
DES	BCOM De-select	H	xxx

6.3.2 Command Sequences

Some commands require more than one clock cycle in order to send additional information needed by the Data Buffer in the execution of the command. This succession of command and its corresponding data transfers is called a command sequence. The command sequence for Read or Write commands requires one additional cycle to transfer the rank number corresponding to the write or read command, the sequence for MRW requires seven clock cycles in addition to the basic command, the sequence for the MRR command uses four additional clock cycles. The MPC requires three additional cycles and the PDE command requires only one additional clock cycle. The RCD will drive BCOM DES on the BUS between commands. The Host must guarantee valid DCA[6:0], DPAR, and DCS input signal levels and timings before using BCOM command sequences, including MPC.

6.4 Command Sequence Descriptions

The timing diagrams in this clause show only the lower nibble of the Data Buffer. The same timing relationships apply independently also for the upper nibble. The timing diagrams only show the case of burst length = 16, preamble = 1 nCK and postamble = 0.5 nCK, but equivalent timing relationships exist for burst length = 8 and 18, preamble = 2 nCK and postamble = 1.5 nCK.

For readability reasons, the timing diagrams use the abbreviations DB_RL and DB_WL for the latency between second cycle of RD command and the first rising edge of MDQS at the Data Buffer inputs and for the latency between second cycle of WR command and the first rising edge of MDQS at the Data Buffer outputs respectively. Both DB_RL and DB_WL include full cycle and fractional cycle delays. The equations for these latencies are as follows:

$$\text{BCOML} = 1 \text{ nCK}$$

$$\text{DB_WL(R)} = \text{CWL} + \text{DWL(R)} - \text{BCOML}$$

$$\text{DB_RL(R)} = \text{CL} + \text{MRE(R)} - \text{BCOML}$$

Fractional cycle delay equations for DQLDWL and MRE for Ranks 0 and 1 are listed below.

xxx[R].l and xxx[R].u are the equations for the Lower and Upper nibbles, respectively.

$$\text{DWL}[0].l = (\text{PG}[0]\text{RWE1}[3:0] * 64 + \text{PG}[0]\text{RWE8}[5:0]) * t_{\text{CK}}/64.$$

6.4 Command Sequence Descriptions (cont'd)

$$DWL[0].u = (PG[0]RWE1[7:4] * 64 + PG[0]RWE9[5:0]) * t_{CK}/64.$$

$$DWL[1].l = (PG[1]RWE1[3:0] * 64 + PG[1]RWE8[5:0]) * t_{CK}/64.$$

$$DWL[1].u = (PG[1]RWE1[7:4] * 64 + PG[1]RWE9[5:0]) * t_{CK}/64.$$

$$MRE[0].l = (PG[0]RWE0[3:0] * 64 + PG[0]RWE2[5:0]) * t_{CK}/64.$$

$$MRE[0].u = (PG[0]RWE0[7:4] * 64 + PG[0]RWE3[5:0]) * t_{CK}/64.$$

$$MRE[1].l = (PG[1]RWE0[3:0] * 64 + PG[1]RWE2[5:0]) * t_{CK}/64.$$

$$MRE[1].u = (PG[1]RWE0[7:4] * 64 + PG[1]RWE3[5:0]) * t_{CK}/64.$$

Table 27 — Data Buffer PG[1:0] Rank Training Control Word Decoding

DB Control Word	Function
RWE0	Lower/Upper Nibble Additional Cycles of DRAM Interface Receive Enable Control Word
RWE1	Lower/Upper Nibble Additional Cycles of DRAM Interface Write Leveling Control Word
RWE2	Lower Nibble DRAM Interface Receive Enable Training Control
RWE3	Upper Nibble DRAM Interface Receive Enable Training Control
RWE8	Lower Nibble DRAM Interface Write Leveling Control
RWE9	Upper Nibble DRAM Interface Write Leveling Control

The Data Buffer delays tPDM_RD and tPDM_WR are defined as the delay between first effective rising edge of MDQS and the first rising edge of DQS for a RD command and as the delay between first effective rising edge of DQS and the first rising edge of MDQS for a WR command respectively. By default these delays are constant per Data Buffer for all ranks and nibbles, however the DB may need to adjust tPDM for non-default register settings in order to maintain the data path fall through time in the device.

Timing parameters that are integer multiples of tCK are shown in **bold blue** letters while timing parameters that are analog non-integer multiples or fractions of tCK are shown in *red italic* letters.

For simplicity the timing diagrams only show one of the possible host interface termination modes and RTT_PARK is never shown even though it is always supported.

6.4.1 WR/RD Burst Length Processing

Table 28 — WR/RD Burst Length Processing

DRAM MR0 Snooped BL Type PG8RWE0[1:0]	DRAM MR50 Snooped WR/RD CRC Enabled PG8RWE8[1:0]	WR/RD Sequence DAT0 BCOM[0]	DB Burst Length Processing (BL)	DB Burst Length Processing Details
00 = BL16 Fixed	Disabled	X	16	
	Enabled	X	18	
01 = BC8 OTF	Disabled	0 = BC8	8	
		1 = BL16	16	
	Enabled	X	18	
10 = BL32 Fixed	Disabled	X	32	The dummy WR/RD command that is 8 tCK after the primary WR/RD command is ignored. ¹
	Enabled	X	36	The dummy WR/RD command that is 9 tCK after the primary WR/RD command is ignored. ¹
11 = BL32 OTF	Disabled	0 = BL32	32	The dummy WR/RD command that is 8 tCK after the primary WR/RD command is ignored. ¹
		1 = BL16	16	
	Enabled	0 = BL32	36	The dummy WR/RD command that is 9 tCK after the primary WR/RD command is ignored. ¹
		1 = BL16	18	
NOTE 1 The dummy WR/RD command is assumed to be to the same rank and of the same burst length type as the primary WR/RD command.				

6.4.2 MRR Burst Length Processing

Table 29 — MRR Burst Length Processing

DRAM MR0 Snooped BL Type PG8RWE0[1:0]	DRAM MR50 Snooped Read CRC Enable (PG8RWE8[0])	Control Register Command Sequence	DB Burst Length Processing (BL)	DB Burst Length Processing Details
XX	Disabled	DRAM-Space MRR	16	
	Enabled		18	
XX	Disabled	DB-Space MRR to Control Word	16	
	Enabled		18	The last two bits of the BL18 burst are Don't Care (X).
XX	Disabled	DB-Space MRR to the Read Training Pattern	16	
	Enabled		Illegal	The DB assumes that the host will ensure that read CRC is disabled since it is necessary for the LFSR pattern to be kept continuous. Read CRC is unnecessary since the known LFSR pattern intrinsically allows for error checking.

6.4.2 MRR Burst Length Processing (cont'd)

Table 30 — PBA Enumerate ID Programming Burst Length Processing

DRAM MR0 Snooped BL Type (PG8RWE0[1:0])	DRAM MR50 Snooped Write CRC Enable (PG8RWE8[1])	Control Register Command Sequence	DB Burst Length Processing (BL)	DB Burst Length Processing Details
XX	Disabled	PBA Enumerate ID Programming MRW	16	Used to capture the Enumerate ID.
	Enabled		Illegal	The DB assumes that the host will ensure that Write CRC is disabled.

6.4.3 MRW Commands

The RCD generates MRW commands to the data buffers for each MR command. Table 31 shows the sequence for MRW commands.

Table 31 — Multi-cycle Sequence for MRW Commands

Time (clock cycle)	BCS_n	BCOM[2:0]	Description
0		Prev Cmd	Previous command or data transfer
1	0	MRW	MRW Command BCOM[2:0] = 000
2	1	DAT0	BCOM[0] = CW; 0 = DRAM space, 1 = DB space BCOM[2:1] = RFU
3	1	DAT1	BCOM[1:0] = MRA[7:6] BCOM[2] = RFU
4	1	DAT2	BCOM[2:0] = MRA[5:3]
5	1	DAT3	BCOM[2:0] = MRA[2:0]
6	1	DAT4	BCOM[1:0] = OP[7:6] BCOM[2] = RFU
7	1	DAT5	BCOM[2:0] = OP[5:3]
8	1	DAT6	BCOM[2:0] = OP[2:0]
9		Next Cmd	Next Command

The sequence for an MRS Write command is shown in Figure 59. The timing diagrams show how the MR Write command is followed by seven data transfer cycles. Since the command sequence uses eight cycles it is necessary to include these cycles as part of the tMRD parameter that indicates the spacing from the MRW command to the following valid command (also shown in the diagrams). The number of additional transfers on the BCOM bus after the MRW command also imposes a limitation on how close consecutive MRW commands can be issued to the Data Buffer.

6.4.3 MRW Commands (cont'd)

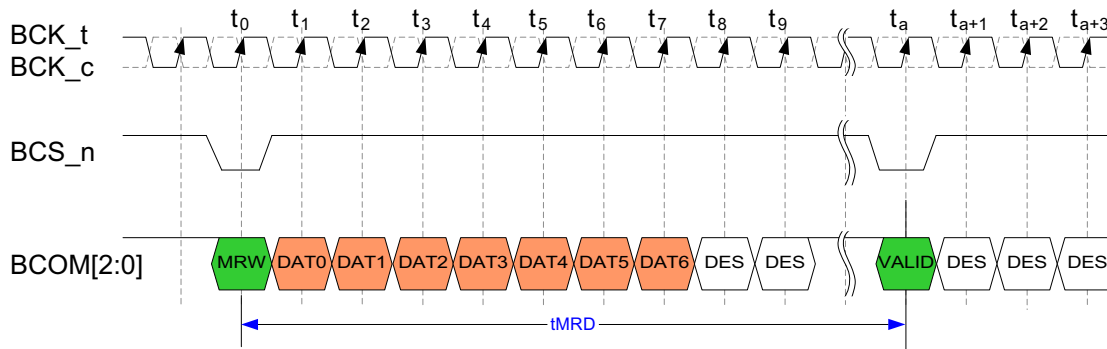


Figure 59 — MRW Command Sequence

6.4.4 MRR Commands

6.4.4.1 Multi-cycle Sequence for MRR Commands

Table 32 shows the sequence for MRR commands.

Table 32 — Multi-cycle Sequence for MRR Commands

Time (Clock Cycle)	BCS_n	BCOM[2:0]	Description
0		Prev Cmd	Previous command or data transfer
1	0	MRR	Mode Register Read Command BCOM[2:0] = 001
2	1	DAT0	BCOM[0] = CW; 0 = DRAM space, 1 = DB space BCOM[1] = RANK_ID; 0 = Rank 0, 1 = Rank 1 BCOM[2] = RFU Note: Burst length is fixed BL16
3	1	DAT1	BCOM[1:0] = MRA[7:6] BCOM[2] = RFU
4	1	DAT2	BCOM[2:0] = MRA[5:3]
5	1	DAT3	BCOM[2:0] = MRA[2:0]
6		Next Cmd	Next Command

MRR commands read data from the MR locations in either the DRAM or the data buffer. MR locations in the RCD must be read by first transferring their contents to a DRAM register then reading that DRAM register. If the CW bit is a 0 the read will come from the DRAMs, and the data buffer will treat this command like a read command. The case CW = 1 MRA7 = 0 is not defined. Data Buffer will treat it as invalid address and Read zeros. If the CW = 1 and MRA7 = 1, then read will come from the Data Buffer's MR location. The Data Buffer will send the same data on all DQs for the second 8 UI and drive High or Low on the first 8 UI as shown in Table 34.

6.4.4.1 Multi-cycle Sequence for MRR Commands (cont'd)

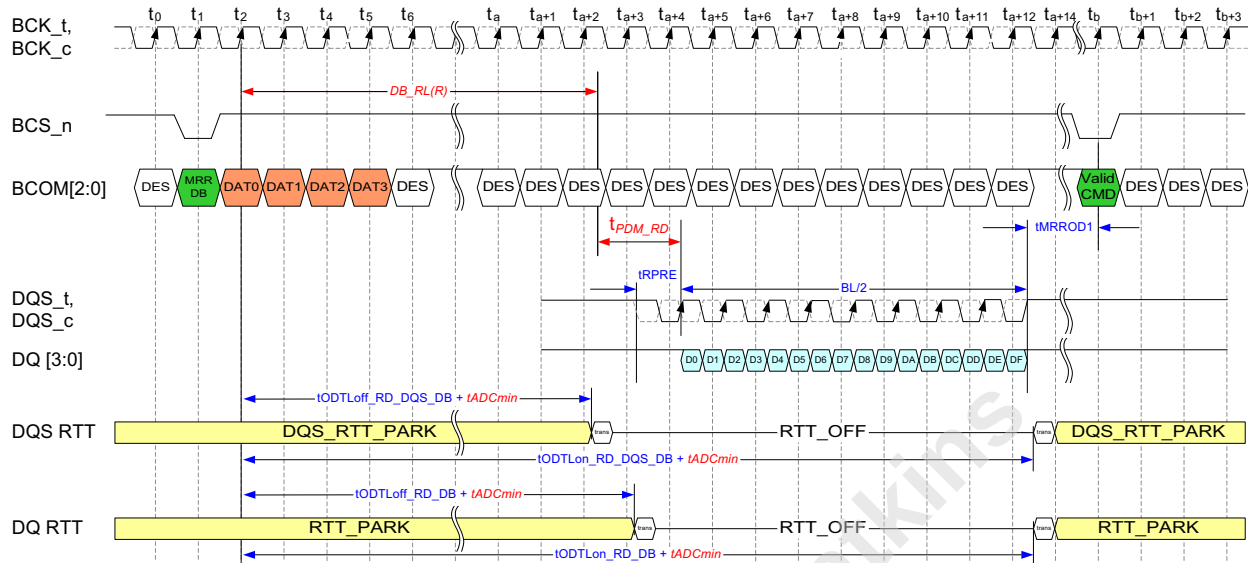


Figure 60 — MRR Command Sequence to Data Buffer Address Space

6.4.4.2 Short MRR Command for SDR Training

Table 33 — Short MRR Command for SDR Training¹

Time (Clock Cycle)	BCS_n	BCOM[2:0]	Description
0		Prev Cmd	Previous command or data transfer
1	0	MRR	Mode Register Read Command BCOM[2:0] = 001
2	1	DAT0	BCOM[0] = CW; 0 = DRAM space, 1 = DB space BCOM[1] = RANK_ID; 0 = Rank 0, 1 = Rank 1 BCOM[2] = RFU Note: Burst length is fixed BL16
3		Next Cmd	Next Command

NOTE 1 Provides a shorter MRR command for SDR mode Host interface Data Buffer DQ training to maintain a continuous read from Data Buffer pattern generator.

6.4.4.2 Short MRR Command for SDR Training (cont'd)**Table 34 — MRR Read Format for Data Buffer Return**

Serial	UI 0-7	UI 8	UI 9	UI 10	UI 11	UI 12	UI 13	UI 14	UI 15
DQ0	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ1	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ2	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ3	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ4	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ5	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ6	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ7	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7

Table 35 — MRR Read Format for Data Buffer Return with Read CRC Enabled

Serial	UI 0-7	UI 8	UI 9	UI 10	UI 11	UI 12	UI 13	UI 14	UI 15	UI 16	UI 17
DQ0	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	x	x
DQ1	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7	x	x
DQ2	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	x	x
DQ3	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7	x	x
DQ4	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	x	x
DQ5	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7	x	x
DQ6	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	x	x
DQ7	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7	x	x

6.4.4.2 Short MRR Command for SDR Training (cont'd)

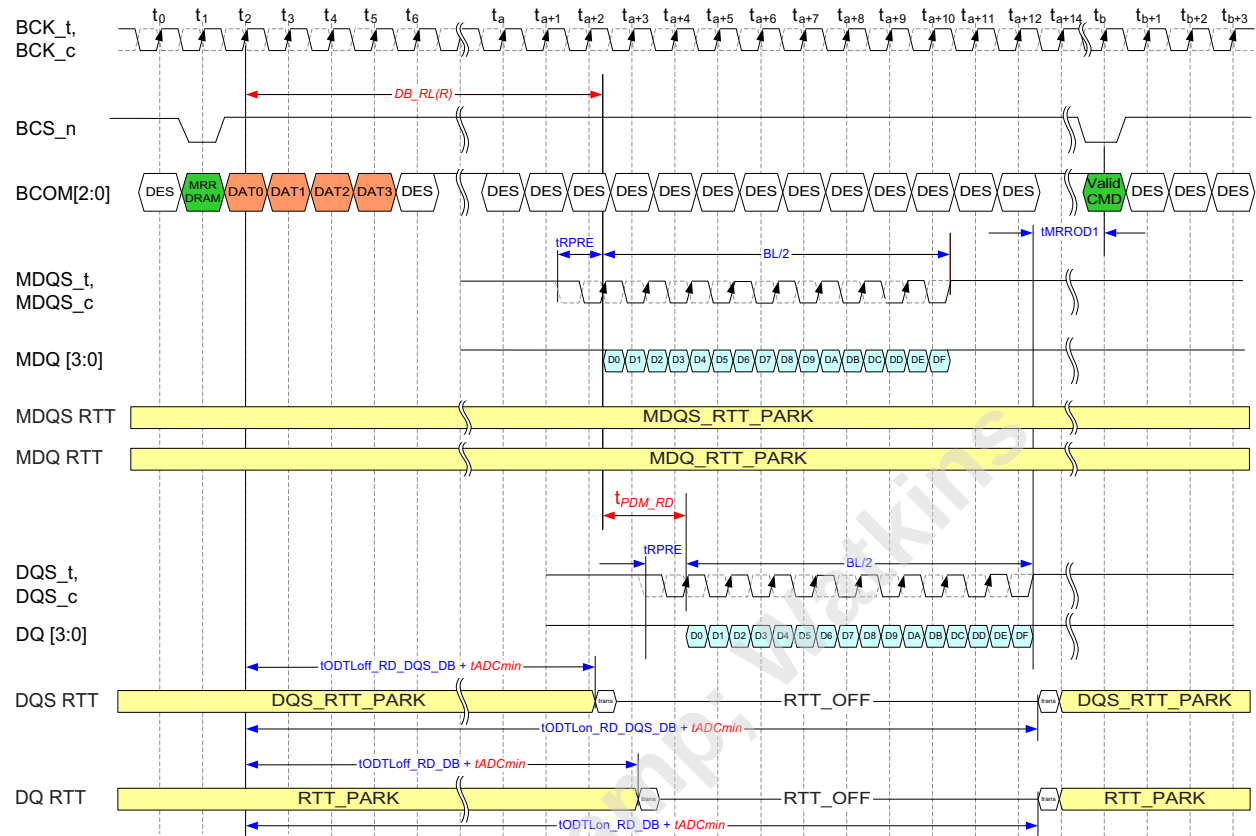


Figure 61 — MRR Command Sequence to DRAM Address Space

6.4.4.2 Short MRR Command for SDR Training (cont'd)

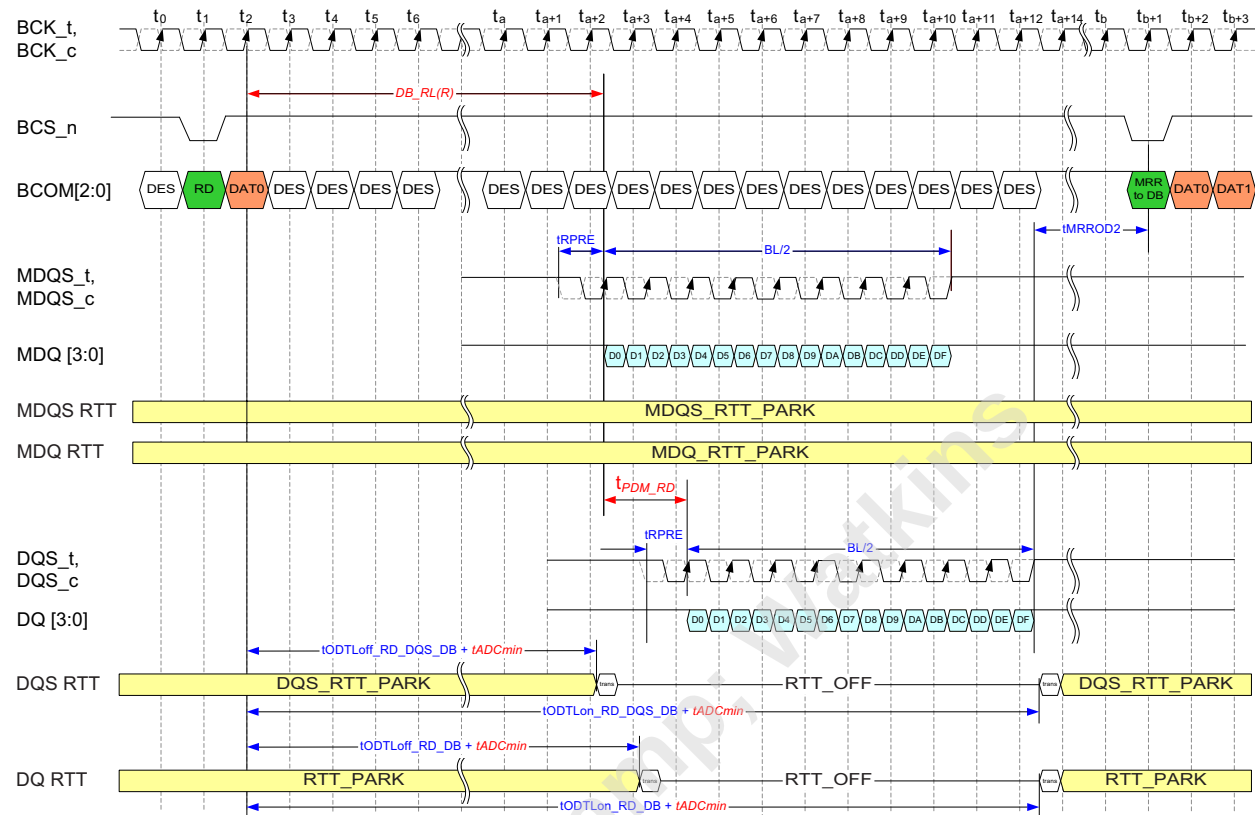
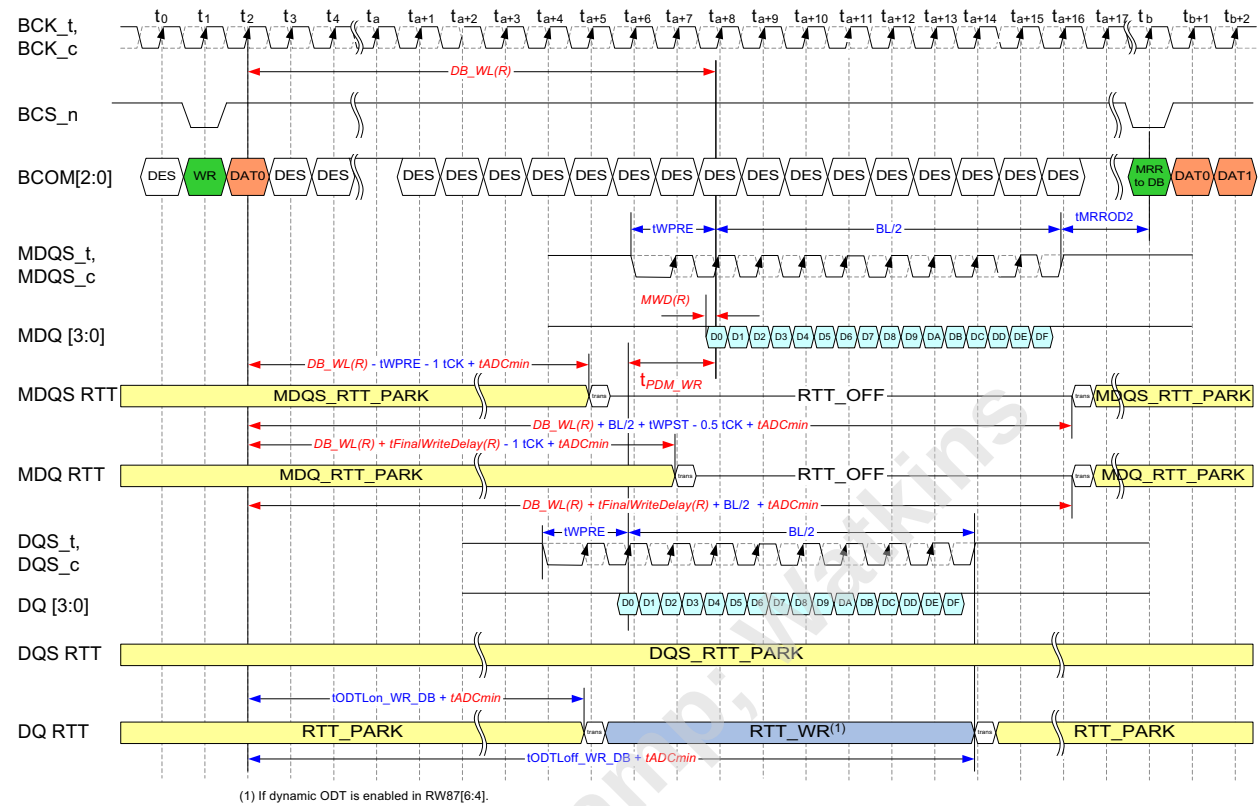


Figure 62 — RD to MRR Command Sequence to Data Buffer Address Space

6.4.4.2 Short MRR Command for SDR Training (cont'd)



Note $t_{FinalWriteDelay}(R) = t_{MdqWriteBaselineDelay}(R) + t_{Dram_DqsDelay_Change}(R)$

Figure 63 — WR to MRR Command Sequence to Data Buffer Address Space

6.4.5 WR Commands

Table 36 shows the sequence for write (WR8, WR16) commands. Each write command is followed by a one-cycle transfer that contains the rank ID that selects 1 rank out of 2 that needs to be accessed and the burst length information for the data transfer. The rank ID may be used to make rank-specific adjustments to the timing controls if necessary. If CRC is enabled in the DRAM and in the Data Buffer, the burst length will always be 18 or 36 UI.

Table 36 — Multi-cycle Sequence for WR Commands

Time (Clock Cycle)	BCS_n	BCOM[2:0]	Description
0		Prev Cmd	Previous command or data transfer
1	0	WR	Write command BCOM[2:0] = 010
2	1	DAT0	Rank ID & BL BCOM[0] = BL; 0=BC8 or BL32, 1=BL16 BCOM[1] = RANK_ID; 0 = Rank 0, 1 = Rank 1 BCOM[2] = RFU
3		Next Cmd	Next Command

If BCS_n is active for 2 clock cycles, it indicates a non-target write termination. The Data Buffer will terminate with RTT_NOM_WR and will not transfer data through the buffer. In this case DB will use Rank 0 timing automatically.

The diagram illustrates the timing relationships for the AD7124-1 ADC. Key signals and their timing parameters are as follows:

- BCK_t, BCK_c**: Clock signal with period t_0 to t_{a+20} .
- BCS_n**: Chip select signal.
- BCOM[2:0]**: Command signals (DES, WR, DATO, DES, DES, ...).
- MDQS_t, MDQS_c**: MDQS clock signal with period t_0 to t_{a+20} .
- MDQ [3:0]**: MDQS data signal (D0, D1, D2, D3, D4, D5, D6, D7, D8, D9, DA, DB, DC, DD, DE, DF).
- MDQS RTT**: MDQS Read Turnaround Time (RTT) signal.
- MDQ RTT**: MDQ Read Turnaround Time (RTT) signal.
- DQS_t, DQS_c**: DQS clock signal with period t_0 to t_{a+20} .
- DQ [3:0]**: DQS data signal (D0, D1, D2, D3, D4, D5, D6, D7, D8, D9, DA, DB, DC, DD, DE, DF).
- DQS RTT**: DQS Read Turnaround Time (RTT) signal.
- DQ RTT**: DQ Read Turnaround Time (RTT) signal.

Timing parameters and delays shown in the diagram include:

- $DB_WL(R)$: Delay from BCK_c to MDQS_c.
- IWP : Internal Write Pulse.
- $BL/2$: Delay from MDQS_c to MDQS_t.
- $MWD(R)$: MDQS Write Delay (R).
- t_{PDM_WR} : Delay from MDQS_c to MDQS_t.
- RTT_OFF : Read Turnaround Time Offset.
- $MDQS_RTT_PARK$: MDQS Read Turnaround Time Park.
- MDQ_RTT_PARK : MDQ Read Turnaround Time Park.
- $DB_WL(R) - IWP - 1\text{ tCK} + t_{ADCmin}$: Delay from BCK_c to MDQS_t.
- $DB_WL(R) + BL/2 + IWPST - 0.5\text{ tCK} + t_{ADCmin}$: Delay from MDQS_c to MDQS_t.
- $DB_WL(R) + t_{FinalWriteDelay}(R) - 1\text{ tCK} + t_{ADCmin}$: Delay from MDQS_c to MDQS_t.
- $DB_WL(R) + t_{FinalWriteDelay}(R) + BL/2 + t_{ADCmin}$: Delay from MDQS_c to MDQS_t.
- $IODT_{Lon_WR_DB} + t_{ADCmin}$: Delay from MDQS_c to MDQS_t.
- $IODT_{Loff_WR_DB} + t_{ADCmin}$: Delay from MDQS_c to MDQS_t.

(1) If dynamic ODT is enabled in RW87[6:4].

Note: $t_{FinalWriteDelay}(R) = t_{MdqWriteBaselineDelay}(R) + t_{DramDqsDelayChange}(R)$

Figure 64 — WRITE Timing

Table 37 shows the sequence for read (RD8, RD16) commands. Each read command is followed by a one-cycle transfer that contains the rank ID that selects 1 rank out of 2 that needs to be accessed and the burst length information for the data transfer. The rank ID may be used to make rank-specific adjustments to the timing controls if necessary. The BCOM[0] bit in DAT0 is only valid for on-the-fly burst length. This bit is ignored by the Data Buffer if the BL field in the snoop register is set for fixed burst length of 16 or 32. If CRC is enabled in the DRAM and in the Data Buffer, the burst length will always be 18 or 36 UI.6.4.6

6.4.6 RD Commands (cont'd)

Table 37 — Multi-cycle Sequence for RD Commands

Time (Clock Cycle)	BCS_n	BCOM[2:0]	Description
0		Prev Cmd	Previous command or data transfer
1	0	RD	Read command BCOM[2:0] = 011
2	1	DAT0	Rank ID & BL BCOM[0] = BL; 0 = BC8 or BL32, 1=BL16 BCOM[1] = RANK_ID; 0 = Rank 0, 1 = Rank 1 BCOM[2] = RFU
3		Next Cmd	Next Command

If BCS_n is active for 2 clock cycles, it indicates a non-target read termination. The Data Buffer will terminate with RTT_NOM_RD and will not transfer data through the buffer. In this case DB will use Rank 0 timing automatically.

Figure 65 shows the timing sequence for a Read command.

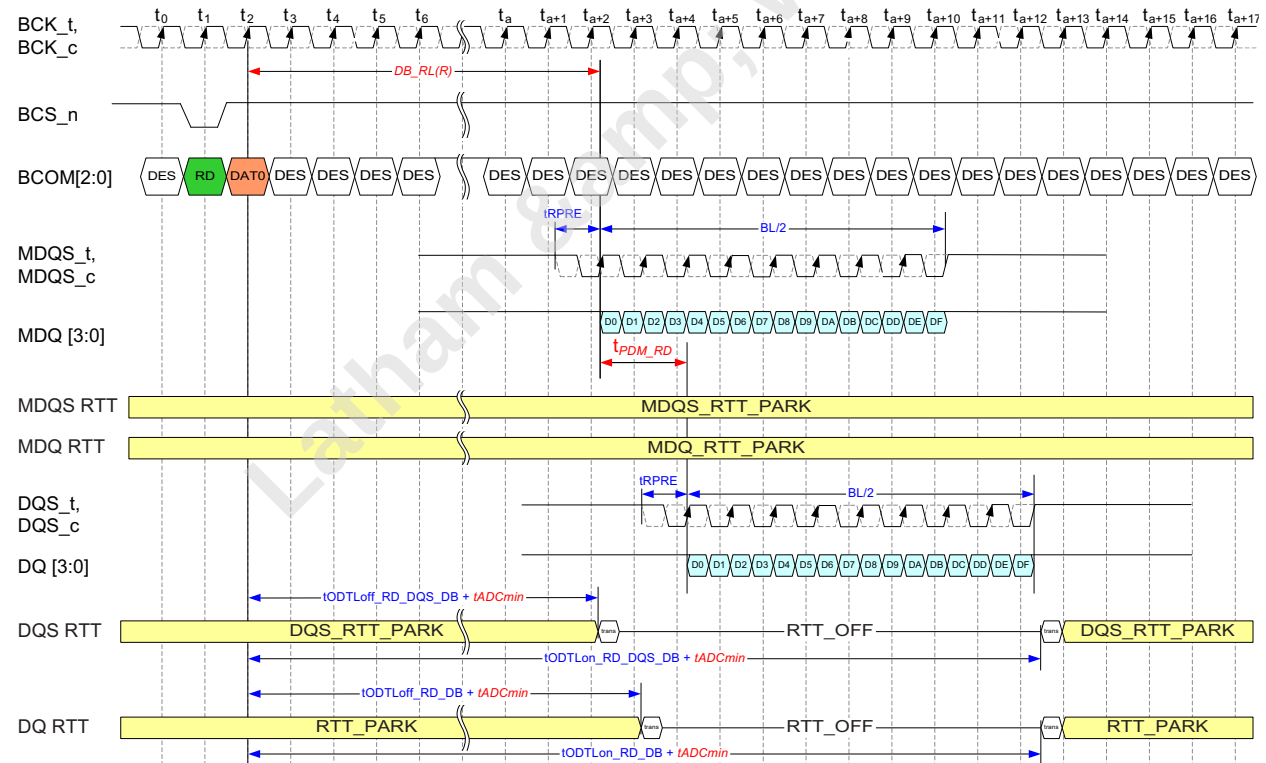


Figure 65 — READ Timing

6.4.7 PDE Commands

The Power Down Entry command is only sent on the BCOM bus when all ranks are being put into or are already in power down. Table 38 shows the sequence for PDE commands.

Table 38 — Multi-cycle Sequence for PDE Commands

Time (Clock Cycle)	BCS_n	BCOM[2:0]	Description
0		Prev Cmd	Previous command or data transfer
1	0	PDE ¹	Power-Down Entry Command BCOM[2:0] = 101
2	1	DAT0	PDE Modes and ODT BCOM [0] = ODT; 0 = Enabled, 1 = Disabled BCOM [1] = SR/PDE; 0 = Self Refresh, 1 = PDE BCOM [2] = RFU
3		Next Cmd	Next Command
NOTE 1 Since the PDE command is a 1-Cycle command on the DRAM bus and a 2-Cycle command on the BCOM bus, the host controller must ensure there is enough spacing between the PDE command and the following command that uses the BCOM bus. When the BCOM interface is enabled (RW09[3] = 0), a PDE command received on the DCA bus shall not be followed immediately (i.e. in the following clock cycle) by any command that is processed in the BCOM interface including MRW, MRR, WR, RD, PDE, SRE, MPC, and NOP/PDX.			

6.4.8 MPC Command

Table 39 shows the sequence for MPC commands.

Table 39 — Multi-cycle Sequence for MPC Commands

Time (Clock Cycle)	BCS_n	BCOM[2:0]	Description
0		Prev Cmd	Previous command or data transfer
1	0	MPC	Multi-purpose Command BCOM[2:0] = 110
2	1	DAT0	BCOM[1:0] = OP[7:6] BCOM[2] = RFU
3	1	DAT1	BCOM[2:0] = OP[5:3]
4	1	DAT2	BCOM[2:0] = OP[2:0]
5		Next Cmd	Next Command

6.4.9 NOP / Power Down Exit

Table 40 shows the sequence for NOP / PDX commands.

Table 40 — Multi-cycle Sequence for NOP/PDX Commands

Time (Clock Cycle)	BCS_n	BCOM[2:0]	Description
0		Prev Cmd	Previous command or data transfer
1	0	NOP/PDX	NOP / PDX Command BCOM[2:0] = 111
2		Next Cmd	Next Command

When in power down, an active BCS_n signal will return the Data Buffer to the active state. The BCOM command will indicate NOP so that no other command is executed.

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7 SidebandBus Interface

For all configuration registers, the DDR5RCD02 supports register access mechanisms through a SidebandBus in addition to In-band Channel commands. The registers may be read by software from the SidebandBus host at any time the RCD is powered on and receiving valid input clock, except in the Self Refresh power down state with Clock Stop or when the device DRST_n pin is asserted.

The SidebandBus interface of the DDR5RCD02 device is compliant to the JEDEC Module SidebandBus Specification, JESD403-1. See JESD403-1 for details. SidebandBus is a two-wire interface based on SDA and SCL. This section describes the minimum subset of the I3C Basic Specification that must be supported to meet the requirements and usage modes of the SidebandBus in the DDR5 DIMM application space.

The DDR5RCD02 SidebandBus Interface must co-exist with other devices using the same bus on an RDIMM or LRDIMM, including Temperature Sensor, SPD, and PMIC devices, and shall not inhibit the operation of the SidebandBus when the module V_{DD} or V_{DDQ} supply rails are disabled (but V_{DDIO} is available) or when the RCD goes through Self Refresh power down state Entry or Exit with Clock Stop.

DRST_n does not reset the SidebandBus Interface of the DDR5RCD02 device. When it is under Clock Stop condition or when DRST_n is asserted (LOW), the DDR5RCD02 device is required to support the following operations from normal operation:

- Timeout Reset (see Section 7.6 on page 109).
- I2C/I3C Basic Common Command Code (CCC) packets (excluding DEVCTRL with RegMod = 1).

I2C and I3C Basic Write/Read packets are not supported in Clock Stop condition or when DRST_n is asserted and they may result in undefined responses.

7.1 SidebandBus Requirements During Initialization of RCD

- DDR5RCD02 is responsible for staying off the bus when V_{DD} is not powered on but V_{DDIO} is enabled. When V_{DD} is not powered on, the RCD does not respond to any commands or events.
- The host is responsible for keeping the SidebandBus idle (i.e., no traffic) during the time V_{DD} is ramping up (e.g., following a PMIC VR Enable command).
- DDR5RCD02 is required to respond to CCC (excepting DEVCTRL with RegMode = 1) and also to Timeout Reset when V_{DD} is valid and stable (as well as V_{DDIO}) even when DRST_n is still LOW during power-up initialization.
- The RCD is required to support register Write/Read access and DEVCTRL CCC with RegMode = 1 only after t_{STAB01} following stable DCK clock (which happens after DCS_n LOW-to-HIGH transition).

7.2 Feature Summary

- I²C Fast-mode Plus (FM+) and I3C Basic supported.
- In-band interrupt supported in I3C Basic mode.
- The DDR5RCD02 SidebandBus Interface shall not initiate clock stretching.
- Multi-controller capability not supported.
- Hot join capability not supported.

The DDR5RCD02 is required to support Read and Write Transactions without requiring clock stretching in order to simplify Host controller requirements. For similar reasons, the RCD shall not control SidebandBus transactions in normal operation.

7.3 Operating Ranges

- Speed: support 1 MHz (I²C FM+) in I²C Mode and up to 12.5 MHz in I3C Basic mode.
- Voltage: V_{DDIO} Supply from 1.0 V - 0.05 V to 1.0 V + 0.05 V for SDA/SCL I/O levels.

7.4 External Pins

The RCD Sideband Interface uses the following two external device pins.

7.4.1 Serial Clock (SCL)

This input signal is used to strobe data in and out of the device. This signal driven by the bus controller using a push-pull driver. In I²C mode only, it is also possible to use an open-drain driver in the bus controller, provided that the board has a pull-up resistor connected between SCL and V_{DDIO}, for example.

7.4.2 Serial Data (SDA)

This bi-directional signal is used to transfer data in or out of the device. It is an output capable of operating in open-drain mode or push-pull mode. A pull up resistor may be connected from Serial Data (SDA) to V_{DDIO}, for example. If the pull-up resistor is not present in the board, the bus controller must provide on-chip termination.

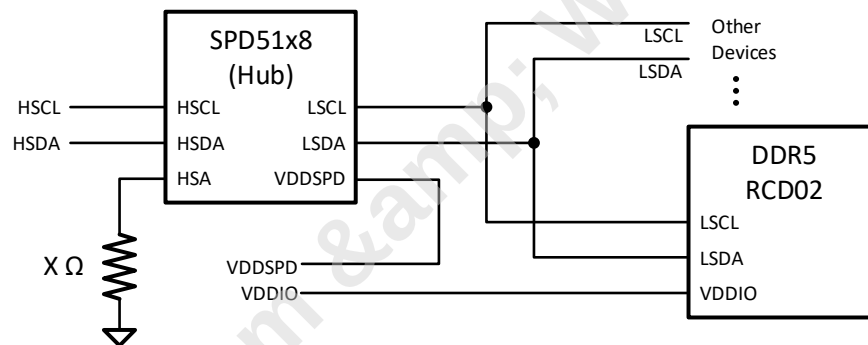


Figure 66 — SidebandBus Wiring Diagram

7.5 System Management Access

System Management software in the platform can initiate system management access to the configuration registers. This can be done through SidebandBus accesses.

DDR5RCD02 components contain a SidebandBus target port and allow access to the configuration registers.

SidebandBus operations are made up of two major steps:

1. Writing information to registers within each component.
2. Reading configuration registers from each component.

The following clauses will describe the protocol for a SidebandBus controller to access an RCD component's internal configuration registers.

7.5.1 Target Address

The 7-bit target address used for each primitive SidebandBus transaction is determined by the 4-bit Local Device address (LID) and the 3-bit Host Device address (HID). The HID bits of the RCD device have a default value of '111' but the value of LID is fixed.

- DDR5RCD02 I²C Bus and I3C Basic Bus address:
 - Target Address[6:3] = 4'b1011 (LID)
 - Target Address[2:0] = HID code configured by SETHID command

7.5.2 Supported SidebandBus Commands

The DDR5RCD02 Sideband target port supports register reads and writes built out of the following Management Bus primitive commands:

- I²C Mode:
 - **Block Write** — **Byte Write**
 - **Block Read** — **Byte Read**
- I3C Basic Mode:
 - **Block Write**
 - **Block Read**

The DDR5RCD02 device is not required to support block accesses larger than a double word.

Each SidebandBus transaction has an 8-bit command driven by the controller. The format for this command is defined in Table 41, “SidebandBus Command Format in I²C Mode” and Table 42, “SidebandBus Command Format in I3C Basic Mode”.

Table 41 — SidebandBus Command Format in I²C Mode

Bit 7	Bit 6	Bit 5	Bit 4	Bits [3:2]	Bits [1:0]
Begin	End	Rsvd	PEC_en ¹	Internal Command: 00 - Read DWord 01 - Write Byte 10 - Write Word 11 - Write DWord	SidebandBus Command: 00 - Byte Mode ² 01 - Reserved 10 - Block Mode 11 - Reserved
NOTE 1 See Table 42. The PEC_en bit only applies in I ² C mode. In I3C Basic mode, Bit 4 of the SidebandBus Command is ignored and PEC is controlled by PEC Enable bit received via DEVCTRL CCC (see Table 87 on page 122). In I ² C mode, the value of PEC Enable bit received via DEVCTRL CCC is ignored and PEC is controlled by Bit 4 of the SidebandBus Command.					
NOTE 2 See Table 42. Byte mode transactions are only supported in I ² C mode. When not in I ² C mode, Bits [1:0] of the SidebandBus Command are ignored and all transactions are all conducted in block mode.					

Table 42 — SidebandBus Command Format in I3C Basic Mode

Bit 7	Bit 6	Bit 5	Bit 4	Bits [3:2]	Bits [1:0]
Begin	End	Rsvd	Rsvd ¹	Internal Command: 00 - Read DWord 01 - Write Byte 10 - Write Word 11 - Write DWord	Rsvd ²
NOTE 1 In I3C Basic mode, Bit 4 of the SidebandBus Command is ignored and PEC is controlled by PEC Enable bit received via DEVCTRL CCC (see Table 87 on page 122).					
NOTE 2 In I3C Basic mode, Bits [1:0] of the SidebandBus Command are ignored and all transactions are all conducted in block mode.					

7.5.2 Supported SidebandBus Commands (cont'd)

The *Begin* bit indicates the first transaction of a read or write sequence.

The *End* bit indicates the last transaction of a read or write sequence.

The *PEC_en* bit enables the 8-bit PEC generation and checking logic when the interface is running in I²C mode.¹

The *Internal Command* field specifies the internal command to be issued by the SidebandBus target logic. Note that the Internal Command must remain consistent (i.e. not change) during a sequence that accesses a configuration register. Operation cannot be guaranteed if it is not consistent when the command setup sequence is done.

The *SidebandBus Command* field specifies the I²C/I3C Basic command to be issued on the bus. This field is used as an indication of the length of transfer so the target knows when to expect the PEC packet (if enabled).

Reserved bits should be written to zero to preserve future compatibility.

Hosts are required to form commands that are internally self consistent. The length indicated by the internal command should be consistent with the byte count and the DDR5RCD02 behavior will be undefined when the internal command is inconsistent with the byte count.

7.5.3 Register Access Protocols

Sequences of these basic commands will initiate internal accesses to the component's configuration registers.

Each Configuration Read or Write consists of an I²C/I3C Basic Write sequence which initializes the register's address. The term sequence is used since these variables may be written with a single Block Write (in I²C or I3C Basic modes) or multiple Byte Writes (in I²C mode). Once these parameters are initialized, the SidebandBus controller can initiate a Read sequence (which performs a Configuration Read) or a Write sequence (which performs a Configuration Write).

The DDR5RCD02 will generate a BCOM MRW command sequence for each configuration Write Byte Internal Command targeted to address range 80h-FFh. The MRW command will be issued only on the BCOM interface channel selected in the Dev/Function byte of the Sideband command (provided that the BCOM interface is not disabled).

1. To keep backwards-compatibility with the DDR4 RCD device, PEC support in I²C mode is required in the DDR5RCD02 device even though this feature is not described in the JESD403-1 specification.

7.5.3 Register Access Protocols (cont'd)

Table 43 — SidebandBus Protocol Addressing Fields

Address Field Name	Bits	Description
Reserved	7:0	Reserved - Device may alias all these bits to 00h
Dev	3:0	Reserved - Device may alias all these bits to 0h
Channel_Num	3:0	Channel Number 0000 = Channel A 0001 = Channel B All others Vendor Specific
Page_Num[15:8]	7:0	Page number when Reg_Num is within 60h-7Fh address range. Ignored if Reg_Num is within 00h-5Fh address range.
Reg_Num[7:0]	7:0	Register Address. Addresses within 60h-7Fh range use the page number above. In case of configuration Write Byte Internal Command, and provided that the Data Buffer interface is enabled in RW09[3] , addresses within the 80h-FFh range result in a BCOM MRW command sequence for the selected channel.

7.5.3.1 Access Mechanism

For all Configuration and Status registers, the DDR5RCD02 supports register access mechanisms through the SidebandBus Interface as well as through In-band Channel commands. The registers may be read by software from the SidebandBus host at any time the DDR5RCD02 is powered on, except in the Self Refresh power down state with Clock Stop or when the device DRST_n pin is asserted.

Table 44 shows the mapping between DDR5RCD02 control words and I²C/I3C Basic register numbers. The entire register space of the DDR5RCD02 is accessible with the Channel_Num, Page_Num, and Reg_Num SidebandBus address fields.

Table 44 — SidebandBus Address Map

RW Type	Page_Num [15:8]	Reg_Num [7:0]	RW Page	RW
Direct Control Words	x	00	NA	RW00
	x	01	NA	RW01
	...	...	...	..
	x	5Fh	NA	RW5F
Paged Control Words	00	60h	PG[00]	RW60
	00	61h	PG[00]	RW61
	...	...	...	...
	00	7Fh	PG[00]	RW7F
	01	60h	PG[01]	RW60
	...	...	...	...
	FFh	7Fh	PG[FF]	RW7F

Each byte of the DDR5RCD02 register space has its own unique address. For CSR Writes, the RCD registers can be accessed in Byte (8-bit), Word (16-bit) or Double Word (32-bit) quantities. For CSR Reads, the device registers can only be accessed in Double Word (32-bit) quantities.

7.5.3.1 Access Mechanism (cont'd)

All multi-byte numeric fields use “little-endian” ordering (i.e., lower addresses contain the least significant parts of the field). As an example Table 45 defines the byte order for double word accesses to the DDR5RCD02 control words.

In this chapter, CSRs are defined as 8-bit registers. Since a byte is the smallest access granularity for CSRs, a single 8-bit CSR contains one RCD Control Word.

Table 45 — Double Word Byte Order for SidebandBus Control Word Access

Description	Data[31:24]	Data[23:16]	Data[15:8]	Data[7:0]
RW03 RW00	RW03	RW02	RW01	RW00
RW07 - RW04	RW07	RW06	RW05	RW04
...	...	...	...	...
RW7F - RW7C	RW7F	RW7E	RW7D	RW7C

7.5.4 Sequence for switching from I²C mode to I3C Basic

The DDR5RCD02 device will start with the Sideband Interface enabled to support I²C protocol by default. When I²C protocol is selected in [RW25\[5\]](#) (SidebandBus Mode status bit controlled by SETAASA and RSTDAA commands), the following limitations apply.

1. Operation speed is limited to 1 MHz compatible with FM+ mode I²C, and
2. In-band interrupts are not supported.
3. CCC support limited to DEVCTRL, SETHID, and SETAASA common command codes.
4. In I²C mode, Start or Repeat Start operation followed by 7'h7E with W = 0 is only allowed for the purpose of issuing CCCs that are supported in I²C mode.

In I3C Basic mode, full functionality including 12.5 MHz operation and IBIs are enabled.

The Host puts the DDR5RCD02 device in I3C Basic mode by issuing SETAASA CCC. The host must issue DEVCTRL and SETHID Commands first (if needed) followed by SETAASA. When SETAASA CCC is registered by the device, the DDR5RCD02 updates [RW25\[5\]](#) to '1'.

The Host puts the DDR5RCD02 device in I²C mode by issuing RSTDAA CCC. When RSTDAA CCC is registered by the device, the DDR5RCD02 updates [RW25\[5\]](#) to '0'. Moreover, the RCD is allowed to disable IBI, disable PEC, and enable I3C Basic Parity Checking functions. The host is expected to send DISEC and DEVCTRL commands to disable IBI, disable PEC, and enable Parity Checking prior to issuing the RSTDAA CCC.

7.5.5 Parity Error Checking

By default, when device is put in I3C Basic mode, parity function is automatically enabled. The host can disable the function after it is enabled. Host can also disable the parity function with DEVCTRL CCC. When parity function is disabled, the device simply ignores the “T” bit information from the Host. The host may actually choose to compute the parity and send that information during “T” bit or simply drive static LOW or HIGH in “T” bit.

The device implements ODD parity. If an odd number of bits in the byte are '1', the parity bit value is '0'. If even number of bits in the byte are '1', the parity bit value is '1'. The host computes the parity for each byte except for the device select code byte and sends it during “T” bit.

7.5.6 PEC Support

DDR5RCD02 devices operate on a standard I3C Basic serial interface. Transactions where the DDR5RCD02 device is the targeted Target device begin with the I3C Basic Host issuing a START condition followed by a 7-bit DDR5RCD02 device type identifier then a Read or Write bit, RW. All I3C Basic data are transmitted with the most significant bit MSB first. During select code transmission, the DDR5RCD02 device typically replies with an ACK unless there are exceptional conditions when it may passively assert a NACK.

The optional appended PEC on the data stream covers the information transmission between the host and the DDR5RCD02 device.

If the optional IBI header (7'h7E and W = 0) byte is present, the DDR5RCD02 device will exclude it in its PEC calculation. Other than that, the RCD target device will include all bytes, including the address byte, in its local PEC checking logic.

All DDR5RCD02 devices implement an 8-bit Packet Error Code (PEC) which is appended at the end of all I3C Basic transactions if PECs are enabled through DEVCTRL CCC. The RCD also supports PEC in I²C mode, if enabled in Bit 4 of the Bus Command. The PEC is a CRC-8 value calculated on all the message bytes excluding START, Repeat START, STOP, T-bits, ACK, NACK, and IBI header (7'h7E followed by W = 0) bits. The algorithm for all CRC-8 calculations is:

$$C(x) = x^8 + x^2 + x^1 + 1$$

The seed value for PEC function is all zero.

PECs are disabled by default on power up or RESET. The host may optionally enable PEC function. If enabled, the host must complete the burst length (or byte count) as indicated. In other words, the host must not interrupt the burst length (or byte count) prematurely for Write or Read operation. It is considered an illegal operation.

7.5.7 Sideband Control Word Read Protocol

Configuration Reads are accomplished through an I²C/I3C Basic Write(s) and later followed by an I²C/I3C Basic Read. The Write sequence is used to initialize the Device, Channel, Page, and Register Numbers for the configuration access. The writing of this information can be accomplished through any combination of the supported I²C/I3C Basic write commands (Block command in I²C and I3C Basic modes or Byte command in I²C mode). The *Internal Command* field for each write should specify Read DWord.

All I²C/I3C Basic Configuration Reads should be DWord aligned. The DDR5RCD02 register will ignore the lowest two bits of the Register Address and return the four bytes within a DWord in the byte order shown in the following examples, i.e. Most Significant Byte (Data[31:24]) first and Least Significant Byte (Data[7:0]) last.

After all the information is set up, the last Write (*End* bit is set) initiates an Internal Configuration Read. If an error occurs during the internal access, the last Write Command will receive a NACK. A status field indicates abnormal termination and contains status information such as target abort. The status field encoding is defined in Table 46.

7.5.7 Sideband Control Word Read Protocol (cont'd)

Table 46 — Status Field Encoding for SidebandBus Reads

Bit	Description
7	Reserved
6	Reserved
5	Reserved
4	Internal Target Abort ¹
3:1	Reserved
0	Successful
NOTE 1 A value of 1 indicates access to invalid internal register address. The only condition for this event is attempting Read access to control/status registers located in Data Buffer space. Access to Reserved locations of the RCD is not considered invalid.	

7.5.7.1 I²C Read Packet Format Definitions

Examples of Configuration Reads are illustrated in Table 47 through Table 51. For I²C Bus read transactions, the last byte of data (or the PEC byte if enabled) is NACKed by the controller to indicate the end of the transaction.

Table 47 — I²C Read Command Packet Example (Block Mode, PEC Disabled, Legacy Format)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop
S	1	0	1	1	HID			W=0	A	
	SidebandBus Command = 11000010								A	
	Byte Count = 4								A	
	Reserved								A	
	Dev/Channel_Num								A	
	Page_Num [7:0]								A	
	Reg_Num [7:0]								A	

S	1	0	1	1	HID			W=0	A		
	SidebandBus Command = 11000010								A		
Sr	1	0	1	1	HID			R=1	A		
	Byte Count = 5								A		
	Status								A		
	Rd Data[31:24]								A		
	Rd Data[23:16]								A		
	Rd Data[15:8]								A		
	Rd Data[7:0]								N		P

The above packet format must be supported in I²C Legacy mode for backwards compatibility with older controllers.

7.5.7.1 I²C Read Packet Format Definitions (cont'd)

Table 48 — I²C Read Command Packet Example (Block Mode, PEC Enabled, Legacy Format)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop
S	1	0	1	1	HID			W=0	A	
	SidebandBus Command = 11010010								A	
	Byte Count = 4								A	
	Reserved								A	
	Dev/Channel_Num								A	
	Page_Num [7:0]								A	
	Reg_Num [7:0]								A	
	PEC(8)								A	P

S	1	0	1	1	HID			W=0	A	
	SidebandBus Command = 11010010								A	
Sr	1	0	1	1	HID			R=1	A	
	Byte Count = 5								A	
	Status								A	
	Rd Data[31:24]								A	
	Rd Data[23:16]								A	
	Rd Data[15:8]								A	
	Rd Data[7:0]								A	
	PEC(8)								N	

The above packet format must be supported in I²C mode for backwards compatibility with older controllers. The table coloring identifies the portion of the packet covered by each PEC byte.

Table 49 — I²C Read Command Packet Example (Block Mode, PEC Disabled, Optimized Format)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop
S	1	0	1	1	HID			W=0	A	
	SidebandBus Command = 11000010								A	
	Byte Count = 4								A	
	Reserved								A	
	Dev/Channel_Num								A	
	Page_Num [7:0]								A	
	Reg_Num [7:0]								A	
Sr	1	0	1	1	HID			R=1	A	
	Byte Count = 5								A	
	Status								A	
	Rd Data[31:24]								A	
	Rd Data[23:16]								A	
	Rd Data[15:8]								A	
	Rd Data[7:0]								N	P

The optimized I²C Read packet format is only supported in Block Mode with PEC disabled. This optimized case without redundant bytes must be supported in I²C mode for better protocol efficiency.

7.5.7.1 I²C Read Packet Format Definitions (cont'd)

Table 50 — I²C Read Command Packet Example (Byte Mode, PEC Disabled, Legacy Format)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop							
S	1	0	1	1	HID			W=0	A								
								SidebandBus Command = 10000000								A	
								Reserved								A	P
S	1	0	1	1	HID			W=0	A								
								SidebandBus Command = 00000000								A	
								Dev/Channel_Num								A	P
S	1	0	1	1	HID			W=0	A								
								SidebandBus Command = 00000000								A	
								Page_Num [7:0]								A	P
S	1	0	1	1	HID			W=0	A								
								SidebandBus Command = 01000000								A	
								Reg_Num [7:0]								A	P
S	1	0	1	1	HID			W=0	A								
								SidebandBus Command = 10000000								A	
								Status								N	P
S	1	0	1	1	HID			W=0	A								
								SidebandBus Command = 00000000								A	
								Sr	1		0	1	1	HID			R=1
Rd Data[31:24]										N							P
S	1	0	1	1	HID			W=0	A								
								SidebandBus Command = 00000000								A	
								Sr	1		0	1	1	HID			R=1
Rd Data[23:16]										N							P
S	1	0	1	1	HID			W=0	A								
								SidebandBus Command = 00000000								A	
								Sr	1		0	1	1	HID			R=1
Rd Data[15:8]										N							P
S	1	0	1	1	HID			W=0	A								
								SidebandBus Command = 01000000								A	
								Sr	1		0	1	1	HID			R=1
Rd Data[7:0]										N							P

The package format shown above must be supported in I²C mode for backwards compatibility with older controllers.

7.5.7.1 I²C Read Packet Format Definitions (cont'd)Table 51 — I²C Read Command Packet Example (Byte Mode, PEC Enabled, Legacy Format)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop				
S	1	0	1	1	HID			W=0	A	P				
					SidebandBus Command = 10010000						A			
					Reserved						A			
					PEC(8)						A			
S	1	0	1	1	HID			W=0	A	P				
					SidebandBus Command = 00010000						A			
					Dev/Channel_Num						A			
					PEC(8)						A			
S	1	0	1	1	HID			W=0	A	P				
					SidebandBus Command = 00010000						A			
					Page_Num [7:0]						A			
					PEC(8)						A			
S	1	0	1	1	HID			W=0	A	P				
					SidebandBus Command = 01010000						A			
					Reg_Num [7:0]						A			
					PEC(8)						A			
S	1	0	1	1	HID			W=0	A	P				
					SidebandBus Command = 10010000						A			
					Status						A			
					PEC(8)						N			
S	1	0	1	1	HID			W=0	A	P				
					SidebandBus Command = 00010000						A			
					Sr	1	0	1	1		HID		R=1	A
											Rd Data[31:24]			
PEC(8)						N								
S	1	0	1	1	HID			W=0	A	P				
					SidebandBus Command = 00010000						A			
					Sr	1	0	1	1		HID		R=1	A
											Rd Data[23:16]			
PEC(8)						N								
S	1	0	1	1	HID			W=0	A	P				
					SidebandBus Command = 00010000						A			
					Sr	1	0	1	1		HID		R=1	A
											Rd Data[15:8]			
PEC(8)						N								
S	1	0	1	1	HID			W=0	A	P				
					SidebandBus Command = 00010000						A			
					Sr	1	0	1	1		HID		R=1	A
											Rd Data[7:0]			
PEC(8)						N								
S	1	0	1	1	HID			W=0	A	P				
					SidebandBus Command = 01010000						A			
					Sr	1	0	1	1		HID		R=1	A
											Rd Data[7:0]			
PEC(8)						N								

The packet format shown above must be supported in I²C mode for backwards compatibility with older controllers. The table coloring identifies the portion of the packet covered by PEC.

The case for Optimized Packet format is not supported in Byte Mode I²C Read transactions in byte mode with optimization does not need to be supported due to the inherent inefficiency of Byte Mode.

7.5.7.2 I3C Basic Mode Read Packet Format Definitions

Prior to the Repeated START in the Read command data packet, the “T” bit carries Parity information from the Host for Address Offset bytes. See Section 7.5.5.

After the Host sends the Repeated START, the “T” bit carries information from the device to the Host (“T” bit semantics are compliant with the I3C Basic Specification). The device sends Continue (‘1’) or Stop (‘0’) information to the host to inform whether it is transmitting the last byte or not. The number of bytes transmitted is predetermined by the contents of the SidebandBus Command (see Table 41).

Table 52 — I3C Basic Read Command Packet Example (PEC Disabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	0	1	1	HID			W=0	A ^{1,2,3}	
	SidebandBus Command = 11000000								T	
	Byte Count = 4								T	
	Reserved								T	
	Dev/Channel_Num								T	
	Page_Num [7:0]								T	
	Reg_Num [7:0]								T	
Sr	1	0	1	1	HID			R=1	A/N ^{4,5}	
	Byte Count = 5								T=1	
	Status								T=1	
	Rd Data[31:24]								T=1	
	Rd Data[23:16]								T=1	
	Rd Data[15:8]								T=1	
	Rd Data[7:0]								T=0 ⁶	
NOTE 1	Figure 124 depicts the transition from Target Open Drain (ACK) to Host Push-Pull Operation (1st bit, SidebandBus Command).									
NOTE 2	The DDR5RCD02 NACKs if there was a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.									
NOTE 3	The DDR5RCD02 device does not check for parity error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The RCD device ignores the entire packet until STOP or next Repeat Start operation.									
NOTE 4	Figure 128 shows the transition from Host Push-Pull Operation to Target Open Drain (ACK).									
NOTE 5	If target device NACKs during Repeat START for any reason, the host may re-try Repeat START again. The host may do the Repeat START as many times as it may desire. If the target device NACKs due to parity error in previous bytes from the Host, it will always NACK regardless of how many times the Host tries Repeat START. If there were no Parity errors, the RCD may eventually ACK (e.g., if delay in access of register caused the first NACK).									
NOTE 6	Figure 126 shows how the RCD ends the operation followed by the Controller STOP bit.									
NOTE 7	Repeat Start or Repeat Start with 7'h7E.									

Table 53 shows the I3C Basic Read Command Packet format for the case PEC is enabled. This packet includes two PEC bytes. The first PEC byte is generated by the Host and it covers from the START bit to the first PEC Byte. The RCD checks the data it receives from the host with respect to the first PEC byte. The second PEC byte is generated by the RCD and it covers from the Repeated START bit to the second PEC byte. The Host checks the data it receives from the RCD with respect to the second PEC byte.

7.5.7.2 I3C Basic Mode Read Packet Format Definitions (cont'd)**Table 53 — I3C Basic Read Command Packet Example (Block Mode, PEC Enabled)**

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	0	1	1	HID			W=0	A ^{1,2,3}	
	SidebandBus Command = 11000000								T	
	Byte Count = 4								T	
	Reserved								T	
	Dev/Channel_Num								T	
	Page_Num [7:0]								T	
	Reg_Num [7:0]								T	
	PEC(8)								T	
Sr	1	0	1	1	HID			R=1	A/N ^{4,5}	
	Byte Count = 5								T=1	
	Status								T=1	
	Rd Data[31:24]								T=1	
	Rd Data[23:16]								T=1	
	Rd Data[15:8]								T=1	
	Rd Data[7:0]								T=1	
	PEC(8)								T=0 ⁶	
NOTE 1	Figure 124 depicts the transition from Target Open Drain (ACK) to Host Push-Pull Operation (1st bit, SidebandBus Command).									
NOTE 2	The DDR5RCD02 NACKs if there was a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.									
NOTE 3	The DDR5RCD02 device does not check for parity or PEC error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The RCD device ignores the entire packet until STOP or next Repeat Start operation.									
NOTE 4	Figure 128 shows the transition from Host Push-Pull Operation to Target Open Drain (ACK).									
NOTE 5	If target device NACKs during Repeat START for any reason, the host may re-try Repeat START again. The host may do the Repeat START as many times as it may desire. If the target device NACKs due to parity error or PEC error in previous bytes from the Host, it will always NACK regardless of how many times the Host tries Repeat START. If there were no Parity or PEC errors, the RCD may eventually ACK (e.g., if delay in access of register caused the first NACK). The PEC calculation by the target device only includes device select code of the ACK response of the Repeat start operation.									
NOTE 6	Figure 126 shows how the RCD ends the operation followed by the Controller STOP bit.									
NOTE 7	Repeat Start or Repeat Start with 7'h7E.									

The table coloring identifies the portion of the packet covered by each PEC byte. In I3C Basic mode, only the block mode optimized format without redundant bytes is supported.

In I3C Basic mode, Bits [4] and [1:0] of the SidebandBus Command are ignored (see Table 41).

The host may optionally allow the DDR5RCD02 device to request IBI. For this case, the transactions to the DDR5RCD02 device begin with the I3C Basic host issuing a START condition followed by 7'h7E and W = 0. If the RCD or any other connected device has a pending IBI, it transmits its 7-bit device select code followed by R = 1. If the RCD has no pending IBI, there is no action taken by the RCD. Table 54 shows the I3C Basic Read Command Packet format from Table 53 with the optional IBI header. As shown in color coding, the IBI header is not included in the calculations to generate PEC.

7.5.7.2 I3C Basic Mode Read Packet Format Definitions (cont'd)

Table 54 — I3C Basic Read Command Packet Example with IBI Header and No Pending Interrupt (Block Mode, PEC Enabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W = 0	A ¹	
Sr	1	0	1	1	HID			W = 0	A ^{1,2,3}	
	SidebandBus Command = 11000000								T	
	Byte Count = 4								T	
	Reserved								T	
	Dev/Channel_Num								T	
	Page_Num [7:0]								T	
	Reg_Num [7:0]								T	
	PEC(8)								T	
Sr	1	0	1	1	HID			R=1	A/N ^{4,5}	
	Byte Count = 5								T=1	
	Status								T=1	
	Rd Data[31:24]								T=1	
	Rd Data[23:16]								T=1	
	Rd Data[15:8]								T=1	
	Rd Data[7:0]								T=1	
	PEC(8)								T=0 ⁶	Sr ⁷ or P
NOTE 1	The DDR5RCD02 NACKs if there was a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.									
NOTE 2	Figure 124 depicts the transition from Target Open Drain (ACK) to Host Push-Pull Operation (1st bit, SidebandBus Command).									
NOTE 3	The DDR5RCD02 device does not check for parity or PEC error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The RCD device ignores the entire packet until STOP or next Repeat Start operation.									
NOTE 4	Figure 128 shows the transition from Host Push-Pull Operation to Target Open Drain (ACK).									
NOTE 5	If target device NACKs during Repeat START for any reason, the host may re-try Repeat START again. The host may do the Repeat START as many times as it may desire. If the target device NACKs due to parity error or PEC error in previous bytes from the Host, it will always NACK regardless of how many times the Host tries Repeat START. If there were no Parity or PEC errors, the RCD may eventually ACK (e.g., if delay in access of register caused the first NACK). The PEC calculation by the target device only includes device select code of the ACK response of the Repeat start operation.									
NOTE 6	Figure 126 shows how the RCD ends the operation followed by the Controller STOP bit.									
NOTE 7	Repeat Start or Repeat Start with 7'h7E.									

7.5.8 Sideband Control Word Write Protocol

Configuration Writes are accomplished through a series of I²C/I3C Basic Writes. As with Configuration Reads, a Write Sequence is first used to initialize the Device, Channel, Page, and Register Numbers for the configuration access. The writing of this information can be accomplished through any combination of the supported I²C/I3C Basic write commands (Block command in I²C and I3C Basic modes or Byte command in I²C mode).

On SidebandBus, there is no concept of Byte Enables. Therefore, the Register Number written to the target is assumed to be aligned to the length of the Internal Command. In other words, for a Write Byte Internal Command, the Register Number specifies the Byte Address. For a Write DWord Internal Command, the two least-significant bits of the Register Number are ignored. This is different from PCI, where Byte Enables are used to indicate the byte of interest.

After all the information is set up, the SidebandBus Controller initiates one or more Writes which sets up the data to be written. The final Write (*End* bit is set) initiates an Internal Configuration Write. If an error occurred, the SidebandBus interface NACKs the last write operation just before the stop bit.

7.5.8.1 I²C Write Packet Format Definitions

Examples of Configuration Writes are shown in Table 55 through Table 60.

Table 55 — I²C Write Command Packet Example (Write DWord, Block Mode, PEC Disabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop
S	1	0	1	1	HID			W=0	A	
	SidebandBus Command = 11001110								A	
	Byte Count = 8								A	
	Reserved								A	
	Dev/Channel_Num								A	
	Page_Num [7:0]								A	
	Reg_Num [7:0]								A	
	Wr Data[31:24]								A	
	Wr Data[23:16]								A	
	Wr Data[15:8]								A	
	Wr Data[7:0]								A	

The above packet format must be supported in I²C mode for backwards compatibility with older controllers.

Table 56 — I²C Write Command Packet Example (Write DWord, Block Mode, PEC Enabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop
S	1	0	1	1	HID			W=0	A	
SidebandBus Command = 11011110									A	
Byte Count = 8									A	
Reserved									A	
Dev/Channel_Num									A	
Page_Num [7:0]									A	
Reg_Num [7:0]									A	
Wr Data[31:24]									A	
Wr Data[23:16]									A	
Wr Data[15:8]									A	
Wr Data[7:0]									A	
PEC(8)									A	

The above packet format must be supported in I²C mode for backwards compatibility with older controllers. The table coloring identifies the portion of the packet covered by PEC.

Table 57 — I²C Write Command Packet Example (Write Byte, Block Mode, PEC Disabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop
S	1	0	1		HID			W=0	A	
	SidebandBus Command = 11000110								A	
	Byte Count = 5								A	
	Reserved								A	
	Dev/Channel_Num								A	
	Page_Num [7:0]								A	
	Reg_Num [7:0]								A	
	Wr Data[7:0]								A	

The above packet format must be supported in I²C mode for backwards compatibility with older controllers.

7.5.8.1 I²C Write Packet Format Definitions (cont'd)

Table 58 — I²C Write Command Packet Example (Write Word, Block Mode, PEC Enabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop
S	1	0	1	1	HID			W=0	A	
	SidebandBus Command = 11011010								A	
	Byte Count = 6								A	
	Reserved								A	
	Dev/Channel_Num								A	
	Page_Num [7:0]								A	
	Reg_Num [7:0]								A	
	Wr Data[15:8]								A	
	Wr Data[7:0]								A	
	PEC(8)								A	

The above packet format must be supported in I²C mode for backwards compatibility with older controllers. The table coloring identifies the portion of the packet covered by PEC.

Table 59 — I²C Write Command Packet Example (Write DWord, Byte Mode, PEC Disabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop							
S	1	0	1	1	HID			W=0	A								
									SidebandBus Command = 10001100							A	
									Reserved							A	P
S	1	0	1	1	HID			W=0	A								
									SidebandBus Command = 00001100							A	
									Dev/Channel Num							A	P
S	1	0	1	1	HID			W=0	A								
									SidebandBus Command = 00001100							A	
									Page_Num [7:0]							A	P
S	1	0	1	1	HID			W=0	A								
									SidebandBus Command = 00001100							A	
									Reg_Num [7:0]							A	P
S	1	0	1	1	HID			W=0	A								
									SidebandBus Command = 00001100							A	
									Wr Data[31:24]							A	P
S	1	0	1	1	HID			W=0	A								
									SidebandBus Command = 00001100							A	
									Wr Data[23:16]							A	P
S	1	0	1	1	HID			W=0	A								
									SidebandBus Command = 00001100							A	
									Wr Data[15:8]							A	P
S	1	0	1	1	HID			W=0	A								
									SidebandBus Command = 01001100							A	
									Wr Data[7:0]							A	P

The above packet format must be supported in I²C mode for backwards compatibility with older controllers.

7.5.8.1 I²C Write Packet Format Definitions (cont'd)Table 60 — I²C Write Command Packet Example (Write Word, Byte Mode, PEC Enabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop
S	1	0	1	1	HID			W=0	A	
	SidebandBus Command = 10011000								A	
	Reserved								A	
	PEC(8)								A	
S	1	0	1	1	HID			W=0	A	
	SidebandBus Command = 00011000								A	
	Dev/Channel_Num								A	
	PEC(8)								A	
S	1	0	1	1	HID			W=0	A	
	SidebandBus Command = 00011000								A	
	Page_Num [7:0]								A	
	PEC(8)								A	
S	1	0	1	1	HID			W=0	A	
	SidebandBus Command = 00011000								A	
	Reg_Num [7:0]								A	
	PEC(8)								A	
S	1	0	1	1	HID			W=0	A	
	SidebandBus Command = 00011000								A	
	Wr Data[15:8]								A	
	PEC(8)								A	
S	1	0	1	1	HID			W=0	A	
	SidebandBus Command = 01011000								A	
	Wr Data[7:0]								A	
	PEC(8)								A	

The above packet format must be supported in I²C mode for backwards compatibility with older controllers. The table coloring identifies the portion of the packet covered by PEC.

7.5.8.2 I3C Basic Write Packet Format Definitions

During Write command data packet, the “T” bit carries Parity information from the Host. See Section 7.5.5.

The number of bytes transmitted is predetermined by the contents of the SidebandBus Command (see Table 41).

Table 61 — I3C Basic Write Command Packet Example (Write DWord, PEC Disabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	0	1	1	HID			W=0	A ^{1,2,3}	
	SidebandBus Command = 110011xx								T	
	Byte Count = 8								T	
	Reserved								T	
	Dev/Channel_Num								T	
	Page_Num [7:0]								T	
	Reg_Num [7:0]								T	
	Wr Data[31:24]								T	
	Wr Data[23:16]								T	
	Wr Data[15:8]								T	
	Wr Data[7:0]								T	Sr ⁴ or P
NOTE 1	Figure 124 depicts the transition from Target Open Drain (ACK) to Host Push-Pull Operation (1st bit, SidebandBus Command).									
NOTE 2	The DDR5RCD02 NACKs if there was a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.									
NOTE 3	The DDR5RCD02 device does not check for parity error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The RCD device ignores the entire packet until STOP or next Repeat Start operation.									
NOTE 4	Repeat Start or Repeat Start with 7'h7E.									

In I3C Basic mode, Bits [4] and [1:0] of the SidebandBus Command are ignored (see Table 41).

Table 62 — I3C Basic Write Command Packet Example (Write DWord, Block Mode, PEC Enabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	0	1	1	HID			W=0	A ^{1,2,3}	
	SidebandBus Command = 110111xx								T	
	Byte Count = 8								T	
	Reserved								T	
	Dev/Channel_Num								T	
	Page_Num [7:0]								T	
	Reg_Num [7:0]								T	
	Wr Data[31:24]								T	
	Wr Data[23:16]								T	
	Wr Data[15:8]								T	
	Wr Data[7:0]								T	
	PEC(8)								T	

NOTE 1

Figure 125 depicts the transition from Target Open Drain (ACK) to Host Push-Pull Operation (1st bit, SidebandBus Command).

NOTE 2

The DDR5RCD02 NACKs if there was a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.

NOTE 3

The DDR5RCD02 device does not check for parity or PEC error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code.The RCD device ignores the entire packet until STOP or next Repeat Start operation.

NOTE 4

Repeat Start or Repeat Start with 7'h7E.

The table coloring identifies the portion of the packet covered by the PEC byte. In I3C Basic mode, Bits [4] and [1:0] of the SidebandBus Command are ignored (see Table 41).

7.5.8.2 I3C Basic Write Packet Format Definitions (cont'd)**Table 63 — I3C Basic Write Command Packet Example (Write Byte, PEC Disabled)**

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	0	1	1	HID			W=0	A ^{1,2,3}	
	SidebandBus Command = 110001xx								T	
	Byte Count = 5								T	
	Reserved								T	
	Dev/Channel_Num								T	
	Page_Num [7:0]								T	
	Reg_Num [7:0]								T	
	Wr Data[7:0]								T	
NOTE 1	Figure 124 depicts the transition from Target Open Drain (ACK) to Host Push-Pull Operation (1st bit, SidebandBus Command).									
NOTE 2	The DDR5RCD02 NACKs if there was a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.									
NOTE 3	The DDR5RCD02 device does not check for parity error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code.The RCD device ignores the entire packet until STOP or next Repeat Start operation.									
NOTE 4	Repeat Start or Repeat Start with 7'h7E.									

In I3C Basic mode, Bits [4] and [1:0] of the SidebandBus Command are ignored (see Table 41).

Table 64 — I3C Basic Write Command Packet Example (Write Word, Block Mode, PEC Enabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	0	1	1	HID			W=0	A ^{1,2,3}	
	SidebandBus Command = 110111xx								T	
	Byte Count = 6								T	
	Reserved								T	
	Dev/Channel_Num								T	
	Page_Num [7:0]								T	
	Reg_Num [7:0]								T	
	Wr Data[31:24]								T	
	Wr Data[7:0]								T	
	PEC(8)								T	
NOTE 1	Figure 124 depicts the transition from Target Open Drain (ACK) to Host Push-Pull Operation (1st bit, SidebandBus Command).									
NOTE 2	The DDR5RCD02 NACKs if there was a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.									
NOTE 3	The DDR5RCD02 device does not check for parity or PEC error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code.The RCD device ignores the entire packet until STOP or next Repeat Start operation.									
NOTE 4	Repeat Start or Repeat Start with 7'h7E.									

The table coloring identifies the portion of the packet covered by the PEC byte. In I3C Basic mode, Bits [4] and [1:0] of the SidebandBus Command are ignored (see Table 41).

The host may optionally allow the DDR5RCD02 device to request IBI. For this case, the transactions to the DDR5RCD02 device begin with the I3C Basic host issuing a START condition followed by 7'h7E and W = 0. If the RCD or any other connected device has a pending IBI, it transmits its 7-bit device select code followed by R = 1. If the RCD has no pending IBI, there is no action taken by the RCD. Table 65 shows the same I3C Basic Write Command Packet format from Table 64 with the optional byte to allow target devices to issue an IBI request. As shown in color coding, the IBI byte is not included in the calculations to generate PEC.

7.5.8.2 I3C Basic Write Packet Format Definitions (cont'd)

Table 65 — I3C Basic Write Command Packet Example with IBI Header and No Pending Interrupt (Write Word, Block Mode, PEC Enabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W = 0	A ^{1,2}	
Sr	1	0	1	1	HID			W = 0	A ^{2,3,4}	
	SidebandBus Command = 110111xx								T	
	Byte Count = 6								T	
	Reserved								T	
	Dev/Channel_Num								T	
	Page_Num [7:0]								T	
	Reg_Num [7:0]								T	
	Wr Data[31:24]								T	
	Wr Data[7:0]								T	
	PEC(8)								T	Sr ⁵ or P
NOTE 1	Figure 124 depicts the transition from Target Open Drain (ACK) to Host Push-Pull Operation (Repeat Start).									
NOTE 2	The DDR5RCD02 NACKs if there was a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.									
NOTE 3	Figure 128 shows the transition from Host Push-Pull Operation to Target Open Drain (ACK) and Figure 124 depicts the transition from Target Open Drain (ACK) to Host Push-Pull Operation (1st bit, SidebandBus Command).									
NOTE 4	The DDR5RCD02 device does not check for parity or PEC error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The RCD device ignores the entire packet until STOP or next Repeat Start operation.									
NOTE 5	Repeat Start or Repeat Start with 7'h7E.									

7.5.9 In-Band Interrupts

In-Band interrupts may be generated by the RCD device if IBI is enabled by receiving a broadcast or directed ENEC CCC. In I²C mode, in-band interrupt function is not supported. Only I3C Basic mode supports the in-band interrupt function.

7.5.9.1 Enabling/Disabling In-Band Event Interrupts

By default, all interrupt sources are disabled (i.e. set to '0'). The host may enable interrupts in the DDR5RCD02 device. Once enabled, the RCD device sends an IBI when that event occurs.

1. When any of the bits in RW28[1:0] (Parity Error or PEC Error Status Bits) get set to '1', the RCD device sets RW28[7] (In-band Interrupt Status) to 1 and updates Pending Interrupt Bits [3:0] = '0001' for GETSTATUS CCC.
 - a. If IBI is enabled by ENEC CCC, the RCD device sends the IBI at next available opportunity.
 - b. If IBI is disabled by DISEC CCC, the device does not send the IBI regardless of the register bit status in RW28.

7.5.9.2 Mechanics of Interrupt Generation

Event interrupts may be generated by the local device if IBI is enabled by receiving a broadcast or directed ENEC CCC. When there is a pending Event In-Band Interrupt (i.e. RW28[7] = '1') and IBI is enabled, the DDR5RCD02 will request an interrupt after detecting a START bit by transmitting its 7-bit binary address (LID bits '1011' followed by HID bits) followed by R/W bit = '1' on the SDA bus serially (synchronized by SCL falling transitions).

If the RCD detects no START bits, but the Host has been inactive (no edges seen) for t_{AVAIL} period, the RCD may assert SDA Low by t_{IBI_ISSUE} time to request an interrupt. The Host will respond by toggling SCL. The RCD device transmits its 7-bit binary address (LID bits '1011' followed by HID bits) followed by R/W bit = '1'. This is shown in Figure 67.

7.5.9.2 Mechanics of Interrupt Generation (cont'd)

When the RCD requests an interrupt, the host may take one of the two actions described below.

- Option 1. The Controller sends ACK on 9th bit to accept the interrupt request. At this point, if the DDR5RCD02 confirms that it has won the arbitration (i.e. the address driven in SDA matches its own Device address) the RCD transmits the IBI payload as shown in Table 66 and Table 67 for PEC disabled and PEC enabled configurations, respectively. Figure 67 depicts this case, but it only shows the first two data bits of the MDB byte (0x00) to illustrate the timing. The host then issues a STOP command. The Controller accepts the IBI payload if it sends an ACK on the 9th bit to accept the interrupt request. The Controller can terminate the IBI payload at T bit as shown in Table 66 or Table 67. If Controller stops the IBI payload after a T bit in the middle of the payload, the RCD will retain the IBI flag internally and will wait for the next opportunity to request an interrupt. If the DDR5RCD02 device successfully transmits the entire IBI payload, it then clears IBI status flag and Pending Interrupt Bits [3:0] = '0000' on its own and does not request for an IBI again unless there is another different event occurs; for another same event, the device does not request for an IBI.
- Option 2. The Controller sends NACK on the 9th bit as shown in Figure 68 followed by a STOP command. In this case, DDR5RCD02 does not transmit the IBI payload and waits for the next opportunity to request an interrupt. At this point, even though the Controller sent an NACK, it does have knowledge of which device sent the IBI request.

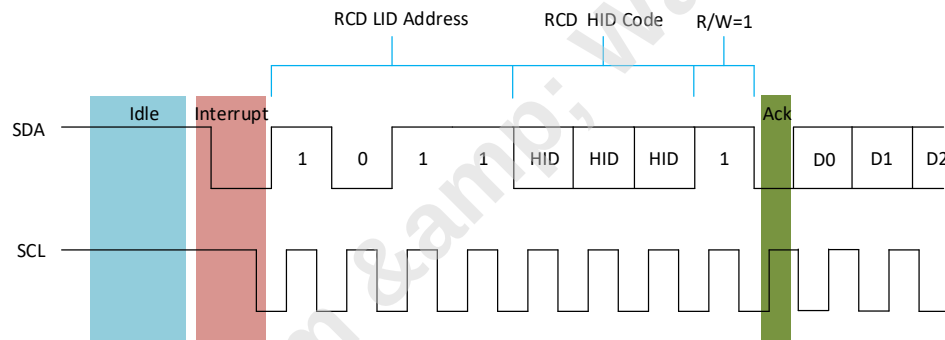


Figure 67 — RCD Interrupt Request - Controller ACK Followed by RCD IBI Payload

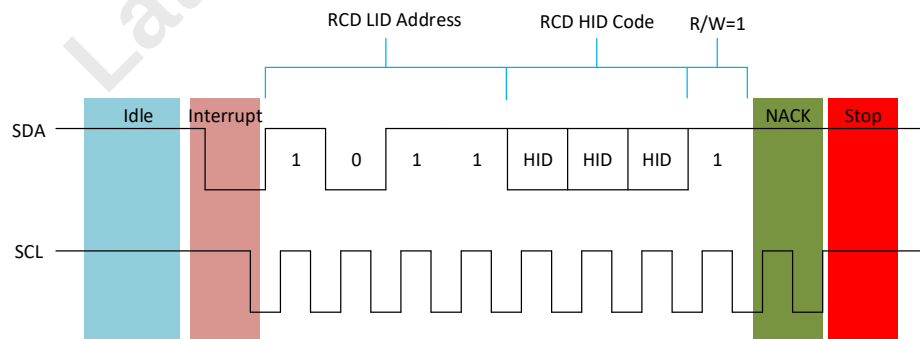


Figure 68 — RCD Interrupt Request - Controller NACK Followed by STOP

7.5.9.2 Mechanics of Interrupt Generation (cont'd)

Table 66 — Interrupt Response (PEC disabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	Device Select Code (7)							RW(1)	Host ACK ¹	
	MDB = 0x00								T = 1	
	RW28[7:0] (last error code byte) (8)								T = 0 ²	
NOTE 1	Figure 125 illustrates the transition from Controller Open Drain (ACK) to Target Push Pull Operation (1st bit of MDB Byte).									
NOTE 2	Figure 126 shows how the RCD ends the operation followed by the Controller STOP bit.									

Table 67 — Interrupt Response (PEC enabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	Device Select Code (7)							RW(1)	Host ACK ¹	
	MDB = 0x00								T = 1	
	RW28[7:0] (last error code byte) (8)								T = 1	
	PEC(8)								T = 0 ²	
NOTE 1	Figure 125 illustrates the transition from Controller Open Drain (ACK) to Target Push Pull Operation (1st bit of MDB Byte).									
NOTE 2	Figure 126 shows how the RCD ends the operation followed by the Controller STOP bit.									

7.5.9.3 Interrupt Arbitration

As there are multiple devices in the I3C Basic bus, more than one device may request an interrupt when the Host I3C Basic bus is inactive for t_{AVAL} period. This makes an arbitration process necessary.

There could be up to 13 different devices, including the RCD, in the I3C Basic bus of a DDR5 DIMM application environment.

On a typical DDR5 DIMM application environment, all devices behind the SPD5 Hub device have the same 3-bit HID code. Hence, the arbitration is always won by the lowest 4-bit LID code. For example, if one local target device has LID code of '0010' and other device (RCD) has a LID code of '1011', through the arbitration process, the LID code of '0010' wins. If a higher priority device than the RCD device is serviced, the RCD must release the bus and wait for the next opportunity to request an interrupt.

It is possible for the controller to start a command transaction at the same time as the RCD requests an interrupt. When this happens, the controller participates in the arbitration process along with the other devices in the bus. During the arbitration phase, there will always be only one winning device and it could be one of the target devices (e.g. RCD) or the controller. If the controller wins during the arbitration phase, it continues with normal operation. The losing devices (e.g. RCD) must wait for next opportunity to send an interrupt. If the controller loses the arbitration, it must let go of the bus. When the controller loses during the arbitration, it must allow the winning device (e.g. RCD) to finish sending its 4-bit LID code followed by the 3-bit HID code followed by R/W = '1'. At this point, during the 9th bit, the controller has the following two options:

- Option 1. The controller sends ACK to accept the interrupt and hence accepts the IBI payload from the winning device (e.g., RCD). After the IBI payload, the controller issues a STOP operation.
- Option 2. Host sends NACK followed by STOP operation.

In the cases when the controller is starting a command transaction to an RCD that is requesting an interrupt at the same time, neither the controller or the RCD know if they are a winner until the 8th bit, and controller always wins. This is because the RCD sends R/W=1 (8th bit) during the interrupt. The controller sets R/W=0 (8th bit) during the operation. As a result, the controller wins and the RCD must let go of the bus and wait for the next opportunity to send an interrupt.

7.5.9.4 Clearing Status Registers

The DDR5RCD02 device provides the device status in Table 135, “RW28: I2C and I3C Basic Error Status Global Word”. When the RCD device generates an IBI condition in RW28[1:0], it sets RW28[7] to ‘1’. The RCD clears RW28[7] to ‘0’ automatically when it sends a complete IBI (including payload and without interruption) the operation gets an ACK from the host. The status information in Table 135 RW28[1:0] are latched and remains valid even after target device sends payload or if the condition that triggered to generate the status is no longer present. The host must explicitly clear the status register through Clear command by writing ‘1’ to register bits in Table 135, “RW28: I2C and I3C Basic Error Status Global Word”, respectively. After the Host Clear command, if the condition is still present, the device will again set the status information in Table 135, “RW28: I2C and I3C Basic Error Status Global Word” registers appropriately and will again generate interrupt at next available opportunity. When RW28[1:0] are cleared to ‘00’ as a result of a Clear command in RW29, the IBI Status Register RW28[7] will also get cleared to ‘0’ by the RCD hardware. IBI will get disabled when the RCD receives DISEC CCC. IBI may also get disabled when the RCD receives RSTDAA CCC since the host is expected to disable Error Interrupts by issuing DISEC CCC prior to sending the RSTDAA CCC.

7.5.10 I/O Operation

The DDR5RCD02 device supports configurable IO operation scheme of either Open Drain or Push-Pull on its Host interface (SCL and SDA).

At power on, by default, DDR5RCD02 comes up in I²C mode of operation with Open Drain I/O for SidebandBus interface. The device shall operate in this mode until put into I3C Basic mode via SETAASA CCC.

After power on, the host may put the SidebandBus interface of the RCD device in I3C Basic mode of operation by issuing a SETAASA command.

In I3C Basic mode, the host may drive the SCL clock input of the RCD device using either Push-Pull output driver or using open-drain output driver.

To support in band interrupt, the RCD device supports dynamic switching between Open Drain mode and Push Pull mode on its SDA I/O for various events. The Table 68 below describes the different mode of operation by the RCD device for each cycle in I3C Basic mode.

Table 68 — RCD Device Dynamic I/O Operation Mode Switching; I3C Basic Mode

Operation	Open Drain Mode	Push-Pull Mode
START + Device Select Codes	Yes	No
START + 7'h7E IBI Header Byte	Yes	No
REPEAT START + Device Select Codes	No	Yes
REPEAT START + 7h'7E Header Byte	No	Yes
CCC Byte (i.e., after 7'h7E + W = 0 + ACK)	No	Yes
STOP	No	Yes
ACK/NACK Responses	Yes	No
Command, Address Operation	No	Yes
Interrupt Request by Target	Yes	No
IBI Payload	No	Yes
Write Data, T-Bit Sequence	No	Yes
Read Data, T-Bit Sequence	No	Yes
PEC, T-Bit Sequence	No	Yes

7.6 I²C and I3C Basic Reset

To prevent a malfunctioning device from locking up the I3C Basic bus, a bus protocol reset mechanism is defined. It uses a timeout mechanism on SCL as shown in Figure 69. All devices (including SPD5 Hub and all target devices behind the hub) on an I²C or I3C Basic bus are RESET simultaneously. Bus reset operation works same way regardless of whether device is operating in I2C or I3C Basic mode.

To guarantee the device resets I²C bus or I3C Basic bus, the SCL clock input LOW time has to be greater than or equal to $t_{\text{TIMEOUT(Max)}}$.

The device does not reset I²C bus or I3C Basic bus if the SCL clock input LOW time is less than $t_{\text{TIMEOUT(Min)}}$.

If the SCL clock input LOW time is between $t_{\text{TIMEOUT(Min)}}$ and $t_{\text{TIMEOUT(Max)}}$, the device does not guaranteed and it may or may not reset the I2C bus or I3C Basic bus.

When a SidebandBus Reset is detected, DDR5RCD02 devices:

1. Interface and any pending command or transaction cleared.
2. All internal register values are preserved unless noted otherwise in Item 3 below.
3. Device mode returns to power-on default conditions as follows: I3C Basic error interrupt disabled, I²C mode enabled (RW25[5] cleared to '0'), I3C Basic parity checking enabled, I3C Basic PEC checking disabled, Parity and PEC Error Status bits (RW28[1:0]) cleared '00', and Device HID code set to '111'.
4. Device floats the SDA pin such that it gets pulled High by the external pull-up.
5. Device treats Bus Reset as STOP operation.

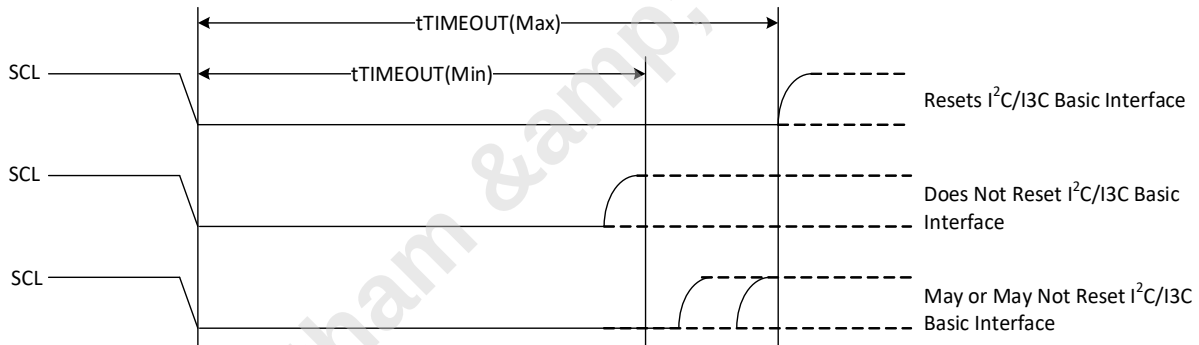


Figure 69 — I²C or I3C Basic Reset

7.6.1 Sideband Interface and DRST_n RCD Reset Events

To maintain consistent configuration state with other devices in the DDR5 module, configuration settings and status registers pertaining to the Sideband Interface in the DDR5RCD02 device will not be affected by DRST_n Reset events. In particular:

1. I²C or I3C Basic protocol selected before DRST_n event will continue in operation after DRST_n Reset event. RW25[5] shall be sticky.
2. I3C Basic error interrupt remains unchanged. IBI remains enabled or disabled after DRST_n Reset event.
3. I3C Basic parity checking and I3C Basic PEC checking remain unchanged after DRST_n Reset event.
4. Parity and PEC Error Status information retained after DRST_n Reset event. RW28 shall be sticky.
5. Device HID code configured before DRST_n event will remain unchanged after DRST_n Reset event.
6. When DRST_n is asserted, the DDR5RCD02 device is required to detect Timeout Reset and to execute I2C/I3C Basic Common Command Code (CCC) packets (excluding DEVCTRL with RegMod = 1).

7.7 I²C Bus Clear

The DDR5RCD02 device supports the following described Bus Clear feature in I²C mode only. Any attempt by host to perform I²C Bus clear on a target device in I³C mode may result in an active drive bus contention on the SDA data line.

There may be abnormal circumstances when the host abruptly stops clocking SCL while the target device is in the middle of outputting data for read operation. For these types of events, the SDA data line may appear as stuck LOW as the device is expecting to receive more clock pulses from the host. Eventually when the host has control of the SCL clock, the host may optionally clear the device that is stuck LOW on the SDA data line by sending continuous 18 clock pulses without driving the SDA data line followed by STOP operation. The device floats the SDA line within 18 clock pulses and returns to the Idle state. The device is ready for normal new transaction with Start condition.

7.8 I²C Bus Error Handling

The I²C Bus target interface handles two types of errors: internal and PEC. These errors manifest as a Not-Acknowledge (NACK) for the read command (End bit is set). If an internal error occurs during a configuration write, the final write command receives a NACK just before the stop bit. If the controller receives a NACK, the entire configuration transaction should be reattempted.

If the controller supports packet error checking (PEC) and the PEC_en bit in the command is set, then the PEC byte is checked in the target interface. If the check indicates a failure, then the target will NACK the PEC packet.

7.9 Error Handling in I³C Basic

There are two types of error checking done by the DDR5RCD02 device. I³C Basic Parity error checking and Packet Error checking. By default, the parity error checking is always enabled (in I³C Basic mode) and packet error checking is disabled. The host may enable the packet error checking at any time. The parity error is calculated for each byte. The packet error is computed for the entire packet. The host sends parity error information in “T” bit.

I³C Basic defines TE0, TE1, TE2, TE3, TE4, TE5, TE6 and CE0, CE1, CE2 errors. Only TE1 and TE2 errors are supported by the DDR5RCD02 device. All other errors are not supported and not applicable. The DDR5RCD02 device will NACK unsupported CCCs.

7.9.1 Write Command Data Packet Error Handling - PEC Is Disabled

When I³C Basic mode is enabled, the DDR5RCD02 device checks for the parity error for each byte that it receives from the host in the Write Command Data Packet as shown in Table 61 through Table 63. The following description applies to I³C Basic mode only.

If no parity error detected in Write command:

- The DDR5RCD02 device executes the command.

If parity error in Write command:

- The DDR5RCD02 device discards the byte that had parity error.
- The DDR5RCD02 device discards all sub-sequent bytes in that packet until STOP operation. The DDR5RCD02 device may or may not check the parity for all sub-sequent bytes in that packet.
- Note that if the host write command was for two-byte write operation and if the parity error occurred in second byte of data, the RCD may or may not execute the first byte write operation but the second byte write operation is discarded.

7.9.1 Write Command Data Packet Error Handling - PEC Is Disabled (cont'd)

- The DDR5RCD02 device sets registers [RW28\[0\]](#) and [RW28\[7\]](#) to '1'; sets P_Err to '1' and Pending Interrupt Bits [3:0] to '0001' in GETSTATUS CCC response; and waits for the next opportunity to send in band interrupt (if IBI is enabled by ENEC CCC).

7.9.2 Read Command Data Packet Error Handling - PEC Is Disabled

For Read Command Data Packet as shown in Table 52, the device checks for the parity error for each byte in a packet except for the device select code byte that it receives from the host.

The DDR5RCD02 device does not compute the parity when it sends the data to the Host. The Host does not check for parity error for the data bytes sent by the RCD. The device sends Continue ('1') or Stop ('0') information during T bit when device is sending the read data.

If no parity error detected in Read command:

- The DDR5RCD02 sends ACK back to the host when Host perform Repeat START operation.
- The DDR5RCD02 device executes the command and sends the data back to the host as shown in Table 52.

If parity error in Read command:

- The DDR5RCD02 device discards the byte that had parity error.
- The DDR5RCD02 device discards all sub-sequent bytes in that packet until STOP operation. The DDR5RCD02 device may or may not check the parity for all sub-sequent bytes in that packet.
- The DDR5RCD02 sends NACK back to the host when the host applies a Repeat START. This is shown in the **RED color** cell in Table 52. The NACK represents either a parity error in one of the six bytes or that DDR5RCD02 is not able to start the read operation. The host may re-try Repeat START again. The Host may do the Repeat START as many times as it may desire. If the RCD NACKs due to parity error in previous bytes from the host, it will always NACK regardless of how many times Host tries Repeat START.
- The DDR5RCD02 does not send Read data back to the host.
- The DDR5RCD02 device sets registers [RW28\[0\]](#) and [RW28\[7\]](#) to '1'; sets P_Err to '1' and Pending Interrupt Bits [3:0] to '0001' in GETSTATUS CCC response; and waits for the next opportunity to send in band interrupt (if IBI is enabled by ENEC CCC).

7.9.3 Write Command Data Packet Error Handling - PEC Is Enabled

When I3C Basic mode is enabled, the DDR5RCD02 device checks for the parity error for each byte and each packet that it receives from the host in the Write Command Data Packet as shown in Table 64. The following description applies to I3C Basic mode only.

If no parity error in Write command:

- The DDR5RCD02 device waits for the entire packet. If no error in packet, the DDR5RCD02 device executes the command. If there is an error in the packet, the DDR5RCD02 device discards the entire packet and does not execute that packet and waits for STOP, sets registers [RW28\[1\]](#) and [RW28\[7\]](#) to '1'; sets PEC_Err to '1' and Pending Interrupt Bits [3:0] to '0001' in GETSTATUS CCC response; and waits for the next opportunity to send in band interrupt (if IBI is enabled by ENEC CCC).

If parity error in Write command:

- The DDR5RCD02 device discards that byte and the entire packet until STOP operation.
- The DDR5RCD02 device sets registers [RW28\[0\]](#) and [RW28\[7\]](#) to '1'; sets P_Err to '1' and Pending Interrupt Bits [3:0] to '0001' in GETSTATUS CCC response; and waits for the next opportunity to send in band interrupt (if IBI is enabled by ENEC CCC).

7.9.3 Write Command Data Packet Error Handling - PEC Is Enabled (cont'd)

- The DDR5RCD02 may or may not check the error for the packet. If the device checks for the packet error, likely it will detect an error in the packet and the device may also set the [RW28\[1\]](#) register and PEC_Err in GETSTATUS CCC response to '1' as well.

7.9.4 Read Command Data Packet Error Handling - PEC Is Enabled

For Read Command Data Packet as shown in Table 53, the DDR5RCD02 device checks for the parity error for each byte in a packet except for the device select code byte that it receives from the host (prior to Repeat Start).

The RCD does not compute the parity when it sends the data to the Host. The Host does not check for parity error for the bytes shown in Table 53. The device sends Continue ('1') or Stop ('0') information during "T" bit when the device is sending the Read data.

The RCD device checks for the PEC error for in a packet that it receives from the host from START condition to Repeat START condition (From first Device Select code followed by the Address Offset).

The device computes the packet error code for the entire packet starting with Repeat START (Device Select code and the data that device transmits back to the Host).

If no parity error and no PEC error in Read command:

- The DDR5RCD02 device sends ACK back to the host in response to Repeat START event.
- The DDR5RCD02 device executes the command and sends the Read data back to the host as shown in Table 53.
- The RCD computes PEC from Repeat START condition until STOP condition (Device Select code followed by the Data) the bytes shown in Table 53.

If parity error or PEC error in Read command:

- The DDR5RCD02 device discards the byte that had a parity error.
- The DDR5RCD02 device discards second byte in that packet if a parity error occurred in first byte. The DDR5RCD02 device may or may not check parity for the second byte in that packet.
- The DDR5RCD02 device sends NACK back to the host in response to Repeat START event. This is shown in the **RED color** cell in Table 53. The NACK represents either parity error in one of the seven bytes or that DDR5RCD02 is not able to start the read operation. The host may re-try Repeat START again. The Host may do the Repeat START as many times as it may desire. If the RCD NACKs due to parity error or PEC error in previous bytes from the host, it will always NACK regardless of how many times Host tries Repeat START.
- The device does not send Read data back to the host and instead expects the host to perform STOP operation.
- The DDR5RCD02 device sets [RW28\[0\]](#) register and P_Err bit in GETSTATUS CCC response to '1' for parity error and [RW28\[1\]](#) register and PEC_Err bit in GETSTATUS CCC response to '1' for PEC error. Further, the RCD sets [RW28\[7\]](#) register to '1' and Pending Interrupt Bits [3:0] to '0001' in GETSTATUS CCC response and waits for the next opportunity to send in band interrupt (if IBI is enabled by ENEC CCC).

7.9.5 CCC Packet Error Handling

Parity error and PEC error detected in a CCC packet are handled the same way as described for normal Read/Write operations.

7.10 Supported I3C Basic Common Command Codes (CCCs)

Table 69 lists all the I3C Basic Common Command Codes that must be supported by the DDR5RCD02 device. The RCD will NACK unsupported CCCs.

7.10 Supported I3C Basic Common Command Codes (CCCs) (cont'd)

The DDR5RCD02 device requires STOP operation in between when switching from CCC operation to private device specific Write or Read operation and vice versa. In other words, any CCC operation must be followed by STOP operation before continuing to any device specific Write or Read. Similarly, any device specific Write or Read operation must be followed by STOP operation before continuing to any CCC operation. The DDR5RCD02 device also requires STOP operation from any Direct CCC to Broadcast CCC.

The DDR5RCD02 device does allow Repeat Start operation between any Direct CCC to any other Direct CCC or between any Broadcast CCC to any other Broadcast CCC or between any private Write or Read operation to any other private Write or Read operation.

Table 69 — Supported I3C Basic Common Command Codes (CCCs)

Command	Mode	Code	Description	Notes
ENEC	Broadcast	0x00	Enable Events Command	
	Direct	0x80		
DISEC	Broadcast	0x01	Disable Events Command	
	Direct	0x81		
RSTDAA	Broadcast	0x06	Put the Device in I ² C Mode (a.k.a. Reset Dynamic Address Assignment Command)	1
SETAASA	Broadcast	0x29	Put the Device in I3C Basic Mode (a.k.a. Set All Addresses to Static Address Command)	
GETSTATUS	Direct	0x90	Get Device Status Command	
DEVCAP	Direct	0xE0	Get Device Capability Command	2
SETHID	Broadcast	0x61	Update HID Code	2
DEVCTRL	Broadcast	0x62	Configure SPD Hub and All Devices Behind Hub	2
NOTE 1 It is illegal for the host to send Direct RSTDAA CCC Code 0x86				
NOTE 2 JEDEC specific CCC				

7.10.1 Enable/Disable Events Command (ENEC/DISEC) Operation

Refer to the I3C Basic Version 1.0 Specification document for details on the encoding and usage of the ENEC/DISEC Command. This command is only supported by the RCD when it is operating in I3C Basic mode. In I²C mode, it is illegal for host to issue this CCC. If host issues this CCC while the RCD is in I²C mode, the response is undefined.

The only event types support by the DDR5RCD02 for ENEC/DISEC Command are **Target Interrupt Requests**. All other event types are not supported and ignored.

When ENEC/DISEC Command is received, it takes effect at the next START operation (i.e. after STOP condition). If PEC function is enabled, the PEC calculation starts with Start or Repeat Start operation but does not include 7'h7E with W=0 byte in PEC calculation.

Table 70 — ENEC/DISEC CCC (I3C Basic, Broadcast, PEC Disabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W = 0	A ¹	
	0x00 (Broadcast ENEC) or 0x01 (Broadcast DISEC)								T	
	0	0	0	0	0 ²	0	0 ²	ENINT/ DISINT	T	Sr ³ or P
NOTE 1 The DDR5RCD02 NACKs if there was a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.										
NOTE 2 DDR5RCD02 ignores HJ (Hot Join) and MR (Control Request) bits.										
NOTE 3 Repeat Start or Repeat Start with 7'h7E.										

7.10.1 Enable/Disable Events Command (ENEC/DISEC) Operation (cont'd)

Table 71 — ENEC/DISEC CCC (I3C Basic, Broadcast, PEC Enabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W = 0	A ¹	
	0x00 (Broadcast ENEC) or 0x01 (Broadcast DISEC)								T	
	0	0	0	0	0 ²	0	0 ²	ENINT/ DISINT	T	
	PEC(8)								T	
NOTE 1	The DDR5RCD02 NACKs if there was a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start									
NOTE 2	DDR5RCD02 ignores HJ (Hot Join) and MR (Control Request) bits.									
NOTE 3	Repeat Start or Repeat Start with 7'h7E.									

Table 72 — ENEC/DISEC CCC (I3C Basic, Direct, PEC Disabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W = 0	A ¹	
	0x80 (Direct ENEC) or 0x81 (Direct DISEC)								T	
Sr	1	0	1	1	HID			W = 0	A ^{1,2}	
	0	0	0	0	0 ³	0	0 ³	ENINT/ DISINT	T	Sr ⁴ or P
NOTE 1 The DDR5RCD02 NACKs if there was a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.										
NOTE 2 The DDR5RCD02 device does not check for parity error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The RCD device ignores the entire packet until STOP or next Repeat Start operation.										
NOTE 3 DDR5RCD02 ignores HJ (Hot Join) and MR (Control Request) bits.										
NOTE 4 Repeat Start or Repeat Start with 7'h7E.										

Table 73 — ENEC/DISEC CCC (I3C Basic, Direct, PEC Enabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W = 0	A ¹	
	0x80 (Direct ENEC) or 0x81 (Direct DISEC)								T	
	PEC(8)								T	
Sr	1	0	1	1	HID			W = 0	A ^{1,2}	
	0	0	0	0	0 ³	0	0 ³	ENINT/ DISINT	T	
	PEC(8)								T	Sr ⁴ or P
NOTE 1	The DDR5RCD02 NACKs if there was a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start									
NOTE 2	The DDR5RCD02 device does not check for parity or PEC error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The RCD device ignores the entire packet until STOP or next Repeat Start operation.									
NOTE 3	DDR5RCD02 ignores HJ (Hot Join) and MR (Control Request) bits.									
NOTE 4	Repeat Start or Repeat Start with 7'h7E.									

7.10.2 Reset Dynamic Address Assignment Command (RSTDAA) Operation

Refer to the I3C Basic Version 1.0 Specification document for details on the encoding and usage of the RSTDAA Command. This command is only supported by the RCD when it is operating in I3C Basic mode. In I²C mode, this CCC is ignored (i.e., the RSTDAA command is not executed internally and any bytes arriving after the 0x86 RSTDAA Direct CCC are not acknowledged and they are ignored for all purposes, including parity checking, until the next STOP operation or Repeat START with 7'h7E is received).

7.10.2 Reset Dynamic Address Assignment Command (RSTDAA) Operation (cont'd)

The Host puts the DDR5RCD02 device in I²C mode by issuing RSTDAA CCC. When RSTDAA CCC is registered by the device, the DDR5RCD02 updates [RW25\[5\]](#) to '0' and it takes effect at the next START operation (i.e. after STOP condition). Moreover, the RCD is allowed to disable IBI, disable PEC, and enable I3C Basic Parity Checking functions. The host is expected to send DISEC and DEVCTRL commands to disable IBI, disable PEC, and enable Parity Checking prior to issuing the RSTDAA CCC.

If PEC function is enabled, the PEC calculation starts with Start or Repeat Start operation but does not include 7'h7E with W=0 byte in PEC calculation.

Table 74 — RSTDAA CCC (I3C Basic, Broadcast, PEC Disabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W = 0	A ¹	
	0x06 (Broadcast RSTDAA)								T	Sr ² or P
NOTE 1	The DDR5RCD02 NACKs if there was a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.									
NOTE 2	Repeat Start or Repeat Start with 7'h7E.									

Table 75 — RSTDAA CCC (I3C Basic, Broadcast, PEC Enabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W = 0	A ¹	
	0x06 (Broadcast RSTDAA)								T	
	PEC(8)								T	
NOTE 1	The DDR5RCD02 NACKs if there was a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start									
NOTE 2	Repeat Start or Repeat Start with 7'h7E.									

7.10.3 Set All Addresses to Static Address Command (SETAASA) Operation

Refer to the I3C Basic Version 1.0 Specification document for details on the encoding and usage of the SETAASA Command. This command is supported by the RCD only when it is operating in I²C mode. In I²C mode, when host issues this CCC, to guarantee that this CCC is registered by the device without any error and to avoid subsequent undesired effects, the host shall limit the maximum speed of operation for this CCC to 1 MHz. In I3C Basic mode, this CCC is ignored (i.e., the SETAASA command is not executed internally).

Upon receiving SETAASA Command, the DDR5RCD02 device will update the setting in [RW25\[5\]](#) to '1' and it takes effect at the next START operation (i.e. after STOP condition).

As the device is in I2C mode when SETAASA CCC is issued, SETAASA CCC does not support PEC function.

Table 76 — SETAASA CCC (I²C, Broadcast)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W = 0	A	
	0x29 (Broadcast SETAASA)								T	P

7.10.4 Get Device Status Command (GETSTATUS) Operation

Refer to the I3C Basic Version 1.0 Specification document for details on the encoding and usage of the GETSTATUS Command. This command is only supported by the RCD when it is operating in I3C Basic mode. In I²C mode, this CCC is ignored (i.e., the GETSTATUS command is not executed internally and GETSTATUS Direct CCC are not acknowledged and host must do STOP operation).

7.10.4 Get Device Status Command (GETSTATUS) Operation (cont'd)

Table 77 shows the format of the MSb and LSb sent by the DDR5RCD02 device in response to a GETSTATUS command. When the DDR5RCD02 device completes a GETSTATUS CCC operation, it does not modify any of the status bits in Table 135, “RW28: I2C and I3C Basic Error Status Global Word”.

If PEC function is enabled, the PEC calculation starts with Start or Repeat Start operation but does not include 7'h7E with W=0 byte in PEC calculation.

Table 77 — GETSTATUS MSb-LSb Format

Bits	Field	Description
15	PEC Error	DDR5RCD02 returns the value of RW28[1]. This bit is cleared when Host issues Clear Command to RW29[1]. 0 = No Error 1 = PEC Error Occurred
14:8	Vendor Reserved	Hard coded to all-zeros in DDR5RCD02.
7:6	Activity Mode	Hard coded to all-zeros in DDR5RCD02.
5	Protocol Error	DDR5RCD02 returns the value of RW28[0]. This bit is cleared when Host issues Clear Command to RW29[0]. 0 = No Error 1 = Protocol Error; Parity Error Occurred
4	Reserved	Hard coded to zero in DDR5RCD02.
3:0	Pending Interrupt	DDR5RCD02 returns the value of RW28[7] in Bit 0. Bit 0 is cleared when Host issues a Clear Command that causes the IBI Status register RW28[7] to get cleared. 0000 = No Pending Interrupt 0001 = Pending Interrupt All other encodings are reserved

Table 78 — GETSTATUS CCC (I3C Basic, Direct, PEC Disabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W = 0	A ¹	
	0x90 (Direct GETSTATUS)								T	
Sr	1	0	1	1	HID			R = 1	A ¹	
	PEC_Err	0	0	0	0	0	0	0	T = 1	
	0	0	Prot_Err	0	Pending Interrupt				T = 0 ²	
NOTE 1	The DDR5RCD02 NACKs if there was a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.									
NOTE 2	Figure 126 shows how the RCD ends the operation followed by the Controller STOP bit.									
NOTE 3	Repeat Start or Repeat Start with 7'h7E.									

Table 79 — GETSTATUS CCC (I3C Basic, Direct, PEC Enabled)¹

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W = 0	A ²	
	0x90 (Direct GETSTATUS)								T	
	PEC(8)								T	
Sr	1	0	1	1	HID			R = 1	A ²	
	PEC_Err	0	0	0	0	0	0	0	T = 1	
	0	0	Prot_Err	0	Pending Interrupt				T = 1	
	PEC(8)								T = 0 ³	
NOTE 1	GETSTATUS CCC with PEC check is only supported in I3C Basic mode.									
NOTE 2	The DDR5RCD02 NACKs if there was a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.									
NOTE 3	Figure 126 shows how the RCD ends the operation followed by the Controller STOP bit.									
NOTE 4	Repeat Start or Repeat Start with 7'h7E.									

When the device responds to GETSTATUS CCC, after it completes the response, the PEC_Err, Prot_Err, and Pending Interrupt Bits [3:0] do not automatically get cleared. The host must explicitly clear the appropriate status register through Clear command by writing '1' to corresponding register or by issuing Global Clear command. Once the DDR5RCD02 device clears the appropriate status register, only then PEC_Err, Prot_Err, and Pending Interrupt Bits [3:0] get cleared.

After the host issues a Clear command, if the condition is still present, the device will again set the appropriate status register, sets the IBI status register to '1' and Pending Interrupt Bits [3:0] to '0001'.

7.10.5 Get Device Capability Command (DEVCAP) Operation

Refer to JESD403-1 for details on the encoding and usage of the DEVCAP Command. This command is only supported by the RCD when it is operating in I3C Basic mode. In I²C mode, it is illegal for host to issue this CCC. If host issues this CCC while the RCD is in I²C mode, the response is undefined.

If PEC function is enabled, the PEC calculation starts with Start or Repeat Start operation but does not include 7'h7E with W=0 byte in PEC calculation.

Table 80 and Table 81 show the format of the DEVCAP CCC Packet.

Table 80 — DEVCAP CCC (I3C Basic, Direct, PEC Disabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W = 0	A ¹	
	0xE0 (Direct DEVCAP)								T	
Sr	1	0	1	1	HID			R = 1	A ¹	
	MSB (each bit defines capability); 1 = support; 0 = no support								T = 1	
	LSB (each bit defines capability); 1 = support; 0 = no support								T = 0 ²	
NOTE 1	The DDR5RCD02 NACKs if there was a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.									
NOTE 2	Figure 126 shows how the RCD ends the operation followed by the Controller STOP bit.									
NOTE 3	Repeat Start or Repeat Start with 7'h7E.									

7.10.5 Get Device Capability Command (DEVCAP) Operation (cont'd)

Table 81 — DEVCAP CCC (I3C Basic, Direct, PEC Enabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W = 0	A ¹	
Sr	0xE0 (Direct DEVCAP)								T	
	PEC(8)								T	
	1	0	1	1	HID			R = 1	A ¹	
	MSB (each bit defines capability); 1 = support; 0 = no support								T = 1	
	LSB (each bit defines capability); 1 = support; 0 = no support								T = 1	
	PEC(8)								T = 0 ²	
NOTE 1	The DDR5RCD02 NACKs if there was a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.									
NOTE 2	Figure 126 shows how the RCD ends the operation followed by the Controller STOP bit.									
NOTE 3	Repeat Start or Repeat Start with 7'h7E.									

Table 82 shows the format of the MSb and LSb sent by the DDR5RCD02 device in response to a DEVCAP command.

Table 82 — DEVCAP MSb-LSb Format

Bits	Field	Description
MSB[7]	Reserved	Hard coded to zero in DDR5RCD02.
MSB[6]	Reserved	Hard coded to zero in DDR5RCD02.
MSB[5]	Reserved	Hard coded to zero in DDR5RCD02.
MSB[4]	Reserved	Hard coded to zero in DDR5RCD02.
MSB[3]	Reserved	Hard coded to zero in DDR5RCD02.
MSB[2]	Timer Based Reset	Hard coded to one (yes support) in DDR5RCD02.
MSB[1:0]	Reserved	Hard coded to zero in DDR5RCD02.
LSB[7:0]	Reserved	Hard coded to zero in DDR5RCD02.

7.10.6 Set HID Command (SETHID) Operation

Refer to JESD403-1 for details on the encoding and usage of the SETHID Command. This command is supported by the RCD only when it is operating in I²C mode. In I²C mode, when host issues this CCC, to guarantee that this CCC is registered by the device without any error and to avoid subsequent undesired effects, the host shall limit the maximum speed of operation for this CCC to 1 MHz. In I3C Basic mode, it is illegal for host to issue this CCC. If host issues this CCC while the RCD is in I3C Basic mode, the response is undefined.

Upon receiving SETHID Command, the DDR5RCD02 device will update the HID code with the HID[2:0] bits it received in the SETHID command packet, and it takes effect at the next START operation (i.e. after STOP condition).

As the device is in I2C mode when SETHID CCC is issued, SETHID CCC does not support PEC function.

Table 83 — SETHID CCC (I²C, Broadcast)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W = 0	A	
	0x61 (Broadcast SETHID)								T	
	0	0	0	0	HID[2:0]			0	T	P

7.10.7 Configure SPD Hub and All Devices Behind Hub Command (DEVCTRL) Operation

Refer to JESD403-1 for details on the encoding and usage of the DEVCTRL Command. This command is supported by the DDR5RCD02 device in both I²C and in I3C Basic modes. In I²C mode, when host issues this CCC, to guarantee that this CCC is registered by the device without any error and to avoid subsequent undesired effects, the host shall limit the maximum speed of operation for this CCC to 1 MHz.

On a typical I²C or I3C Basic bus there can be up to 120 devices. For certain operation such as selecting the I3C Basic mode of operation or to enable or disable functions that are common to all devices such as Packet Error Check, the host must go through one device at a time which takes significant amount of time at initial power up. Further, it requires additional complexity on the host because it must speak different protocol depending on how it may access the device until all devices are configured identically.

To help expedite this configuration operation and to simplify the host complexity, the device supports the DEVCTRL CCC. The DEVCTRL CCC is supported either in I²C mode or I3C Basic mode of operation. In I²C mode, when host issues this CCC, to guarantee that this CCC is registered by the device without any error and to avoid subsequent undesired implication, the host shall limit the maximum speed of operation for this CCC to 1 MHz. Table 84 to Table 85 show examples of DEVCTRL CCC command packets.

If PEC function is enabled, the PEC calculation starts with Start or Repeat Start operation but does not include 7'h7E with W = 0 byte in PEC calculation.

The host must pay attention to DEVCTRL CCC. If DEVCTRL CCC is used to access device specific registers (i.e. RegMod = '1'), the host must still follow any device specific register restriction. For instance, if any device specific register requires STOP operation for device to take in the effect of the setting, the host must also use STOP operation when using DEVCTRL CCC to access device specific register.

Table 84 — DEVCTRL Command Operation (I²C or I3C Basic, Broadcast, PEC Disabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W = 0	A ¹	
	0	1	1	0	0	0	1	0	T	
	AddrMask[2:0]			StartOffset[1:0]		0	0	RegMod	T	
	DevID[6:0]							0	T ²	
	Byte 0 Data Payload								T	
	Byte 1 Data Payload								T	
	Byte 2 Data Payload								T	
	Byte 3 Data Payload								T	
NOTE 1	The DDR5RCD02 NACKs if there was a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.									
NOTE 2	As an exception, the DDR5RCD02 device will check parity in this byte even when the 7-bit Device ID code issued by the host does not match with its own device code. If the Device ID code is unmatched, the RCD will not check for Parity errors in subsequent bytes and it will ignore the entire packet until STOP or Repeat START operation.									
NOTE 3	Repeat Start or Repeat Start with 7'h7E.									

7.10.7 Configure SPD Hub and All Devices Behind Hub Command (DEVCTRL) Operation (cont'd)**Table 85 — DEVCTRL Command Operation (I3C Basic. Broadcast, PEC Enabled)¹**

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W = 0	A ²	
	0	1	1	0	0	0	1	0	T	
	AddrMask[2:0]			StartOffset[1:0]		PEC_BL		RegMod	T	
	DevID[6:0]							0	T ³	
	Byte 0 Data Payload								T	
	Byte 1 Data Payload								T	
	Byte 2 Data Payload								T	
	Byte 3 Data Payload								T	
	PEC(8)								T	

NOTE 1 DEVCTRL CCC with PEC check is only supported in I3C Basic mode.

NOTE 2 The DDR5RCD02 NACKs if there was a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.

NOTE 3 As an exception, the DDR5RCD02 device will check parity in this byte even when the 7-bit Device ID code issued by the host does not match with its own device code. If the Device ID code is unmatched, the RCD will not check for Parity or PEC errors in subsequent bytes and it will ignore the entire packet until STOP or Repeat START operation.

NOTE 4 Repeat Start or Repeat Start with 7'h7E.

7.10.7 Configure SPD Hub and All Devices Behind Hub Command (DEVCTRL) Operation (cont'd)

Table 86 — DEVCTRL Command Definition

Parameter	Definition
AddrMask[2:0]	Broadcast, Uni-cast or Multi-cast Command Selection 000 = Uni-Cast Command; Target device responds if DevID[6:0] field matches with target device's own 7-bit address (4-bit LID + 3-bit HID) 011 = Multi-Cast Command; Target and possible other device responds if DevID[6:3] field matches with target's own 4-bit LID address 111 = Broadcast Command; All devices responds to this command All other encodings are reserved
StartOffset[1:0]	Only applicable if RegMod = '0' Identifies the starting Byte. (Byte 0 or Byte 1 or Byte 2 or Byte 3) in DEVCTRL command data packet. Host can start any Byte and can have continuous access to next byte until STOP operation. If Byte 3 is reached, the host is responsible for applying STOP operation. 00 = Byte 0 01 = Byte 1 10 = Byte 2 11 = Byte 3
PEC_BL[1:0]	Only applicable if RegMod = '0' Identifies the burst length just for the DEVCTRL command data packet. The device uses the setting in this field to know when the PEC byte is expected after the data bytes. 00 = 1 Byte 01 = 2 Byte 10 = 3 Byte 11 = 4 Byte
RegMod	Identifies if DEVCTRL command is going to be used for device specific offset register or general registers as identified in Byte 0 to Byte 3. 0 = Access to General Registers in Byte 0 to Byte 3 (i.e. StartOffset[1:0] = Valid) 1 = Device Specific Offset Address (i.e. StartOffset[1:0] & PECBL[1:0] is a don't care and does not apply). The host shall not use RegMod = '1' with Broadcast Command for DDR5 DIMM because there are different types of devices on the I3C Basic bus. The DDR5RCD02 device is not required to support the RegMod = '1' option when the interface is running in I²C mode.¹
DevID[6:0]	Identifies 7-bit device address. Device responds to DEVCTRL command data packet depending on AddrMask[2:0]. If AddrMask[2:0] = '111', DevID[6:0] is a don't care and device always responds. If AddrMask[2:0] = '000', DevID[6:0] must match for device to respond If AddrMask[2:0] = '011', DevID[6:3] must match for device to respond. DevID[2:0] is don't care. For any other codes for AddrMask[2:0], the device always NACKs.
NOTE 1 This is a deviation from the JESD403-1 Specification for JEDEC Module SidebandBus.	

7.10.7 Configure SPD Hub and All Devices Behind Hub Command (DEVCTRL) Operation (cont'd)**Table 87 — DEVCTRL Command Definition**

Byte	Bit	Function	Definition	Comment
Byte 0	[7]	PEC Enable	0 = Disable 1 = Enable ¹	
	[6]	Parity Disable	0 = Enable 1 = Disable	
	[5]	RFU	RFU	
	[4:2]	RFU	RFU	
	[1]	RSVD - Reserved	X (0 or 1) = No action	This bit is used by other devices and must be ignored by the RCD.
	[0]	RFU	RFU	
Byte 1	[7:5]	RFU	RFU	
	[4]	RFU	RFU	
	[3]	Global and IBI Clear	0 = No action 1 = Clear All Event and pending IBI ^{1,2}	RW28[1:0] are updated
	[2]	RFU	RFU	RFU
	[1:0]	RFU	RFU	
Byte2	[7:0]	RFU	RFU	
Byte 3	[7:4]	RFU	RFU	
	[3:1]	RFU	RFU	
	[0]	RFU	RFU	

NOTE 1 Global Clear and PEC Enable only applicable after device is configured in I3C Basic mode.

NOTE 2 After DDR5RCD02 clears all events, the RCD can still have certain registers set to '1' if the event is still present in which case, the RCD will generate an IBI again at the next opportunity.

7.10.7.1 DEVCTRL Command Examples

The following DEVCTRL Command configures frequency information to 4560 MT/s (RW05 = 04h and RW06 = 0Ch) in all RCD devices simultaneously.

Table 88 — DEVCTRL Command Example (I3C Basic, Broadcast, PEC Enabled)

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W = 0	A	
	0	1	1	0	0	0	1	0	T	
AddrMask[2:0] = '011' (Mcast)				StartOffset = xx		PEC_BL = xx		RegMod=1	T	
DevID[6:0] = 1011_xxx (Match LID code for RCD Device)								0	T	
SidebandBus Command = 11xx_11xx (Double Word Write) ¹									T	
Byte Count = 08h									T	
Reserved									T	
Dev/Channel_Num = 00h (Channel A)									T	
Page_Num [7:0] = 00h (Ignored for RW05 & RW06)									T	
Reg_Num [7:0] = 04h (points at RW04)									T	
Wr Data[31:24] = 00h (RW07 default)									T	
Wr Data[23:16] = 0Ch (fbin - 240 MT/s)									T	
Wr Data[15:8] = 04 h (4400 MT/s to 4800 MT/s)									T	
Wr Data[7:0] = 00h (Command Space CMD 0 - NOP)									T	
PEC(8)									T	

NOTE 1

In DEVCTRL, Bits [3:2] of SidebandBus Command can be set to 01 (Write Byte), 10 (Write Word), or 11 (Write DWord). In other words, all Write Internal Commands are supported in the DEVCTRL CCC packet.

8 Control Words

The DDR5RCD02 device features a set of 8-bit control words, which allow the optimization of the device properties for different raw card designs. DDR5RCD02 control word writes appear as MRW commands but with the CW bit HIGH. DRAMs will ignore MRW commands with the CW bit set HIGH. Each Register Word (RW) write generates an MRW command to the DRAMs behind the register, unless there is a parity error when parity checking is enabled or the DRAM interface Forward All Commands feature is disabled in [RW01\[1\]](#). The different control words and settings are described below. Any change to these control words require some time for the device to settle. For changes to the control word setting, except for [RW05](#) and [RW06](#), the controller needs to wait t_{MRD} after the last control word access before further access to the DRAM can take place. For any changes to the clock timing ([RW05](#) and [RW06](#)) this settling may take up to t_{STAB02} time. All chip select inputs for the accessed channel, DCS[1:0]_n, must be kept HIGH during that time.

The control words are independent between the two channels. Each channel has a separate control word space that is accessible via the input chip selects for that channel. Some control words are only accessible through Channel A. These are control words that pertain to the clock frequency, etc., that are global in nature. Control word writes may occur independently on the two channels, so it is possible that both channels will be processing control word accesses at the same time, possibly offset by any number of DCKs from each other. The DDR5RCD02 must support control word access for any of the DDR5RCD02 specified frequency ranges.

RCD Control Word attributes can be:

- Reserved¹
- Read Only
- Write Only
- RD/WR
- One Time Programmable
- Sticky - Cleared by power cycle not Reset

The control word space is divided as follows:

Table 89 — Control Word Addressing

CW bit	MRA7	Description
0	x	DRAM MR space.
1	0	RCD CW space. 128 registers decoded via MRA[6:0]
1	1	Data Buffer CW space. 128 registers decoded via MRA[6:0]

Within the RCD control word space, 96 registers are accessed directly, while 32 registers are addressed through an 8-bit paging system, providing 8192 registers. [RW5F](#) is the page register while addresses 60-7F are used to access the registers within each page. Pages 00-7F are for JEDEC use while pages 80-FF are vendor specific.

1. Reserved control bits may not be physically implemented and they shall be written to zero to ensure forward compatibility.

8.0 Control Words (cont'd)

Table 90 — RCD and DB Control Word Spaces

MRA[7:0]	Location	Page Pointer RW ¹	Page Pointer Value	Definition	Size (bytes)	Description
00h to 5Fh	RCD	RW5F	X	JEDEC	96	96 directly addressed registers
60h to 7Fh	RCD	RW5F	00h-7Fh	JEDEC	4096	32 paged registers
60h to 7Fh	RCD	RW5F	80h-FFh	Vendor	4096	32 paged registers
80h to DFh	Data Buffer	RWDF	X	JEDEC	96	96 directly addressed registers
E0h to FFh	Data Buffer	RWDF	00h-7Fh	JEDEC	4096	32 paged registers
E0h to FFh	Data Buffer	RWDF	80h-FFh	Vendor	4096	32 paged registers
NOTE 1 RW5F is the Page Pointer located in the DDR5 Register and RWDF is the Page Pointer located in the DDR5 Data Buffer						

All registers are 8-bit. If enabled in RCD, all MRW accesses are sent across the BCOM bus to the data buffers, allowing the data buffers to snoop values sent to the DRAM or RCD.

Control Word Naming Convention:

PG[x]RWyy[z:z]
 x = Page
 yy = RW Number
 z = OP

8.1 Reading Control Words

Control words are read by passing the data onto the DRAM devices, where they are read by the host controller by doing an MRR (Mode Register Read). The sequence is as follows:

Table 91 — Control Word Read Sequence

Function	Cmd	CW bit	Address (hex)	Description
Select which control word to read	MRW	1	5Eh	MRW command to RCD control word RW5E which points to which RCD control word will be read. ¹
Transfer the RCD control word content to the DRAMs	MRW	0	3Fh	MRW command to the DRAM MR63 (3Fh). When the RCD detects address MR63 in an MRW command in the 1st UI, it will substitute the data from the control word pointed to by RW5E onto CA[7:0] of the 2nd UI (replacing the data from the host) ² . This data is written into the scratch pad register MR63 of the DRAM.
Read the data into the host controller	MRR	0	3Fh	MRR command to the DRAM MR63 to read the data from the DRAM.
NOTE 1 If the RCD Control Word to be read is in the paged space, then the host must set the page register properly before issuing the MRW write to the DRAM scratch-pad register.				
NOTE 2 When the MR63 command is issued only CA[7:0] get replaced by the control word data and all other bits are forwarded to the DRAM.				

NOTE: It is required that the DRAM has an 8-bit scratch-pad register at location MR63. MR63 is read and written in the same manner as all other mode registers. The RCD will check MRA[7:0] when determining whether to replace the host data with the selected RW register content.

8.2 Control Word Decoding

The control words are programmed by the host controller using the MRW command function. The DDR5RCD02 forwards the MRW command to the DRAM. The DRAM is required to ignore the MRW Command when CW = '1'.

Table 92 — DDR MRW Command to RCD¹

MRW	DCS_n	CA Pins						
		CA0	CA1	CA2	CA3	CA4	CA5	CA6
UI1	L	H	L	H	L	L	MRA0	MRA1
UI2	H	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V
UI3	H	OP0	OP1	OP2	OP3	OP4	OP5	OP6
UI4	H	OP7	V	V	CW	V	V	V
NOTE 1 V means H or L (a defined logic level).								

The power-on reset default state of all control words is '0' unless specified otherwise. Every time the device is reset, its default state is restored with the exception of sticky RWs. Cycling the device in and out of low-power mode will not alter the control word settings.

8.3 Control Word Decoding

Table 93 — Control Word Decoding

Register Control Word	MRA [7:0] HEX ¹	Meaning	Global Only (Requires Channel A Access) ²
RW00	0x00	Global Features	Yes
RW01	0x01	Parity, CMD Blocking, and Alert	Yes
RW02	0x02	Host Interface Training Mode	Yes
RW03	0x03	DRAM & DB Interface Training Mode	Yes
RW04	0x04	Command Space	Yes
RW05	0x05	DIMM Operating Speed	Yes
RW06	0x06	Fine Granularity DIMM Operating Speed	Yes
RW07	0x07	Validation Pass-Through and Lockout Protection	Yes
RW08	0x08	Clock Driver Enable	No
RW09	0x09	Output Address and Control Enable	No
RW0A	0x0A	QCK Signals Driver Characteristics	No
RW0B	0x0B	Reserved	No
RW0C	0x0C	QCA and QxCS_n Signals Characteristics	No
RW0D	0x0D	Data Buffer Interface Driver Characteristics	No
RW0E	0x0E	QCK, QCA and QCS Output Slew Rate	No
RW0F	0x0F	BCK, BCOM and BCS Output Slew Rate	No
RW10	0x10	IBT	Yes
RW11	0x11	Command Latency Adder	No
RW12	0x12	QACK Output Delay	No
RW13	0x13	QBCK Output Delay	No
RW14	0x14	QCCK Output Delay Control	No
RW15	0x15	QDCK Output Delay Control	No
RW16	0x16	Reserved	No
RW17	0x17	QACS0 Output Delay	No
RW18	0x18	QACS1 Output Delay	No
RW19	0x19	QBCS0 Output Delay	No
RW1A	0x1A	QBCS1 Output Delay	No
RW1B	0x1B	QACA Output Delay	No
RW1C	0x1C	QBCA Output Delay	No
RW1D	0x1D	BCS and BCOM Output Delay	No
RW1E	0x1E	BCK Output Delay	No
RW1F	0x1F	Reserved	No
RW20	0x20	Error Log Register	No
RW21	0x21	Error Log Register	No
RW22	0x22	Error Log Register	No
RW23	0x23	Error Log Register	No
RW24	0x24	Error Log Register	No
RW25	0x25	SidebandBus	Yes
RW26	0x26	Loopback	Yes
RW27	0x27	Loop-back I/O	Yes
RW28	0x28	I ² C & I3C Basic Error Status	Yes
RW29	0x29	I ² C & I3C Basic Clear Error Status	Yes
RW2A	0x2A	Vendor Specific	Yes
RW2B~RW2F	0x2B~0x2F	Reserved	No
RW30	0x30	DFE_Vref Range Limit	Yes
RW31	0x31	DFE Configuration	No
RW32	0x32	DPAR and DCA[6:0] DFE Training Mode	No

Table 93 — Control Word Decoding (cont'd)

Register Control Word	MRA [7:0] HEX ¹	Meaning	Global Only (Requires Channel A Access) ²
RW33	0x33	Additional Filtering for DFE Training Mode	No
RW34	0x34	LFSR DFE Training Mode	No
RW35	0x35	LFSR State for DFE Training Mode	No
RW36	0x36	DFETM Inner Loop Start Value	No
RW37	0x37	DFETM Outer Loop Start Value	No
RW38	0x38	DFETM Inner Loop Current Value	No
RW39	0x39	DFETM Outer Loop Current Value	No
RW3A	0x3A	DFETM Inner and Outer Loop Step Size	No
RW3B	0x3B	DFETM Inner Loop Number of Increments	No
RW3C	0x3C	DFETM Outer Loop Number of Increments	No
RW3D	0x3D	DFETM Inner Loop Current Increment	No
RW3E	0x3E	DFETM Outer Loop Current Increment	No
RW3F	0x3F	DFE Vref Range Selection	No
RW40	0x40	DCA0 Internal Vref	No
RW41	0x41	DCA1 Internal Vref	No
RW42	0x42	DCA2 Internal Vref	No
RW43	0x43	DCA3 Internal Vref	No
RW44	0x44	DCA4 Internal Vref	No
RW45	0x45	DCA5 Internal Vref	No
RW46	0x46	DCA6 Internal Vref	No
RW47	0x47	DPAR Internal Vref	No
RW48	0x48	DCS0 Internal Vref	No
RW49	0x49	DCS1 Internal Vref	No
RW4A	0x4A	DERROR_IN_n Vref	No
RW4B	0x4B	Loop-Back Vref	No
RW4C	0x4C	Loop-Back Input Delay	No
RW4D~RW4F	0x4D~0x4F	Reserved	No
RW50	0x50	CTLE Configuration Global Control Word	Yes
RW51	0x51	CTLE per-pin Disable	No
RW52	0x52	CTLE Parameter Set A	No
RW53	0x53	CTLE Parameter Set B	No
RW54	0x54	CTLE Parameter Set Selection	No
RW55~RW5D	0x55~0x5D	Reserved	No
RW5E	0x5E	CW Read Pointer	No
RW5F	0x5F	CW Page ³	No

NOTE 1 MRA7 must be 0 for RCD Control Word accesses and must be 1 for Data Buffer (DB) Control Words accesses.

NOTE 2 This column indicates global Register Control Words that apply to both channels but are written and read from Channel A only. Any attempts to write to CH_B will do nothing.

NOTE 3 All paged Control Words RW60 to RW7F are not global control words which means they are defined per sub channel and can be accessed through CH_A and CH_B unless otherwise stated.

8.4 Paged Control Word Decoding

8.4.1 Page 0 DFE Paged Control Word Decoding

Table 94 — Page 0 DFE Control Word Decoding

Page	Register Control Word	MR [7:0] HEX ¹	Meaning
0x00	RW60	0x60	DCA0 Rx DFE Gain Coefficients
	RW61	0x61	DCA0 Rx DFE Tap 1 Coefficients
	RW62	0x62	DCA0 Rx DFE Tap 2 Coefficients
	RW63	0x63	DCA0 Rx DFE Tap 3 Coefficients
	RW64	0x64	DCA0 Rx DFE Tap 4 Coefficients
	RW65	0x65	RFU
	RW66	0x66	RFU
	RW67	0x67	RFU
	RW68	0x68	DCA1 Rx DFE Gain Coefficients
	RW69	0x69	DCA1 Rx DFE Tap 1 Coefficients
	RW6A	0x6A	DCA1 Rx DFE Tap 2 Coefficients
	RW6B	0x6B	DCA1 Rx DFE Tap 3 Coefficients
	RW6C	0x6C	DCA1 Rx DFE Tap 4 Coefficients
	RW6D	0x6D	RFU
	RW6E	0x6E	RFU
	RW6F	0x6F	RFU
	RW70	0x70	DCA2 Rx DFE Gain Coefficients
	RW71	0x71	DCA2 Rx DFE Tap 1 Coefficients
	RW72	0x72	DCA2 Rx DFE Tap 2 Coefficients
	RW73	0x73	DCA2 Rx DFE Tap 3 Coefficients
	RW74	0x74	DCA2 Rx DFE Tap 4 Coefficients
	RW75	0x75	RFU
	RW76	0x76	RFU
	RW77	0x77	RFU
	RW78	0x78	DCA3 Rx DFE Gain Coefficients
	RW79	0x79	DCA3 Rx DFE Tap 1 Coefficients
	RW7A	0x7A	DCA3 Rx DFE Tap 2 Coefficients
	RW7B	0x7B	DCA3 Rx DFE Tap 3 Coefficients
	RW7C	0x7C	DCA3 Rx DFE Tap 4 Coefficients
	RW7D	0x7D	RFU
	RW7E	0x7E	RFU
	RW7F	0x7F	RFU
NOTE 1 MRA7 must be 0 for RCD accesses and must be 1 for Data Buffer (DB) Control Words accesses.			

8.4.2 Page 1 DFE Paged Control Word Decoding

Table 95 — Page 1 DFE Control Word Decoding

Page	Register Control Word	MR [7:0] HEX ¹	Meaning
0x01	RW60	0x60	DCA4 Rx DFE Gain Coefficients
	RW61	0x61	DCA4 Rx DFE Tap 1 Coefficients
	RW62	0x62	DCA4 Rx DFE Tap 2 Coefficients
	RW63	0x63	DCA4 Rx DFE Tap 3 Coefficients
	RW64	0x64	DCA4Rx DFE Tap 4 Coefficients
	RW65	0x65	RFU
	RW66	0x66	RFU
	RW67	0x67	RFU
	RW68	0x68	DCA5 Rx DFE Gain Coefficients
	RW69	0x69	DCA5 Rx DFE Tap 1 Coefficients
	RW6A	0x6A	DCA5 Rx DFE Tap 2 Coefficients
	RW6B	0x6B	DCA5 Rx DFE Tap 3 Coefficients
	RW6C	0x6C	DCA5 Rx DFE Tap 4 Coefficients
	RW6D	0x6D	RFU
	RW6E	0x6E	RFU
	RW6F	0x6F	RFU
	RW70	0x70	DCA6 Rx DFE Gain Coefficients
	RW71	0x71	DCA6 Rx DFE Tap 1 Coefficients
	RW72	0x72	DCA6 Rx DFE Tap 2 Coefficients
	RW73	0x73	DCA6 Rx DFE Tap 3 Coefficients
	RW74	0x74	DCA6 Rx DFE Tap 4 Coefficients
	RW75	0x75	RFU
	RW76	0x76	RFU
	RW77	0x77	RFU
	RW78	0x78	DPAR Rx DFE Gain Coefficients
	RW79	0x79	DPAR Rx DFE Tap 1 Coefficients
	RW7A	0x7A	DPAR Rx DFE Tap 2 Coefficients
	RW7B	0x7B	DPAR Rx DFE Tap 3 Coefficients
	RW7C	0x7C	DPAR Rx DFE Tap 4 Coefficients
	RW7D	0x7D	RFU
	RW7E	0x7E	RFU
	RW7F	0x7F	RFU
NOTE 1 MRA7 must be 0 for RCD accesses and must be 1 for Data Buffer (DB) Control Words accesses.			

8.4.3 Page 2 DFE Paged Control Word Decoding

Table 96 — Page 2 DFE Control Word Decoding

Page	Register Control Word	MR [7:0] HEX ¹	Meaning
0x02	RW60	0x60	DCA0 Rx DFE Error Counter Lower 8 Bits
	RW61	0x61	DCA0 Rx DFE Error Counter Upper 8 Bits
	RW62	0x62	DCA0 DFE_Vref
	RW63	0x63	DCA0 LFSR Seed
	RW64	0x64	DCA1 Rx DFE Error Counter Lower 8 Bits
	RW65	0x65	DCA1 Rx DFE Error Counter Upper 8 Bits
	RW66	0x66	DCA1 DFE_Vref
	RW67	0x67	DCA1 LFSR Seed
	RW68	0x68	DCA2 Rx DFE Error Counter Lower 8 Bits
	RW69	0x69	DCA2 Rx DFE Error Counter Upper 8 Bits
	RW6A	0x6A	DCA2 DFE_Vref
	RW6B	0x6B	DCA2 LFSR Seed
	RW6C	0x6C	DCA3 Rx DFE Error Counter Lower 8 Bits
	RW6D	0x6D	DCA3 Rx DFE Error Counter Upper 8 Bits
	RW6E	0x6E	DCA3 DFE_Vref
	RW6F	0x6F	DCA3 LFSR Seed
	RW70	0x70	DCA4 Rx DFE Error Counter Lower 8 Bits
	RW71	0x71	DCA4 Rx DFE Error Counter Upper 8 Bits
	RW72	0x72	DCA4 DFE_Vref
	RW73	0x73	DCA4 LFSR Seed
	RW74	0x74	DCA5 Rx DFE Error Counter Lower 8 Bits
	RW75	0x75	DCA5 Rx DFE Error Counter Upper 8 Bits
	RW76	0x76	DCA5 DFE_Vref
	RW77	0x77	DCA5 LFSR Seed
	RW78	0x78	DCA6 Rx DFE Error Counter Lower 8 Bits
	RW79	0x79	DCA6 Rx DFE Error Counter Upper 8 Bits
	RW7A	0x7A	DCA6 DFE_Vref
	RW7B	0x7B	DCA6 LFSR Seed
	RW7C	0x7C	DPAR Rx DFE Error Counter Lower 8 Bits
	RW7D	0x7D	DPAR Rx DFE Error Counter Upper 8 Bits
	RW7E	0x7E	DPAR DFE_Vref
	RW7F	0x7F	DPAR LFSR Seed
NOTE 1 MRA7 must be 0 for RCD accesses and must be 1 for Data Buffer (DB) Control Words accesses.			

8.4.4 Page 3 Manufacturing ID Control Word Decoding

Page 3 RW60 ~ RW6E control words in the DDR5RCD02 contain detailed RCD manufacturing information provided by RCD vendor. This information is programmed in OTP. The purpose is for improved manufacturing logistics management and not meant to be used by host platform to control normal operation. These registers must be accessible during run time.

Table 97 — Page 3 Manufacturing ID Control Word Decoding

Page	Register Control Word	MR [7:0] HEX ¹	Meaning
0x03	RW60	0x60	Date Code Byte 0
	RW61	0x61	Date Code Byte 1
	RW62	0x62	Date Code Byte 2
	RW63	0x63	Vendor Specific Unique Unit Code Byte 0
	RW64	0x64	Vendor Specific Unique Unit Code Byte 1
	RW65	0x65	Vendor Specific Unique Unit Code Byte 2
	RW66	0x66	Vendor Specific Unique Unit Code Byte 3
	RW67	0x67	Vendor Specific Unique Unit Code Byte 4
	RW68	0x68	Vendor Specific Unique Unit Code Byte 5
	RW69	0x69	Vendor Specific Unique Unit Code Byte 6
	RW6A	0x6A	Vendor ID Byte 0
	RW6B	0x6B	Vendor ID Byte 1
	RW6C	0x6C	Device ID Byte 0
	RW6D	0x6D	Device ID Byte 1
	RW6E	0x6E	Vendor Revision ID
	RW6F	0x6F	Reserved
	RW70	0x70	Reserved
	RW71	0x71	Reserved
	RW72	0x72	Reserved
	RW73	0x73	Reserved
	RW74	0x74	Reserved
	RW75	0x75	Reserved
	RW76	0x76	Reserved
	RW77	0x77	Reserved
	RW78	0x78	Reserved
	RW79	0x79	Reserved
	RW7A	0x7A	Reserved
	RW7B	0x7B	Reserved
	RW7C	0x7C	Reserved
	RW7D	0x7D	Reserved
	RW7E	0x7E	Reserved
	RW7F	0x7F	Reserved

NOTE 1 MRA7 must be 0 for RCD accesses and must be 1 for Data Buffer (DB) Control Words accesses.

8.4.5 Page 4 VHost Paged Control Word Decoding

Table 98 — Page 4 VHost Control Word Decoding ^{1,2,3}

Page	Register Control Word	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0	Encoding	
0x04	RW60	cmd0a_CA07	cmd0a_CA06	cmd0a_CA05	cmd0a_CA04	cmd0a_CA03	cmd0a_CA02	cmd0a_CA01	cmd0a_CA00	cmd0 1st UI	
	RW61	cmd0a_CS1	cmd0a_CS0	cmd0a_CA13	cmd0a_CA12	cmd0a_CA11	cmd0a_CA10	cmd0a_CA09	cmd0a_CA08		
	RW62	cmd0b_CA07	cmd0b_CA06	cmd0b_CA05	cmd0b_CA04	cmd0b_CA03	cmd0b_CA02	cmd0b_CA01	cmd0b_CA00	cmd0 2nd UI	
	RW63	cmd0b_CS1	cmd0b_CS0	cmd0b_CA13	cmd0b_CA12	cmd0b_CA11	cmd0b_CA10	cmd0b_CA09	cmd0b_CA08		
	RW64	cmd1a_CA07	cmd1a_CA06	cmd1a_CA05	cmd1a_CA04	cmd1a_CA03	cmd1a_CA02	cmd1a_CA01	cmd1a_CA00	cmd1 1st UI	
	RW65	cmd1a_CS1	cmd1a_CS0	cmd1a_CA13	cmd1a_CA12	cmd1a_CA11	cmd1a_CA10	cmd1a_CA09	cmd1a_CA08		
	RW66	cmd1b_CA07	cmd1b_CA06	cmd1b_CA05	cmd1b_CA04	cmd1b_CA03	cmd1b_CA02	cmd1b_CA01	cmd1b_CA00	cmd1 2nd UI	
	RW67	cmd1b_CS1	cmd1b_CS0	cmd1b_CA13	cmd1b_CA12	cmd1b_CA11	cmd0a_CA10	cmd1b_CA09	cmd1b_CA08		
	RW68	cmd2a_CA07	cmd2a_CA06	cmd2a_CA05	cmd2a_CA04	cmd2a_CA03	cmd2a_CA02	cmd2a_CA01	cmd2a_CA00	cmd2 1st UI	
	RW69	cmd2a_CS1	cmd2a_CS0	cmd2a_CA13	cmd2a_CA12	cmd2a_CA11	cmd2a_CA10	cmd2a_CA09	cmd2a_CA08		
	RW6A	cmd2b_CA07	cmd2b_CA06	cmd2b_CA05	cmd2b_CA04	cmd2b_CA03	cmd2b_CA02	cmd2b_CA01	cmd2b_CA00	cmd2 2nd UI	
	RW6B	cmd2b_CS1	cmd2b_CS0	cmd2b_CA13	cmd2b_CA12	cmd2b_CA11	cmd2b_CA10	cmd2b_CA09	cmd2b_CA08		
	RW6C	cmd3a_CA07	cmd3a_CA06	cmd3a_CA05	cmd3a_CA04	cmd3a_CA03	cmd3a_CA02	cmd3a_CA01	cmd3a_CA00	cmd3 1st UI	
	RW6D	cmd3a_CS1	cmd3a_CS0	cmd3a_CA13	cmd3a_CA12	cmd3a_CA11	cmd3a_CA10	cmd3a_CA09	cmd3a_CA08		
	RW6E	cmd3b_CA07	cmd3b_CA06	cmd3b_CA05	cmd3b_CA04	cmd3b_CA03	cmd3b_CA02	cmd3b_CA01	cmd3b_CA00	cmd3 2nd UI	
	RW6F	cmd3b_CS1	cmd3b_CS0	cmd3b_CA13	cmd3b_CA12	cmd3b_CA11	cmd3b_CA10	cmd3b_CA09	cmd3b_CA08		
	RW70	Reserved	cmd1 to cmd2 Delay				Reserved	cmd0 to cmd1 Delay			Delay between commands
	RW71	Reserved	cmd3 to cmdn Delay				Reserved	cmd2 to cmd3 Delay			
	RW72	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Repeat Enable	Repeat Control		Repeat
	RW73	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	VHost Start/Stop	VHost Selection		Start
	RW74	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
	RW75	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
	RW76	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
	RW77	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
	RW78	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
	RW79	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
	RW7A	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
	RW7B	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
	RW7C	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
	RW7D	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
	RW7E	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
	RW7F	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
NOTE 1	For LRDIMM, the BCOM interface interprets the relevant commands from VHost and sends them out as BCOM command sequences similar to the way it operates in normal mode. Irrelevant commands result in BCOM Deselect (idle) cycles.										
NOTE 2	For LRDIMM, the user is responsible for programming long enough command intervals in PG[4]RW70/PG[4]RW71 so that there is no chance of chopping any BCOM sequences or dropping commands that occur when the BCOM interface is busy transmitting the previous command. This includes avoiding sending another command requiring the BCOM bus one clock after a 1UI DRAM command which requires 2 clocks on the BCOM bus (PDE command).										
NOTE 3	For LRDIMM, the BCOM command sequences will be delayed one cycle with respect to the QCA commands similar to the way things work in normal operation.										

8.4.6 Page 5 Per-bit QCA Output Delay Control Word Decoding

Table 99 — Page 5 Per-bit QCA Output Delay Control Word Decoding¹

Page	Register Control Word	MR[7:0] HEX	Meaning
0x05	RW60	0x60	QACA0 Output Delay
	RW61	0x61	QACA1 Output Delay
	RW62	0x62	QACA2 Output Delay
	RW63	0x63	QACA3 Output Delay
	RW64	0x64	QACA4 Output Delay
	RW65	0x65	QACA5 Output Delay
	RW66	0x66	QACA6 Output Delay
	RW67	0x67	QACA7 Output Delay
	RW68	0x68	QACA8 Output Delay
	RW69	0x69	QACA9 Output Delay
	RW6A	0x6A	QACA10 Output Delay
	RW6B	0x6B	QACA11 Output Delay
	RW6C	0x6C	QACA12 Output Delay
	RW6D	0x6D	QACA13 Output Delay
	RW6E	0x6E	QBCA0 Output Delay
	RW6F	0x6F	QBCA1 Output Delay
	RW70	0x70	QBCA2 Output Delay
	RW71	0x71	QBCA3 Output Delay
	RW72	0x72	QBCA4 Output Delay
	RW73	0x73	QBCA5 Output Delay
	RW74	0x74	QBCA6 Output Delay
	RW75	0x75	QBCA7 Output Delay
	RW76	0x76	QBCA8 Output Delay
	RW77	0x77	QBCA9 Output Delay
	RW78	0x78	QBCA10 Output Delay
	RW79	0x79	QBCA11 Output Delay
	RW7A	0x7A	QBCA12 Output Delay
	RW7B	0x7B	QBCA13 Output Delay
	RW7C	0x7C	Reserved
	RW7D	0x7D	Reserved
	RW7E	0x7E	Reserved
	RW7F	0x7F	Reserved
NOTE 1 The corresponding group fractional delay feature must also be enabled to enable per-pin physical delay for a given output. For example, in order to change the QACA0 per-pin physical delay, the control bit RW1B[7] must be set to 1 in addition to the per-pin enable bit PG[5]RW60[7].			

8.5 Sticky Bits

8.5.1 Direct

1. [RW00\[1:0\]](#): Command Address Rate & SDR Modes
2. [RW05\[3:0\]](#): Operating Speed
3. [RW05\[6:5\]](#): V_{DD} Operating Voltage & Frequency Context
4. [RW06](#): Fine Granularity DIMM Operating Speed
5. [RW08](#): Clock Driver Enable
6. [RW09](#): Output Address and Control Enable
7. [RW0A](#): QCK Signal Driver Characteristics
8. [RW0C](#): QxCA and QxCS Signal Driver Characteristics
9. [RW0D](#): Data Buffer Interface Driver Characteristics
10. [RW0E](#): QCK, QCA and QCS Output Slew Rate
11. [RW0F](#): BCK, BCOM and BCS Output Slew Rate
12. [RW10](#): IBT
13. [RW11](#): Command Latency Adder
14. [RW12](#): QACK Output Delay
15. [RW13](#): QBCK Output Delay
16. [RW14](#): QCK Output Delay
17. [RW15](#): QDCK Output Delay
18. [RW17](#): QACS0_n Output Delay
19. [RW18](#): QACS1_n Output Delay
20. [RW19](#): QBCS0_n Output Delay
21. [RW1A](#): QBCS1_n Output Delay
22. [RW1B](#): QACA Output Delay
23. [RW1C](#): QBCA Output Delay
24. [RW1D](#): BCS_n and BCOM[2:0] Output Delay
25. [RW1E](#): BCK Output Delay
26. [RW25\[5\]](#): SidebandBus Mode
27. [RW28](#): I²C & I3C Basic Error Status
28. [RW31](#): DFE Configuration
29. [RW\[47:40\]](#): Internal VrefCA
30. [RW\[49:48\]](#): Internal VrefCS
31. [RW50\[0\]](#), [RW51](#), [RW52\[3:0\]](#), [RW53\[3:0\]](#), [RW54](#): CTLE Control Words

8.5.2 Paged

1. [PG\[1:0\]RW\[61, 69, 71, 79\]](#): DPAR and DCA[6:0] Receiver DFE Tap 1 Coefficients
2. [PG\[1:0\]RW\[62, 6A, 72, 7A\]](#): DPAR and DCA[6:0] Receiver DFE Tap 2 Coefficients
3. [PG\[1:0\]RW\[63, 6B, 73, 7B\]](#): DPAR and DCA[6:0] Receiver DFE Tap 3 Coefficients
4. [PG\[1:0\]RW\[64, 6C, 74, 7C\]](#): DPAR and DCA[6:0] Receiver DFE Tap 4 Coefficients
5. [PG\[1:0\]RW\[60, 68, 70, 78\]](#): DPAR and DCA[6:0] Receiver DFE Gain Offset Adjustment
6. [PG\[5\]RW\[7B:60\]](#): Per-bit QCA Output Delay Control Words

8.6 Mode and Command Control Words

8.6.1 RW00 - Global Features Control Word

Table 100 — RW00 - Global Features Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	Command Address Rate ¹	Single Data Rate (SDR)
x	x	x	x	x	x	x	1		Double Data Rate (DDR)
x	x	x	x	x	x	0	x	SDR Modes ^{1,2}	SDR1 (additional setup time)
x	x	x	x	x	x	1	x		SDR2
x	x	x	x	x	0	x	x	CA Pass Through mode Enable	Normal operation
x	x	x	x	x	1	x	x		Enabled - Rank selected in OP[3] ^{3,4}
x	x	x	x	0	x	x	x	CA Pass Through mode Rank Selection	CA Pass Through Rank 0 ⁵
x	x	x	x	1	x	x	x		CA Pass Through Rank 1 ⁶
x	x	x	0	x	x	x	x	BCOM Pass-Through mode	Normal operation
x	x	x	1	x	x	x	x		Enabled Pass-Through Mode
x	x	0	x	x	x	x	x	Output Inversion ^{7,8}	Enabled
x	x	1	x	x	x	x	x		Disabled
x	0	x	x	x	x	x	x	Power Down Mode ⁹	Disabled
x	1	x	x	x	x	x	x		Enabled
0	x	x	x	x	x	x	x	Transparent Mode	Disabled
1	x	x	x	x	x	x	x		Enabled

NOTE 1 RW00[1:0] will be sticky, cleared by power cycle not reset.

NOTE 2 This bit does not apply for DDR CA when RW00[0] = 1.

NOTE 3 Internal RCD logic is allowed to block command processing by the BCOM interface in this mode. However, the Host must write RW01[3] = 0 to ensure BCOM commands are being blocked when CA Pass Through mode is enabled in RW00[2].

NOTE 4 If DRAM commands are blocked by RW01[1] = 0, DCSy_n pulses are not forwarded to QCSy_n outputs for any rank even if CA Pass Through Mode is enabled in RW00[2].

NOTE 5 When Pass through mode is enable on Rank 0 the RCD will process MRW commands on Rank 1.

NOTE 6 When Pass through mode is enable on Rank 1 the RCD will process MRW commands on Rank 0

NOTE 7 For normal operation, output inversion is always enabled. For DIMM vendor test purpose, output inversion can be disabled. When disabled, register tPDM is not guaranteed to be met.

NOTE 8 Both output copies must be driven HIGH between CMD, regardless of the setting in RW00[5].

NOTE 9 This bit will disable vendor specific circuitry if any, for power reduction during PDE mode.

8.6.2 RW01 - Parity, CMD Blocking, and Alert Control Word Global Control Word CA

Table 101 — RW01 - Parity, CMD Blocking, and Alert Global Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	Parity Checking	Parity checking disabled ¹
x	x	x	x	x	x	x	1		Parity checking enabled
x	x	x	x	x	x	0	x	DRAM Interface Forward All CMDS	Block
x	x	x	x	x	x	1	x		Do not block
x	x	x	x	x	0	x	x	Reserved	Reserved
x	x	x	x	x	1	x	x		Reserved
x	x	x	x	0	x	x	x	Data Buffer Interface Forward All CMDS	Block
x	x	x	x	1	x	x	x		Do not block
x	x	x	0	x	x	x	x	Reserved	Reserved
x	x	x	1	x	x	x	x		Reserved
x	x	0	x	x	x	x	x	Host Interface Training Feedback ²	Both Sub channels feedback on ALERT _n
x	x	1	x	x	x	x	x		Sub CH_A feedback on QLBD; Sub CH_B feedback on QLBS
x	0	x	x	x	x	x	x	ALERT _n Assertion ^{3,4}	Static ALERT _n Assertion Mode ⁵
x	1	x	x	x	x	x	x		Pulsed ALERT _n Assertion Mode ⁶
0	x	x	x	x	x	x	x	ALERT _n Re-enable ⁷	Parity checking remains disabled after ALERT _n pulse
1	x	x	x	x	x	x	x		Parity checking is re-enabled after ALERT _n pulse

NOTE 1 Register does not check for parity including control word programming.

NOTE 2 Includes DCS, DCA, and DFE Training Mode

NOTE 3 This bit only affects ALERT_n assertion that is a result of a CA parity error. In case of a LOW level on the DERROR_IN_n input, ALERT_n stays asserted as long as DERROR_IN_n remains LOW unless DRST_n is LOW or the device is in clock stopped power down mode.

NOTE 4 For details on ALERT_n output driver strength settings, see Table 229, “Output Driver DC Electrical Characteristics, Entire Operating Temperature Range”.

NOTE 5 ALERT_n stays asserted until ‘Clear CA Parity Error’ command is sent.

NOTE 6 ALERT_n pulse width according to Table 102.

NOTE 7 CA Parity Error Status bit in Error Log Register remains set until cleared by sending a Clear CA Parity Error command.

Figure 70, “DxCSx_n Assertion During ALERT_n Pulse Width” shows DxCSx_n assertion are not permitted t_{CSALT} prior and after of ALERT_n de-assertion when RW01[0] = 1.

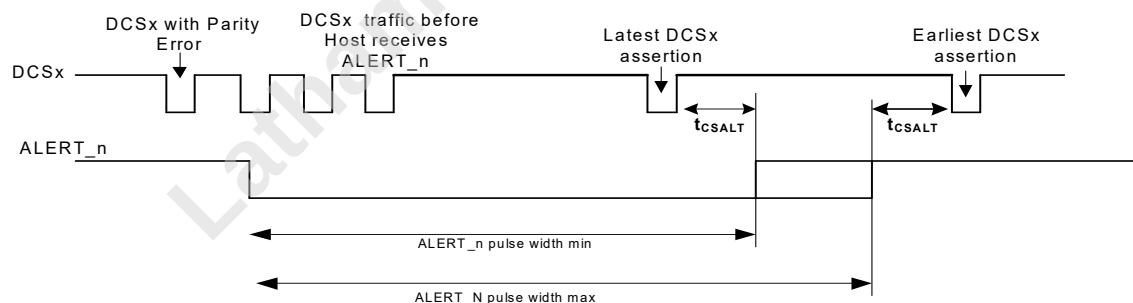


Figure 70 — DxCSx_n Assertion During ALERT_n Pulse Width

Table 102 — ALERT_n Pulse Width for Parity Error

ALERT _n Pulse Width ¹														
DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		DDR5-5200		DDR5-5600		Units
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
96	192	108	216	120	240	132	264	144	288	156	312	168	336	nCK

NOTE 1 The values in this table correspond to parity error in one of the channels. If parity errors occur in both CA channels, the resulting pulse width could be wider than the Maximum values shown in this table due to overlapping of the error pulse signals generated from each channel.

8.6.3 RW02 - Host Interface Training Mode Global Control Word

Table 103 — RW02: Host Interface Training Mode Global Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	0	0	Host Interface Training Mode selection CH_A	Normal operating mode
x	x	x	x	x	x	0	1		Clock-to-DCAy_A training mode
x	x	x	x	x	x	1	0		DCS0_A_n training mode
x	x	x	x	x	x	1	1		DCS1_A_n training mode
x	x	x	x	0	0	x	x	Host Interface Training Mode selection CH_B	Normal operating mode
x	x	x	x	0	1	x	x		Clock-to-DCAy_B training mode
x	x	x	x	1	0	x	x		DCS0_B_n training mode
x	x	x	x	1	1	x	x		DCS1_B_n training mode
x	x	0	0	x	x	x	x	DCA Training XOR sampling edge ^{1,2}	Both Rising and Falling Edges
x	x	0	1	x	x	x	x		Rising Edge
x	x	1	0	x	x	x	x		Falling Edge
x	x	1	1	x	x	x	x		Reserved
x	0	x	x	x	x	x	x	VrefCA Broadcast ³	Disabled
x	1	x	x	x	x	x	x		Enabled ⁴
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1 The DCA sampling edge bits only applies when Clock-to-DCAy_A or Clock-to-DCAy_B training modes are selected in RW02[3:2] or RW02[1:0].

NOTE 2 When SDR mode is enabled (i.e., RW00[0] = 0), only code 01 'Rising Edge' is legal for this field.

NOTE 3 The VrefCA Broadcast Enable mode bit is global and is enabled on both sub-channels by setting RW02[6] = 1 on channel A only. The actual broadcasting of the VrefCA control word writes is independent per sub-channel.

NOTE 4 When VrefCA Broadcast mode is enabled, writes to RW[47:41] will be ignored by the RCD on both sub-channels. Vref DCA[6:0]_A and Vref DPAR_A are configured by writing to RW[40] on channel A. Vref DCA[6:0]_B and Vref DPAR_B are configured by writing to RW40 on channel B.

8.6.4 RW03 - DRAM and DB Interface Training Modes Global Control Word

Table 104 — RW03: DRAM and DB Interface Training Modes Global Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	QCS Training Mode Enable	Disabled
x	x	x	x	x	x	x	1		Enabled - Rank selected in OP[1]
x	x	x	x	x	x	0	x	QCS Training Rank Selection	[Rank 0] QACS0 and QBCS0 Training mode
x	x	x	x	x	x	1	x		[Rank 1] QACS1 and QBCS1 Training mode
x	x	x	x	x	0	x	x	Short MRR command for Data Buffer Training in SDR mode ¹	Disabled
x	x	x	x	x	1	x	x		Enable
x	x	x	x	0	x	x	x	Static Control Value for BCOM0 ²	Output driven statically Low
x	x	x	x	1	x	x	x		Output driven statically High
x	x	x	0	x	x	x	x	Static Control Value for BCOM1 ²	Output driven statically Low
x	x	x	1	x	x	x	x		Output driven statically High
x	x	0	x	x	x	x	x	Static Control Value for BCOM2 ²	Output driven statically Low
x	x	1	x	x	x	x	x		Output driven statically High
x	0	x	x	x	x	x	x	Static Control Value for BCS_n ²	Output driven statically Low
x	1	x	x	x	x	x	x		Output driven statically High
0	x	x	x	x	x	x	x	Static BCOM Control Enable ^{3,4}	Normal
1	x	x	x	x	x	x	x		Enabled; BCOM and BCS driven statically according to OP[6:3]

NOTE 1 When Enabled the DDR5RCD02 will forward a Short MRR Command on the BCOM Bus instead of normal MRR when it receives an MRR CMD from Host.

NOTE 2 Static Control Value, when RW03[7] = 1

NOTE 3 BCOM[2:0] and BCS_n outputs driven statically by the RCD with the values contained in RW03[6:3].

NOTE 4 If RW03[7] = 1 the RCD will apply the static values selected in RW03[6:3] on BCS_n and BCOM[2:0] regardless of the setting in RW01[3] (Data Buffer Interface Block All Commands).

8.6.5 RW04 - Command Space Global Control Word

After issuing a register command via writes to **RW04**, t_{MRC} is the minimum waiting time required by the RCD before the next DRAM command or MRW write can be issued. Some of the commands in **RW04** may require additional waiting times (longer than t_{MRC}) to meet timing parameters specified for the DRAM or Data Buffers.

Table 105 — RW04: Command Space Global Control Word Definition

Command								Command No	Command Name	Result
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0			
0	0	0	0	0	0	0	0	CMD 0	NOP	NOP ¹
0	0	0	0	0	0	0	1	CMD 1	CH_A DB Reset ²	Switches BRST_A_n to active which is LOW. Does NOT reset RCD.
0	0	0	0	0	0	1	0	CMD 2	Clear CH_A DB Reset ²	Switches BRST_A_n to inactive which is HIGH.
0	0	0	0	0	0	1	1	CMD 3	CH_B DB Reset ²	Switches BRST_B_n to active which is LOW. Does NOT reset RCD
0	0	0	0	0	1	0	0	CMD 4	Clear CH_B DB Reset ²	Switches BRST_B_n to inactive which is HIGH
0	0	0	0	0	1	0	1	CMD 5	CH_A DRAM Reset	Switches QRST_A_n & QCSx_A_n to active which is LOW
0	0	0	0	0	1	1	0	CMD 6	Clear CH_A DRAM Reset	Switches QRST_A_n to inactive which is HIGH
0	0	0	0	0	1	1	1	CMD 7	CH_B DRAM Reset	Switches QRST_B_n & QCSx_B_n to active which is LOW
0	0	0	0	1	0	0	0	CMD 8	Clear CH_B DRAM Reset	Switches QRST_B_n to inactive which is HIGH
0	0	0	0	1	0	0	1	CMD 9	CH_A Clear parity Error	Clear 'CA Parity Error Status' bit and '> 1 Error' bit and re-enable parity checking (if not already enabled)
0	0	0	0	1	0	1	0	CMD 10	CH_B Clear parity Error	Clear 'CA Parity Error Status' bit and '> 1 Error' bit and re-enable parity checking (if not already enabled)
0	0	0	0	1	0	1	1	CMD 11	CH_A DFE ERROR Counter Reset	Resets All of CH_A DFE ERROR Counters to zero
0	0	0	0	1	1	0	0	CMD 12	CH_B DFE ERROR Counter Reset	Resets All of CH_B DFE ERROR Counters to zero
0	0	0	0	1	1	0	1	CMD 13	ALERT_n Toggle	Sends one 6 to 10-cycle LOW pulse on ALERT_n
0	0	0	0	1	1	1	0	CMD 14	CH_A QCS HIGH ³	QCSx_A_n Released
0	0	0	0	1	1	1	1	CMD 15	CH_B QCS HIGH ⁴	QCSx_B_n Released
1	1	1	1	0	0	0	0	Reserved	Reserved	Reserved
1	1	1	1	0	0	0	1	Reserved	Reserved	Reserved
1	1	1	1	0	0	1	0	Reserved	Reserved	Reserved
1	1	1	1	0	0	1	1	Reserved	Reserved	Reserved
1	1	1	1	0	1	0	0	Reserved	Reserved	Reserved
1	1	1	1	0	1	0	1	Reserved	Reserved	Reserved
1	1	1	1	0	1	1	0	Reserved	Reserved	Reserved
1	1	1	1	0	1	1	1	Reserved	Reserved	Reserved
1	1	1	1	1	0	0	0	Reserved	Reserved	Reserved
1	1	1	1	1	0	0	1	Reserved	Reserved	Reserved
1	1	1	1	1	0	1	0	Reserved	Reserved	Reserved
1	1	1	1	1	0	1	1	Reserved	Reserved	Reserved
1	1	1	1	1	1	0	0	Reserved	Reserved	Reserved
1	1	1	1	1	1	0	1	Reserved	Reserved	Reserved
1	1	1	1	1	1	1	0	Reserved	Reserved	Reserved
1	1	1	1	1	1	1	1	Reserved	Reserved	Reserved

NOTE 1 This operation was added specifically for hosts that do not support byte writes over the SidebandBus, i.e. that have to write more than one CW at the same time.

NOTE 2 The RCD is not responsible for blocking command traffic in the BCOM[2:0] and BCS_n outputs unless Static BCOM/BCS_n Control is enabled in **RW03**[7].

NOTE 3 CMD 14 can be used during initialization sequences to start the QCK clocks and enable the QCA outputs in CH_A even when the RCD is not provided with the DCS input pulse shown in the Reset initialization timing diagram.

NOTE 4 CMD 15 can be used during initialization sequences to start the QCK clocks and enable the QCA outputs in CH_B even when the RCD is not provided with the DCS input pulse shown in the Reset initialization timing diagram.

8.7 Clocking Control Words

8.7.1 RW05 - DIMM Operating Speed Global Control Word

Table 106 — RW05: DIMM Operating Speed Global Control Word¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	0	0	0	0	Operating Speed ²	DDR5-3200 (2800 MT/s < f ≤ 3200 MT/s)
x	x	x	x	0	0	0	1		DDR5-3600 (3200 MT/s < f ≤ 3600 MT/s)
x	x	x	x	0	0	1	0		DDR5-4000 (3600 MT/s < f ≤ 4000 MT/s)
x	x	x	x	0	0	1	1		DDR5-4400 (4000 MT/s < f ≤ 4400 MT/s)
x	x	x	x	0	1	0	0		DDR5-4800 (4400 MT/s < f ≤ 4800 MT/s)
x	x	x	x	0	1	0	1		DDR5-5200 (4800 MT/s < f ≤ 5200 MT/s)
x	x	x	x	0	1	1	0		DDR5 5600 (5200 MT/s < f ≤ 5600 MT/s)
x	x	x	x	0	1	1	1		Reserved for DDR5 6000 (5600 MT/s < f ≤ 6000 MT/s)
x	x	x	x	1	0	0	0		Reserved for DDR5 6400 (6000 MT/s < f ≤ 6400 MT/s)
x	x	x	x	1	0	0	1		Reserved for DDR5 6800 (6400 MT/s < f ≤ 6800 MT/s)
x	x	x	x	1	0	1	0		Reserved for DDR5 7200 (6800 MT/s < f ≤ 7200 MT/s)
x	x	x	x	1	0	1	1		Reserved
x	x	x	x	1	1	0	0		Reserved
x	x	x	x	1	1	0	1		Reserved
x	x	x	x	1	1	1	0		Down-bin Data Rate (1980 MT/s ≤ f ≤ 2100 MT/s) ³
x	x	x	x	1	1	1	1		PLL bypass mode enabled
x	x	x	0	x	x	x	x	Reserved	Reserved
x	x	x	1	x	x	x	x		Reserved
x	x	0	x	x	x	x	x	Register V _{DD} Operating Voltage ⁴	1.1 V
x	x	1	x	x	x	x	x		Reserved for lower voltage
x	0	x	x	x	x	x	x	Context for operation training	Default; Context 1 operation
x	1	x	x	x	x	x	x		Context 2 operation ⁵
0	x	x	x	x	x	x	x	Frequency Band Select	Operation (Frequency Band 1)
1	x	x	x	x	x	x	x		Test Mode (Frequency Band 2) ⁶

NOTE 1 The encoding value is used to inform the register the operating speed that it is being run at in a system. It is not an indicator of how fast or slow a register can run.

NOTE 2 RW05[6:5] and RW05[3:0] will be sticky, cleared by power cycle not Reset.

NOTE 3 This setting may or may not support fractional delay settings, Fine Granularity DIMM Operating Speed Control, and will require a longer clock stabilization time.

NOTE 4 RW05[5] will be used to inform DDR5RCD02 under what operating voltage VDD will be used. The RCD can use the information to optimize their functionality and performance at low-voltage conditions.

NOTE 5 The control words listed in Table 23 on page 68 are duplicated by the DDR5RCD02 for the 2nd frequency context.

NOTE 6 RW05[7] shall be set to '1' in order to guarantee PLL Lock when DCK is running in the Band-2 frequency range (i.e., 140 MHz < f_{TEST} < 990 MHz).

8.7.2 RW06 - Fine Granularity DIMM Operating Speed Global Control Word

Table 107 — RW06: Fine Granularity DIMM Operating Speed Global Control Word¹

Setting								Definition	Encoding ²
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	0	0	0	0	0	Fine Granularity Operating Speed in terms of f_{bin} , where f_{bin} is the top speed for the speed range selected in RW05 OP[3:0] ³	$(f_{bin} - 20 \text{ MT/s}) < f \leq f_{bin}$
x	x	x	0	0	0	0	1		$(f_{bin} - 40 \text{ MT/s}) < f \leq (f_{bin} - 20 \text{ MT/s})$
x	x	x	0	0	0	1	0		$(f_{bin} - 60 \text{ MT/s}) < f \leq (f_{bin} - 40 \text{ MT/s})$
x	x	x	0	0	0	1	1		$(f_{bin} - 80 \text{ MT/s}) < f \leq (f_{bin} - 60 \text{ MT/s})$
x	x	x	0	0	1	0	0		$(f_{bin} - 100 \text{ MT/s}) < f \leq (f_{bin} - 80 \text{ MT/s})$
x	x	x	0	0	1	0	1		$(f_{bin} - 120 \text{ MT/s}) < f \leq (f_{bin} - 100 \text{ MT/s})$
x	x	x	0	0	1	1	0		$(f_{bin} - 140 \text{ MT/s}) < f \leq (f_{bin} - 120 \text{ MT/s})$
x	x	x	0	0	1	1	1		$(f_{bin} - 160 \text{ MT/s}) < f \leq (f_{bin} - 140 \text{ MT/s})$
x	x	x	0	1	0	0	0		$(f_{bin} - 180 \text{ MT/s}) < f \leq (f_{bin} - 160 \text{ MT/s})$
x	x	x	0	1	0	0	1		$(f_{bin} - 200 \text{ MT/s}) < f \leq (f_{bin} - 180 \text{ MT/s})$
x	x	x	0	1	0	1	0		$(f_{bin} - 220 \text{ MT/s}) < f \leq (f_{bin} - 200 \text{ MT/s})$
x	x	x	0	1	0	1	1		$(f_{bin} - 240 \text{ MT/s}) < f \leq (f_{bin} - 220 \text{ MT/s})$
x	x	x	0	1	1	0	0		$(f_{bin} - 260 \text{ MT/s}) < f \leq (f_{bin} - 240 \text{ MT/s})$
x	x	x	0	1	1	0	1		$(f_{bin} - 280 \text{ MT/s}) < f \leq (f_{bin} - 260 \text{ MT/s})$
x	x	x	0	1	1	1	0		$(f_{bin} - 300 \text{ MT/s}) < f \leq (f_{bin} - 280 \text{ MT/s})$
x	x	x	0	1	1	1	1		$(f_{bin} - 320 \text{ MT/s}) < f \leq (f_{bin} - 300 \text{ MT/s})$
x	x	x	1	0	0	0	0		$(f_{bin} - 340 \text{ MT/s}) < f \leq (f_{bin} - 320 \text{ MT/s})$
x	x	x	1	0	0	0	1		$(f_{bin} - 360 \text{ MT/s}) < f \leq (f_{bin} - 340 \text{ MT/s})$
x	x	x	1	0	0	1	0		$(f_{bin} - 380 \text{ MT/s}) < f \leq (f_{bin} - 360 \text{ MT/s})$
x	x	x	1	0	0	1	1		$(f_{bin} - 400 \text{ MT/s}) < f \leq (f_{bin} - 380 \text{ MT/s})$
x	x	x	1	0	1	0	0	Reserved	Reserved
x	x	x	...						Reserved
x	x	x	1	1	1	1	1	Reserved	Reserved
x	x	0	x	x	x	x	x	Reserved	Reserved
x	x	1	x	x	x	x	x	Reserved	Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x	Reserved	Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x	Reserved	Reserved

NOTE 1 This control word defines the frequency of the DCK_t - DCK_c input reference clock during normal operation (i.e., when not in test frequency range) in units of 20 MT/s.

NOTE 2 The frequency ranges shown in this column are for the base frequency of the input clock and they do not include SSC modulation effects. In some cases, due to SSC modulation, the actual frequency of the input clock can straddle two adjacent frequency ranges.

NOTE 3 RW06[4:0] will be sticky, cleared by power cycle not Reset.

8.8 Validation and Lockout Control Words

8.8.1 RW07 - Validation Pass-Through and Lockout Protection Global Control Word

Table 108 — RW07: Validation Pass-Through and Lockout Protection Global Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	Lockout Protection Enable ¹	No effect ²
x	x	x	x	x	x	x	1		Protection enabled ^{3,4}
x	x	x	x	x	x	0	x	Reserved	Reserved
x	x	x	x	x	x	1	x		Reserved
x	x	x	x	x	0	x	x	Reserved	Reserved
x	x	x	x	x	1	x	x		Reserved
x	x	x	x	0	x	x	x	Reserved	Reserved
x	x	x	x	1	x	x	x		Reserved
x	x	x	0	x	x	x	x	Validation Pass-Through Feature	Feature disabled
x	x	x	1	x	x	x	x		Feature enabled
x	x	0	x	x	x	x	x	Validation Pass-Through Mode Selection	Process MRW
x	x	1	x	x	x	x	x		Ignore MRW ⁵
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1 This feature is intended to protect the RCD against lockout state that is unrecoverable with DRST_n Reset and is caused by invalid DCA commands associated with incorrect input signal voltage levels or timing.

NOTE 2 This control bit is type “Write-1 Only” and cannot be written to 0 by the user. It only gets cleared to zero by DRST_n Reset or by internal Power-On Reset.

NOTE 3 When OP[0] = 1, the RCD prevents Writes to vendor specific CW pages (i.e., Pages 80h to FFh) triggered by commands received at the Host DCA interface of the RCD including MRW commands and CW in-band updates enabled when RW32[6] = 1.

NOTE 4 This feature does not affect access performed through SMBus.

NOTE 5 SMBus access to the RCD registers will not be blocked in Validation Pass-Through mode.

8.9 Output Enable Control Words

8.9.1 RW08 - Clock Driver Enable Control Word

Table 109 — RW08: Clock Driver Enable Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	Disable QACK_t/QACK_c clock ^{1,2}	QACK_t/QACK_c clock enabled
x	x	x	x	x	x	x	1		QACK_t/QACK_c clock disabled
x	x	x	x	x	x	0	x	Disable QBCK_t/QBCK_c clock ^{1,2}	QBCK_t/QBCK_c clock enabled
x	x	x	x	x	x	1	x		QBCK_t/QBCK_c clock disabled
x	x	x	x	x	0	x	x	Disable QCCK_t/QCCK_c clock ^{1,2}	QCCK_t/QCCK_c clock enabled
x	x	x	x	x	1	x	x		QCCK_t/QCCK_c clock disabled
x	x	x	x	0	x	x	x	Disable QDCK_t/QDCK_c clock ^{1,2}	QDCK_t/QDCK_c clock enabled
x	x	x	x	1	x	x	x		QDCK_t/QDCK_c clock disabled
x	x	x	0	x	x	x	x	Reserved	Reserved
x	x	x	1	x	x	x	x		Reserved
x	x	0	x	x	x	x	x	BCK_t/BCK_c ²	BCK_t/BCK_c clock enabled
x	x	1	x	x	x	x	x		BCK_t/BCK_c clock disabled
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1 Output clocks may be individually turned on or off to conserve power. The system must read the module SPD to determine which clock outputs are used by the module. The PLL remains locked on DCK_t/DCK_c unless the system stops the clock inputs to the DDR5RCD02 to enter the lowest power mode.

NOTE 2 RW08[5] and RW08[3:0] will be sticky, cleared by power cycle not Reset.

8.9.2 RW09 - Output Address and Control Enable Control Word

Table 110 — RW09: Output Address and Control Enable Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	QACA outputs	Enabled
x	x	x	x	x	x	x	1		Disabled
x	x	x	x	x	x	0	x	QBCA outputs	Enabled
x	x	x	x	x	x	1	x		Disabled
x	x	x	x	x	0	x	x	DCS1_n Input Buffer & QxCS1_n	DCS1_n & QxCS1_n pins Enabled
x	x	x	x	x	1	x	x	Outputs ¹ (single rank)	DCS1_n & QxCS1_n pins Disabled ²
x	x	x	x	0	x	x	x	BCS_n, BCOM[2:0] & BRST_n	Enabled (LRDIMM)
x	x	x	x	1	x	x	x	outputs ³	Disabled (RDIMM)
x	x	x	0	x	x	x	x	Q[B:A]CA13 output driver disable	Enabled
x	x	x	1	x	x	x	x		Disabled
x	x	0	x	x	x	x	x	QACS[1:0]_n output	Enable
x	x	1	x	x	x	x	x		Disabled
x	0	x	x	x	x	x	x	QBCS[1:0]_n output	Enabled
x	1	x	x	x	x	x	x		Disabled
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1 DCS1 is a required signal for training modes and must be routed even on single rank DIMMs. RW09[2] can be written to 1 to save power in single rank DIMMs during normal operation after initialization and training sequences have been completed.

NOTE 2 The RCD will enter Power Down mode on a Single Rank DIMM when PDE command sent to Rank 0.

NOTE 3 RW09[6:0] will be sticky, cleared by power cycle not Reset.

8.10 Output Driver Strengths

For details on RCD output driver strengths in this section, see Table 229, “Output Driver DC Electrical Characteristics, Entire Operating Temperature Range”.

8.10.1 RW0A - QCK Signals Driver Characteristics Control Word

Table 111 — RW0A: QCK Signals Driver Characteristics Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	0	0	QACK_t/QACK_c ¹	Light Drive
x	x	x	x	x	x	0	1		Moderate Drive
x	x	x	x	x	x	1	0		Strong Drive
x	x	x	x	x	x	1	1		Reserved
x	x	x	x	0	0	x	x	QBCK_t/QBCK_c ¹	Light Drive
x	x	x	x	0	1	x	x		Moderate Drive
x	x	x	x	1	0	x	x		Strong Drive
x	x	x	x	1	1	x	x		Reserved
x	x	0	0	x	x	x	x	QCCK_t/QCCK_c ¹	Light Drive
x	x	0	1	x	x	x	x		Moderate Drive
x	x	1	0	x	x	x	x		Strong Drive
x	x	1	1	x	x	x	x		Reserved
0	0	x	x	x	x	x	x	QDCK_t/QDCK_c ¹	Light Drive
0	1	x	x	x	x	x	x		Moderate Drive
1	0	x	x	x	x	x	x		Strong Drive
1	1	x	x	x	x	x	x		Reserved

NOTE 1 RW0A[7:0] will be sticky, cleared by power cycle not Reset.

8.10.2 RW0C - QxCA and QxCS_n Signals Driver Characteristics Control Word

Table 112 — RW0C: QxCA and QxCS_n Signals Driver Characteristics Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	0	0	Address/Command Address for both A&B copies ¹	Light Drive
x	x	x	x	x	x	0	1		Moderate Drive
x	x	x	x	x	x	1	0		Strong Drive
x	x	x	x	x	x	1	1		Reserved
x	x	x	x	0	0	x	x	Reserved	Reserved
x	x	x	x	0	1	x	x		Reserved
x	x	x	x	1	0	x	x		Reserved
x	x	x	x	1	1	x	x		Reserved
x	x	0	0	x	x	x	x	QxCS0_n, QxCS1_n, Outputs ¹	Light Drive
x	x	0	1	x	x	x	x		Moderate Drive
x	x	1	0	x	x	x	x		Strong Drive
x	x	1	1	x	x	x	x		Reserved
0	0	x	x	x	x	x	x	Reserved	Reserved
0	1	x	x	x	x	x	x		Reserved
1	0	x	x	x	x	x	x		Reserved
1	1	x	x	x	x	x	x		Reserved

NOTE 1 RW0C[1:0] and RW0C[5:4] will be sticky, cleared by power cycle not Reset.

8.10.3 RW0D - Data Buffer Interface Driver Characteristics Control Word

Table 113 — RW0D: Data Buffer Interface Driver Characteristics Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	0	0	BCOM[2:0], BCS_n driver strength ¹	Light Drive
x	x	x	x	x	x	0	1		Moderate Drive
x	x	x	x	x	x	1	0		Reserved
x	x	x	x	x	x	1	1		Reserved
x	x	x	x	x	0	x	x	Reserved	Reserved
x	x	x	x	x	1	x	x		Reserved
x	x	x	0	0	x	x	x	BCK_t/BCK_c driver strength ¹	Light Drive
x	x	x	0	1	x	x	x		Moderate Drive
x	x	x	1	0	x	x	x		Strong Drive
x	x	x	1	1	x	x	x		Reserved
x	x	0	x	x	x	x	x	Reserved	Reserved
x	x	1	x	x	x	x	x		Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1 RW0D[1:0] and RW0D[4:3] will be sticky, cleared by power cycle not Reset.

8.11 Output Slew Rate Control

The DDR5RCD02 will support output slew rate control, per channel, as described in the following control words.

8.11.1 RW0E - QCK, QCA, and QCS Output Slew Rate Control Word

Table 114 — RW0E - QCK, QCA, and QCS Output Slew Rate Control Word¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	0	0	QCK[D:A] t / QCK[D:A] c	Moderate
x	x	x	x	x	x	0	1	Differential Slew Rate Setting ²	Fast
x	x	x	x	x	x	1	0		Slow
x	x	x	x	x	x	1	1		Reserved
x	x	x	x	x	0	0	x		Reserved
x	x	x	x	0	1	x	x	Q[B:A]CA[13:0] Single Ended Slew Rate setting ²	Moderate
x	x	x	x	1	0	x	x		Fast
x	x	x	x	1	1	x	x		Slow
x	x	x	x	1	1	x	x		Reserved
x	x	0	0	x	x	x	x	Q[B:A]CS[1:0] Single Ended Slew Rate setting ²	Moderate
x	x	0	1	x	x	x	x		Fast
x	x	1	0	x	x	x	x		Slow
x	x	1	1	x	x	x	x		Reserved
0	0	x	x	x	x	x	x	Reserved	Reserved
0	1	x	x	x	x	x	x		Reserved
1	0	x	x	x	x	x	x		Reserved
1	1	x	x	x	x	x	x		Reserved

NOTE 1 RW0E[5:0] will be sticky, cleared by power cycle not reset.

NOTE 2 Slew Rate Control encodings of Moderate, Fast and Slow apply to all driver strength settings. The corresponding base range values specified in Table 227 are applicable for only Ron = RZQ/24, VDD = 1.1 V, and 25 °C. Slew rates will be less for other Ron values. The Output slew rate is verified by design and characterization, and may not be subject to production test

8.11.2 RW0F - BCK, BCOM, and BCS Output Slew Rate Control Word

Table 115 — RW0F - BCK, BCOM, and BCS Output Slew Rate Control Word¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	0	0	BCOM[2:0], BCS_n Single Ended Slew Rate setting ²	Moderate
x	x	x	x	x	x	0	1		Fast
x	x	x	x	x	x	1	0		Slow
x	x	x	x	x	x	1	1		Reserved
x	x	x	x	0	0	x	x	BCK_t / BCK_c Differential Slew Rate Setting ³	Moderate
x	x	x	x	0	1	x	x		Fast
x	x	x	x	1	0	x	x		Slow
x	x	x	x	1	1	x	x		Reserved
x	x	0	0	x	x	x	x	Reserved	Reserved
x	x	0	1	x	x	x	x		Reserved
x	x	1	0	x	x	x	x		Reserved
x	x	1	1	x	x	x	x		Reserved
0	0	x	x	x	x	x	x	Reserved	Reserved
0	1	x	x	x	x	x	x		Reserved
1	0	x	x	x	x	x	x		Reserved
1	1	x	x	x	x	x	x		Reserved

NOTE 1 RW0F[3:0] will be sticky, cleared by power cycle not reset.

NOTE 2 Slew Rate Control encodings of Moderate, Fast and Slow apply to all driver strength settings. The corresponding base range values specified in Table 227 are applicable for only Ron = RZQ/17, VDD = 1.1 V, and 25 °C. Slew rates will be less for other Ron values. The Output slew rate is verified by design and characterization, and may not be subject to production test.

NOTE 3 Slew Rate Control encodings of Moderate and Fast apply to all driver strength settings. The corresponding base range values specified in Table 227 on page 235 are applicable for only Ron = RZQ/24, VDD = 1.1 V, and 25 °C. Slew rates will be less for other Ron values. The Output slew rate is verified by design and characterization, and may not be subject to production test.

8.12 Input Bus Termination Strengths

8.12.1 RW10 - IBT Global Control Word

Table 116 — RW10: IBT Global Control Word

Setting								Definition ¹	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	0	0	DCA[6:0]_ [B:A] and DPAR_ [B:A] Input Bus Termination ²	60 Ω
x	x	x	x	x	x	0	1		Reserved
x	x	x	x	x	x	1	0		Reserved
x	x	x	x	x	x	1	1		OFF ³
x	x	x	x	0	0	x	x	DCS[1:0]_ [B:A]_n Input Bus ² Termination	60 Ω
x	x	x	x	0	1	x	x		Reserved
x	x	x	x	1	0	x	x		Reserved
x	x	x	x	1	1	x	x		OFF ³
x	x	0	0	x	x	x	x	DCK Input Bus Termination ²	60 Ω
x	x	0	1	x	x	x	x		Reserved
x	x	1	0	x	x	x	x		Reserved
x	x	1	1	x	x	x	x		OFF ³
0	0	x	x	x	x	x	x	DERROR_IN_ [B:A]_n Input Bus Termination ²	60 Ω
0	1	x	x	x	x	x	x		Reserved
1	0	x	x	x	x	x	x		Reserved
1	1	x	x	x	x	x	x		OFF ³

NOTE 1 These are target IBT values. Acceptable actual values are determined based on tolerances defined in electrical section.

NOTE 2 RW10[7:0] will be sticky, cleared by power cycle not Reset.

NOTE 3 Not intended for normal operation. For normal operation, Input Bus Termination is always enabled.

8.13 Timing Control Words

8.13.1 RW11 - Command Latency Adder Configuration Control Word

Table 117 — RW11: Command Latency Adder Configuration Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	0	0	0	0	Latency adder nLadd to all DRAM commands ¹	0 nCK latency adder
x	x	x	x	0	0	0	1		1 nCK ² latency adder to Qn ³ , Bn ⁴
x	x	x	x	0	0	1	0		2 nCK ² latency adder to Qn ³ , Bn ⁴
x	x	x	x	0	0	1	1		3 nCK ² latency adder to Qn ³ , Bn ⁴
x	x	x	x	0	1	0	0		4 nCK ² latency adder to Qn ³ , Bn ⁴
x	x	x	x	0	1	0	1		Reserved for 5 nCK ² latency adder to Qn ³ , Bn ⁴
x	x	x	x	0	1	1	0		Reserved for 6 nCK ² latency adder to Qn ³ , Bn ⁴
x	x	x	x	0	1	1	1		Reserved for 7 nCK ² latency adder to Qn ³ , Bn ⁴
x	x	x	x	1	0	0	0		Reserved for 8 nCK ² latency adder to Qn ³ , Bn ⁴
x	x	x	x	1	0	0	1		Reserved
x	x	x	x						Reserved
x	x	x	x	1	1	1	1		Reserved
x	x	x	0	x	x	x	x	Reserved	Reserved
x	x	x	1	x	x	x	x	Reserved	Reserved
x	x	0	x	x	x	x	x	Reserved	Reserved
x	x	1	x	x	x	x	x	Reserved	Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x	Reserved	Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x	Reserved	Reserved

NOTE 1 RW11[3:0] will be sticky, cleared by power cycle not Reset.

NOTE 2 The correct frequency range has to be programmed in RW05 before any changes to RW11 from the power-on default are allowed.

NOTE 3 Qn = QxCA[13:0]. QxCS[1:0]_n.

NOTE 4 Bn = BCOM[2:0]. BCS_n.

8.13.2 RW12 - QACK Output Delay Control Word

Table 118 — RW12: QACK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QACK_t/ QACK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64) * t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64)* t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1	Delay Outputs by +(63/64) * t _{CK}	
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACK_t/QACK_c ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1

These control bits do not have any effect unless RW12[7] = 1.

NOTE 2

RW12[7:0] will be sticky, cleared by power cycle not Reset.

NOTE 3

When feature is enabled the delay settings in RW12[5:0] require a time of t_{ODU} for the delay to become stable on the outputs.

8.13.3 RW13 - QBCK Output Delay Control Word

Table 119 — RW13: QBCK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QBCK_t/ QBCK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64)* t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64) * t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x	Reserved	Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for	(Default) Feature Disabled
1	x	x	x	x	x	x	x	QBCK_t/QBCK_c ²	Feature Enabled ³

NOTE 1

These control bits do not have any effect unless [RW13\[7\]](#) = 1.

NOTE 2

[RW13\[7:0\]](#) will be sticky, cleared by power cycle not Reset.

NOTE 3

When feature is enabled the delay settings in [RW13\[5:0\]](#) require a time of t_{ODU} for the delay to become stable on the outputs.

8.13.4 RW14 - QCCK Output Delay Control Word

Table 120 — RW14: QCCK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QCCK_t/ QCCK_c Output Signals ^{1,2}	Delay Outputs by $+(0/64) * t_{CK}$ (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by $+(1/64) * t_{CK}$
x	x	0	0	0	0	1	0		Delay Outputs by $+(2/64) * t_{CK}$
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by $+(61/64) * t_{CK}$
x	x	1	1	1	1	1	0		Delay Outputs by $+(62/64) * t_{CK}$
x	x	1	1	1	1	1	1		Delay Outputs by $+(63/64) * t_{CK}$
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x	Reserved	Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for	(Default) Feature Disabled
1	x	x	x	x	x	x	x	QCCK_t/QCCK_c ²	Feature Enabled ³

NOTE 1

These control bits do not have any effect unless [RW14\[7\]](#) = 1.

NOTE 2

[RW14\[7:0\]](#) will be sticky, cleared by power cycle not Reset.

NOTE 3

When feature is enabled the delay settings in [RW14\[5:0\]](#) require a time of t_{ODU} for the delay to become stable on the outputs.

8.13.5 RW15 - QDCK Output Delay Control Word

Table 121 — RW15: QDCK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QDCK_t/ QDCK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64)* t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64) * t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x	Reserved	Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for	(Default) Feature Disabled
1	x	x	x	x	x	x	x	QDCK_t/QDCK_c ²	Feature Enabled ³

NOTE 1

These control bits do not have any effect unless RW15[7] = 1.

NOTE 2

RW15[7:0] will be sticky, cleared by power cycle not Reset.

NOTE 3

When feature is enabled the delay settings in RW15[5:0] require a time of t_{ODU} for the delay to become stable on the outputs.

8.13.6 RW17 - QACS0_n Output Delay Control Word

Table 122 — RW17: QACS0_n Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QACSO_n ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64)* t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64) * t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Full cycle delay ¹	Disabled
x	1	x	x	x	x	x	x		Enabled adds 1t _{CK} additional delay to OP[5:0]
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACSO_n ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1

These control bits do not have any effect unless RW17[7] = 1.

NOTE 2

RW17[7:0] will be sticky, cleared by power cycle not Reset.

NOTE 3

When feature is enabled the delay settings in RW17[6:0] require a time of t_{ODU} for the delay to become stable on the outputs.

8.13.7 RW18 - QACS1_n Output Delay Control Word

Table 123 — RW18: QACS1_n Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QACS1_n ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64)* t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64) * t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Full cycle delay ¹	Disabled
x	1	x	x	x	x	x	x		Enabled adds 1tCK additional delay to OP[5:0]
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACS1_n ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1 These control bits do not have any effect unless RW18[7] = 1.

NOTE 2 RW18[7:0] will be sticky, cleared by power cycle not Reset.

NOTE 3 When feature is enabled the delay settings in RW18[6:0] require a time of t_{ODU} for the delay to become stable on the outputs.

8.13.8 RW19 - QBCS0_n Output Delay Control Word

Table 124 — RW19: QBCS0_n Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QBCS0_n ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64)* t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64) * t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Full cycle delay ¹	Disabled
x	1	x	x	x	x	x	x		Enabled adds 1tCK additional delay to OP[5:0]
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QBCS0_n output signals ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1 These control bits do not have any effect unless RW19[7] = 1.

NOTE 2 RW19[7:0] will be sticky, cleared by power cycle not Reset.

NOTE 3 When feature is enabled the delay settings in RW19[6:0] require a time of t_{ODU} for the delay to become stable on the outputs.

8.13.9 RW1A - QBCS1_n Output Delay Control Word

Table 125 — RW1A: QBCS1_n Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QBCS1_n ^{1,2}	Delay Outputs by $+(0/64) \cdot t_{CK}$ (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by $+(1/64) \cdot t_{CK}$
x	x	0	0	0	0	1	0		Delay Outputs by $+(2/64) \cdot t_{CK}$
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by $+(61/64) \cdot t_{CK}$
x	x	1	1	1	1	1	0		Delay Outputs by $+(62/64) \cdot t_{CK}$
x	x	1	1	1	1	1	1		Delay Outputs by $+(63/64) \cdot t_{CK}$
x	0	x	x	x	x	x	x	Full cycle delay ¹	Disabled
x	1	x	x	x	x	x	x		Enabled adds 1tCK additional delay to OP[5:0]
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QBCS1_n output signals ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³
NOTE 1 These control bits do not have any effect unless RW1A[7] = 1.									
NOTE 2 RW1A[7:0] will be sticky, cleared by power cycle not Reset.									
NOTE 3 When feature is enabled the delay settings in RW1A[6:0] require a time of t _{ODU} for the delay to become stable on the outputs.									

8.13.10 RW1B - QACA Output Delay Control Word

Table 126 — RW1B: QACA Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QACA Output Signals ^{1,2}	Delay Outputs by $+(0/64) * t_{CK}$ (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by $+(1/64) * t_{CK}$
x	x	0	0	0	0	1	0		Delay Outputs by $+(2/64) * t_{CK}$
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by $+(61/64) * t_{CK}$
x	x	1	1	1	1	1	0		Delay Outputs by $+(62/64) * t_{CK}$
x	x	1	1	1	1	1	1		Delay Outputs by $+(63/64) * t_{CK}$
x	0	x	x	x	x	x	x	Full cycle delay ¹	Disabled
x	1	x	x	x	x	x	x		Enabled adds 1tCK additional delay to OP[5:0]
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACA ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³
NOTE 1 These control bits do not have any effect unless RW1B[7] = 1									
NOTE 2 RW1B[7:0] will be sticky, cleared by power cycle not Reset.									
NOTE 3 When feature is enabled the delay settings in RW1B[6:0] require a time of t_{OPU} for the delay to become stable on the outputs.									

8.13.11 RW1C - QBCA Output Delay Control Word

Table 127 — RW1C: QBCA Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QBCA Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64)* t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64) * t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Full cycle delay ¹	Disabled
x	1	x	x	x	x	x	x		Enabled adds 1t _{CK} additional delay to OP[5:0]
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QBCA ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1

These control bits do not have any effect unless **RW1C[7]** = 1

NOTE 2

RW1C[7:0] will be sticky, cleared by power cycle not Reset.

NOTE 3

When feature is enabled the delay settings in **RW1C[6:0]** require a time of t_{ODU} for the delay to become stable on the outputs.

8.13.12 RW1D - BCS_n and BCOM[2:0] Output Delay Control Word

Table 128 — RW1D: BCS_n and BCOM[2:0] Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for BCS_n and BCOM[2:0] Output Signals ^{1,2}	Delay Outputs by $+(0/64) * t_{CK}$ (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by $+(1/64) * t_{CK}$
x	x	0	0	0	0	1	0		Delay Outputs by $+(2/64) * t_{CK}$
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by $+(61/64) * t_{CK}$
x	x	1	1	1	1	1	0		Delay Outputs by $+(62/64) * t_{CK}$
x	x	1	1	1	1	1	1		Delay Outputs by $+(63/64) * t_{CK}$
x	0	x	x	x	x	x	x	Full cycle delay ¹	Disabled
x	1	x	x	x	x	x	x		Enabled adds 1tCK additional delay to OP[5:0]
0	x	x	x	x	x	x	x	Output Delay Feature Enable for BCS_n and BCOM[2:0] ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1

These control bits do not have any effect unless [RW1D\[7\]](#) = 1.

NOTE 2

[RW1D\[7:0\]](#) will be sticky, cleared by power cycle not Reset.

NOTE 3

When feature is enabled the delay settings in [RW1D\[6:0\]](#) require a time of t_{ODU} for the delay to become stable on the outputs.

8.13.13 RW1E - BCK Output Delay Control Word

Table 129 — RW1E: BCK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for BCK_t/BCK_c Output Signals ^{1,2}	Delay Outputs by $+(0/64) \cdot t_{CK}$ (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by $+(1/64) \cdot t_{CK}$
x	x	0	0	0	0	1	0		Delay Outputs by $+(2/64) \cdot t_{CK}$
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by $+(61/64) \cdot t_{CK}$
x	x	1	1	1	1	1	0		Delay Outputs by $+(62/64) \cdot t_{CK}$
x	x	1	1	1	1	1	1		Delay Outputs by $+(63/64) \cdot t_{CK}$
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x	Reserved	Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for BCK_t/BCK_c ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1

These control bits do not have any effect unless [RW1E\[7\]](#) = 1.

NOTE 2

[RW1E\[7:0\]](#) will be sticky, cleared by power cycle not Reset.

NOTE 3

When feature is enabled the delay settings in [RW1E\[5:0\]](#) require a time of t_{ODU} for the delay to become stable on the outputs.

8.14 RW[24:20] - Error Log Register

Control word locations [RW\[24:20\]](#) function as a 32-bit error log register. Upon occurrence of a CA parity error, the device logs the following sampled command and address bits in the Error Log Register.

Table 130 — RW[24:20]: Error Log Register Encoding for DDR Mode^{1,2}

Control Word	Setting (OP[7:0])							
RW20	CA13 2nd UI	CA12 2nd UI	CA11 2nd UI	CA10 2nd UI	CA9 2nd UI	CA8 2nd UI	CA7 2nd UI	DPAR 2nd UI
RW21	CA6 1st UI	CA5 1st UI	CA4 1st UI	CA3 1st UI	CA2 1st UI	CA1 1st UI	CA0 1st UI	DPAR 1st UI
RW22	CA13 4th UI	CA12 4th UI	CA11 4th UI	CA10 4th UI	CA9 4th UI	CA8 4th UI	CA7 4th UI	DPAR 4th UI
RW23	CA6 3rd UI	CA5 3rd UI	CA4 3rd UI	CA3 3rd UI	CA2 3rd UI	CA1 3rd UI	CA0 3rd UI	DPAR 3rd UI
RW24	> 1 Error ^{3,4}	CA Parity Error Status ^{3,5}	Reserved	CS1_n 1st cycle	CS0_n 1st cycle	CS1_n 2nd cycle	CS0_n 2nd Cycle	Reserved

NOTE 1 The encoding shown in this table is valid when the DCA interface is operating in DDR mode ([RW00\[0\]](#) = 1).

NOTE 2 When command and address bits are logged in the Error Log Register upon occurrence of a CA parity error, all bits in [RW20](#) to [RW24](#) are updated with the corresponding values from the offending command. Four consecutive UI's are always captured because, in some cases, it is not possible to distinguish between 1UI and 2UI commands due to the parity error. [RW20](#) and [RW21](#) always contain the start of the command (qualified by DCS_n assertion) regardless of the UI or cycle within the command where the parity error has been detected.

NOTE 3 This bit will get reset to '0' when "Clear CA Parity Error" command is sent. The Clear CA Parity Error command does not affect any bits in [RW\[23:20\]](#) and [RW24\[4:1\]](#).

NOTE 4 With [RW01\[7\]](#) = 1 and [RW01\[6\]](#) = 1, the device will forward commands and re-enable parity after the ALERT_n pulse. The CA Parity Error Status bit will remain set. If a subsequent parity error is detected, the device will re-enter the parity error state and set the '> 1 Error' bit.

NOTE 5 The DDR5RCD02 will set this bit upon occurrence of a CA parity error.

8.14 RW[24:20] - Error Log Register (cont'd)

Table 131 — RW[24:20]: Error Log Register Encoding for SDR Mode^{1,2}

Control Word	Setting (OP[7:0])							
RW20	CA13 2nd Cycle	CA12 2nd Cycle	CA11 2nd Cycle	CA10 2nd Cycle	CA9 2nd Cycle	CA8 2nd Cycle	CA7 2nd Cycle	DPAR 2nd Cycle
RW21	CA6 1st Cycle	CA5 1st Cycle	CA4 1st Cycle	CA3 1st Cycle	CA2 1st Cycle	CA1 1st Cycle	CA0 1st Cycle	DPAR 1st Cycle
RW22	CA13 4th Cycle	CA12 4th Cycle	CA11 4th Cycle	CA10 4th Cycle	CA9 4th Cycle	CA8 4th Cycle	CA7 4th Cycle	DPAR 4th Cycle
RW23	CA6 3rd Cycle	CA5 3rd Cycle	CA4 3rd Cycle	CA3 3rd Cycle	CA2 3rd Cycle	CA1 3rd Cycle	CA0 3rd Cycle	DPAR 3rd Cycle
RW24	> 1 Error ^{3,4}	CA Parity Error Status ^{3,5}	Reserved	CS1_n 1st cycle	CS0_n 1st cycle	CS1_n 3rd cycle	CS0_n 3rd Cycle	Reserved

NOTE 1 The encoding shown in this table is valid when the DCA interface is operating in DDR mode (RW00[0] = 1).

NOTE 2 When command and address bits are logged in the Error Log Register upon occurrence of a CA parity error, all bits in RW20 to RW24 are updated with the corresponding values from the offending command. Four consecutive UI's are always captured because, in some cases, it is not possible to distinguish between 1UI and 2UI commands due to the parity error. RW20 and RW21 always contain the start of the command (qualified by DCS_n assertion) regardless of the UI or cycle within the command where the parity error has been detected.

NOTE 3 This bit will get reset to '0' when "Clear CA Parity Error" command is sent. The Clear CA Parity Error command does not affect any bits in RW[23:20] and RW24[4:1].

NOTE 4 With RW01[7] = 1 and RW01[6] = 1, the device will forward commands and re-enable parity after the ALERT_n pulse. The CA Parity Error Status bit will remain set. If a subsequent parity error is detected, the device will re-enter the parity error state and set the '> 1 Error' bit.

NOTE 5 The DDR5RCD02 will set this bit upon occurrence of a CA parity error.

8.15 SidebandBus

8.15.1 RW25 - SidebandBus Global Control Word

Table 132 — RW25: SidebandBus Global Control Word¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	SidebandBus Interface Disabled	SidebandBus interface is enabled
x	x	x	x	x	x	x	1		SidebandBus interface is disabled. DDR5RCD02 will not claim or acknowledge any access to its SidebandBus address space.
x	x	x	x	x	x	0	x	Vendor Specific ²	Vendor Specific.
x	x	x	x	x	x	1	x		Vendor Specific.
x	x	x	x	x	0	x	x	Reserved	Reserved
x	x	x	x	x	1	x	x		Reserved
x	x	x	x	0	x	x	x	Reserved	Reserved
x	x	x	x	1	x	x	x		Reserved
x	x	x	0	x	x	x	x	Vendor Specific ²	Vendor Specific
x	x	x	1	x	x	x	x		Vendor Specific
x	x	0	x	x	x	x	x	SidebandBus Mode ³	I ² C (Max speed of 1 MHz) Enabled
x	x	1	x	x	x	x	x		I ³ C Basic (Max speed of 12.5 MHz) Enabled
x	0	x	x	x	x	x	x	Vendor Specific ²	Vendor Specific
x	1	x	x	x	x	x	x		Vendor Specific
0	x	x	x	x	x	x	x	Vendor Specific ²	Vendor Specific
1	x	x	x	x	x	x	x		Vendor Specific

NOTE 1 The write (or update) transaction to this register must be followed by STOP operation to allow the DDR5RCD02 device to update the setting.

NOTE 2 This register is vendor specific and must be written to '0' by the host whenever RW25 is accessed for Writes.

NOTE 3 This is a Read-Only Status register. This is a Sticky register (not cleared by DRST_n). This register is automatically updated to '1' or to '0', respectively, when SETAASA CCC or RSTDAA CCC is registered by the DDR5RCD02 device. This register is automatically cleared to '0' when the Timeout Reset condition described in Section 7.6, "I²C and I³C Basic Reset" is detected.

8.16 RX Loopback

8.16.1 RW26 - RX Loopback Global Control Word

Table 133 — RW26: RX Loopback Global Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	0	0	0	RCD RX Loopback Port Selection	Loopback Disabled ¹
x	x	x	x	x	0	0	1		Internal RCD CH_A Port ²
x	x	x	x	x	0	1	0		Internal RCD CH_B Port ²
x	x	x	x	x	0	1	1		External CH_A Port ³
x	x	x	x	x	1	0	0		External CH_B Port ³
x	x	x	x	x	1	0	1		Internal QCK Loopback ⁴
x	x	x	x	x	1	1	0		Reserved
x	x	x	x	x	1	1	1		Reserved
x	x	0	0	0	x	x	x	Internal RCD bit selection ⁵	DCA0
x	x	0	0	1	x	x	x		DCA1
x	x	0	1	0	x	x	x		DCA2
x	x	0	1	1	x	x	x		DCA3
x	x	1	0	0	x	x	x		DCA4
x	x	1	0	1	x	x	x		DCA5
x	x	1	1	0	x	x	x		DCA6
x	x	1	1	1	x	x	x		DPAR
x	0	x	x	x	x	x	x	Loopback Data Rate Applied to both Internal and External Loopback Port ^{6,7,8}	Incoming and outgoing signals are 1/2 rate ⁹
x	1	x	x	x	x	x	x		Incoming and outgoing signals are 1/4 rate
0	x	x	x	x	x	x	x	Internal Loopback Select Phase, for 1/2 Rate Mode (Lower Select for 1/4 Rate Mode)	Loopback Select Phase A or C ¹⁰
1	x	x	x	x	x	x	x		Loopback Select Phase B or D ¹¹

NOTE 1 Loopback circuitry is disabled.

NOTE 2 Internal Port contains input Clock and DCA. Loopback signal operation can be configured as 1/4 rate or 1/2 rate modes selected in RW26[6].

NOTE 3 External Port contains input Strobe and Data. A waiting time $t_{EXT_LB_Entry}$ applies when External Loopback is enabled in RW26[2:0].

NOTE 4 The DDR5RCD02 can output its internal PLL clock to the two loopback output pins QLBS and QLBD. The delay from input differential clock DCK_t and DCK_c to the output differential clock on QLBS and QLBD is vendor specific. The slew rate of the output differential clock on QLBS and QLBD is vendor specific. The mode is for 1-UI Rj without BUJ test purpose only. Any other parameters including Tj, Dj and N-UI Rj tests are not supported by the device.

NOTE 5 This field only applies when Internal RCD CH_A or Internal RCD CH_B ports are selected in RW26[2:0].

NOTE 6 For Internal RCD CH A and Internal RCD CHB ports, this bit selects whether 1/2 rate or 1/4 rate Loopback mode is enabled.

NOTE 7 For External CH_A and External CH_B ports, this bit selects whether the incoming and outgoing signals are running in 1/2 rate or 1/4 rate Loopback mode.

NOTE 8 For External CH_A and External CH_B ports, the host must ensure this bit's setting is consistent with the Loopback signal rate received at the DLBD_N/DLBS_N pins selected in RW26[2:0]. The DDR5RCD02 device is allowed to use the information contained in this bit to configure its External Loopback circuitry.

NOTE 9 The 1/2 rate loopback feature is supported up to DDR5-4800. It is not required above DDR5-4800.

NOTE 10 When RW27[7]=0, Phase A is selected; and when RW27[7]=1, Phase C is selected. Phase C is only supported in 1/4 rate mode (i.e., RW26[6] = '1')

NOTE 11 When RW27[7]=0, Phase B is selected; and when RW27[7]=1, Phase D is selected. Phase D is only supported in 1/4 rate mode (i.e., RW26[6] = '1')

8.16.2 RW27 - Loop-back I/O Global Control Word

Table 134 — RW27: Loop-back I/O Global Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	0	0	RX DLBS Termination	60 Ω
x	x	x	x	x	x	0	1		80 Ω
x	x	x	x	x	x	1	0		40 Ω
x	x	x	x	x	x	1	1		OFF
x	x	x	x	0	0	x	x	RX DLBD Termination	60 Ω
x	x	x	x	0	1	x	x		80 Ω
x	x	x	x	1	0	x	x		40 Ω
x	x	x	x	1	1	x	x		OFF
x	x	0	0	x	x	x	x	TX QLBS and QLBD Drive Strength	RZQ/5 (48 Ω)
x	x	0	1	x	x	x	x		RZQ/7 (34 Ω)
x	x	1	0	x	x	x	x		Reserved
x	x	1	1	x	x	x	x		Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Internal Loopback upper Phase Select for 1/4 Rate ¹	RW26[7] selects between lower phases A and B
1	x	x	x	x	x	x	x		RW26[7] selects between upper phases C and D

NOTE 1 This control bit has no effect unless Internal Loopback is running in 1/4 rate mode (i.e., RW26[2:0] is programmed to '001' or '010' and RW26[6] to 1). This bit must be programmed to 0 in 1/2 Rate mode.

8.17 SidebandBus Error Status

8.17.1 RW28: I²C and I3C Basic Error Status Global Word

Table 135 — RW28: I²C and I3C Basic Error Status Global Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	Error in Parity Check ^{1,2,3}	No PARITY error has been detected
x	x	x	x	x	x	x	1		PARITY error has been detected ⁴
x	x	x	x	x	x	0	x	Error in Packet Error Code Check ^{1,2,5}	No PEC error has been detected
x	x	x	x	x	x	1	x		PEC error has been detected ⁶
x	x	x	x	x	0	x	x	Reserved	Reserved
x	x	x	x	x	1	x	x		Reserved
x	x	x	x	0	x	x	x	Reserved	Reserved
x	x	x	x	1	x	x	x		Reserved
x	x	x	0	x	x	x	x	Reserved	Reserved
x	x	x	1	x	x	x	x		Reserved
x	x	0	x	x	x	x	x	Reserved	Reserved
x	x	1	x	x	x	x	x		Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Device Event In-Band Interrupt Status ¹	No Pending IBI ⁷
1	x	x	x	x	x	x	x		Pending IBI ⁸

NOTE 1 This is a Read-Only Status bit. This is a Sticky register (not cleared by DRST_n).

NOTE 2 This register is automatically cleared to '0' when the Timeout Reset condition described in Section 7.8, "I2C Bus Error Handling" is detected.

NOTE 3 This status bit only applies when I3C Basic mode is enabled or for CCC supported in I²C mode, provided that the Parity Checking function is not disabled.

NOTE 4 The RCD device detected a PARITY error in one or more bytes received.

NOTE 5 This status bit only applies when I3C Basic mode is enabled.

NOTE 6 The RCD device detected a PEC error for one or more data packets received. When a PEC error is detected in I²C mode, this status field will not be updated.

NOTE 7 The IBI status bit gets cleared to '0' when the RCD device processes a complete IBI operation (including uninterrupted IBI payload). This status bit also gets cleared when RW28[0] and RW28[1] become '00' after a Clear command in RW29.

NOTE 8 The IBI status bit gets set to '1' when the RCD sets RW28[0] or RW28[1] to '1' after detecting a Parity Error or a PEC Error.

8.17.2 RW29: I²C and I3C Basic Clear Error Status Global Word

Table 136 — RW29: I²C and I3C Basic Clear Error Status Global Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	Clear Parity Error Status ¹	No effect.
x	x	x	x	x	x	x	1		Clear RW28[0].
x	x	x	x	x	x	0	x	Clear Packet Error Status ¹	No effect.
x	x	x	x	x	x	1	x		Clear RW28[1].
x	x	x	x	x	0	x	x	Reserved	Reserved
x	x	x	x	1	x	x	x		Reserved
x	x	x	x	0	x	x	x	Reserved	Reserved
x	x	x	x	1	x	x	x		Reserved
x	x	x	0	x	x	x	x	Reserved	Reserved
x	x	x	1	x	x	x	x		Reserved
x	x	0	x	x	x	x	x	Reserved	Reserved
x	x	1	x	x	x	x	x		Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1 This is a Write-1 Only register bit, and it self clears after the target status register has been cleared.

8.17.3 RW2A: Vendor Specific Global Word

Table 137 — RW2A: Vendor Specific Global Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	Reserved	Reserved
x	x	x	x	x	x	x	1		Reserved
x	x	x	x	0	0	0	x	Vendor Specific	Vendor Specific
x	x	x	x	0	0	1	x		Vendor Specific
x	x	x	x	0	1	0	x		Vendor Specific
x	x	x	x	0	1	1	x		Vendor Specific
x	x	x	x	1	0	0	x		Vendor Specific
x	x	x	x	1	0	1	x		Vendor Specific
x	x	x	x	1	1	0	x		Vendor Specific
x	x	x	x	1	1	1	x		Vendor Specific
x	x	x	0	x	x	x	x	Reserved	Reserved
x	x	x	1	x	x	x	x		Reserved
x	x	0	x	x	x	x	x	Reserved	Reserved
x	x	1	x	x	x	x	x		Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

8.18 DFE Control Words

8.18.1 RW30 - DFE_Vref Range Limit Global Status Word

Table 138 — RW30: DFE_Vref Range Limit Global Status Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	DFE_Vref Configuration Limit ^{1,2}	Applies to 8 LSB portion of DFE Vref control in PG[2]RW[62,66,6A,6E,72,76,7A,7E] (Maximum Setting Supported by Device)
0	0	0	0	0	0	0	1		
0	0	0	0	0	0	1	0		
...									
1	1	1	1	1	1	0	1		
1	1	1	1	1	1	1	0		
1	1	1	1	1	1	1	1		
NOTE 1 Read only register. Returns positive limit. DFE_Vref must have a symmetrical range.									
NOTE 2 This is a global status word. The status code can only be accessed from CH_A.									

8.18.2 RW31 - DFE Configuration Control Word

Table 139 — RW31: DFE Configuration Control Word¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	DFE Feature Enable Control ²	(Default) DFE and gain features disabled
x	x	x	x	x	x	x	1		DFE and gain features enabled
x	x	x	x	x	x	0	x	DFE_Vref Enable ³	DFE_Vref Circuitry Disabled
x	x	x	x	x	x	1	x		DFE_Vref Circuitry Enabled
x	x	x	x	x	0	x	x	ERROR Counter Enable ^{4,5}	All Error Counters Disabled
x	x	x	x	x	1	x	x		All Error Counters Enabled
x	x	x	x	0	x	x	x	RW Control Word Writes Broadcast	RW Write to selected DCAn or DPAR ⁶
x	x	x	x	1	x	x	x		RW Writes Broadcast to DCA[6:0] and DPAR ⁷
x	x	x	0	x	x	x	x	Tap 1 Enable ⁸	(Default) Tap 1 disabled
x	x	x	1	x	x	x	x		Tap 1 enabled
x	x	0	x	x	x	x	x	Tap 2 Enable ⁸	(Default) Tap 2 disabled
x	x	1	x	x	x	x	x		Tap 2 enabled
x	0	x	x	x	x	x	x	Tap 3 Enable ⁸	(Default) Tap 3 disabled
x	1	x	x	x	x	x	x		Tap 3 enabled
0	x	x	x	x	x	x	x	Tap 4 Enable ⁸	(Default) Tap 4 disabled
1	x	x	x	x	x	x	x		Tap 4 enabled
NOTE 1 RW31[7:0] will be sticky, cleared by power cycle not Reset.									
NOTE 2 This control bit enables DFE circuitry in the RCD.									
NOTE 3 To save power, the host can keep DFE_VREF circuitry disabled when not in use.									
NOTE 4 Error Counter Reset is located RW04 . CMD 11 for CH_A and CMD 12 for CH_B.									
NOTE 5 Avoid changing DFE settings when the error counters are enabled, or reset the error counters after DFE settings are updated.									
NOTE 6 When RW31[3] = 0, RW Writes to PG[2:0] are written to a specific receiver selected in MRW command.									
NOTE 7 When RW31[3] = 1, RW Writes to any of the DCn Gain, Tap, and DFE_Vref registers in PG[2:0] are written to all DCA receivers DCA[6:0] and DPAR.									
NOTE 8 Enables Tap in DPAR and DCA[6:0] receivers.									

8.18.3 RW32 - DPAR and DCA[6:0] DFE Training Mode Control Word

Table 140 — RW32 DPAR and DCA[6:0] DFE Training Mode Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	Global DFE Training Mode Enable ¹	DFE Training Mode disabled
x	x	x	x	x	x	x	1		DFE Training Mode enabled ^{2,3}
x	x	x	x	x	x	0	0	Training Source	Monitor
x	x	x	x	x	x	0	1		Monitor XOR Slicer Output
x	x	x	x	x	x	1	0		LFSR XOR Monitor
x	x	x	x	x	x	1	1		Reserved
x	x	0	0	0	x	x	x	DCn DFE pin selection ⁴	DCA0
x	x	0	0	1	x	x	x		DCA1
x	x	0	1	0	x	x	x		DCA2
x	x	0	1	1	x	x	x		DCA3
x	x	1	0	0	x	x	x		DCA4
x	x	1	0	1	x	x	x		DCA5
x	x	1	1	0	x	x	x		DCA6
x	x	1	1	1	x	x	x		DPAR
x	0	x	x	x	x	x	x		(Default) Feature disabled ⁶
x	1	x	x	x	x	x	x		Feature enabled ⁷
0	x	x	x	x	x	x	x	Exit From RW In-Band Updated Mode ⁵	No action
1	x	x	x	x	x	x	x		(Default) Take the RCD out from RWUPD state ⁸

NOTE 1 RW32[5:3] determines the target receiver for DFE training.

NOTE 2 DFE circuits are configured into training mode for selected pin RW32[5:3].

NOTE 3 Vref generator circuits are configured so that a DFE training reference voltage (DFE_Vref) is controlled by PG[2]RW[62, 66, 6A, 6E, 72, 76, 7A, 7E].

NOTE 4 Logic in DDR5RCD02 device ensures that only one DCAn receiver is configured in training mode at any given time.

NOTE 5 Default state of RW32[6] is 0, and the default state of RW32[7] is 1.

NOTE 6 DCS1_n ignored during DFE training.

NOTE 7 DCS1_n Low pulse received in idle state of DFE training triggers entry into RW Update Mode.

NOTE 8 Writing a 1 in this bit location takes the RCD out from RWUPD state. The RCD hardware clears this register bit upon entry to RWUPD mode when DCS1_n Low pulse is received. Therefore, a setting RW32[7] = 1 indicates the channel is not in RWUPD mode.

8.18.4 RW33 - Additional Filtering for DFE Training Mode Control Word

Table 141 — RW33 Additional Filtering for DFE Training Mode Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	0	0	UI Filtering Enable	UI Filter Disabled
x	x	x	x	x	x	0	1		Enable Per UI Filtering according to OP[4:2]
x	x	x	x	x	x	1	0		Enable Even UI Filtering ¹
x	x	x	x	x	x	1	1		Enable Odd UI Filtering ¹
x	x	x	0	0	0	x	x	Per-UI filtering ²	UI 0
x	x	x	0	0	1	x	x		UI 1
x	x	x	0	1	0	x	x		UI 2
x	x	x	0	1	1	x	x		UI 3
x	x	x	1	0	0	x	x		UI 4
x	x	x	1	0	1	x	x		UI 5
x	x	x	1	1	0	x	x		UI 6
x	x	x	1	1	1	x	x		UI 7
x	x	0	x	x	x	x	x		Reserved
x	x	1	x	x	x	x	x		Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1 When selected OP[4:2] is ignored by the DDR5RCD02.

NOTE 2 When selected OP[4:2] is ignored by the DDR5RCD02.

8.18.5 RW34 - LFSR DFE Training Mode Control Word

Table 142 — RW34 LFSR DFE Training Mode Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	DFE Training Accelerator	Stop DFE Training Accelerator
x	x	x	x	x	x	x	1		Start DFE Training Accelerator
x	x	x	x	x	x	0	x	Reserved	Reserved
x	x	x	x	x	x	1	x		Reserved
x	x	x	x	0	0	x	x	Select Inner Loop Parameter	Select Inner Loop Parameter - NULL ¹
x	x	x	x	0	1	x	x		Select Inner Loop Parameter - DFE Vref
x	x	x	x	1	0	x	x		Select Inner Loop Parameter - VrefCA
x	x	x	x	1	1	x	x		Reserved
x	0	0	0	x	x	x	x	Select Outer Loop Parameter	Select Outer Loop Parameter - NULL
x	0	0	1	x	x	x	x		Select Outer Loop Parameter - Tap 1
x	0	1	0	x	x	x	x		Select Outer Loop Parameter - Tap 2
x	0	1	1	x	x	x	x		Select Outer Loop Parameter - Tap 3
x	1	0	0	x	x	x	x		Select Outer Loop Parameter - Tap 4
x	1	0	1	x	x	x	x		Select Outer Loop Parameter - UI
x	1	1	0	x	x	x	x		Reserved
x	1	1	1	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1 NULL parameter choice means no change. NULL parameter loop can have multiple iterations.

8.18.6 RW35 - LFSR State for DFE Training Mode Control Word

Table 143 — RW35 LFSR State for DFE Training Mode Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	LFSR State (Read Only) ¹	LFSR State = 0x00*
0	0	0	0	0	0	0	1		LFSR State = 0x01
0	0	0	0	0	0	1	0		LFSR State = 0x02
...									...
1	1	1	1	1	1	0	0		LFSR State = 0xFC
1	1	1	1	1	1	0	1		LFSR State = 0xFD
1	1	1	1	1	1	1	0		LFSR State= 0xFE
1	1	1	1	1	1	1	1		LFSR State = 0xFF
NOTE 1 The LFSR State is the internal state of the Galois LFSR. This read only register is used for debug, and tracks the state of the LFSR associated with the selected DCA signal for training.									

NOTE 1 The LFSR State is the internal state of the Galois LFSR. This read only register is used for debug, and tracks the state of the LFSR associated with the selected DCA signal for training.

8.18.7 RW36 - DFETM Inner Loop Start Value Control Word

Table 144 — RW36 DFETM Inner Loop Start Value Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Inner Loop Start Value ^{1,2}	Inner Loop Start Value = 0x00
0	0	0	0	0	0	0	1		Inner Loop Start Value = 0x01
0	0	0	0	0	0	1	0		Inner Loop Start Value = 0x02
...									...
1	1	1	1	1	1	0	0		Inner Loop Start Value = 0xFC
1	1	1	1	1	1	0	1		Inner Loop Start Value = 0xFD
1	1	1	1	1	1	1	0		Inner Loop Start Value = 0xFE
1	1	1	1	1	1	1	1		Inner Loop Start Value = 0xFF
NOTE 1	Depending on the Inner Loop Parameter Selection, the range may be limited. The DFE Accelerator will only apply the OP bits that are within range for the specific Inner Loop Parameter.								
NOTE 2	Even though the tap setting parameter registers are defined as signed magnitude fields, this start value is a two's complement encoding. The RCD must convert this value to a signed magnitude format prior to updating the DFE_Vref when DFE_Vref is chosen for the Inner Loop Parameter.								

8.18.8 RW37 - DFETM Outer Loop Start Value Control Word

Table 145 — RW37 DFETM Outer Loop Start Value Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	0	0	0	0	0	0	0	Outer Loop Start Value ^{1,2}	Outer Loop Start Value = 0x00
x	0	0	0	0	0	0	1		Outer Loop Start Value = 0x01
x	0	0	0	0	0	1	0		Outer Loop Start Value = 0x02
x	...								...
x	1	1	1	1	1	0	0		Outer Loop Start Value = 0x7C
x	1	1	1	1	1	0	1		Outer Loop Start Value = 0x7D
x	1	1	1	1	1	1	0		Outer Loop Start Value = 0x7E
x	1	1	1	1	1	1	1		Outer Loop Start Value = 0x7F
0	x	x	x	x	x	x	x	RW[36] Inner Loop Start Value ^{3,4}	Inner Loop Start Value[8] = 0
1	x	x	x	x	x	x	x		Inner Loop Start Value[8] = 1
NOTE 1	Depending on the Outer Loop Parameter Selection, the range may be limited. The DFE Accelerator will only apply the OP bits that are within range for the specific Outer Loop Parameter.								
NOTE 2	Even though the tap setting parameter registers are defined as signed magnitude fields, this start value is a two's complement encoding. The RCD must convert this value to a signed magnitude format prior to updating the tap setting when Tap1, Tap2, Tap3, or Tap4 is chosen for the Outer Loop Parameter.								
NOTE 3	Depending on the Inner Loop Parameter Selection, the range may be limited. The DFE Accelerator will only apply the OP bits that are within range for the specific Inner Loop Parameter.								
NOTE 4	Even though the tap setting parameter registers are defined as signed magnitude fields, this start value is a two's complement encoding. The RCD must convert this value to a signed magnitude format prior to updating the DFE_Vref when DFE_Vref is chosen for the Inner Loop Parameter.								

8.18.9 RW38 - DFETM Inner Loop Current Value Control Word

Table 146 — RW38 DFETM Inner Loop Current Value Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0		
0	0	0	0	0	0	0	1	Inner Loop Current Value (Read Only)	Inner Loop Current Value = 0x00
0	0	0	0	0	0	0	1		Inner Loop Current Value = 0x01
0	0	0	0	0	0	1	0		Inner Loop Current Value = 0x02
...									...
1	1	1	1	1	1	0	0		Inner Loop Current Value = 0xFC
1	1	1	1	1	1	0	1		Inner Loop Current Value = 0xFD
1	1	1	1	1	1	1	0		Inner Loop Current Value = 0xFE
1	1	1	1	1	1	1	1		Inner Loop Current Value = 0xFF

8.18.10 RW39 - DFETM Outer Loop Current Value Control Word**Table 147 — RW39 DFETM Outer Loop Current Value Control Word**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	0	0	0	0	0	0	0	Outer Loop Current Value (Read Only)	Outer Loop Current Value = 0x00
x	0	0	0	0	0	0	1		Outer Loop Current Value = 0x01
x	0	0	0	0	0	1	0		Outer Loop Current Value = 0x02
x	...								...
x	1	1	1	1	1	0	0		Outer Loop Current Value = 0x7C
x	1	1	1	1	1	0	1		Outer Loop Current Value = 0x7D
x	1	1	1	1	1	1	0		Outer Loop Current Value = 0x7E
x	1	1	1	1	1	1	1		Outer Loop Current Value = 0x7F
0	x	x	x	x	x	x	x	RW38[8] Inner Loop Current Value (Read Only)	Inner Loop Current Value[8] = 0
1	x	x	x	x	x	x	x		Inner Loop Current Value[8] = 1

8.18.11 RW3A - DFETM Inner and Outer Loop Step Size Control Word**Table 148 — RW3A DFETM Inner and Outer Loop Step Size Control Word**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	0	0	0	0	Inner Loop Step Size ¹	Inner Loop Step Size = 1
x	x	x	x	0	0	0	1		Inner Loop Step Size = 2
x	x	x	x	0	0	1	0		Inner Loop Step Size = 3
x	x	x	x	0	0	1	1		Inner Loop Step Size = 4
x	x	x	x	0	1	0	0		Inner Loop Step Size = 5
x	x	x	x	0	1	0	1		Inner Loop Step Size = 6
x	x	x	x	0	1	1	0		Inner Loop Step Size = 7
x	x	x	x	0	1	1	1		Inner Loop Step Size = 8
x	x	x	x	1	0	0	0		Inner Loop Step Size = 9
x	x	x	x	1	0	0	1		Inner Loop Step Size = 10
x	x	x	x	1	0	1	0		Inner Loop Step Size = 11
x	x	x	x	1	0	1	1		Inner Loop Step Size = 12
x	x	x	x	1	1	0	0		Inner Loop Step Size = 13
x	x	x	x	1	1	0	1		Inner Loop Step Size = 14
x	x	x	x	1	1	1	0		Inner Loop Step Size = 15
x	x	x	x	1	1	1	1		Inner Loop Step Size = 16
0	0	0	0	x	x	x	x	Outer Loop Step Size ¹	Outer Loop Step Size = 1
0	0	0	1	x	x	x	x		Outer Loop Step Size = 2
0	0	1	0	x	x	x	x		Outer Loop Step Size = 3
0	0	1	1	x	x	x	x		Outer Loop Step Size = 4
0	1	0	0	x	x	x	x		Outer Loop Step Size = 5
0	1	0	1	x	x	x	x		Outer Loop Step Size = 6
0	1	1	0	x	x	x	x		Outer Loop Step Size = 7
0	1	1	1	x	x	x	x		Outer Loop Step Size = 8
1	0	0	0	x	x	x	x		Outer Loop Step Size = 9
1	0	0	1	x	x	x	x		Outer Loop Step Size = 10
1	0	1	0	x	x	x	x		Outer Loop Step Size = 11
1	0	1	1	x	x	x	x		Outer Loop Step Size = 12
1	1	0	0	x	x	x	x		Outer Loop Step Size = 13
1	1	0	1	x	x	x	x		Outer Loop Step Size = 14
1	1	1	0	x	x	x	x		Outer Loop Step Size = 15
1	1	1	1	x	x	x	x		Outer Loop Step Size = 16

NOTE 1 The step size is always positive, and thus the increment is always from lowest to highest value in the sweep. Note that the actual field setting indicates the Step Size – 1. The RCD must account for this in the increment logic.

8.18.12 RW3B - DFETM Inner Loop Number of Increments Control Word**Table 149 — RW3B DFETM Inner Loop Number of Increments Control Word**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Inner Loop Number of Increments ¹	Inner Loop Number of Increments = 0x00
0	0	0	0	0	0	0	1		Inner Loop Number of Increments = 0x01
0	0	0	0	0	0	1	0		Inner Loop Number of Increments = 0x02
...									...
1	1	1	1	1	1	0	0		Inner Loop Number of Increments = 0xFC
1	1	1	1	1	1	0	1		Inner Loop Number of Increments = 0xFD
1	1	1	1	1	1	1	0		Inner Loop Number of Increments = 0xFE
1	1	1	1	1	1	1	1		Inner Loop Number of Increments = 0xFF
NOTE 1 Depending on the Inner Loop Parameter Selection and the Inner Loop Step Size, the range may be limited. The DFE Accelerator will only apply the OP bits that are within range for the specific Inner Loop Parameter. When the Number of Increments = 0x00, the Start Value will be applied for a single iteration of the loop.									

8.18.13 RW3C - DFETM Outer Loop Number of Increments Control Word**Table 150 — RW3C DFETM Outer Loop Number of Increments Control Word**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	0	0	0	0	0	0	0	Outer Loop Number of Increments ¹	Outer Loop Number of Increments = 0x00
x	0	0	0	0	0	0	1		Outer Loop Number of Increments = 0x01
x	0	0	0	0	0	1	0		Outer Loop Number of Increments = 0x02
x	...								...
x	1	1	1	1	1	0	0		Outer Loop Number of Increments = 0x7C
x	1	1	1	1	1	0	1		Outer Loop Number of Increments = 0x7D
x	1	1	1	1	1	1	0		Outer Loop Number of Increments = 0x7E
x	1	1	1	1	1	1	1		Outer Loop Number of Increments = 0x7F
0	x	x	x	x	x	x	x	RW3B[8] Inner Loop Number of Increments ²	Inner Loop Number of Increments = 0
1	x	x	x	x	x	x	x		Inner Loop Number of Increments = 1
NOTE 1	Depending on the Outer Loop Parameter Selection and the Outer Loop Step Size, the range may be limited. The DFE Accelerator will only apply the OP bits that are within range for the specific Outer Loop Parameter. When the Number of Increments = 0x00, the Start Value will be applied for a single iteration of the loop.								
NOTE 2	Depending on the Inner Loop Parameter Selection and the Inner Loop Step Size, the range may be limited. The DFE Accelerator will only apply the OP bits that are within range for the specific Inner Loop Parameter. When the Number of Increments = 0x00, the Start Value will be applied for a single iteration of the loop.								

8.18.14 RW3D - DFETM Inner Loop Current Increment Control Word**Table 151 — RW3D DFETM Inner Loop Current Increment Control Word**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Inner Loop Current Increment (Read Only)	Inner Loop Current Increment = 0x00
0	0	0	0	0	0	0	1		Inner Loop Current Increment = 0x01
0	0	0	0	0	0	1	0		Inner Loop Current Increment = 0x02
...									...
1	1	1	1	1	1	0	0		Inner Loop Current Increment = 0xFC
1	1	1	1	1	1	0	1		Inner Loop Current Increment = 0xFD
1	1	1	1	1	1	1	0		Inner Loop Current Increment = 0xFE
1	1	1	1	1	1	1	1		Inner Loop Current Increment = 0xFF

8.18.15 RW3E - DFETM Outer Loop Current Increment Control Word**Table 152 — RW3E DFETM Outer Loop Current Increment Control Word**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	0	0	0	0	0	0	0	Outer Loop Current Increment (Read Only)	Outer Loop Current Increment = 0x00
x	0	0	0	0	0	0	1		Outer Loop Current Increment = 0x01
x	0	0	0	0	0	1	0		Outer Loop Current Increment = 0x02
x				...					...
x	1	1	1	1	1	0	0		Outer Loop Current Increment = 0x7C
x	1	1	1	1	1	0	1		Outer Loop Current Increment = 0x7D
x	1	1	1	1	1	1	0		Outer Loop Current Increment = 0x7E
x	1	1	1	1	1	1	1		Outer Loop Current Increment = 0x7F
0	x	x	x	x	x	x	x	RW3E[8] Inner Loop Current Increment (Read Only)	Inner Loop Current Increment = 0
1	x	x	x	x	x	x	x		Inner Loop Current Increment = 1

8.18.16 RW3F - DFE Vref Range Selection Control Word**Table 153 — RW3F DFE Vref Range Selection Control Word¹**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	DCA0	Positive
x	x	x	x	x	x	x	1		Negative
x	x	x	x	x	x	0	x	DCA1	Positive
x	x	x	x	x	x	1	x		Negative
x	x	x	x	x	0	x	x	DCA2	Positive
x	x	x	x	x	1	x	x		Negative
x	x	x	x	0	x	x	x	DCA3	Positive
x	x	x	x	1	x	x	x		Negative
x	x	x	0	x	x	x	x	DCA4	Positive
x	x	x	1	x	x	x	x		Negative
x	x	0	x	x	x	x	x	DCA5	Positive
x	x	1	x	x	x	x	x		Negative
x	0	x	x	x	x	x	x	DCA6	Positive
x	1	x	x	x	x	x	x		Negative
0	x	x	x	x	x	x	x	DPA0	Positive
1	x	x	x	x	x	x	x		Negative

NOTE 1 RW3F selects the sign for DFE Vref registers located in PG[2]RW[62, 66, 6A, 6E, 72, 76, 7A, 7E].

8.19 Vref Control Words

8.19.1 RW[47:40] - Internal VrefCA Control Word

Table 154 — RW[47:40]: Internal VrefCA Control Word^{1,2}

Setting								Definition	Encoding VrefCA as% of V _{DD} ³
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	0	0	0	0	0	0	0	Internal VrefCA Control	97.5%
x	0	0	0	0	0	0	1		97.0%
x	0	0	0	0	0	1	0		96.5%
x	0	0	0	0	0	1	1		96.0%
x	0	0	0	0	1	0	0		95.5%
x	0	0	0	0	1	0	1		95.0%
x	0	0	0	0	1	1	0		94.5%
x	0	0	0	0	1	1	1		94.0%
x	...								...
x	0	1	0	1	1	0	1		75% (Default)
x	...								...
x	1	1	0	0	0	0	1		49.0%
x	1	1	0	0	0	1	0		48.5%
x	1	1	0	0	0	1	1		48.0%
x	1	1	0	0	1	0	0		47.5%
x	1	1	0	0	1	0	1		47.0%
x	1	1	0	0	1	1	0		46.5%
x	1	1	0	0	1	1	1		46.0%
x	1	1	0	1	0	0	0		45.5%
x	1	1	0	1	0	0	1		45.0%
x	1	1	0	1	0	1	0		Reserved
x	1	1	0	1	0	1	1		Reserved
x	...								Reserved
x	1	1	1	1	1	1	1		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1

RW[47:40] OP[6:0] will be sticky, cleared by power cycle not Reset.

NOTE 2

Table 93 illustrates the assignment of each control word in the RW[47:40] group to the corresponding input pin it controls.

NOTE 3

These are target VrefCA values. Acceptable actual values are determined based on tolerances defined in electrical section.

8.19.2 RW[49:48] - Internal VrefCS Control Word

Table 155 — RW[49:48]: Internal VrefCS Control Word^{1,2}

Setting								Definition	Encoding VrefCS as% of V_{DD}^3
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	0	0	0	0	0	0	0	Internal VrefCS Control	97.5%
x	0	0	0	0	0	0	1		97.0%
x	0	0	0	0	0	1	0		96.5%
x	0	0	0	0	0	1	1		96.0%
x	0	0	0	0	1	0	0		95.5%
x	0	0	0	0	1	0	1		95.0%
x	0	0	0	0	1	1	0		94.5%
x	0	0	0	0	1	1	1		94.0%
x	...								...
x	0	1	0	1	1	0	1		75% (Default)
x	...								...
x	1	1	0	0	0	0	1		49.0%
x	1	1	0	0	0	1	0		48.5%
x	1	1	0	0	0	1	1		48.0%
x	1	1	0	0	1	0	0		47.5%
x	1	1	0	0	1	0	1		47.0%
x	1	1	0	0	1	1	0		46.5%
x	1	1	0	0	1	1	1		46.0%
x	1	1	0	1	0	0	0		45.5%
x	1	1	0	1	0	0	1		45.0%
x	1	1	0	1	0	1	0		Reserved
x	1	1	0	1	0	1	1		Reserved
x	...								Reserved
x	1	1	1	1	1	1	1		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1

RW[49:48] OP[6:0] will be sticky, cleared by power cycle not Reset.

NOTE 2

Table 93 illustrates the assignment of each control word in the RW[49:48] group to the corresponding input pin it controls.

NOTE 3

These are target VrefCS values. Acceptable actual values are determined based on tolerances defined in electrical section.

8.19.3 RW4A - DERROR_IN_n Vref Control Word

Table 156 — RW4A: DERROR_IN_n Vref Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	0	0	0	DERROR_IN_n Vref Control	0.80 x V_{DD} (Default)
x	x	x	x	x	0	0	1		0.75 x V_{DD}
x	x	x	x	x	0	1	0		0.70 x V_{DD}
x	x	x	x	x	0	1	1		0.60 x V_{DD}
x	x	x	x	x	1	0	0		Reserved
x	x	x	x	x	1	0	1		Reserved
x	x	x	x	x	1	1	0		Reserved
x	x	x	x	x	1	1	1		Reserved
x	x	x	x	0	x	x	x	Reserved	Reserved
x	x	x	x	1	x	x	x		Reserved
x	x	x	0	x	x	x	x	Reserved	Reserved
x	x	x	1	x	x	x	x		Reserved
x	x	0	x	x	x	x	x	Reserved	Reserved
x	x	1	x	x	x	x	x		Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

8.20 Loop-Back Vref and Input Delay Control

8.20.1 RW4B - Loop-Back Vref Control Word

Table 157 — RW4B: Loop-Back Vref Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	0	0	0	0	DLBS Vref Control	0.80 x V _{DD} (Default)
x	x	x	x	0	0	0	1		0.75 x V _{DD}
x	x	x	x	0	0	1	0		0.70 x V _{DD}
x	x	x	x	0	0	1	1		0.60 x V _{DD}
x	x	x	x	0	1	0	0		0.875 x VDD
x	x	x	x	0	1	0	1		0.85 x VDD
x	x	x	x	0	1	1	0		0.825 x VDD
x	x	x	x	0	1	1	1		0.775 x VDD
x	x	x	x	1	0	0	0		0.725 x VDD
x	x	x	x	1	0	0	1		0.675 x VDD
x	x	x	x	1	0	1	0		0.65 x VDD
x	x	x	x	1	0	1	1		0.625 x VDD
x	x	x	x	1	1	0	0		0.575 x VDD
x	x	x	x	1	1	0	1		0.55 x VDD
x	x	x	x	1	1	1	0		0.525 x VDD
x	x	x	x	1	1	1	1		0.5 x VDD
0	0	0	0	x	x	x	x	DLBD Vref Control	0.80 x V _{DD} (Default)
0	0	0	1	x	x	x	x		0.75 x V _{DD}
0	0	1	0	x	x	x	x		0.70 x V _{DD}
0	0	1	1	x	x	x	x		0.60 x V _{DD}
0	1	0	0	x	x	x	x		0.875 x VDD
0	1	0	1	x	x	x	x		0.85 x VDD
0	1	1	0	x	x	x	x		0.825 x VDD
0	1	1	1	x	x	x	x		0.775 x VDD
1	0	0	0	x	x	x	x		0.725 x VDD
1	0	0	1	x	x	x	x		0.675 x VDD
1	0	1	0	x	x	x	x		0.65 x VDD
1	0	1	1	x	x	x	x		0.625 x VDD
1	1	0	0	x	x	x	x		0.575 x VDD
1	1	0	1	x	x	x	x		0.55 x VDD
1	1	1	0	x	x	x	x		0.525 x VDD
1	1	1	1	x	x	x	x		0.5 x VDD

8.20.2 RW4C - Loop-Back Input Delay Control Word

Table 158 — RW4C: Loop-Back Input Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	0	0	DLBS Input Delay Control	No delay (Default)
x	x	x	x	x	x	0	1		Delay DLBS input signal by 20 ps ¹
x	x	x	x	x	x	1	0		Delay DLBS input signal by 40 ps ¹
x	x	x	x	x	x	1	1		Delay DLBS input signal by 60 ps ¹
x	x	x	x	x	0	x	x	Reserved	Reserved
x	x	x	x	x	1	x	x		Reserved
x	x	x	x	0	x	x	x	Reserved	Reserved
x	x	x	x	1	x	x	x		Reserved
x	x	0	0	x	x	x	x	DLBD Input Delay Control	No delay (Default)
x	x	0	1	x	x	x	x		Delay DLBD input signal by 20 ps ¹
x	x	1	0	x	x	x	x		Delay DLBD input signal by 40 ps ¹
x	x	1	1	x	x	x	x		Delay DLBD input signal by 60 ps ¹
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1 Allowed tolerance of the delay is +/-35% of the set delay value.

8.21 CTLE Control Words

8.21.1 RW50 - CTLE Configuration Global Control Word

Table 159 — RW50 - CTLE Configuration Global Control Word¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	CTLE Feature Enable Control	(Default) CTLE feature disabled in all pins
x	x	x	x	x	x	x	1		CTLE feature enabled ²
x	x	x	x	x	x	0	x	Reserved	Reserved
x	x	x	x	x	x	1	x		Reserved
x	x	x	x	x	0	x	x	Reserved	Reserved
x	x	x	x	x	1	x	x		Reserved
x	x	x	x	0	x	x	x	Reserved	Reserved
x	x	x	x	1	x	x	x		Reserved
x	x	x	0	x	x	x	x	Reserved	Reserved
x	x	x	1	x	x	x	x		Reserved
0	0	0	x	x	x	x	x	CTLE Configuration Range Status ³ (Read Only Register)	Range A
0	0	1	x	x	x	x	x		Range B
0	1	0	x	x	x	x	x		Range C
0	1	1	x	x	x	x	x		Range D
1	0	0	x	x	x	x	x		Reserved
1	0	1	x	x	x	x	x		Reserved
1	1	0	x	x	x	x	x		Reserved
1	1	1	x	x	x	x	x		Reserved

NOTE 1 RW50[0] will be sticky, cleared by power cycle not reset.

NOTE 2 The CTLE feature can be enabled per pin based on the settings in RW51[7:0].

NOTE 3 The device can only support one of the CTLE Configuration ranges, A, B, C or D. The Host can read out the supported range from RW50[7:5].

8.21.2 RW51 - CTLE Per-Pin Disable Control Word

Table 160 — RW51 - CTLE Per-Pin Disable Control Word¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	DCA0 CTLE Feature Disable	(Default) CTLE enabled in this pin ²
x	x	x	x	x	x	x	1		CTLE disabled in this pin
x	x	x	x	x	x	0	x	DCA1 CTLE Feature Disable	(Default) CTLE enabled in this pin ²
x	x	x	x	x	x	1	x		CTLE disabled in this pin
x	x	x	x	x	0	x	x	DCA2 CTLE Feature Disable	(Default) CTLE enabled in this pin ²
x	x	x	x	x	1	x	x		CTLE disabled in this pin
x	x	x	x	0	x	x	x	DCA3 CTLE Feature Disable	(Default) CTLE enabled in this pin ²
x	x	x	x	1	x	x	x		CTLE disabled in this pin
x	x	x	0	x	x	x	x	DCA4 CTLE Feature Disable	(Default) CTLE enabled in this pin ²
x	x	x	1	x	x	x	x		CTLE disabled in this pin
x	x	0	x	x	x	x	x	DCA5 CTLE Feature Disable	(Default) CTLE enabled in this pin ²
x	x	1	x	x	x	x	x		CTLE disabled in this pin
x	0	x	x	x	x	x	x	DCA6 CTLE Feature Disable	(Default) CTLE enabled in this pin ²
x	1	x	x	x	x	x	x		CTLE disabled in this pin
0	x	x	x	x	x	x	x	DPAR CTLE Feature Disable	(Default) CTLE enabled in this pin ²
1	x	x	x	x	x	x	x		CTLE disabled in this pin

NOTE 1 RW51 will be sticky, cleared by power cycle not reset.

NOTE 2 CTLE will be enabled in the corresponding input pin provided that RW50[0] is set to 1.

8.21.3 RW52 - CTLE Parameter Set A Control Word

Table 161 — RW52 - CTLE Parameter Set A Control Word¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	0	0	0	0	CTLE Parameter Set A Control ²	Select Setting 0 for Parameter Set A
x	x	x	x	0	0	0	1		Select Setting 1 for Parameter Set A
x	x	x	x	0	0	1	0		Select Setting 2 for Parameter Set A
x	x	x	x	...	...	...	...		...
x	x	x	x	1	1	0	1		Select Setting 13 for Parameter Set A
x	x	x	x	1	1	1	0		Select Setting 14 for Parameter Set A
x	x	x	x	1	1	1	1		Select Setting 15 for Parameter Set A
x	x	x	0	x	x	x	x	Reserved	Reserved
x	x	x	1	x	x	x	x		Reserved
x	x	0	x	x	x	x	x	Reserved	Reserved
x	x	1	x	x	x	x	x		Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1 RW52[3:0] will be sticky, cleared by power cycle not reset.

NOTE 2 Refer to Table 13, “Supported CTLE Parameter Value Combinations” for detailed encoding description.

8.21.4 RW53 - CTLE Parameter Set B Control Word

Table 162 — RW53 - CTLE Parameter Set B Control Word¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	0	0	0	0	CTLE Parameter Set B Control ²	Select Setting 0 for Parameter Set B
x	x	x	x	0	0	0	1		Select Setting 1 for Parameter Set B
x	x	x	x	0	0	1	0		Select Setting 2 for Parameter Set B
x	x	x	x	...					...
x	x	x	x	1	1	0	1		Select Setting 13 for Parameter Set B
x	x	x	x	1	1	1	0		Select Setting 14 for Parameter Set B
x	x	x	x	1	1	1	1		Select Setting 15 for Parameter Set B
x	x	x	0	x	x	x	x	Reserved	Reserved
x	x	x	1	x	x	x	x		Reserved
x	x	0	x	x	x	x	x	Reserved	Reserved
x	x	1	x	x	x	x	x		Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1

RW53[3:0] will be sticky, cleared by power cycle not reset.

NOTE 2

Refer to Table 13, “Supported CTLE Parameter Value Combinations,” on page 30 for detailed encoding description.

NOTE 1 RW53[3:0] will be sticky, cleared by power cycle not reset.

NOTE 2 Refer to Table 13, “Supported CTLE Parameter Value Combinations,” on page 30 for detailed encoding description.

8.21.5 RW54 - Per-Pin CTLE Parameter Set Selection Control Word

Table 163 — RW54 - Per-Pin CTLE Parameter Set Selection Control Word¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	DCA0 Parameter Set Selection	(Default) Set A Selected
x	x	x	x	x	x	x	1		Set B Selected
x	x	x	x	x	x	0	x	DCA1 Parameter Set Selection	(Default) Set A Selected
x	x	x	x	x	x	1	x		Set B Selected
x	x	x	x	x	0	x	x	DCA2 Parameter Set Selection	(Default) Set A Selected
x	x	x	x	x	1	x	x		Set B Selected
x	x	x	x	0	x	x	x	DCA3 Parameter Set Selection	(Default) Set A Selected
x	x	x	x	1	x	x	x		Set B Selected
x	x	x	0	x	x	x	x	DCA4 Parameter Set Selection	(Default) Set A Selected
x	x	x	1	x	x	x	x		Set B Selected
x	x	0	x	x	x	x	x	DCA5 Parameter Set Selection	(Default) Set A Selected
x	x	1	x	x	x	x	x		Set B Selected
x	0	x	x	x	x	x	x	DCA6 Parameter Set Selection	(Default) Set A Selected
x	1	x	x	x	x	x	x		Set B Selected
0	x	x	x	x	x	x	x	DPA0 Parameter Set Selection	(Default) Set A Selected
1	x	x	x	x	x	x	x		Set B Selected

NOTE 1 RW54 will be sticky, cleared by power cycle not reset.

8.22 Read and Paging Control Words

8.22.1 RW5E - CW Read Pointer Control Word

Table 164 — RW5E - CW Read Pointer Control Word

Read Pointer ¹							
OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
MRA7 ²	MRA6	MRA5	MRA4	MRA3	MRA2	MRA1	MRA0

NOTE 1 Points to the RCD Register Control Word to be read.
NOTE 2 MRA7 will always be '0' for RCD Reads.

8.22.2 RW5F - CW Page Control Word

Table 165 — RW5F - CW Page Control Word

Page Control Register ^{1,2}								Definition	Encoding
O P 7	O P 6	O P 5	O P 4	O P 3	O P 2	O P 1	O P 0		
0	0	0	0	0	0	0	0	Page selection control	PG[0] selected
0	0	0	0	0	0	0	1		PG[1] selected
0	0	0	0	0	0	1	0		PG[2] selected
.....									
1	1	1	1	1	1	0	1		PG[253] selected
1	1	1	1	1	1	1	0		PG[254] selected
1	1	1	1	1	1	1	1		PG[255] selected

NOTE 1 Power on Default is OP[7:0] = 0. This 8-bit control field uses binary encoding and it can be programmed with settings 0x00 (for CW Page 0) through 0xFF (for CW Page 255).
NOTE 2 The host is responsible to set the page register properly before issuing the MRW write to the DRAM scratch-pad register.

8.23 DFE Paged Control Words

8.23.1 PG[1:0]RW[60, 68, 70, 78] - DPAR and DCA[6:0] Receiver DFE Gain Offset

Table 166 — PG[1:0]RW[60, 68, 70, 78]: DPAR and DCA[6:0] Receiver DFE Gain Offset Adjustment¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	0	0	0	Flat-band (DC) gain adjustment ^{2,3,4,5,6}	Gain Adjustment = 0 dB (default)
x	x	x	x	x	0	0	1		Gain Adjustment = +6 dB
x	x	x	x	x	0	1	0		Gain Adjustment = +4 dB
x	x	x	x	x	0	1	1		Gain Adjustment = +2 dB
x	x	x	x	x	1	0	0		Gain Adjustment = 0 dB (same as default)
x	x	x	x	x	1	0	1		Gain Adjustment = -2 dB
x	x	x	x	x	1	1	0		Gain Adjustment = -4 dB
x	x	x	x	x	1	1	1		Gain Adjustment = -6 dB
x	x	x	x	0	x	x	x	Reserved	Reserved
x	x	x	x	1	x	x	x		Reserved
x	x	x	0	x	x	x	x		Reserved
x	x	x	1	x	x	x	x		Reserved
x	x	0	x	x	x	x	x		Reserved
x	x	1	x	x	x	x	x		Reserved
x	0	x	x	x	x	x	x		Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x		Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1 Table 94 and Table 95 illustrate the assignment of each control word in the PG[1:0]RW[60, 68, 70, 78] group to the corresponding input pin it controls.

NOTE 2 Flat-band (DC) gain adjustment (up to the Nyquist rate) adjustment control from I/O die pad to latching element in DCAn.

NOTE 3 The Gain Adjustment is applied to the baseline (default) inherent gain implemented in the receiver.

NOTE 4 Gain Adjustment values shown in Table 3 are verified by design and the measurement from device pins is defined in a separate specification.

NOTE 5 Allowable Differential nonlinearity (DNL) and the allowable Integral nonlinearity (INL) are defined in Table 12, “DFE Gain and Tap Coefficient Step Parameters Required per Speed Bin,” on page 26.

NOTE 6 PG[1:0]RW[60, 68, 70, 78] will be sticky, cleared by power cycle not reset.

8.23.2 PG[1:0]RW[61, 69, 71, 79] - DPAR and DCA[6:0] Receiver DFE Tap 1 Coefficients**Table 167 — PG[1:0]RW[61, 69, 71, 79]: DPAR and DCA[6:0] Receiver DFE Tap 1 Coefficients¹**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Tap 1 DFE Coefficient ^{2,3,4}	(Default) Tap 1 DFE bias = 0 mV
x	x	0	0	0	0	0	1		Tap 1 DFE bias +1 Tap Step
x	x	0	0	0	0	1	0		Tap 1 DFE bias +2 Tap Steps
x	x	0	0	0	0	1	1		Tap 1 DFE bias +3 Tap Step
x	x	0	0	0	1	0	0		Tap 1 DFE bias +4 Tap Steps
x	x	0	0	0	1	0	1		Tap 1 DFE bias +5 Tap Steps
x	x	0	0	0	1	1	0		Tap 1 DFE bias +6 Tap Steps
x	x	0	0	0	1	1	1		Tap 1 DFE bias +7 Tap Steps
x	x	0	0	1	0	0	0		Tap 1 DFE bias +8 Tap Steps
x	x	0	0	1	0	0	1		Tap 1 DFE bias +9 Tap Steps
x	x	0	0	1	0	1	0		Tap 1 DFE bias +10 Tap Steps
x	x	0	0	1	0	1	1		Tap 1 DFE bias +11 Tap Steps
x	x	0	0	1	1	0	0		Tap 1 DFE bias +12 Tap Steps
x	x	0	0	1	1	0	1		Tap 1 DFE bias +13 Tap Steps
x	x	0	0	1	1	1	0		Tap 1 DFE bias +14 Tap Steps
x	x	0	0	1	1	1	1		Tap 1 DFE bias +15 Tap Steps
x	x	0	1	0	0	0	0		Tap 1 DFE bias +16 Tap Steps
x	x	0	1	0	0	0	1		Tap 1 DFE bias +17 Tap Steps
x	x	0	1	0	0	1	0		Tap 1 DFE bias +18 Tap Steps
x	x	0	1	0	0	1	1		Tap 1 DFE bias +19 Tap Steps
x	x	0	1	0	1	0	0		Tap 1 DFE bias +20 Tap Steps
x	x	0	1	0	1	0	1		Tap 1 DFE bias +21 Tap Steps
x	x	0	1	0	1	1	0		Tap 1 DFE bias +22 Tap Steps
x	x	0	1	0	1	1	1		Tap 1 DFE bias +23 Tap Steps
x	x	0	1	1	0	0	0		Tap 1 DFE bias +24 Tap Steps
x	x	0	1	1	0	0	1		Tap 1 DFE bias +25 Tap Steps
x	x	0	1	1	0	1	0		Tap 1 DFE bias +26 Tap Steps
x	x	0	1	1	0	1	1		Tap 1 DFE bias +27 Tap Steps
x	x	0	1	1	1	0	0		Tap 1 DFE bias +28 Tap Steps
x	x	0	1	1	1	0	1		Tap 1 DFE bias +29 Tap Steps
x	x	0	1	1	1	1	0		Tap 1 DFE bias +30 Tap Steps
x	x	0	1	1	1	1	1		Tap 1 DFE bias +31 Tap Steps
x	x	1	0	0	0	0	0		Tap 1 DFE bias +32 Tap Steps
x	x	1	0	0	0	0	1		Tap 1 DFE bias +33 Tap Steps
x	x	1	0	0	0	1	0		Tap 1 DFE bias +34 Tap Steps
x	x	1	0	0	0	1	1		Tap 1 DFE bias +35 Tap Steps
x	x	1	0	0	1	0	0		Tap 1 DFE bias +36 Tap Steps
x	x	1	0	0	1	0	1		Tap 1 DFE bias +37 Tap Steps
x	x	1	0	0	1	1	0		Tap 1 DFE bias +38 Tap Steps
x	x	1	0	0	1	1	1		Tap 1 DFE bias +39 Tap Steps
x	x	1	0	1	0	0	0		Tap 1 DFE bias +40 Tap Steps
x	x	1	0	1	0	0	1		Tap 1 DFE bias +41 Tap Steps
x	x	1	0	1	0	1	0		Tap 1 DFE bias +42 Tap Steps
x	x	1	0	1	0	1	1		Tap 1 DFE bias +43 Tap Steps
x	x	1	0	1	1	0	0		Tap 1 DFE bias +44 Tap Steps
x	x	1	0	1	1	0	1		Tap 1 DFE bias +45 Tap Steps
x	x	1	0	1	1	1	0		Tap 1 DFE bias +46 Tap Steps
x	x	1	0	1	1	1	1		Tap 1 DFE bias +47 Tap Steps
x	x	1	1	0	0	0	0		Tap 1 DFE bias +48 Tap Steps
x	x	1	1	0	0	0	1		Tap 1 DFE bias +49 Tap Steps
x	x	1	1	0	0	1	0		Tap 1 DFE bias +50 Tap Steps
x	x	1	1	0	0	1	1		Reserved
x	x	1	1	0	1	0	0		Reserved
x	x	1	1	0	1	0	1		Reserved
x	x	1	1	0	1	1	0		Reserved

Table 167 — PG[1:0]RW[61, 69, 71, 79]: DPAR and DCA[6:0] Receiver DFE Tap 1 Coefficients¹ (cont'd)

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	1	1	0	1	1	1	Tap 1 DFE Coefficient ^{1,2,3}	Reserved
x	x	1	1	1	0	0	0		Reserved
x	x	1	1	1	0	0	1		Reserved
x	x	1	1	1	0	1	0		Reserved
x	x	1	1	1	0	1	1		Reserved
x	x	1	1	1	1	0	0		Reserved
x	x	1	1	1	1	0	1		Reserved
x	x	1	1	1	1	1	0		Reserved
x	x	1	1	1	1	1	1		Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Tap 1 Coefficient Sign Bit	(Default) Positive Tap 1 DFE bias when Tap 1 post-cursor is Logic 1 (Negative bias for Logic 0 Tap 1 post-cursor)
1	x	x	x	x	x	x	x		Negative Tap 1 DFE bias when Tap 1 post-cursor is Logic 1 (Positive bias for Logic 0 Tap 1 post-cursor)
NOTE 1 Table 94 and Table 95 illustrate the assignment of each control word in the PG[1:0]RW[61, 69, 71, 79] group to the corresponding input pin it controls.									
NOTE 2 Tap coefficient values shown in Table 168 are verified by design and the measurement from device pins is defined in a separate specification.									
NOTE 3 Allowable Differential nonlinearity (DNL) and the allowable Integral nonlinearity (INL) are defined in Table 12, “DFE Gain and Tap Coefficient Step Parameters Required per Speed Bin,” on page 26.									
NOTE 4 PG[1:0]RW[61, 69, 71, 79]OP[7:0] will be sticky, cleared by power cycle not reset.									

8.23.3 PG[1:0]RW[62, 6A, 72, 7A]- DPAR and DCA[6:0] Receiver DFE Tap 2 Coefficients

Table 168 — PG[1:0]RW[62, 6A, 72, 7A]: DPAR and DCA[6:0] Receiver DFE Tap 2 Coefficients¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Tap 2 DFE Coefficient ^{2,3,4}	(Default) Tap 2 DFE bias = 0 mV
x	x	0	0	0	0	0	1		Tap 2 DFE bias +1 Tap Steps
x	x	0	0	0	0	1	0		Tap 2 DFE bias +2 Tap Steps
x	x	0	0	0	0	1	1		Tap 2 DFE bias +3 Tap Steps
x	x	0	0	0	1	0	0		Tap 2 DFE bias +4 Tap Steps
x	x	0	0	0	1	0	1		Tap 2 DFE bias +5 Tap Steps
x	x	0	0	0	1	1	0		Tap 2 DFE bias +6 Tap Steps
x	x	0	0	0	1	1	1		Tap 2 DFE bias +7 Tap Steps
x	x	0	0	1	0	0	0		Tap 2 DFE bias +8 Tap Steps
x	x	0	0	1	0	0	1		Tap 2 DFE bias +9 Tap Steps
x	x	0	0	1	0	1	0		Tap 2 DFE bias +10 Tap Steps
x	x	0	0	1	0	1	1		Tap 2 DFE bias +11 Tap Steps
x	x	0	0	1	1	0	0		Tap 2 DFE bias +12 Tap Steps
x	x	0	0	1	1	0	1		Tap 2 DFE bias +13 Tap Steps
x	x	0	0	1	1	1	0		Tap 2 DFE bias +14 Tap Steps
x	x	0	0	1	1	1	1		Tap 2 DFE bias +15 Tap Steps
x	x	0	1	0	0	0	0		Tap 2 DFE bias +16 Tap Steps
x	x	0	1	0	0	0	1		Tap 2 DFE bias +17 Tap Steps
x	x	0	1	0	0	1	0		Tap 2 DFE bias +18 Tap Steps
x	x	0	1	0	0	1	1		Tap 2 DFE bias +19 Tap Steps
x	x	0	1	0	1	0	0	Reserved	Tap 2 DFE bias +20 Tap Steps
x	x	0	1	0	1	0	1		Reserved
x	x	0	1	0	1	1	0		Reserved
x	x	0	1	0	1	1	1		Reserved
x	x	0	1	1	0	0	0		Reserved
x	x	0	1	1	0	0	1		Reserved
x	x	0	1	1	0	1	0		Reserved
x	x	0	1	1	0	1	1		Reserved
x	x	0	1	1	1	0	0		Reserved
x	x	0	1	1	1	0	1		Reserved
x	x	0	1	1	1	1	0	Reserved	Reserved
x	x	0	1	1	1	1	1		Reserved
x	x	1	0	0	0	0	0		Reserved
x	x	1	0	0	0	0	1		Reserved
x	x	1	0	0	0	1	0		Reserved
x	x	1	0	0	0	1	1		Reserved
x	x	1	0	0	1	0	0		Reserved
x	x	1	0	0	1	0	1		Reserved
x	x	1	0	0	1	1	0		Reserved
x	x	1	0	0	1	1	1		Reserved
x	0	x	x	x	x	x	x	Tap 2 Coefficient Sign Bit	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Tap 2 Coefficient Sign Bit	(Default) Positive Tap 2 DFE bias when Tap 2 post-cursor is Logic 1 (Negative bias for Logic 0 Tap 2 post-cursor)
1	x	x	x	x	x	x	x		Negative Tap 2 DFE bias when Tap 2 post-cursor is Logic 1 (Positive bias for Logic 0 Tap 2 post-cursor)

NOTE 1 Table 94 and Table 95 illustrate the assignment of each control word in the PG[1:0]RW[62, 6A, 72, 7A] group to the corresponding input pin it controls.

NOTE 2 Tap coefficient values shown in Table 168 are verified by design and the measurement from device pins is defined in a separate specification.

NOTE 3 Allowable Differential nonlinearity (DNL) and the allowable Integral nonlinearity (INL) are defined in Table 12, “DFE Gain and Tap Coefficient Step Parameters Required per Speed Bin,” on page 26.

NOTE 4 PG[1:0]RW[62, 6A, 72, 7A] OP[7:0] will be sticky, cleared by power cycle not reset.

8.23.4 PG[1:0]RW[63, 6B, 73, 7B] - DPAR and DCA[6:0] Receiver DFE Tap 3 Coefficients

Table 169 — PG[1:0]RW[63, 6B, 73, 7B]: DPAR and DCA[6:0] Receiver DFE Tap 3 Coefficients¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	0	0	0	0	0	Tap 3 DFE Coefficient ^{2,3,4}	(Default) Tap 3 DFE bias = 0 mV
x	x	x	0	0	0	0	1		Tap 3 DFE bias +1 Tap Step
x	x	x	0	0	0	1	0		Tap 3 DFE bias +2 Tap Steps
x	x	x	0	0	0	1	1		Tap 3 DFE bias +3 Tap Steps
x	x	x	0	0	1	0	0		Tap 3 DFE bias +4 Tap Steps
x	x	x	0	0	1	0	1		Tap 3 DFE bias +5 Tap Steps
x	x	x	0	0	1	1	0		Tap 3 DFE bias +6 Tap Steps
x	x	x	0	0	1	1	1		Tap 3 DFE bias +7 Tap Steps
x	x	x	0	1	0	0	0		Tap 3 DFE bias +8 Tap Steps
x	x	x	0	1	0	0	1		Tap 3 DFE bias +9 Tap Steps
x	x	x	0	1	0	1	0		Tap 3 DFE bias +10 Tap Steps
x	x	x	0	1	0	1	1		Tap 3 DFE bias +11 Tap Steps
x	x	x	0	1	1	0	0		Tap 3 DFE bias +12 Tap Steps
x	x	x	0	1	1	0	1		Tap 3 DFE bias +13 Tap Steps
x	x	x	0	1	1	1	0		Tap 3 DFE bias +14 Tap Steps
x	x	x	0	1	1	1	1		Tap 3 DFE bias +15 Tap Steps
x	x	x	1	0	0	0	0		Reserved
x	x	x	1	0	0	0	1		Reserved
x	x	x	1	0	0	1	0		Reserved
x	x	x	1	0	0	1	1		Reserved
x	x	x	1	0	1	0	0		Reserved
x	x	x	1	0	1	0	1		Reserved
x	x	x	1	0	1	1	0		Reserved
x	x	x	1	0	1	1	1		Reserved
x	x	x	1	1	0	0	0		Reserved
x	x	x	1	1	0	0	1		Reserved
x	x	x	1	1	0	1	0		Reserved
x	x	x	1	1	0	1	1		Reserved
x	x	x	1	1	1	0	0		Reserved
x	x	x	1	1	1	0	1		Reserved
x	x	x	1	1	1	1	0		Reserved
x	x	x	1	1	1	1	1		Reserved
x	x	0	x	x	x	x	x	Reserved	Reserved
x	x	1	x	x	x	x	x		Reserved
x	0	x	x	x	x	x	x		Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Tap 3 Coefficient Sign Bit	(Default) Positive Tap 3 DFE bias when Tap 3 post-cursor is Logic 1 (Negative bias for Logic 0 Tap 3 post-cursor)
1	x	x	x	x	x	x	x		Negative Tap 3 DFE bias when Tap 3 post-cursor is Logic 1 (Positive bias for Logic 0 Tap 3 post-cursor)

NOTE 1 Table 94 and Table 95 illustrate the assignment of each control word in the PG[1:0]RW[63, 6B, 73, 7B] group to the corresponding input pin it controls.

NOTE 2 Tap coefficient values shown in Table 169 are verified by design and the measurement from device pins is defined in a separate specification.

NOTE 3 Allowable Differential nonlinearity (DNL) and the allowable Integral nonlinearity (INL) are defined in Table 12, “DFE Gain and Tap Coefficient Step Parameters Required per Speed Bin,” on page 26.

NOTE 4 PG[1:0]RW[63, 6B, 73, 7B] OP[7:0] will be sticky, cleared by power cycle not reset.

8.23.5 PG[1:0]RW[64, 6C, 74, 7C]- DPAR and DCA[6:0] Receiver DFE Tap 4 Coefficients**Table 170 — PG[1:0]RW[64, 6C, 74, 7C]: DPAR and DCA[6:0] Receiver DFE Tap 4 Coefficients¹**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	0	0	0	0	0	Tap 4 DFE Coefficient ^{2,3,4}	(Default) Tap 4 DFE bias = 0 mV
x	x	x	0	0	0	0	1		Tap 4 DFE bias +1 Tap Steps
x	x	x	0	0	0	1	0		Tap 4 DFE bias +2 Tap Steps
x	x	x	0	0	0	1	1		Tap 4 DFE bias +3 Tap Steps
x	x	x	0	0	1	0	0		Tap 4 DFE bias +4 Tap Steps
x	x	x	0	0	1	0	1		Tap 4 DFE bias +5 Tap Steps
x	x	x	0	0	1	1	0		Tap 4 DFE bias +6 Tap Steps
x	x	x	0	0	1	1	1		Tap 4 DFE bias +7 Tap Steps
x	x	x	0	1	0	0	0		Tap 4 DFE bias +8 Tap Steps
x	x	x	0	1	0	0	1		Tap 4 DFE bias +9 Tap Steps
x	x	x	0	1	0	1	0		Tap 4 DFE bias +10 Tap Steps
x	x	x	0	1	0	1	1		Tap 4 DFE bias +11 Tap Steps
x	x	x	0	1	1	0	0		Tap 4 DFE bias +12 Tap Steps
x	x	x	0	1	1	0	1		Tap 4 DFE bias +13 Tap Steps
x	x	x	0	1	1	1	0		Tap 4 DFE bias +14 Tap Steps
x	x	x	0	1	1	1	1		Tap 4 DFE bias +15 Tap Steps
x	x	x	1	0	0	0	0	Reserved	Reserved
x	x	x	1	0	0	0	1		Reserved
x	x	x	1	0	0	1	0		Reserved
x	x	x	1	0	0	1	1		Reserved
x	x	x	1	0	1	0	0		Reserved
x	x	x	1	0	1	0	1		Reserved
x	x	x	1	0	1	1	0		Reserved
x	x	x	1	0	1	1	1		Reserved
x	x	x	1	1	0	0	0		Reserved
x	x	x	1	1	0	0	1		Reserved
x	x	x	1	1	0	1	0		Reserved
x	x	x	1	1	0	1	1		Reserved
x	x	x	1	1	1	0	0		Reserved
x	x	x	1	1	1	0	1		Reserved
x	x	x	1	1	1	1	0		Reserved
x	x	x	1	1	1	1	1		Reserved
x	x	0	x	x	x	x	x	Reserved	Reserved
x	x	1	x	x	x	x	x		Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Tap 4 Coefficient Sign Bit	(Default) Positive Tap 4 DFE bias when Tap 4 post-cursor is Logic 1 (Negative bias for Logic 0 Tap 4 post-cursor)
1	x	x	x	x	x	x	x		Negative Tap 4 DFE bias when Tap 4 post-cursor is Logic 1 (Positive bias for Logic 0 Tap 4 post-cursor)

NOTE 1 Table 94 and Table 95 illustrate the assignment of each control word in the PG[1:0]RW[64, 6C, 74, 7C] group to the corresponding input pin it controls.

NOTE 2 Tap coefficient values shown in Table 170 are verified by design and the measurement from device pins is defined in a separate specification.

NOTE 3 Allowable Differential nonlinearity (DNL) and the allowable Integral nonlinearity (INL) are defined in Table 12, “DFE Gain and Tap Coefficient Step Parameters Required per Speed Bin,” on page 26.

NOTE 4 PG[1:0]RW[64, 6C, 74, 7C] OP[7:0] will be sticky, cleared by power cycle not reset.

8.23.6 PG[2]RW[60, 64, 68, 6C, 70, 74, 78, 7C] DFE Error Counter Lower 8 Bits**Table 171 — PG[2][60, 64, 68, 6C, 70, 74, 78, 7C]- DFE Error Counter Lower 8 Bit**

DFE Error Counter Lower 8 Bit Register ^{1,2,3,4}							
OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
EC7	EC6	EC5	EC4	EC3	EC2	EC1	EC0
NOTE 1 Table 96 illustrates the assignment of each control word in the PG[2]RW[60, 64, 68, 6C, 70, 74, 78, 7C] group to the corresponding input pin it controls.							
NOTE 2 Power on Default is OP[7:0] = 0.							
NOTE 3 Read Only Register.							
NOTE 4 ERROR counters will run regardless of DCn selection.							

8.23.7 PG[2]RW[61, 65, 69, 6D, 71, 75, 79, 7D] - DFE Error Counter Upper 8 Bits**Table 172 — PG[2][61, 65, 69, 6D, 71, 75, 79, 7D] DFE Error Counter Upper 8 Bit**

DFE Error Counter Upper 8 Bit Register ^{1,2,3,4}							
OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
EC 15	EC 14	EC 13	EC 12	EC 11	EC 10	EC 9	EC 8
NOTE 1 Table 96 illustrates the assignment of each control word in the PG[2]RW[61, 65, 69, 6D, 71, 75, 79, 7D] group to the corresponding input pin it controls.							
NOTE 2 Power on Default is OP[7:0] = 0.							
NOTE 3 Read Only Register.							
NOTE 4 ERROR counters will run regardless of DCn selection.							

8.23.8 PG[2]RW[62, 66, 6A, 6E, 72, 76,7A, 7E] - DFE Vref

Table 173 — PG[2]RW[62, 66, 6A, 6E, 72, 76,7A, 7E]: DFE Vref

PG02RW[62, 66 .. 7E]						DFE Training VREF in mV ^{1,2,3,4,5}							
OP7	OP6	OP5	OP4	OP3	OP2	RW3F[x ⁶] = 1 (NEG Range)				RW3F[x ⁴] = 0 (POS Range)			
						OP[1:0] = 00	OP[1:0] = 01	OP[1:0] = 10	OP[1:0] = 11	OP[1:0] = 00	OP[1:0] = 01	OP[1:0] = 10	OP[1:0] = 11
0	0	0	0	0	0	0.0	- 2.5	- 5.0	- 7.5	0.0	+ 2.5	+ 5.0	+ 7.5
0	0	0	0	0	1	- 10.0	- 12.5	- 15.0	- 17.5	+ 10.0	+ 12.5	+ 15.0	+ 17.5
0	0	0	0	1	0	- 20.0	- 22.5	- 25.0	- 27.5	+ 20.0	+ 22.5	+ 25.0	+ 27.5
0	0	0	0	1	1	- 30.0	- 32.5	- 35.0	- 37.5	+ 30.0	+ 32.5	+ 35.0	+ 37.5
0	0	0	1	0	0	- 40.0	- 42.5	- 45.0	- 47.5	+ 40.0	+ 42.5	+ 45.0	+ 47.5
0	0	0	1	0	1	- 50.0	- 52.5	- 55.0	- 57.5	+ 50.0	+ 52.5	+ 55.0	+ 57.5
0	0	0	1	1	0	- 60.0	- 62.5	- 65.0	- 67.5	+ 60.0	+ 62.5	+ 65.0	+ 67.5
0	0	0	1	1	1	- 70.0	- 72.5	- 75.0	- 77.5	+ 70.0	+ 72.5	+ 75.0	+ 77.5
0	0	1	0	0	0	- 80.0	- 82.5	- 85.0	- 87.5	+ 80.0	+ 82.5	+ 85.0	+ 87.5
0	0	1	0	0	1	- 90.0	- 92.5	- 95.0	- 97.5	+ 90.0	+ 92.5	+ 95.0	+ 97.5
0	0	1	0	1	0	- 100.0	- 102.5	- 105.0	- 107.5	+ 100.0	+ 102.5	+ 105.0	+ 107.5
0	0	1	0	1	1	- 110.0	- 112.5	- 115.0	- 117.5	+ 110.0	+ 112.5	+ 115.0	+ 117.5
0	0	1	1	0	0	- 120.0	- 122.5	- 125.0	- 127.5	+ 120.0	+ 122.5	+ 125.0	+ 127.5
0	0	1	1	0	1	- 130.0	- 132.5	- 135.0	- 137.5	+ 130.0	+ 132.5	+ 135.0	+ 137.5
0	0	1	1	1	0	- 140.0	- 142.5	- 145.0	- 147.5	+ 140.0	+ 142.5	+ 145.0	+ 147.5
0	0	1	1	1	1	- 150.0	- 152.5	- 155.0	- 157.5	+ 150.0	+ 152.5	+ 155.0	+ 157.5
0	1	0	0	0	0	- 160.0	- 162.5	- 165.0	- 167.5	+ 160.0	+ 162.5	+ 165.0	+ 167.5
0	1	0	0	0	1	- 170.0	- 172.5	- 175.0	- 177.5	+ 170.0	+ 172.5	+ 175.0	+ 177.5
0	1	0	0	1	0	- 180.0	- 182.5	- 185.0	- 187.5	+ 180.0	+ 182.5	+ 185.0	+ 187.5
0	1	0	0	1	1	- 190.0	- 192.5	- 195.0	- 197.5	+ 190.0	+ 192.5	+ 195.0	+ 197.5
0	1	0	1	0	0	- 200.0	- 202.5	- 205.0	- 207.5	+ 200.0	+ 202.5	+ 205.0	+ 207.5
0	1	0	1	0	1	- 210.0	- 212.5	- 215.0	- 217.5	+ 210.0	+ 212.5	+ 215.0	+ 217.5
0	1	0	1	1	0	- 220.0	- 222.5	- 225.0	- 227.5	+ 220.0	+ 222.5	+ 225.0	+ 227.5
0	1	0	1	1	1	- 230.0	- 232.5	- 235.0	- 237.5	+ 230.0	+ 232.5	+ 235.0	+ 237.5
0	1	1	0	0	0	- 240.0	- 242.5	- 245.0	- 247.5	+ 240.0	+ 242.5	+ 245.0	+ 247.5
0	1	1	0	0	1	- 250.0	- 252.5	- 255.0	- 257.5	+ 250.0	+ 252.5	+ 255.0	+ 257.5
0	1	1	0	1	0	- 260.0	- 262.5	- 265.0	- 267.5	+ 260.0	+ 262.5	+ 265.0	+ 267.5
0	1	1	0	1	1	- 270.0	- 272.5	- 275.0	- 277.5	+ 270.0	+ 272.5	+ 275.0	+ 277.5
0	1	1	1	0	0	- 280.0	- 282.5	- 285.0	- 287.5	+ 280.0	+ 282.5	+ 285.0	+ 287.5
0	1	1	1	0	1	- 290.0	- 292.5	- 295.0	- 297.5	+ 290.0	+ 292.5	+ 295.0	+ 297.5
0	1	1	1	1	0	- 300.0	- 302.5	- 305.0	- 307.5	+ 300.0	+ 302.5	+ 305.0	+ 307.5
0	1	1	1	1	1	- 310.0	- 312.5	- 315.0	- 317.5	+ 310.0	+ 312.5	+ 315.0	+ 317.5
1	0	0	0	0	0	- 320.0	- 322.5	- 325.0	- 327.5	+ 320.0	+ 322.5	+ 325.0	+ 327.5
1	0	0	0	0	1	- 330.0	- 332.5	- 335.0	- 337.5	+ 330.0	+ 332.5	+ 335.0	+ 337.5
1	0	0	0	1	0	- 340.0	- 342.5	- 345.0	- 347.5	+ 340.0	+ 342.5	+ 345.0	+ 347.5
1	0	0	0	1	1	- 350.0	- 352.5	- 355.0	- 357.5	+ 350.0	+ 352.5	+ 355.0	+ 357.5
1	0	0	1	0	0	- 360.0	- 362.5	- 365.0	- 367.5	+ 360.0	+ 362.5	+ 365.0	+ 367.5
1	0	0	1	0	1	- 370.0	- 372.5	- 375.0	- 377.5	+ 370.0	+ 372.5	+ 375.0	+ 377.5
1	0	0	1	1	0	- 380.0	- 382.5	- 385.0	- 387.5	+ 380.0	+ 382.5	+ 385.0	+ 387.5
1	0	0	1	1	1	- 390.0	- 392.5	- 395.0	- 397.5	+ 390.0	+ 392.5	+ 395.0	+ 397.5
1	0	1	0	0	0	- 400.0	- 402.5	- 405.0	- 407.5	+ 400.0	+ 402.5	+ 405.0	+ 407.5
1	0	1	0	0	1	- 410.0	- 412.5	- 415.0	- 417.5	+ 410.0	+ 412.5	+ 415.0	+ 417.5
1	0	1	0	1	0	- 420.0	- 422.5	- 425.0	- 427.5	+ 420.0	+ 422.5	+ 425.0	+ 427.5
1	0	1	0	1	1	- 430.0	- 432.5	- 435.0	- 437.5	+ 430.0	+ 432.5	+ 435.0	+ 437.5
1	0	1	1	0	0	- 440.0	- 442.5	- 445.0	- 447.5	+ 440.0	+ 442.5	+ 445.0	+ 447.5
1	0	1	1	0	1	- 450.0	- 452.5	- 455.0	- 457.5	+ 450.0	+ 452.5	+ 455.0	+ 457.5
1	0	1	1	1	0	- 460.0	- 462.5	- 465.0	- 467.5	+ 460.0	+ 462.5	+ 465.0	+ 467.5
1	0	1	1	1	1	- 470.0	- 472.5	- 475.0	- 477.5	+ 470.0	+ 472.5	+ 475.0	+ 477.5
1	1	0	0	0	0	- 480.0	- 482.5	- 485.0	- 487.5	+ 480.0	+ 482.5	+ 485.0	+ 487.5
1	1	0	0	0	1	- 490.0	- 492.5	- 495.0	- 497.5	+ 490.0	+ 492.5	+ 495.0	+ 497.5
1	1	0	0	1	0	- 500.0	Reserved	Reserved	Reserved	+ 500.0	Reserved	Reserved	Reserved

Table 173 — PG[2]RW[62, 66, 6A, 6E, 72, 76, 7A, 7E]: DFE Vref (cont'd)

PG02RW[62, 66 .. 7E]							DFE Training VREF in mV ^{1,2,3,4,5}							
OP7	OP6	OP5	OP4	OP3	OP2		RW3F[x ⁶] = 1 (NEG Range)				RW3F[x ⁴] = 0 (POS Range)			
							OP[1:0] = 00	OP[1:0] = 01	OP[1:0] = 10	OP[1:0] = 11	OP[1:0] = 00	OP[1:0] = 01	OP[1:0] = 10	OP[1:0] = 11
1	1	0	0	1	1		Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1	1	0	1	0	0		Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1	1	0	1	0	1		Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1	1	0	1	1	0		Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1	1	0	1	1	1		Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1	1	1	0	0	0		Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1	1	1	0	0	1		Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1	1	1	0	1	0		Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1	1	1	0	1	1		Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1	1	1	1	0	0		Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1	1	1	1	0	1		Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1	1	1	1	1	0		Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1	1	1	1	1	1		Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

NOTE 1 Table 96 illustrates the assignment of each control word in the PG[2]RW[62, 66, 6A, 6E, 72, 76, 7A, 7E] group to the corresponding input pin it controls.

NOTE 2 These are target DFE_Vref values. Acceptable actual values are determined based on tolerances defined in electrical section.

NOTE 3 The target DFE_Vref values shown in this table are input referred.

NOTE 4 Even though each input receiver has a dedicated DFE_Vref control word, only one DFE_Vref monitor is allowed to be enabled at the same time in each sub-channel.

NOTE 5 VrefCA and DFE_Vref are independent. External VrefCA must be adjusted for large input swings.

NOTE 6 The range selection per Dx is controlled by writing to RW3F.

8.23.9 PG[2]RW[63, 67, 6B, 6F, 73, 77, 7B, 7F] - LFSR Seed for DFE Training Mode Control Word

Table 174 — PG[2]RW[63, 67, 6B, 6F, 73, 77, 7B, 7F] LFSR Seed for DFE Training Mode Control Word¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	LFSR Seed	LFSR Seed = 0x00*
0	0	0	0	0	0	0	1		LFSR Seed = 0x01
0	0	0	0	0	0	1	0		LFSR Seed = 0x02
...									...
1	1	1	1	1	1	0	0		LFSR Seed = 0xFC
1	1	1	1	1	1	0	1		LFSR Seed = 0xFD
1	1	1	1	1	1	1	0		LFSR Seed = 0xFE
1	1	1	1	1	1	1	1		LFSR Seed = 0xFF
NOTE 1	Table 96 illustrates the assignment of each control word in the PG[2]RW[63, 67, 6B, 6F, 73, 77, 7B, 7F] group to the corresponding input pin it controls.								

8.24 OTP Paged Global Control Words

Page 3 contains Global Read-Only Words that must be accessed through Channel A only.

8.24.1 PG[3]RW60 - Date Code Byte 0 Global Word

Table 175 — PG[3]RW60: Date Code Byte 0 Global Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	0	0	0	0	Date Code Digit 0 ^{1,2} Year Information - Ones Digit (Read Only)	Digit = 0
x	x	x	x	0	0	0	1		Digit = 1
x	x	x	x	0	0	1	0		Digit = 2
x	x	x	x	...	...	...	...		...
x	x	x	x	0	1	1	1		Digit = 7
x	x	x	x	1	0	0	0		Digit = 8
x	x	x	x	1	0	0	1		Digit = 9
x	x	x	x	1	0	1	0		Codes 10 to 15 Reserved
x	x	x	x	...	...	...	...		
x	x	x	x	1	1	1	1		
0	0	0	0	x	x	x	x	Date Code Digit 1 ^{1,2} Year Information - Tens Digit (Read Only)	Digit = 0
0	0	0	1	x	x	x	x		Digit = 1
0	0	1	0	x	x	x	x		Digit = 2
...	...	...	...	x	x	x	x		...
0	1	1	1	x	x	x	x		Digit = 7
1	0	0	0	x	x	x	x		Digit = 8
1	0	0	1	x	x	x	x		Digit = 9
1	0	1	0	x	x	x	x		Codes 10 to 15 Reserved
...	...	...	...	x	x	x	x		
1	1	1	1	x	x	x	x		

NOTE 1 Programmed and locked in one time programmable memory by DDR5RCD02 vendor.

NOTE 2 This is year date code byte for RCD. It must be represented in Binary Coded Decimal (BCD). For example, year 2015 would be coded as 0x15 (0001 0101).

8.24.2 PG[3]RW61 - Date Code Byte 1 Global Word

Table 176 — PG[3]RW61: Date Code Byte 1 Global Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	0	0	0	0	Date Code Digit 2 ^{1,2} Work Week Information - Ones Digit ISO 8601 compliant (Read Only)	Digit = 0
x	x	x	x	0	0	0	1		Digit = 1
x	x	x	x	0	0	1	0		Digit = 2
x	x	x	x	...	...	...	...		...
x	x	x	x	0	1	1	1		Digit = 7
x	x	x	x	1	0	0	0		Digit = 8
x	x	x	x	1	0	0	1		Digit = 9
x	x	x	x	1	0	1	0		Codes 10 to 15 Reserved
x	x	x	x	...	...	...	...		
x	x	x	x	1	1	1	1		
0	0	0	0	x	x	x	x	Date Code Digit 3 ^{1,2} Work Week Information - Tens Digit ISO 8601 compliant (Read Only)	Digit = 0
0	0	0	1	x	x	x	x		Digit = 1
0	0	1	0	x	x	x	x		Digit = 2
...	...	...	...	x	x	x	x		...
0	1	1	1	x	x	x	x		Digit = 7
1	0	0	0	x	x	x	x		Digit = 8
1	0	0	1	x	x	x	x		Digit = 9
1	0	1	0	x	x	x	x		Codes 10 to 15 Reserved
...	...	...	...	x	x	x	x		
1	1	1	1	x	x	x	x		

NOTE 1 Programmed and locked in one time programmable memory by DDR5RCD02 vendor.

NOTE 2 This is work week date code byte for RCD. It must be represented in Binary Coded Decimal (BCD). For example, week 47 would be coded as 0x47 (0100 0111).

8.24.3 PG[3]RW62 - Date Code Byte 2 Global Word

Table 177 — PG[3]RW62: Date Code Byte 2 Global Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	0	0	0	0	Date Code Digit 4 ¹ Reserved (Read Only)	Digit = 0
x	x	x	x	0	0	0	1		Digit = 1
x	x	x	x	0	0	1	0		Digit = 2
x	x	x	x	...					...
x	x	x	x	0	1	1	1		Digit = 7
x	x	x	x	1	0	0	0		Digit = 8
x	x	x	x	1	0	0	1		Digit = 9
x	x	x	x	1	0	1	0		Codes 10 to 15 Reserved
x	x	x	x	...					
x	x	x	x	1	1	1	1		
0	0	0	0	x	x	x	x	Date Code Digit 5 ¹ Reserved (Read Only)	
0	0	0	1	x	x	x	x		Digit = 1
0	0	1	0	x	x	x	x		Digit = 2
...				x	x	x	x		...
0	1	1	1	x	x	x	x		Digit = 7
1	0	0	0	x	x	x	x		Digit = 8
1	0	0	1	x	x	x	x		Digit = 9
1	0	1	0	x	x	x	x		Codes 10 to 15 Reserved
...				x	x	x	x		
1	1	1	1	x	x	x	x		
NOTE 1 Programmed and locked in one time programmable memory by DDR5RCD02 vendor.									

NOTE 1 Programmed and locked in one time programmable memory by DDR5RCD02 vendor.

8.24.4 PG[3]RW63 - Vendor Specific Unique Unit Code Byte 0 Global Word

Table 178 — PG[3]RW63: Vendor Specific Unique Unit Code Byte 0 Global Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Byte 0 of Unique Unit Code ¹ (Read Only)	Code 0
0	0	0	0	0	0	0	1		Code 1
0	0	0	0	0	0	1	0		Code 2
...									...
1	1	1	1	1	1	0	1		Code 253
1	1	1	1	1	1	1	0		Code 254
1	1	1	1	1	1	1	1		Code 255
NOTE 1 Programmed and locked in one time programmable memory by DDR5RCD02 vendor.									

NOTE 1 Programmed and locked in one time programmable memory by DDR5RCD02 vendor.

8.24.5 PG[3]RW64 - Vendor Specific Unique Unit Code Byte 1 Global Word

Table 179 — PG[3]RW64: Vendor Specific Unique Unit Code Byte 1 Global Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Byte 1 of Unique Unit Code ¹ (Read Only)	Code 0
0	0	0	0	0	0	0	1		Code 1
0	0	0	0	0	0	1	0		Code 2
...									...
1	1	1	1	1	1	0	1		Code 253
1	1	1	1	1	1	1	0		Code 254
1	1	1	1	1	1	1	1		Code 255
NOTE 1									Programmed and locked in one time programmable memory by DDR5RCD02 vendor.

NOTE 1 Programmed and locked in one time programmable memory by DDR5RCD02 vendor.

8.24.6 PG[3]RW65 - Vendor Specific Unique Unit Code Byte 2 Global Word

Table 180 — PG[3]RW65: Vendor Specific Unique Unit Code Byte 2 Global Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Byte 2 of Unique Unit Code ¹ (Read Only)	Code 0
0	0	0	0	0	0	0	1		Code 1
0	0	0	0	0	0	1	0		Code 2
...									...
1	1	1	1	1	1	0	1		Code 253
1	1	1	1	1	1	1	0		Code 254
1	1	1	1	1	1	1	1		Code 255
NOTE 1 Programmed and locked in one time programmable memory by DDR5RCD02 vendor.									

8.24.7 PG[3]RW66 - Vendor Specific Unique Unit Code Byte 3 Global Word

Table 181 — PG[3]RW66: Vendor Specific Unique Unit Code Byte 3 Global Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Byte 3 of Unique Unit Code ¹ (Read Only)	Code 0
0	0	0	0	0	0	0	1		Code 1
0	0	0	0	0	0	1	0		Code 2
...									...
1	1	1	1	1	1	0	1		Code 253
1	1	1	1	1	1	1	0		Code 254
1	1	1	1	1	1	1	1		Code 255
NOTE 1 Programmed and locked in one time programmable memory by DDR5RCD02 vendor.									

8.24.8 PG[3]RW67 - Vendor Specific Unique Unit Code Byte 4 Global Word

Table 182 — PG[3]RW67: Vendor Specific Unique Unit Code Byte 4 Global Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Byte 4 of Unique Unit Code ¹ (Read Only)	Code 0
0	0	0	0	0	0	0	1		Code 1
0	0	0	0	0	0	1	0		Code 2
...									...
1	1	1	1	1	1	0	1		Code 253
1	1	1	1	1	1	1	0		Code 254
1	1	1	1	1	1	1	1		Code 255
NOTE 1 Programmed and locked in one time programmable memory by DDR5RCD02 vendor.									

8.24.9 PG[3]RW68 - Vendor Specific Unique Unit Code Byte 5 Global Word

Table 183 — PG[3]RW68: Vendor Specific Unique Unit Code Byte 5 Global Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Byte 5 of Unique Unit Code ¹ (Read Only)	Code 0
0	0	0	0	0	0	0	1		Code 1
0	0	0	0	0	0	1	0		Code 2
...									...
1	1	1	1	1	1	0	1		Code 253
1	1	1	1	1	1	1	0		Code 254
1	1	1	1	1	1	1	1		Code 255
NOTE 1 Programmed and locked in one time programmable memory by DDR5RCD02 vendor.									

8.24.10 PG[3]RW69 - Vendor Specific Unique Unit Code Byte 6 Global Word**Table 184 — PG[3]RW69: Vendor Specific Unique Unit Code Byte 6 Global Word**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Byte 6 of Unique Unit Code ¹ (Read Only)	Code 0
0	0	0	0	0	0	0	1		Code 1
0	0	0	0	0	0	1	0		Code 2
...									...
1	1	1	1	1	1	0	1		Code 253
1	1	1	1	1	1	1	0		Code 254
1	1	1	1	1	1	1	1		Code 255
NOTE 1 Programmed and locked in one time programmable memory by DDR5RCD02 vendor.									

8.24.11 PG[3]RW6A - Vendor ID Byte 0 Global Word**Table 185 — PG[3]RW6A: Vendor ID Byte 0 Global Word¹**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Byte 0 of Vendor ID (Read Only)	VID[7:0] = 0x00h
0	0	0	0	0	0	0	1		VID[7:0] = 0x01h
0	0	0	0	0	0	1	0		VID[7:0] = 0x02h
...									...
1	1	1	1	1	1	0	1		VID[7:0] = 0xFDh
1	1	1	1	1	1	1	0		VID[7:0] = 0xFEh
1	1	1	1	1	1	1	1		VID[7:0] = 0xFFh
NOTE 1 Each vendor will have a specific Vendor ID value assigned by JEDEC.									

8.24.12 PG[3]RW6B - Vendor ID Byte 1 Global Word**Table 186 — PG[3]RW6B: Vendor ID Byte 1 Global Word¹**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Byte 1 of Vendor ID (Read Only)	VID[15:8] = 0x00h
0	0	0	0	0	0	0	1		VID[15:8] = 0x01h
0	0	0	0	0	0	1	0		VID[15:8] = 0x02h
...									...
1	1	1	1	1	1	0	1		VID[15:8] = 0xFDh
1	1	1	1	1	1	1	0		VID[15:8] = 0xFEh
1	1	1	1	1	1	1	1		VID[15:8] = 0xFFh
NOTE 1 Each vendor will have a specific Vendor ID value assigned by JEDEC.									

8.24.13 PG[3]RW6C - Device ID Byte 0 Global Word**Table 187 — PG[3]RW6C: Device ID Byte 0 Global Word¹**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Byte 0 of Device ID (Read Only)	DID[7:0] = 0x00h
0	0	0	0	0	0	0	1		DID[7:0] = 0x01h
0	0	0	0	0	0	1	0		DID[7:0] = 0x02h
...									...
1	1	1	1	1	1	0	1		DID[7:0] = 0xFDh
1	1	1	1	1	1	1	0		DID[7:0] = 0xFEh
1	1	1	1	1	1	1	1		DID[7:0] = 0xFFh
NOTE 1 The device ID value is 0x0052.									

8.24.14 PG[3]RW6D - Device ID Byte 1 Global Word**Table 188 — PG[3]RW6D: Device ID Byte 1 Global Word¹**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Byte 1 of Device ID (Read Only)	DID[15:8] = 0x00h
0	0	0	0	0	0	0	1		DID[15:8] = 0x01h
0	0	0	0	0	0	1	0		DID[15:8] = 0x02h
...									...
1	1	1	1	1	1	0	1		DID[15:8] = 0xFDh
1	1	1	1	1	1	1	0		DID[15:8] = 0xFEh
1	1	1	1	1	1	1	1		DID[15:8] = 0xFFh
NOTE 1 The device ID value is 0x0052.									

8.24.15 PG[3]RW6E - Vendor Revision ID Global Word**Table 189 — PG[3]RW6E: Vendor Revision ID Global Word¹**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
0	0	0	0	0	0	0	0	Vendor Revision ID (Read Only)	RID[7:0] = 0x00h
0	0	0	0	0	0	0	1		RID[7:0] = 0x01h
0	0	0	0	0	0	1	0		RID[7:0] = 0x02h
...									...
1	1	1	1	1	1	0	1		RID[7:0] = 0xFDh
1	1	1	1	1	1	1	0		RID[7:0] = 0xFEh
1	1	1	1	1	1	1	1		RID[7:0] = 0xFFh
NOTE 1 This is a fixed vendor specific register.									

8.25 VHost Paged Control Words

8.25.1 PG[4]RW[6E,6C,6A,68,66,64,62,60] - VHost Cmd[3:0] CA[7:0] Control Word

Table 190 — PG[4]RW[6E,6C,6A,68,66,64,62,60] - VHost Cmd[3:0] CA[7:0] Control Word¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	Cmd[3:0] 1st/2nd UI CA0	QCA0 = 0
x	x	x	x	x	x	x	1		QCA0 = 1
x	x	x	x	x	x	0	x	Cmd[3:0] 1st/2nd UI CA1	QCA1 = 0
x	x	x	x	x	x	1	x		QCA1 = 1
x	x	x	x	x	0	x	x	Cmd[3:0] 1st/2nd UI CA2	QCA2 = 0
x	x	x	x	x	1	x	x		QCA2 = 1
x	x	x	x	0	x	x	x	Cmd[3:0] 1st/2nd UI CA3	QCA3 = 0
x	x	x	x	1	x	x	x		QCA3 = 1
x	x	x	0	x	x	x	x	Cmd[3:0] 1st/2nd UI CA4	QCA4 = 0
x	x	x	1	x	x	x	x		QCA4 = 1
x	x	0	x	x	x	x	x	Cmd[3:0] 1st/2nd UI CA5	QCA5 = 0
x	x	1	x	x	x	x	x		QCA5 = 1
x	0	x	x	x	x	x	x	Cmd[3:0] 1st/2nd UI CA6	QCA6 = 0
x	1	x	x	x	x	x	x		QCA6 = 1
0	x	x	x	x	x	x	x	Cmd[3:0] 1st/2nd UI CA7	QCA7 = 0
1	x	x	x	x	x	x	x		QCA7 = 1

NOTE 1 Table 98 illustrates the assignment of each control word in the PG[4]RW[6E, 6C, 6A, 68, 66, 64, 62, 60] group to the corresponding input pin it controls.

8.25.2 PG[4]RW[6F,6D,6B,69,67,65,63,61] - VHost Cmd[3:0] CS[1:0]/CA[13:8] Control Word

Table 191 — PG[4]RW[6F,6D,6B,69,67,65,63,61] - VHost Cmd[3:0] CA[7:0] Control Word¹

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	x	0	Cmd[3:0] 1st/2nd UI CA8	QCA8 = 0
x	x	x	x	x	x	x	1		QCA8 = 1
x	x	x	x	x	x	0	x	Cmd[3:0] 1st/2nd UI CA9	QCA9 = 0
x	x	x	x	x	x	1	x		QCA9 = 1
x	x	x	x	x	0	x	x	Cmd[3:0] 1st/2nd UI CA10	QCA10 = 0
x	x	x	x	x	1	x	x		QCA10 = 1
x	x	x	x	0	x	x	x	Cmd[3:0] 1st/2nd UI CA11	QCA11 = 0
x	x	x	x	1	x	x	x		QCA11 = 1
x	x	x	0	x	x	x	x	Cmd[3:0] 1st/2nd UI CA12	QCA12 = 0
x	x	x	1	x	x	x	x		QCA12 = 1
x	x	0	x	x	x	x	x	Cmd[3:0] 1st/2nd UI CA13	QCA13 = 0
x	x	1	x	x	x	x	x		QCA13 = 1
x	0	x	x	x	x	x	x	Cmd[3:0] 1st/2nd UI CS0_n	QCS0_n = 0
x	1	x	x	x	x	x	x		QCS0_n = 1
0	x	x	x	x	x	x	x	Cmd[3:0] 1st/2nd UI CS1_n	QCS1_n = 0
1	x	x	x	x	x	x	x		QCS1_n = 1

NOTE 1 Table 98 illustrates the assignment of each control word in the PG[4]RW[6F, 6D, 6B, 69, 67, 65, 63, 61] group to the corresponding input pin it controls.

8.25.3 PG[4]RW70 - VHost [1:0] Cmd to Cmd Delay Control Word**Table 192 — PG[4]RW70 - VHost [1:0] Cmd to Cmd Delay Control Word**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	0	0	0	cmd0 to cmd1 Delay	2 nCK
x	x	x	x	x	0	0	1		4 nCK
x	x	x	x	x	0	1	0		8 nCK
x	x	x	x	x	0	1	1		16 nCK
x	x	x	x	x	1	0	0		64 nCK
x	x	x	x	x	1	0	1		256 nCK
x	x	x	x	x	1	1	0		1024 nCK
x	x	x	x	x	1	1	1		Reserved
x	x	x	x	0	x	x	x	Reserved	Reserved
x	x	x	x	1	x	x	x		Reserved
x	0	0	0	x	x	x	x	cmd1 to cmd2 Delay	2 nCK
x	0	0	1	x	x	x	x		4 nCK
x	0	1	0	x	x	x	x		8 nCK
x	0	1	1	x	x	x	x		16 nCK
x	1	0	0	x	x	x	x		64 nCK
x	1	0	1	x	x	x	x		256 nCK
x	1	1	0	x	x	x	x		1024 nCK
x	1	1	1	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

8.25.4 PG[4]RW71 - VHost [3:2]Cmd to Cmd Delay Control Word**Table 193 — PG[4]RW71 - VHost [3:2] Cmd to Cmd Delay Control Word**

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	0	0	0	cmd2 to cmd3 Delay	2 nCK
x	x	x	x	x	0	0	1		4 nCK
x	x	x	x	x	0	1	0		8 nCK
x	x	x	x	x	0	1	1		16 nCK
x	x	x	x	x	1	0	0		64 nCK
x	x	x	x	x	1	0	1		256 nCK
x	x	x	x	x	1	1	0		1024 nCK
x	x	x	x	x	1	1	1		Reserved
x	x	x	x	0	x	x	x	Reserved	Reserved
x	x	x	x	1	x	x	x		Reserved
x	0	0	0	x	x	x	x	cmd3 to cmdn Delay	2 nCK
x	0	0	1	x	x	x	x		4 nCK
x	0	1	0	x	x	x	x		8 nCK
x	0	1	1	x	x	x	x		16 nCK
x	1	0	0	x	x	x	x		64 nCK
x	1	0	1	x	x	x	x		256 nCK
x	1	1	0	x	x	x	x		1024 nCK
x	1	1	1	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

8.25.5 PG[4]RW72 - VHost Repeat Mode Control Word

Table 194 — PG[4]RW72 - VHost Repeat Mode Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	0	0	Repeat Control	Repeat to cmd0 (repeat all 4)
x	x	x	x	x	x	0	1		Repeat to cmd1 (repeat last 3)
x	x	x	x	x	x	1	0		Repeat to cmd2 (repeat last 2)
x	x	x	x	x	x	1	1		Repeat to cmd3 (repeat last 1)
x	x	x	x	x	0	x	x	Repeat Enable	Do not repeat
x	x	x	x	x	1	x	x		Repeat as defined in RW72[1:0]
x	x	x	x	0	x	x	x	Reserved	Reserved
x	x	x	x	1	x	x	x		Reserved
x	x	x	0	x	x	x	x	Reserved	Reserved
x	x	x	1	x	x	x	x		Reserved
x	x	0	x	x	x	x	x	Reserved	Reserved
x	x	1	x	x	x	x	x		Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

8.25.6 PG[4]RW73 - VHost Start and Stop Global Control Word

Table 195 — PG[4]RW73 - VHost Start and Stop Global Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	x	x	x	0	0	VHost Selection	CH A only
x	x	x	x	x	x	0	1		CH B only
x	x	x	x	x	x	1	0		CH A and CH B Start and Stop at the same time
x	x	x	x	x	x	1	1		Reserved
x	x	x	x	x	0	x	x	VHost Start/Stop ¹	Stop VHost
x	x	x	x	x	1	x	x		Start VHost
x	x	x	x	0	x	x	x	Reserved	Reserved
x	x	x	x	1	x	x	x		Reserved
x	x	x	0	x	x	x	x	Reserved	Reserved
x	x	x	1	x	x	x	x		Reserved
x	x	0	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Reserved	Reserved
1	x	x	x	x	x	x	x		Reserved

NOTE 1 PG[4]RW73 is only accessible from CH A.

8.26 PG[5]RW[7B:60] Per-bit QCA Output Delay Control Words

Table 196 — PG[5]RW[7B:60] Per-bit QCA Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	x	0	0	0	0	0	Per-bit QCA Output Delay ¹	(Default) Delay output by 0 ps
x	x	x	0	0	0	0	1		Delay output by 2.5 ps
x	x	x	0	0	0	1	0		Delay output by 5 ps
x	x	x	0	0	0	1	1		Delay output by 7.5 ps
x	x	x	0	0	1	0	0		Delay output by 10 ps
x	x	x	0	0	1	0	1		Delay output by 12.5 ps
x	x	x	0	0	1	1	0		Delay output by 15 ps
x	x	x	0	0	1	1	1		Delay output by 17.5 ps
x	x	x	0	1	0	0	0		Delay output by 20 ps
x	x	x	0	1	0	0	1		Delay output by 22.5 ps
x	x	x	0	1	0	1	0		Delay output by 25 ps
x	x	x	0	1	0	1	1		Delay output by 27.5 ps
x	x	x	0	1	1	0	0		Delay output by 30 ps
x	x	x	0	1	1	0	1		Delay output by 32.5 ps
x	x	x	0	1	1	1	0		Delay output by 35 ps
x	x	x	0	1	1	1	1		Delay output by 37.5 ps
x	x	x	1	0	0	0	0		Delay output by 40 ps
x	x	x	1	0	0	0	1		Codes 0x11 to 0x15 Reserved
x	x	x	.	.	.	.	.		
x	x	x	1	1	1	1	1		
x	x	0	x	x	x	x	x	Reserved	Reserved
x	x	1	x	x	x	x	x	Reserved	Reserved
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x	Reserved	Reserved
0	x	x	x	x	x	x	x	Per-bit QCA Output Delay feature enable	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ²

NOTE 1 PG[5]RW[7B:60] will be sticky, cleared by power cycle not Reset

NOTE 2 When feature is enabled the delay settings in PG[5]RW[7B:60][6:0] require a time of t_{ODU} for the delay to become stable on the outputs.

9 Absolute Maximum Ratings

Table 197 — Absolute Maximum Ratings Over Operating Free-air Temperature Range¹

Symbol	Parameter	Conditions	Min	Max	Unit
V_{DD}	Supply voltage		-0.3	1.4	V
V_{IN}	Receiver input voltage ²	See Notes 4 and 5	-0.3	$V_{DD} + 0.5$	V
V_{OUT}	Driver output voltage ³	See Notes 4 and 5	-0.3	$V_{DD} + 0.5$	V
I_{IK}	Input clamp current	$V_{IN} < 0$ or $V_{IN} > V_{DD}$	-	-50	mA
I_{OK}	Output clamp current	$V_{OUT} < 0$ or $V_{OUT} > V_{DD}$	-	± 50	mA
I_{OUT}	Continuous output current	$0 < V_{OUT} < V_{DD}$	-	± 50	mA
I_{CCC}	Continuous current through each V_{DD} or V_{SS} pin		-	± 100	mA
T_{stg}	Storage temperature		-55	+100	°C
NOTE 1 Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under “operating conditions” is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability. NOTE 2 This parameter does not apply to SCL. See Management Bus Interface chapter for voltage specifications for these inputs. NOTE 3 This parameter does not apply to SDA. See Management Bus Interface chapter for voltage specifications for these inputs. NOTE 4 The input and output negative voltage ratings may be exceeded if the input and output clamp-current ratings are observed. NOTE 5 This value is limited to 1.4 V maximum.					

10 Electrical - Timing Requirements

10.1 Operating Electrical Characteristics

The DDR5RCD02 parametric values are specified for the device default control word settings, unless otherwise stated.

Table 198 — Operating Electrical Characteristics

Symbol	Parameter	Condition	Min	Nom	Max	Unit
V _{DD}	DC Supply voltage ¹	1.1 V Operation	1.067 (-3%)	1.1	1.166 (+6%)	V
V _{DDIO}	DDR5RCD02 Sideband Interface I/O Supply Voltage		0.95	1.0	1.05	V
T _j	Junction temperature ²		0	-	125	°C
T _{case}	Case temperature	Measurement procedure JESD51-2	-	-	103 ³	°C
NOTE 1 DC bandwidth limited to 20 MHz.						
NOTE 2 For operation beyond T _j min and max datasheet values are not guaranteed and may de-rate. For operation above T _j max lifetime could be affected. All parametric measurements are performed at 0 °C, 25 °C, and 95 °C.						
NOTE 3 This spec is meant to guarantee a T _j of 125 °C by the DDR5RCD02. Since T _j cannot be measured or observed by users, T _{case} is specified instead. Under all thermal condition, the T _j of a DDR5RCD02 shall not be higher than 125 °C.						

10.2 Input Timing Requirements

Table 199 — Input Timing Requirements

Symbol	Parameter	Conditions	DDR5-3200/ 3600/4000		DDR5-4400/ 4800		DDR5-5200/ 5600		Unit
			Min	Max	Min	Max	Min	Max	
f_{CLOCK}	Input clock frequency ¹	Application frequency ²	1400	2030	1400	2436	1400	2842	MHz
f_{TEST}	Input clock frequency	Test frequency ³	140	990	140	990	140	990	MHz
t_{MRD}	Control word to control word delay	Number of clock cycles between two control word accesses or one control word access and any DRAM command.	8	-	8	-	10	-	t_{CK}
$t_{\text{MRD_L}}$	Control word to control word delay	Number of clock cycles between an access to RW0A , RW0C , RW0D , RW0E , RW0F ⁴ , RW10 , RW25 , RW26 , RW27 , RW31[7:2] , RW4B:40 ⁵ , RW5E ⁶ , RW5F ⁶ and the next control word access or DRAM command.	16	-	16	-	20	-	t_{CK}
$t_{\text{MRD_L2}}$	Control word to control word delay	Number of clock cycles between an access to RW00 ⁷ , RW01 , RW02 , RW03 , RW05 ⁷ , RW06 ⁷ , RW07 , RW09 , RW0E , RW0F , RW4C , RW11[2:0] , RW32[7:6, 2:1] ⁸ and the next control word access or DRAM command ⁹	32	-	32	-	40	-	t_{CK}
$t_{\text{MRD_DFE_Tap}}$	Control word to control word delay	Number of clock cycles between an access to PG[1:0]RW[7C:79,74:71,6C:69,64:61] (Rx DFE Tap coefficients) and the next control word access or DRAM command.	64	-	64	-	64	-	t_{CK}
$t_{\text{MRD_L3}}$	Control word to control word delay for DFE Gain Adjustment	Waiting time between an access to RW32[5:3, 0] , PG[1:0]RW[60, 68, 70, 78] (Rx DFE Gain coefficients) and the next input DCS_n assertion.	300	-	300	-	300	-	ns
$t_{\text{MRD_L4}}$	Control word to control word delay for DFE Vref Settling Time	Waiting time between an access to RW31[1:0] , RW3F , PG02RW[62, 66, 6A, 6E, 72, 76, 7A, 7E] and the next input DCS_n assertion.	300	-	300	-	300	-	ns
$t_{\text{MRD_CTLE}}$	Control Word to Control Word or DRAM Command delay for CTLE control and setting updates	Number of clock cycles between an access to RW[54:50] and the next control word access or DRAM command ¹⁰	64	-	64	-	64	-	t_{CK}
t_{MRC}	Register command word to CW or DRAM command delay	Number of clock cycles between register command word (RW04) and CW or any DRAM command	32	-	32	-	40	-	t_{CK}
$t_{\text{Ext_LB_Entry}}$	External Loopback Entry Time	Delay from MRW (RW26) that enables External Loopback and start of valid QLBD/QLBS forwarded from selected DLBD/DLBS.	-	300	-	300	-	300	ns
t_{MRD2N}	Control word to control word delay in SDR/2N mode.	Number of clock cycles between two control word accesses or one control word access and any DRAM command.	16	-	16	-	20	-	
$t_{\text{MRD2N_L}}$	Control word to control word delay in SDR/2N mode	Number of clock cycles between an access to RW0A , RW0B , RW0C , RW0D ⁴ , RW10 , RW25 , RW26 , RW27 , RW[4B:40] ⁵ , RW5E ⁶ , RW5F ⁶ and the next control word access or DRAM command.	24	-	24	-	30	-	t_{CK}
$t_{\text{MRD2N_L2}}$	Control word to control word delay in SDR/2N mode	Number of clock cycles between an access to RW00 ⁷ , RW01 , RW02 , RW03 , RW05 ⁷ , RW06 ⁷ , RW07 , RW09 , RW0E , RW0F , RW4C , RW11[2:0] , RW32[7:6, 2:1] ⁸ and the next control word access or DRAM command ⁹	50	-	50	-	65	-	t_{CK}

Table 199 — Input Timing Requirements (cont'd)

Symbol	Parameter	Conditions	DDR5-3200/ 3600/4000		DDR5-4400/ 4800		DDR5-5200/ 5600		Unit
			Min	Max	Min	Max	Min	Max	
t_{MRC2N}	Register command word to CW or DRAM command delay in SDR/2N mode	Number of clock cycles between register command word (RW04) and CW or any DRAM command	40	-	40	-	50	-	t_{CK}
$t_{RWUPDInit}$	Time needed from enter into RWUPD to being able to receive in band RW updates		500	-	500	-	500	-	ns
$t_{RWUPDA2D}$	Time between address transfer and data transfer		8	-	8	-	8	-	t_{CK}
$t_{RWUPDD2A}$	Time between data transfer and next operation address transfer		8	-	8	-	8	-	t_{CK}
$t_{RWUPDExit}$	Time needed between last RWUPD CS1_n de-assertion and the first CS0_n assertion of DFE training mode or any other operation. ¹¹		500	-	500	-	500	-	ns
t_{DCSTM_Entry}	Registration of DCSTM entry from SMBus RW command to start of training samples time		-	250	-	250	-	250	ns
t_{DCSTM_Exit}	Registration of DCSTM exit from SMBus RW command to end of training mode		-	250	-	250	-	250	ns
t_{DCSTM_Valid}	Time from sample evaluation to output on ALERT_n or QLBx		-	20	-	20	-	20	t_{CK}
$t_{DCSTM_ALERT_Window}$	Time output is available ¹² on ALERT_n or QLBx		2	-	2	-	2	-	t_{CK}
t_{QCSTM_Entry}	Registration of QCSTM entry from MRW command to start of QCS training pattern		-	72	-	72	-	72	t_{CK}
t_{QCSTM_Exit}	Registration of QCSTM exit from SMBus RW command to end of QCS training pattern		-	72	-	72	-	72	t_{CK}
t_{DCATM_Entry}	Registration of DCATM entry from SMBus RW command to start of training samples time		-	250	-	250	-	250	ns
$t_{DCATM_Exit}^{13}$	Registration of DCATM exit from SMBus RW command to end of training mode		-	250	-	250	-	250	ns
$t_{DCATM_CS_Exit}^{13}$	Time DCS0_n must be held LOW to exit from DCATM training mode		2	8	2	8	2	8	t_{CK}
t_{DCATM_Valid}	Time from sample evaluation to output on ALERT_n or QLBx		-	20	-	20	-	20	t_{CK}
$t_{DCATM_ALERT_Window}$	Time output is available ¹² on ALERT_n or QLBx		2	-	2	-	2	-	t_{CK}
$t_{DFETM_Interval}$	Number of cycles for CA DFE Training data comparison for each output result.		4	4	4	4	4	4	t_{CK}
t_{DFETM_Valid}	Time from end of CA DFE Training data comparison to valid result output.		-	20	-	20	-	20	t_{CK}
$t_{DFETM_RSP_Window}$	Duration of CA DFE Training output result signal pulse.		2	-	2	-	2	-	t_{CK}
$t_{DFETM_Pat_tern_Wait_Accel}^{14}$	Time between de-assertion of DCS0_n for one pattern to DCS0_n assertion of the next pattern.	DFE Training Accelerator enabled.	64	-	64	-	64	-	t_{CK}
$t_{DFETM_Pat_tern_Wait}^{15}$	Time between de-assertion of DCS0_n for one pattern to DCS0_n assertion of the next pattern.	DFE Training Accelerator disabled.	16	-	16	-	16	-	t_{CK}

Table 199 — Input Timing Requirements (cont'd)

Symbol	Parameter	Conditions	DDR5-3200/ 3600/4000		DDR5-4400/ 4800		DDR5-5200/ 5600		Unit
			Min	Max	Min	Max	Min	Max	
t_{CSALT}	Chip select assertion before and after ALERT_n Pulse Width deassertion	Number of clock cycles between DCSx assertion and rising edge of ALERT_n when RW01[6] = 1	8	-	8	-	8	-	t_{CK}
t_{CPDED}	Minimum time between SRE and DCS pulse to drive QCS LOW.	DCS[1:0]_n = HIGH; DRST_n = HIGH; DCK_t/DCK_c = Toggling.	Min: max(8 nCK, 5 ns); Max: (-)						t_{CK}
t_{PDEX}	Minimum time between PDE and PDX in the same rank.		Min: max(16 nCK, 10 ns); Max: (-)						t_{CK}
t_{RPDX}	Minimum time between RCD PDX and the next valid command other than DRAM Exit from PD. ¹⁶		Min: max(16 nCK, 10 ns); Max: (-)						t_{CK}
$t_{CPDED2SRX}$	Minimum time from single DCS LOW pulse following SRE command to NOP (SRX) command	DCS[1:0]_n = LOW; DRST_n = HIGH.	Min: max(16 nCK, 10 ns); Max: (-)						t_{CK}
t_{CSSR}	Minimum DCS[1:0]_n LOW time (after SRE) during Self Refresh with DCK stop	DCS[1:0]_n = LOW; DRST_n = HIGH; DCK_t/DCK_c = Z or H or L or Toggling.	Min: max(24 nCK, 15 ns); Max: (-)						ns
t_{CKoff}	Minimum time required after DCS[1:0]_n goes LOW (after SRE) to DCK stop	DCS[1:0]_n = LOW; DRST_n = HIGH; DCK_t/DCK_c = Toggling.	Min: max(24 nCK, 15 ns); Max: (-)						ns
$t_{SRX2SRX}$	Minimum number of valid input clock cycles between SRX command to release QCS and SRX to take DRAM out of Self Refresh	DB control interface disabled (RW09[3] = 0).	Min: max(16 nCK, 10 ns); Max: (-)						t_{CK}
t_{DFE_PRD}	Minimum time from static HIGH or Valid DCA[6:0]/DPAR before initial DCS_n assertion	DFE Enabled during Reset with Stable Power (Figure 33) or Self Refresh Exit (Figure 35).	Min: max(8 nCK, 5 ns); Max: (-)						t_{CK}
$t_{DCSLHSK1}$	Maximum delay between any DCSx_n in LOW-to-HIGH transition from Initialization or Exit from Self Refresh With Clock Stop		200						ns
$t_{DCSLHSK2}$	Maximum delay between DCS0_n and DCS1_n (within the same sub-channel) in LOW-to-HIGH transition from Initialization or Exit from Self Refresh With Clock Stop		500						ps
t_{XS}	DRAM Exit Self Refresh to next valid command not requiring DLL		See DDR5 DRAM Specification						
t_{RINIT1}	Minimum DRST_n LOW time after completion of voltage ramp.		200	-	200	-	200	-	μ s
	Minimum DRST_n LOW time with stable power.		1	-	1	-	1	-	μ s
t_{RINIT2}	Min DCS_n LOW time to DRST_n HIGH.		20	-	20	-	20	-	ns
t_{RINIT3}	Minimum DCS LOW time after DRST_n HIGH.		20	-	20	-	20	-	ns
t_{CKACT}	Maximum time from DCS_n transition HIGH to start of valid DCK.		-	16	-	16	-	16	t_{CK}
t_{XPR}	DRAM minimum time from Exit Reset to first valid Configuration Command		Min: max(5 nCK, $t_{RFC}(\text{min}) + 10 \text{ ns}$); Max: (-); See DDR5 DRAM Specification						ns
t_{ZQCAL}	ZQ Calibration Time		1	-	1	-	1	-	μ s

Table 199 — Input Timing Requirements (cont'd)

Symbol	Parameter	Conditions	DDR5-3200/ 3600/4000		DDR5-4400/ 4800		DDR5-5200/ 5600		Unit
			Min	Max	Min	Max	Min	Max	
t _{ZLAT}	ZQ Calibration Latch Time		Min: max(30 ns, 8 nCK); Max: (-)						ns
NOTE 1	Including SSC according Table 206, “SSC and PLL Loop Filter Characteristics”.								
NOTE 2	All specified timing parameters apply								
NOTE 3	Timing parameters specified for frequency band 2 apply								
NOTE 4	Since RW0A, RW0C, RW0D, RW0E, RW0F affect output driver operation, it is necessary for these settings to be updated during initialization when Bits OP[1] and OP[3] in RCD control word RW01 are still configured to block all commands at the DRAM and Data Buffer interfaces, respectively.								
NOTE 5	When Vref settings for DLBD, DLBS, DERROR_IN_n, DCA[6:0], DPAR, and DCS[1:0]_n are updated, it is necessary to meet the Vref settling times defined in Table 201 in addition to meeting tMRD_L.								
NOTE 6	Applies to the timing between MRW to RW5E or RW5F in RCD space and an MRW to MR63 in DRAM space only during an RCD Control Word Read sequence. Also applies to the timing between an MRW to any control word selected for read via RW5E and RW5F and an MRW to MR63 during an RCD Control Word Read sequence. Otherwise this parameter does not apply to RW5E or RW5F.								
NOTE 7	A t _{STAB} waiting time is required when changing the setting in RW00[0] (Command Address SDR/DDR Rate selection), RW05[3:0] (RDIMM Operating Speed), RW05[7] (Frequency Band Select), and RW06 (Fine Granularity Operating Speed).								
NOTE 8	When exiting from RW In-Band Update mode (i.e., when writing a 1 to RW32[7]), it is necessary to meet the t _{RWUPDExit} timing parameter defined in Table 199, “Input Timing Requirements” in addition to meeting tMRD_L2.								
NOTE 9	This parameter also applies to control words inside the DDR5DB02 device. Refer to JESD82-522 for details.								
NOTE 10	If DFE Gain is enabled by RW31[0]=1, then tMRD_L3 must be met.								
NOTE 11	This parameter also covers the exit latency if exit is triggered through SMBus operation.								
NOTE 12	As shown in Figure 131 on page 260, this is the minimum guaranteed LOW pulse (i.e., below [VOH + VOL]/2, where VOH and VOL are the measured Logic HIGH and LOW DC voltage levels) into the test load. Logically, the minimum pulse width out of the RCD in DCS or DCA training modes is four DCK clock cycles wide.								
NOTE 13	For the double CS Exit the corresponding Host Interface TM selection in RW02 will be reset to default (Normal Operating Mode).								
NOTE 14	When the DFE Training Acceleration is enabled, the minimum waiting time between de-assertion of DCS0_n for one pattern and assertion of DCS0_n of the next pattern is determined by the larger of tDFETM_Pattern_Wait_Accel [min] specified in this table and the minimum settling time for the parameter setting updated by the Training Accelerator logic (e.g. for DFE_Vref or VrefCA, the minimum settling time could be much larger than tDFETM_Pattern_Wait_Accel [min]).								
NOTE 15	When the DFE Training Acceleration is disabled, there is a different minimum spacing between patterns, since there is no need to account for automated parameter changes.								
NOTE 16	During the t _{RPDX} window, the logic levels for QACA[13:0] and QBCA[13:0] outputs are not deterministic as the RCD transitions from disabled DCA[6:0] & DPAR receivers to normal operation. During this time, the host is allowed to take a memory rank still in PDE out of power-down state using one DCS_n pulse, but it must refrain from sending any additional commands to any memory rank that has already received its first PDX CS_n pulse. By the end of the t _{RPDX} window, the DDR5RCD02 device is required to forward any legal command from the DCA[6:0] inputs to the QxCA[13:0] and BCOM[2:0] outputs as defined in normal operation requirements.								

10.3 Output Timing Requirements

Table 200 — Output Timing Requirements¹

Symbol	Parameter	Conditions	DDR5-3200 to 4800		DDR5-5200 to 5600		Unit	Note
			Min	Max	Min	Max		
t_{PDM}	Propagation delay, single bit switching; DCK_t/DCK_c falling edge crosspoint to output	1.1 V Operation, DDR Mode (RW00[0] = 1)	0.95	1.25	0.95	1.25	ns	2
t_{PDM2N}	Propagation delay, single bit switching; DCK_t/DCK_c rising edge crosspoint to output	1.1 V Operation, SDR Mode (RW00[0] = 0)	0.95	1.25	0.95	1.25	ns	3
t_{PDM_LB}	Propagation delay; External Loopback signals	1.1 V Operation	-	2	-	2	ns	4
$t_{QLBSL_2PH_ext}$	1/2 rate Loopback QLBS Output Low Time	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 0	TBD	-	N/A	N/A	t_{CK}	4
$t_{QLBSH_2PH_ext}$	1/2 rate Loopback QLBS Output High Time	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 0	TBD	-	N/A	N/A	t_{CK}	4
$t_{QLBS2Q_2PH_ext}$	1/2 rate Loopback QLBS to QLBD Skew	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 0	-	0.15	N/A	N/A	t_{CK}	4
$t_{QLBDH_2PH_ext}$	1/2 rate Loopback QLBD Output Hold Time	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 0	TBD	-	N/A	N/A	t_{CK}	4
$t_{QLBDVW_2PH_ext}$	1/2 rate Loopback QLBD Valid Window (tQLBDH - tQLBS2Q)	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 0	TBD	-	N/A	N/A	t_{CK}	4
$t_{QLBSL_2PH_int}$	1/2 rate Loopback QLBS Output Low Time	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 0	0.4	-	N/A	N/A	t_{CK}	5
$t_{QLBSH_2PH_int}$	1/2 rate Loopback QLBS Output High Time	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 0	0.4	-	N/A	N/A	t_{CK}	5
$t_{QLBDST_2PH_int}$	1/2 rate Loopback QLBD Output Setup time	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 0	0.35	-	N/A	N/A	t_{CK}	5
$t_{QLBDHD_2PH_int}$	1/2 rate Loopback QLBD Output Hold Time	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 0	0.35	-	N/A	N/A	t_{CK}	5
$t_{QLBDVW_2PH_int}$	1/2 rate Loopback QLBD Valid Window	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 0	0.7	-	N/A	N/A	t_{CK}	5
t_{QLBSL_4PH}	1/4 rate Loopback QLBS Output Low Time	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 1	0.4	-	0.4	-	t_{CK}	6
t_{QLBSH_4PH}	1/4 rate Loopback QLBS Output High Time	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 1	0.4	-	0.4	-	t_{CK}	6
t_{QLBDST_4PH}	1/4 rate Loopback QLBD Output Setup Time	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 1	0.35	-	0.35	-	t_{CK}	6
t_{QLBDHD_4PH}	1/4 rate Loopback QLBD Output Hold Time	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 1	0.35	-	0.35	-	t_{CK}	6
t_{QLBDVW_4PH}	1/4 rate Loopback QLBD Valid Window (tQLBDH - tQLBS2Q)	1.1 V Operation, DDR Mode (RW00[0] = 1), RW26[6] = 1	1.6	-	1.6	-	t_{CK}	6
$t_{Int_LB_Entry}$	Delay from MRW enabling internal loopback mode to valid data driven on QLBS and QLBD	1.1V Operation	-	32	-	32	t_{CK}	
$t_{DCK2QLBS}$	Internal Loopback Propagation Delay from Rising (RW26[7] = 0) or Falling (RW26[7] = 1) DCK crossing point to the Corresponding QLBS Rising Edge	1.1 V Operation, DDR Mode (RW00[0] = 1), Applies to Internal Loopback only (RW26[2:0] = '001' or '010')	-	10	-	16	t_{CK}	
t_{QSK}	Total magnitude of Qn Output to Output Clock tolerance within the same A or B group (in the same unit) $t_{QSK} = t_{QSK_static} + t_{QSK_dynamic}$	Across the supported voltage-temperature range and with SSO conditions and pattern dependence included	0	0.385	0	0.39	t_{CK}	7,8
t_{QSK_static}	Magnitude of Qn Output to Output Clock tolerance within the same A or B group (in the same unit) under quiet conditions	Measured with quiet conditions, only one output switching, and nominal V-T conditions of $V_{DD} = 1.1$ V and 25°C	0	0.3	0	0.3	t_{CK}	8,9
$t_{QSK_dynamic}$	Magnitude of t_{QSK} change (additive to t_{QSK_static}) under noisy conditions	Across the supported voltage-temperature range and with SSO conditions and pattern dependence included	0	0.085	0	0.09	t_{CK}	8,10, 11

Table 200 — Output Timing Requirements¹ (cont'd)

Symbol	Parameter	Conditions	DDR5-3200 to 4800		DDR5-5200 to 5600		Unit	Note
			Min	Max	Min	Max		
t_{Q2Q}	Earliest Qn Output to Latest Qn Output Skew Tolerance, using the DCK differential signal at 0.0 V as the common reference, within the same A or B group (in the same unit)	Measured with quiet conditions, only one output switching, and nominal V-T conditions of $V_{DD} = 1.1$ V and 25°C	0	0.15	0	0.1	t_{CK}	8,11, 12
t_{Qerr}	Total magnitude of t_{QSK} error $t_{Qerr} = 2 \times t_{QSK_dynamic} + t_{Q2Q}$	Across the supported voltage-temperature range and with SSO conditions and pattern dependence included	0	0.25	0	0.25	t_{CK}	8,11
p.pin DNL	Per-Pin deskew deviation from ideal value	Across the supported voltage-temperature range under quiet conditions with single-output switching.	-2	2	-2	2	ps	
p.pin INL	Per-Pin deskew deviation from ideal value	Across the supported voltage-temperature range under quiet conditions with single-output switching.	-8	8	-8	8	ps	
p.group DNL	Per-Group tQSK adjustment step deviation	Across the supported voltage-temperature range under quiet conditions with single-output switching.	-0.75	0.75	-0.75	0.75	LSB	
p.group INL	Per-Group tQSK adjustment deviation with respect to the ideal line based on two endpoints, where the endpoints are the min and max values.	Across the supported voltage-temperature range under quiet conditions with single-output switching.	-1.5	1.5	-1.5	1.5	LSB	
t_{ERR_tot}	total magnitude of tQSK error including training error and all tolerances. $t_{ERR_tot} = 2 \times t_{QSK_dynamic} + \text{per-pin_step_size} + \text{p.pin_DNL} + \text{per-group_step_size} + \text{p.group_DNL} + \text{p.group_INL}$	Across the supported voltage-temperature range and with SSO conditions and pattern dependence included.	-	0.25	-	0.22	t_{CK}	13
t_{ALERT}	ALERT_n propagation delay	DERROR_IN_n LOW to ALERT_n LOW	-	2	-	2	ns	14
t_{ALERT_HL}	ALERT_n assertion delay from DCK_t/DCK_c	DCK_t/DCK_c to ALERT_n LOW	-	1.5	-	1.5	ns	15
t_{RST}	RESET propagation delay	DRST_n LOW to QRST_n LOW	-	20	-	20	ns	
t_{QDIS}	Output buffers (except for QxCK_t/QxCK_c, QxCsn) hi-z after QCS[1:0]_n goes LOW following SRE command.	QCS[1:0]_n = LOW; DRST_n = HIGH; DCK_t/DCK_c = toggling.	2	-	2	-	t_{CK}	
$t_{SRX2QCK}$	Number of cycles from NOP command to release QCS to QCK clocks running and QCS driven HIGH.	DRST_n = HIGH; DCK_t/DCK_c = Toggling	-	16 + t_{PDM}	-	16 + t_{PDM}	t_{CK}	
t_{CASRX}	QCA HIGH to QCS HIGH when exiting Self-Refresh		0	-	0	-	ns	

Table 200 — Output Timing Requirements¹ (cont'd)

Symbol	Parameter	Conditions	DDR5-3200 to 4800		DDR5-5200 to 5600		Unit	Note
			Min	Max	Min	Max		
t _{ODU}	Output Delay Update time	OP7 = 1 for RW12 to RW1E	-	100	-	100	ns	16
NOTE 1	See diagram (Figure 132, “Test Load,” on page 261).							
NOTE 2	See diagram (Figure 7, “One UI DRAM Command Timing Diagram” on page 42). tPDM is defined for 0 nCK command latency adder (RW11[2:0] = ‘000’), default (Moderate) output slew rate control settings in RW0E and RW0F, and output delay features are all enabled in RW12 - RW1A but set to zero delay.							
NOTE 3	tPDM for SDR mode is measured from DCK_t/DCK_c rising edge cross point to output.							
NOTE 4	Applies to External Loopback Only (RW26[2:0] = ‘011’ or ‘100’) only.							
NOTE 5	Applies to Internal Loopback (RW26[2:0] = ‘001’ or ‘010’) only.							
NOTE 6	Applies to both Internal Loopback (RW26[2:0] = ‘001’ or ‘010’) and External Loopback (RW26[2:0] = ‘011’ or ‘100’).							
NOTE 7	This skew represents the absolute Qn skew compared to the output clock QxCK and contains the register pad skew, clock skew, package routing skew and SSO noise (See Figure 10, “Qn Output Skew,” on page 43). The output clock jitter is not included in this skew. The measurement is averaged. The Qn output can be either early or late relative to the output clocks in the channel. For outputs QxCA[13:0]_N, QxCS[1:0]_N_n, this parameter applies to each of the two A/B output groups independently per channel. This parameter also applies to the BCOM[2:0]_N and BCS_N_n outputs w.r.t. to the output clock BCK_N.							
NOTE 8	The values for this parameter are defined for default settings in the output delay control words RW12 to RW1E and default settings in the driver characteristics and slew rate control words RW0A through RW0F.							
NOTE 9	This skew represents the absolute Qn skew compared to the output clock QxCK and contains the register pad skew, clock skew, and package routing skew at nominal V-T conditions of VDD = 1.1 V and 25°C (See Figure 10, “Qn Output Skew,” on page 43). The output clock jitter is not included in this skew. The measurement is averaged. The Qn output can be either early or late relative to the output clocks in the channel. For outputs QxCA[13:0]_N, QxCS[1:0]_N_n, this parameter applies to each of the two A/B output groups independently per channel. This parameter also applies to the BCOM[2:0]_N and BCS_N_n outputs w.r.t. to the output clock BCK_N. In this test, a single output signal switches with clock-like pattern of repeating 6 “1s” and 6 “0s” while all other outputs remaining stable (HIGH).							
NOTE 10	tQSK_dynamic represents the largest change in output skew (relative to tQSK_static) within an output signal group in either direction (i.e., Early signals moving to an earlier time or Late signals moving to a later time). The output clock jitter is not included in this skew. The measurement is averaged. In this test, all outputs send patterns simultaneously to excite realistic on-die noise and crosstalk.							
NOTE 11	tQerr imposes an additional constraint on the maximum value for the total of 2 x tQSK_dynamic + tQ2Q measured per individual unit. The device shall meet tQerr(max) in addition to meeting tQSK_dynamic(max) and tQ2Q(max).							
NOTE 12	This parameter represents the absolute Qn skew between earliest and latest signals within the same A/B group and contains the register pad skew and package routing skew (See Figure 10, “Qn Output Skew,” on page 43). The measurement is averaged. For outputs QxCA[13:0]_N, this parameter applies to each of the two A/B output groups independently per channel. Similarly, for outputs QxCS[1:0]_N_n, this parameter applies to each of the two A/B output groups independently per channel. This parameter also applies to the {BCOM[2:0]_N, BCS_N_n} output groups independently per channel. In this test, a single output signal switches with clock-like pattern of repeating 6 “1s” and 6 “0s” while all other outputs remaining stable (HIGH).							
NOTE 13	tERR_tot imposes an additional constraint on the maximum value for the total of (2 x tQSK_dynamic + per-pin_step_size + p.pin_DNL + per-group_step_size + p.group_DNL + p.group_INL) measured per individual unit. The device shall meet tERR_tot(max) in addition to meeting tQSK_dynamic(max), tQ2Q(max), p.pin_DNL(max), p.pin_INL(max), p.group_DNL(max) and p.group_INL(max).							
NOTE 14	See Figure 129, “Voltage Waveforms, tALERT Measurement,” on page 260.							
NOTE 15	See Figure 132, “Test Load,” on page 261 and Figure 130, “Voltage Waveforms, tALERT_HL Measurement,” on page 260							
NOTE 16	This is the time the RCD requires to update delay timing setting to the output after it receives RW write.							

11 VrefCA and Vref CS Specifications

The internal VrefCA and VrefCS specifications parameters are Vref operating range, Vref step size, Vref set tolerance, Vref step time, and Vref valid tolerance.

The Vref operating range specifies the minimum required Vref setting range for DDR5RCD02 devices and is specific by a Vref min operating point and a Vref max operating point, as depicted in Figure 71.

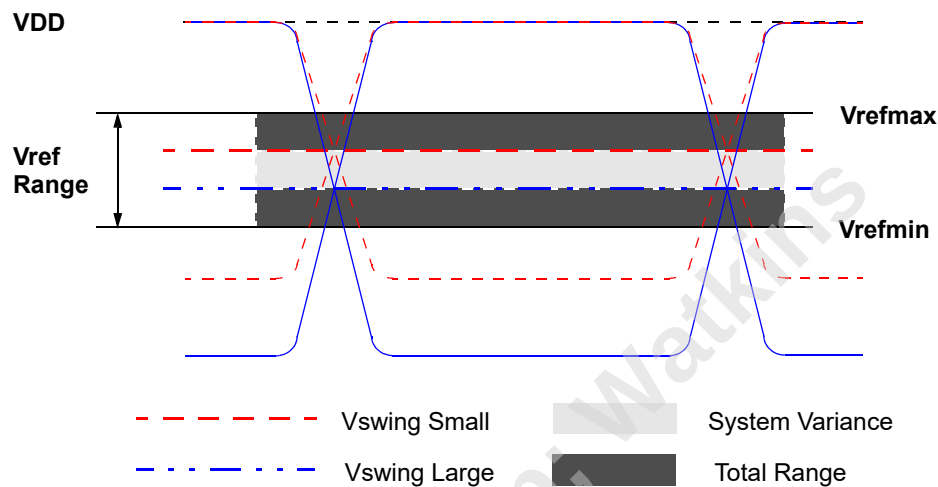


Figure 71 — Vref Operating Range (Vrefmin, Vrefmax)

The Vref step size is defined as the target voltage between adjacent steps. For a given design, the DDR5RCD02 VrefCA and VrefCS step size must be within the range specified.

The Vref set tolerance is the variation in the Vref voltage from the ideal setting. This accounts for accumulated error over multiple steps. There are two parameters for Vref set tolerance uncertainty for different numbers of steps n . The Vref set tolerance is measured with respect to the ideal line which is based on two endpoints, where the endpoints are at the min and max Vref values for a specified range. An illustration depicting an example of the step size and Vref set tolerance is shown in Figure 72.

11 VrefCA and Vref CS Specifications (cont'd)

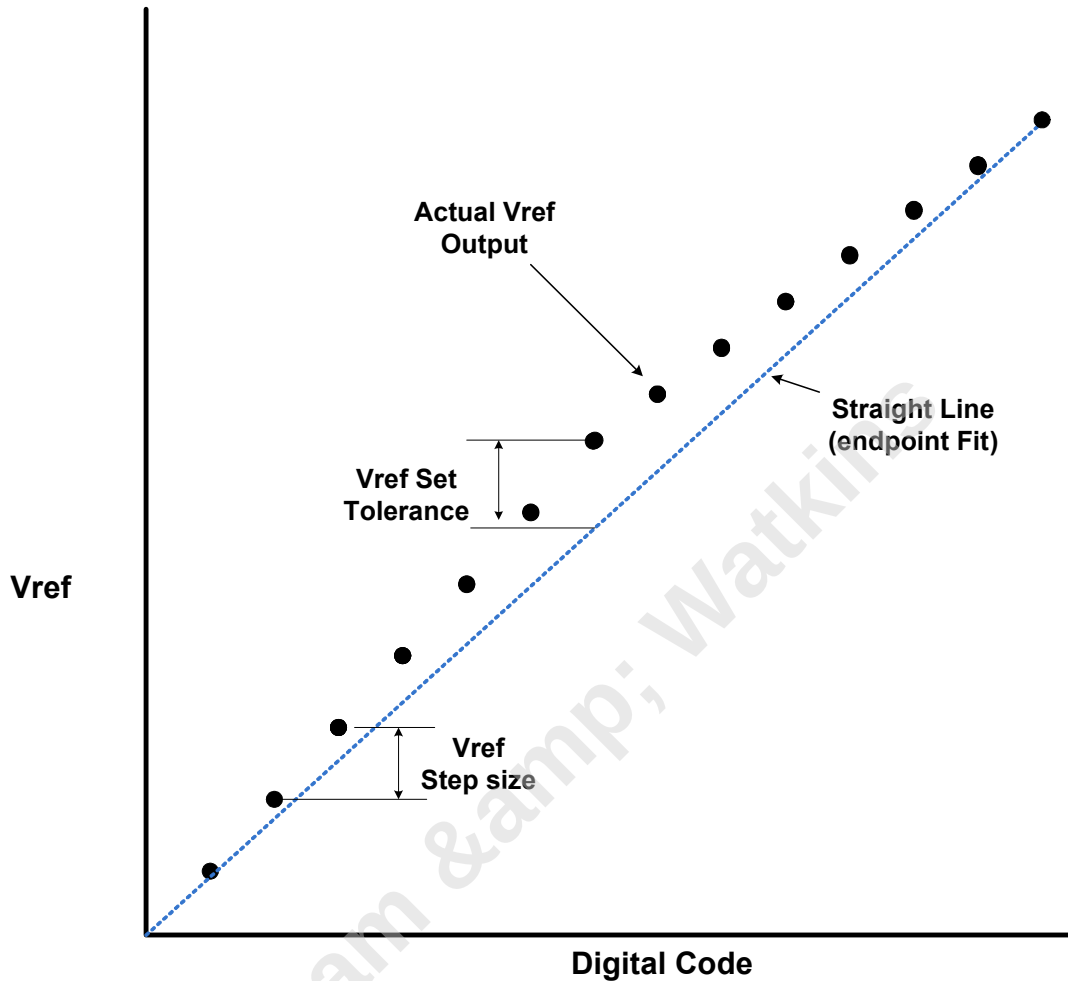


Figure 72 — Example of Vref Set Tolerance (Only Max Case is Shown) and Step Size

The Vref increment/decrement step times are defined by Vref_time. Vref_time is defined from t0 to t1 as shown in Figure 73 where t0 is referenced to when the RW write occurs and t1 is referenced to when the Vref voltage is at the final DC level within the Vref valid tolerance (Vref_val_tol).

The Vref valid level is defined by Vref_val_tol to qualify the step time t1 as shown in Figure 74. This parameter is used to insure an adequate RC time constant behavior of the voltage level change after any Vref increment/decrement adjustment. This parameter is only applicable for component level validation/characterization.

Vref_time is the time including up to Vrefmin to Vrefmax or Vrefmax to Vrefmin change in Vref voltage.

t0 - is referenced to RW write command clock

t1 - is referenced to the Vref_val_tol

11 VrefCA and Vref CS Specifications (cont'd)

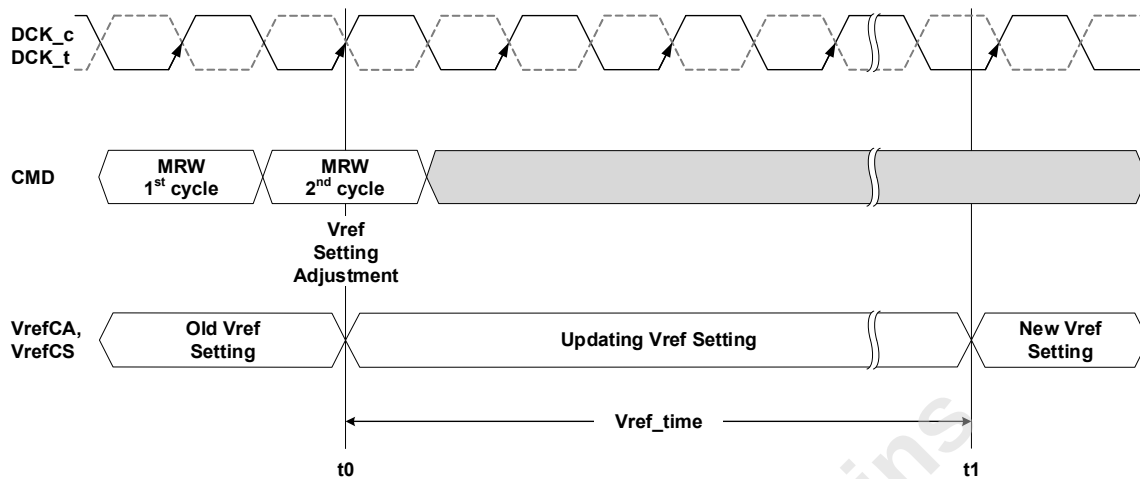


Figure 73 — Vref_time Timing Diagram

An MRW write to the (Internal Vref Control Word) is used to program the Vref value.

The minimum time required between two Vref MRW commands is Vref_time for single step and Vref_time for a full voltage range step.

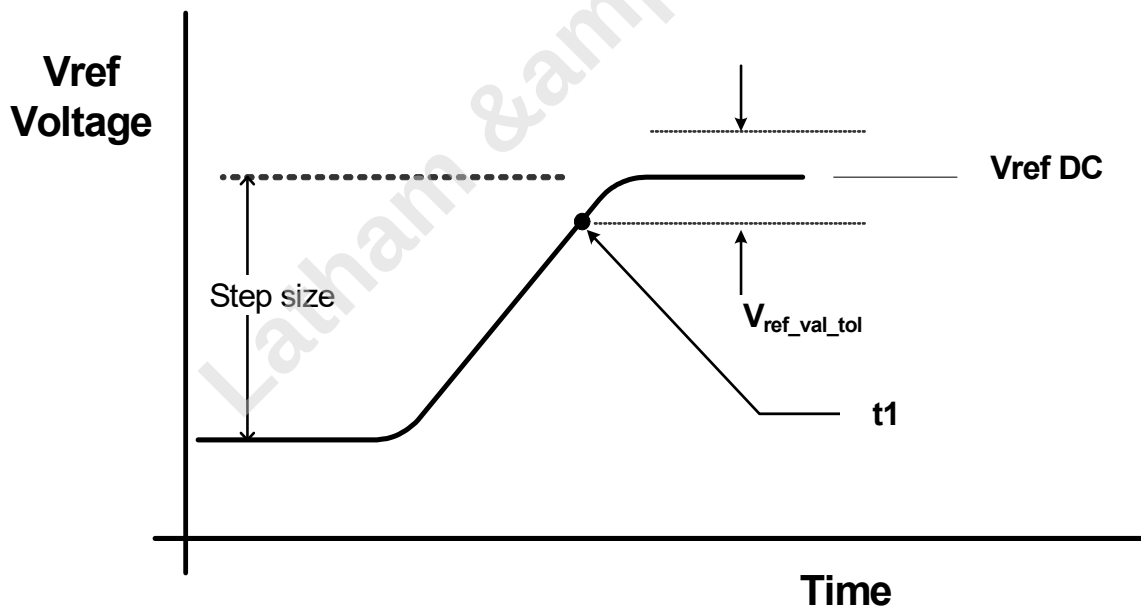


Figure 74 — Vref Step Single Step Size Increment Case

11 VrefCA and Vref CS Specifications (cont'd)

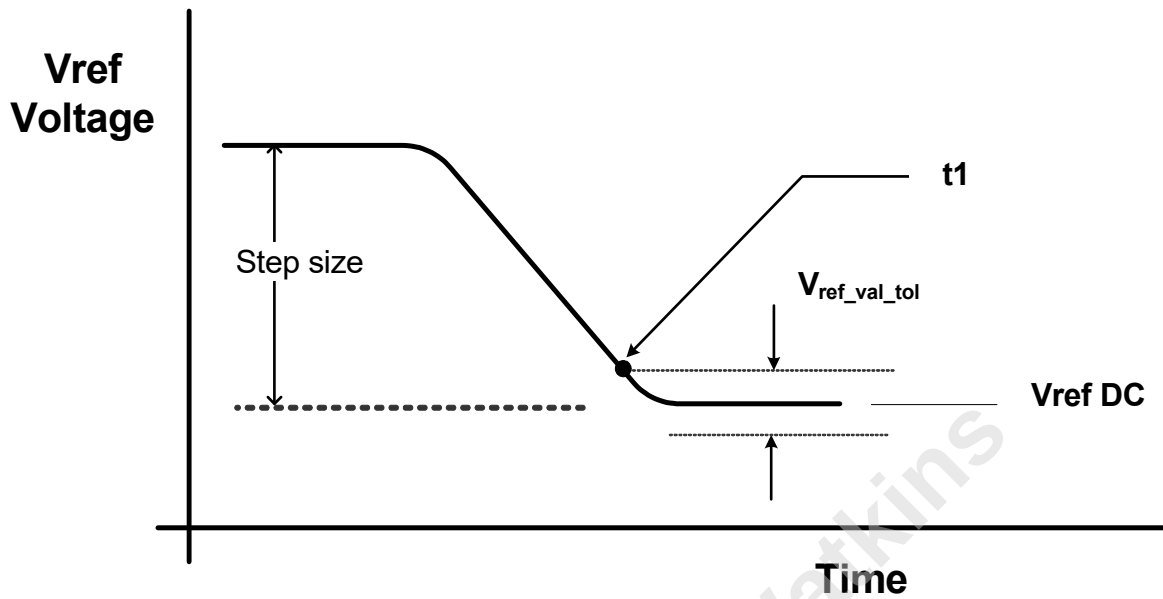


Figure 75 — Vref Step Single Step Size Decrement Case

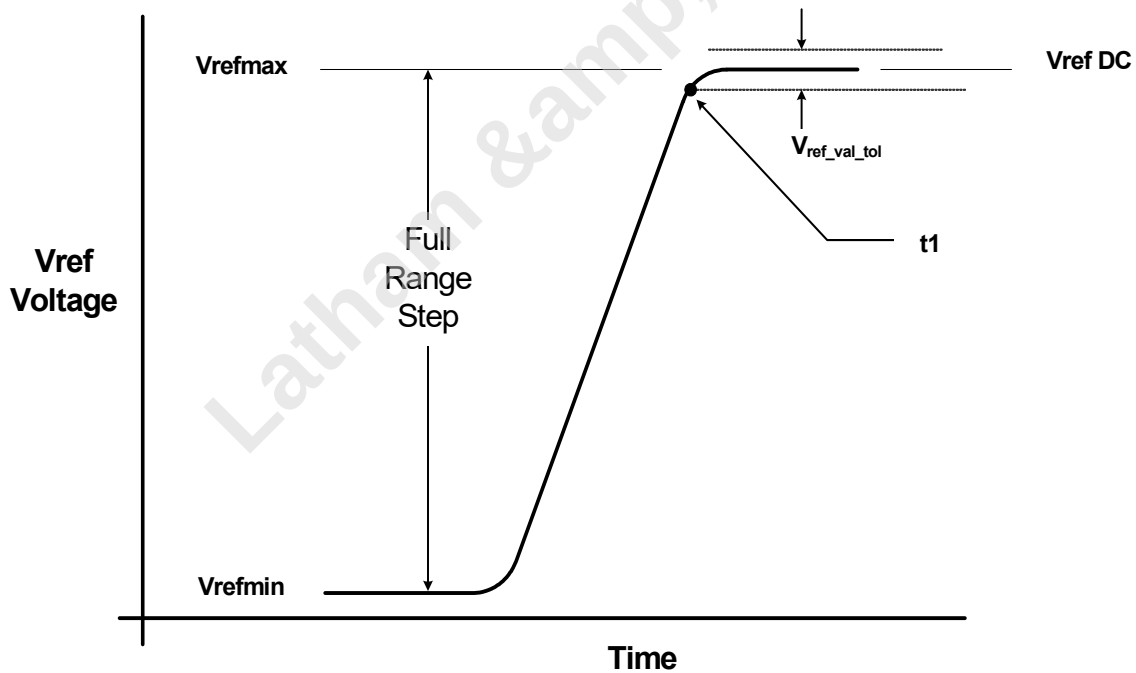


Figure 76 — Vref Full Step from Vrefmin to Vrefmax Case

11 VrefCA and Vref CS Specifications (cont'd)

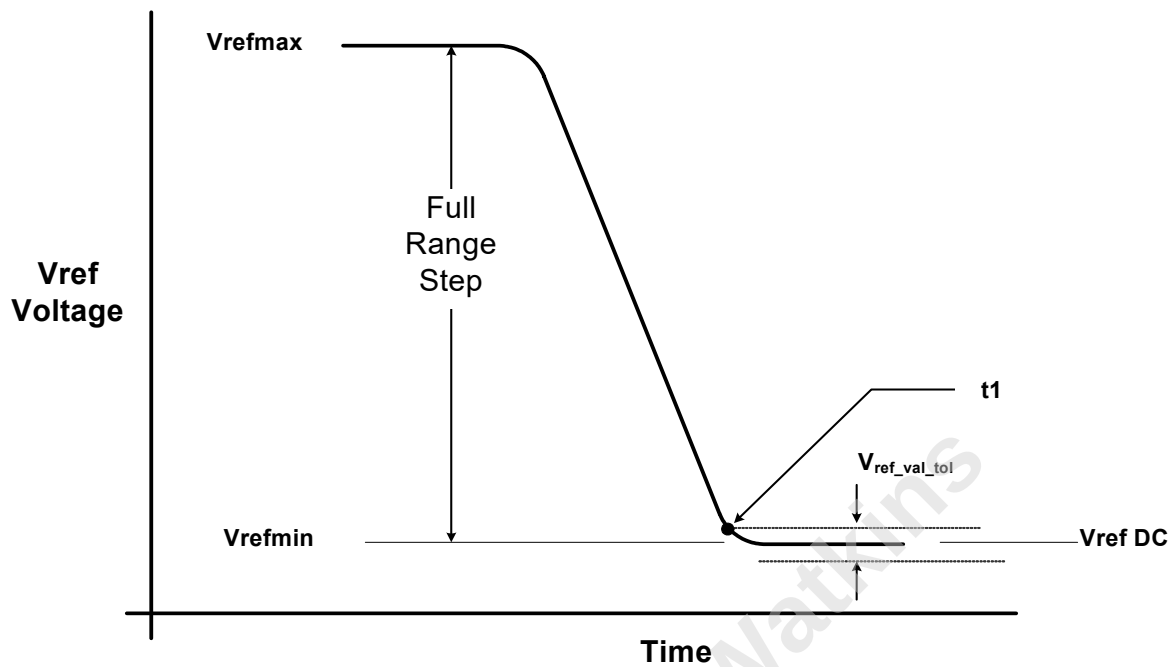


Figure 77 — Vref Full Step from Vrefmax to Vrefmin Case

Table 201, “Internal VrefCA and VrefCS Specifications” below contains the internal VrefCA specifications that will be characterized at the component level for compliance. The characterization method is defined in a separate specification.

Table 201 — Internal VrefCA and VrefCS Specifications

Parameter	Symbol	Min	Typ	Max	Unit	NOTE
Vref Max Operating Point	Vref_max	97.5%	-	-	V _{DD}	1
Vref Min Operating Point	Vref_min	-	-	45%	V _{DD}	1
Vref Step size	Vref_step	0.41%	0.50%	0.59%	V _{DD}	2
Vref Set Tolerance	Vref_set_tol_long	-1.625%	0.0%	1.625%	V _{DD}	3, 4, 6
	Vref_set_tol_short	- 0.15%	0.0%	0.15%	V _{DD}	3, 5, 7
Vref Step Time	Vref_time_short	-	-	300	ns	8, 9
	Vref_time_long	-	-	500	ns	8, 9
Vref Valid Tolerance	Vref_val_tol	- 0.15%	0.0%	0.15%	V _{DD}	

NOTE 1 Vref DC voltage referenced to VDD.

NOTE 2 Vref step size increment/decrement range. Vref at DC level. $Vref_step(avg) = (Vref_max - Vref_min) / 105$.

NOTE 3 $Vref_new = Vref_old + n * Vref_step$; n= number of steps; if increment use “+”; If decrement use “-”.

NOTE 4 The minimum value of Vref setting tolerance = $Vref_new - 1.625\% * VDD$. The maximum value of Vref setting tolerance = $Vref_new + 1.625\% * VDD$. For $n > 4$.

NOTE 5 The minimum value of Vref setting tolerance = $Vref_new - 0.15\% * VDD$. The maximum value of Vref setting tolerance = $Vref_new + 0.15\% * VDD$. For $n < 4$.

NOTE 6 Measured by recording the min and max values of the Vref output over the full range, drawing a straight line between those points and comparing all other Vref output settings to that line

NOTE 7 Measured by recording the min and max values of the Vref output across 4 consecutive steps ($n=4$), drawing a straight line between those points and comparing all other Vref output settings to that line

NOTE 8 Time from MRW command to Vref stable. Applicable for component level test/characterization purposes. Vref valid is to qualify the step times which will be characterized at the component level.

NOTE 9 The maximum value of VrefCA or VrefCS step time = 300 ns for $n < 16$ and 500 ns for $n \geq 16$, where n = number of steps.

11.1 DFE_Vref INL Tolerance

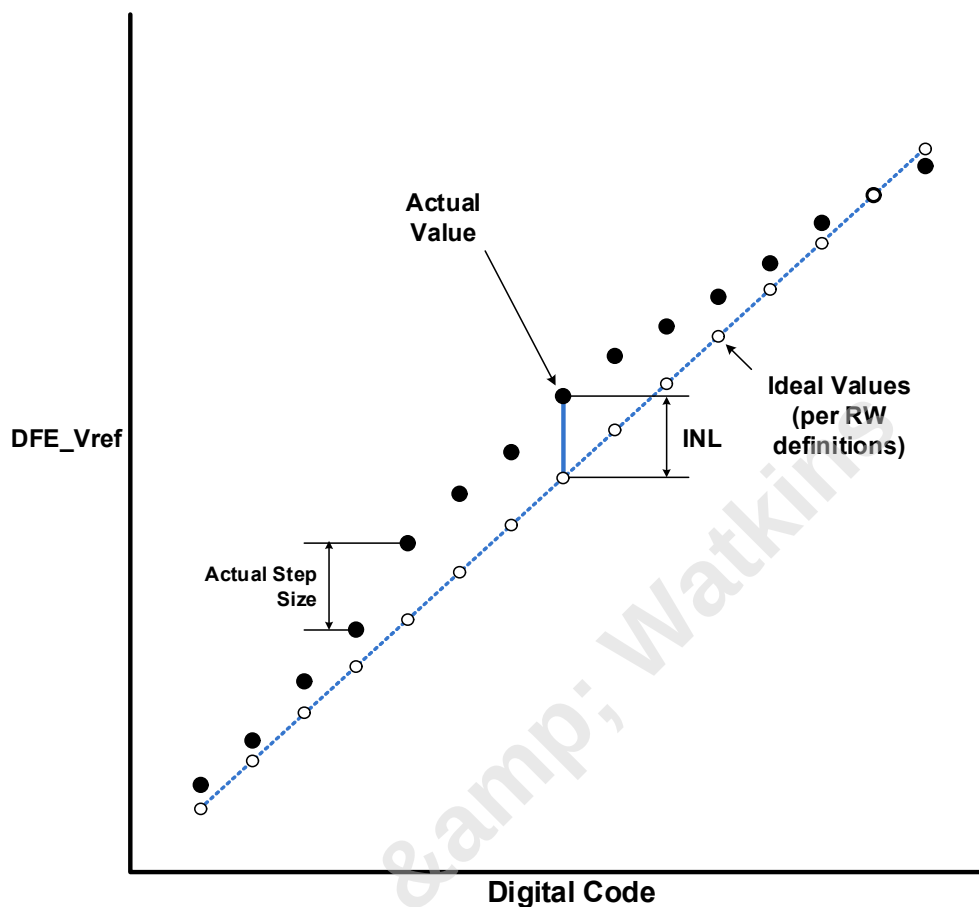


Figure 78 — Example of DFE_Vref INL Tolerance

Integral nonlinearity (INL) is the deviation between effective analog values and expected ideal values for the corresponding settings in Receiver DFE_Vref Control words as defined in [RW3F](#) and [PG02RW](#)[62, 66, 6A,6E, 72, 76,7A, 7E].

11.1 DFE_Vref INL Tolerance (cont'd)

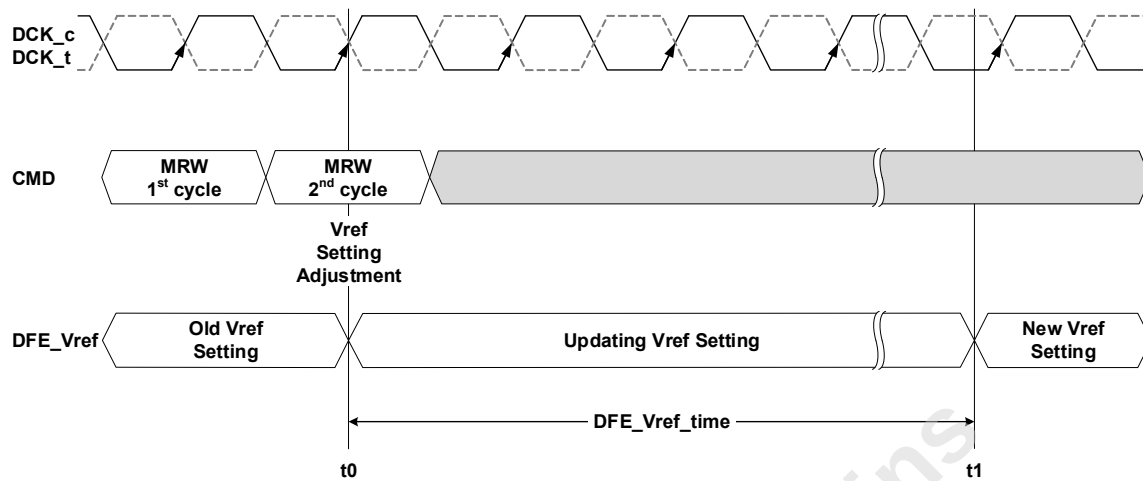


Figure 79 — DFE_Vref_step Time Timing Diagram

An MRW write to the (Internal Vref Control Word) is used to program the Vref value.

The minimum time required between two DFE_Vref MRW commands is DFE_Vref Step Time.

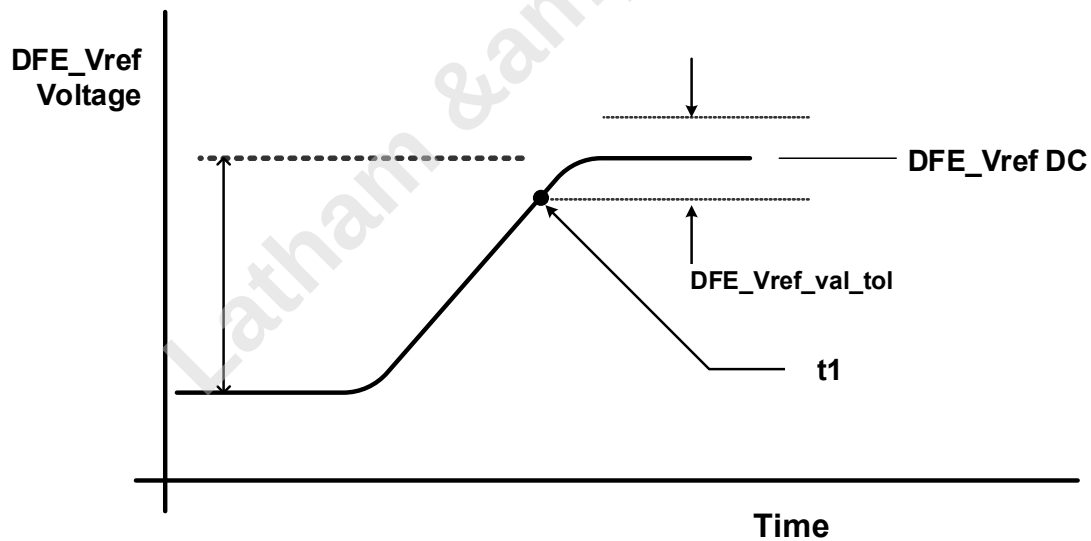


Figure 80 — DFE_Vref Increment Case

11.1 DFE_Vref INL Tolerance (cont'd)

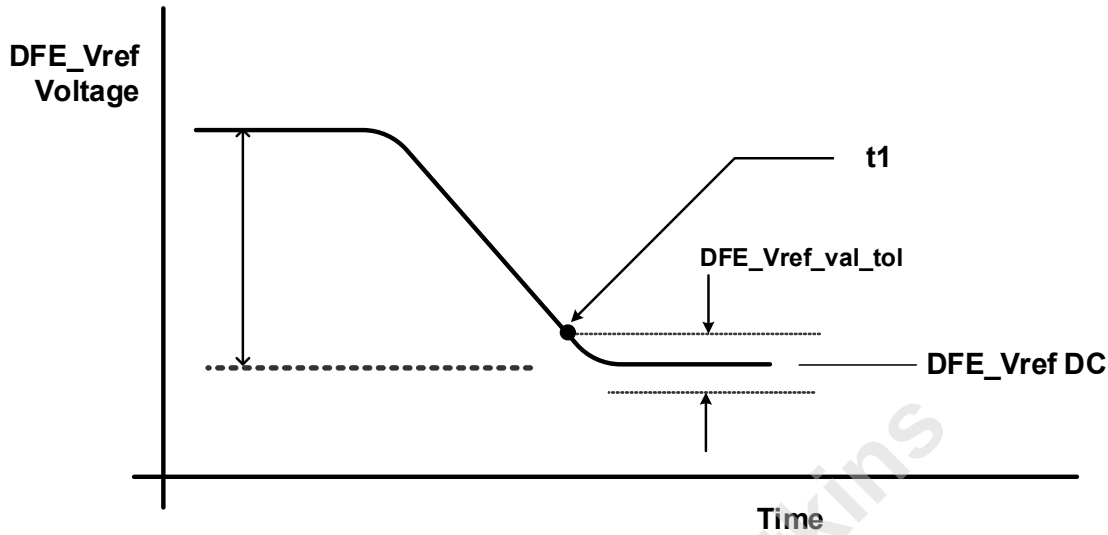


Figure 81 — DFE_Vref Decrement Case

Table 202 contains the DFE Training Vref specifications that will be characterized at the component level for compliance. The characterization method is defined in a separate specification.

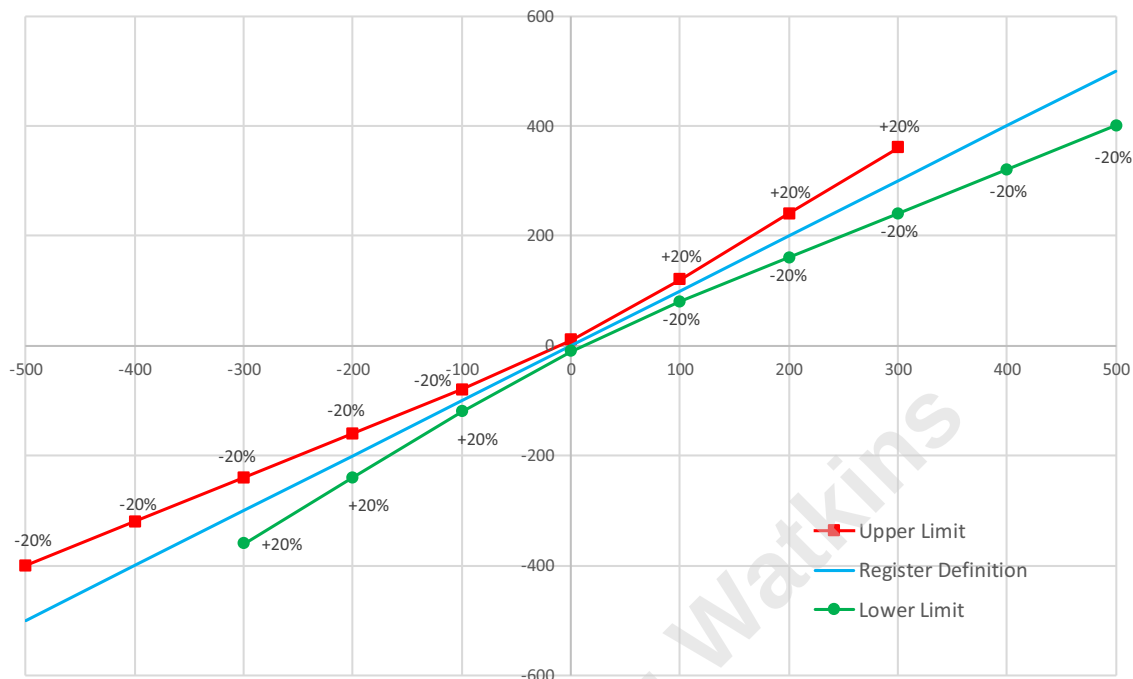
Table 202 — DFE_Vref Specification

Parameter	Symbol	Min	Typ	Max	Unit	NOTE
DFE_Vref Step Time Short	DFE_Vref_Short_16	-	-	200	ns	1, 2
	DFE_Vref_Short_32	-	-	300	ns	1, 3
DFE_Vref Step Time Long	DFE_Vref_Long	-	-	600	ns	1, 4
DFE_Vref Valid Tolerance	DFE_Vref_val_tol	-1.65	0	1.65	mV	1
NOTE 1 Only applicable for component level test/characterization purpose. Not applicable for normal mode of operation. Vref valid is used to qualify the step times which will be characterized at the component level.						
NOTE 2 The maximum value of DFE_Vref step time = 200 ns for $n < 16$, where n = number of steps.						
NOTE 3 The maximum value of DFE_Vref step time = 300 ns for $16 \leq n < 32$, where n = number of steps.						
NOTE 4 The maximum value of DFE_Vref step time = 600 ns for $n \geq 32$, where n = number of steps.						

Table 203 — DFE_Vref INL Tolerance

DFE_Vref Setting	Lower Limit	Upper Limit	NOTE
- 500 mV ~ 300 mV	-	$V_{ideal} * 80\%$	1,3,4
- 300 mV ~ 0 mV	$\text{Min}(V_{ideal} * 120\%, V_{ideal} - 4 * \text{LSB})$	$\text{Max}(V_{ideal} * 80\%, V_{ideal} + 4 * \text{LSB})$	1,2,3,4
0 mV ~ + 300 mV	$\text{Min}(V_{ideal} * 80\%, V_{ideal} - 4 * \text{LSB})$	$\text{Max}(V_{ideal} * 120\%, V_{ideal} + 4 * \text{LSB})$	1,2,3,4
+300 mV ~ + 500 mV	$V_{ideal} * 80\%$	-	1,3,4
NOTE 1 V_{ideal} refers to the ideal DFE_Vref value based on the setting.			
NOTE 2 LSB is 2.5 mV.			
NOTE 3 DFE_Vref must be monotonic.			
NOTE 4 The range values specified in the table are applicable for default VrefCA and DFE Gain Offset setting, under $V_{DD} = 1.1 \text{ V}$ and 25°C ambient temperature.			

11.1 DFE_Vref INL Tolerance (cont'd)



X axis = ideal control word setting value

Y axis = actual implementation value

Figure 82 — Illustration Example of DFE_Vref INL Tolerance

11.2 Measurement Steps

1. Make sure VrefCA are at default setting, DFE_Vref setting is 0, power supply is 1.1 V. Configure Margin Monitor RW32[2:0] to 3'b001.
2. Sweep input signal from 1.1 V to 0 V to find the lowest level for HIGH and record this input signal HIGH level as V1 when HIGH-to-LOW transition starts.
3. Sweep input signal from 0 V to 1.1 V to find the highest level for LOW and record this input signal LOW level as V2 when LOW-to-HIGH transition starts.
4. Transition level $V = (V1 + V2)/2$.
5. Set DFE_Vref, wait for DFE_Vref_Step_Time to let the new setting settle.
6. Repeat steps 2 ~ 4, to get a new transition level V'.
7. Validate $(V' - V)$ is within specification.
8. Repeat steps 2 ~ 7 until all the settings of DFE_Vref are measured.

11.3 DFE Tap Coefficient and Gain Adjustment Tolerances (DNL and INL)

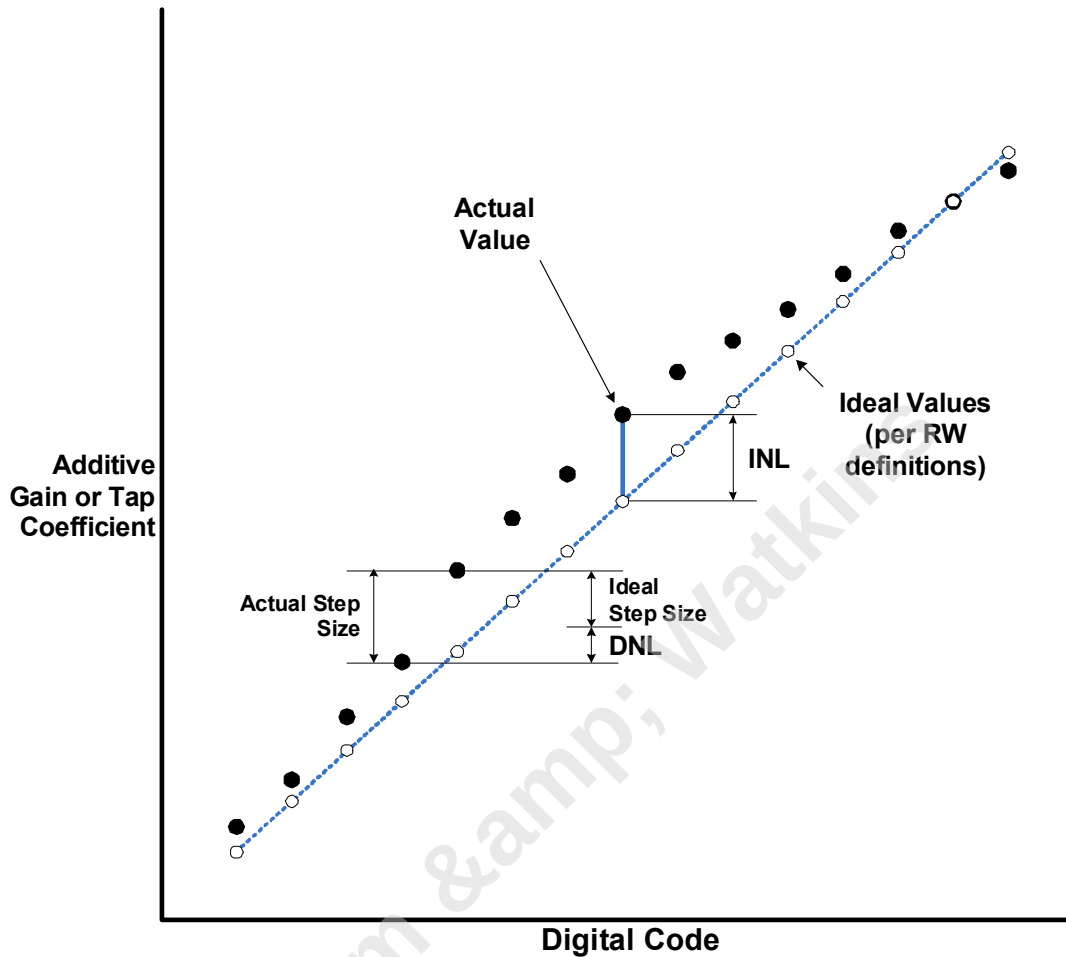


Figure 83 — Example of Tap Coefficient and Gain Adjustment Tolerances (DNL and INL)

Differential nonlinearity (DNL) is the deviation between effective and ideal control step sizes for any pair of analog values resulting from adjacent settings in Receiver DFE Tap Coefficient and Gain Adjustment control words as defined in [PG\[1:0\]RW\[7F:60\]](#).

Integral nonlinearity (INL) is the deviation between effective analog values and expected ideal values for the corresponding settings in Receiver DFE Tap Coefficient and Gain Adjustment control words as defined in [PG\[1:0\]RW\[7F:60\]](#).

12 AC and DC Global Definitions

12.1 Transmitter (Tx), Receiver (Rx), and Channel Definitions

Transmitter (Tx): Input to the transmitter is the DCA from the logical portion of the RCD Core and output is at the package pins. The normal components contained in the transmitter are, but not limited to, the pre-driver and the output driver with the transmitter package.

The DDR5 RCD uses a differential re driven clock-source and single-ended CA.

The transmitter for a forwarded clock based source synchronous system must include specification and test methodology for low and high frequency random and deterministic jitter and duty cycle error and must take into account Bounded Uncorrelated Jitter (BUJ), Duty cycle error, etc.

For single ended signaling cases, crosstalk and Simultaneous switching noise often impact the measurement of the Transmitter. Therefore, the Transmitter's decimated jitter component parameters and the BER must be specified.

Receiver (Rx): The normal components contained in the clock receiver are: (1) receiver package, (2) input amplifier (to receive the clock), and (3) optionally, a PLL and time adjustment circuitry. The data input receives the single ended data and measures it against a reference voltage. The difference between the received signal and reference voltage is then sampled. However, the link may need some form of receiver equalization at the speeds that the DDR5 link needs to operate. Since there may be no input eye margin at the receiver pin, the receiver equalization specification may be defined based on a virtual receive sampler/slicer. This means that components of the receiver, which have nothing to do with inter symbol interference (ISI), will have to be specified in such a way as to avoid the impact of ISI. Hence items like receiver jitter, receiver jitter sensitivity, receiver amplitude sensitivity, which are typically orthogonal to ISI, will have to be specified in an ISI free environment. Either the infrastructure, or the test methodology will ensure that the correct amount of ISI, or no ISI, as the case might be, for that specific Rx parameter is used. The validation of the receiver is also dependent on either loop-back of data being enabled or restricting to receiver testing to be dependent on a pre-specified set of patterns and the accompanying pattern checker and error counters. The Receiver equalization will be evaluated in the presence of a set of pre-specified receiver testing golden channels. These receiver testing golden channels will attempt to span the ISI range to which the receiver needs to be designed. The key measurement parameter in this test is the stressed eye and the receiver's error sensitivity to the stressed eye.

Channel: Channel is defined from the transmitter package pin to the receiver package pin and includes all the interconnect topology components included in the channel definition (connectors, sockets, and so on). However, this spec will not specify system platform level channel models, but restrict itself to specifying the Golden Reference Transmitter and Receiver evaluation channels.

Applicability:

The parameters defined in this section apply to the appropriate sections of the RCD irrespective of the system, platform or the topology of the system where the RCD is situated. The parameters, however, are defined separately for each of the speeds that the DDR5RCD02 is expected to operate at.

12.1.1 Bit Error Rate

Introduction

This section provides an overview of the Bit Error Rate (BER) and the desired Statistical Level of Confidence.

12.1.2 General Equation

$$n = \left(\frac{1}{BER} \right) \left[-\ln(1 - SLC) + \ln \left(\sum_{k=0}^N \frac{(n \cdot BER)^k}{k!} \right) \right]$$

Where:

n = number of bits in a trial

SLC = statistical level of confidence

BER = Bit Error Rate

k = intermediate number of specific errors found in trial

N = number of errors recorded during trial

If no errors are assumed in a given test period, the second term drops out and the equation becomes:

$$n = \left(\frac{1}{BER} \right) [-\ln(1 - SLC)]$$

Testing to 99.5% confidence levels is recommended. However, one may choose a number that is viable for their own manufacturing levels. To determine how many bits of data should be sent (again, assuming zero errors, or $N = 0$), using $BER = 10^{-9}$ and confidence level $SLC = 99.5\%$, the result is $n = (1/BER)(-\ln(1 - 0.995)) = 5.298 \times 10^9$.

Results for commonly used confidence levels of 99.5% down to 70% are shown in Table 204.

Table 204 — Estimated Number of Transmitted Bits (n) for Confidence Level of 70% to 99.5%

Number of Error	$n = \ln(1 - SLC)/BER$							
	99.5%	99%	95%	90%	85%	80%	75%	70%
0	5.298/BER	4.61/BER	2.99/BER	2.3/BER	1.90/BER	1.61/BER	1.39/BER	1.20/BER

12.1.3 Minimum Bit Error Rate (BER) Requirements

Table 205, “Minimum BER Requirements for RX and TX Timing and Voltage Tests” specifies the UI_{AVG} and Bit Error Rate requirements over which certain receiver timing and voltage specifications need to be validated assuming a 99.5% confidence level at $BER = 10^{-9}$.

Table 205 — Minimum BER Requirements for RX and TX Timing and Voltage Tests

Symbol	Parameter	DDR5-3200 - 5600			Unit	NOTE
		Min	Nom	Max		
UI_{AVG}	Average UI	0.999* nominal	1.000/f	1.001* nominal	ps	1
$N_{MIN_UI_Validation}$	Number of UI (min)	5.3×10^9		-	UI	2
BER_{Lane}	Bit Error Rate	-		10^{-16}	Events	3, 4, 5, 6
NOTE 1 Average UI size, “f” is data rate. NOTE 2 # of UI over which certain Rx/Tx timing and voltage specifications need to be validated assuming a 99.5% confidence level at $BER = 10^{-9}$. NOTE 3 This is a system parameter. It is the raw bit error rate for every lane before any logical PHY or link layer based correction. It may not be possible to have a validation methodology for this parameter for a standalone transmitter or standalone receiver, therefore, this parameter has to be validated in selected systems using a suitable methodology as deemed by the platform. NOTE 4 Bit Error Rate per lane. This is a raw bit error rate before any correction. This parameter is primarily used to determine electrical margins during electrical analysis and measurements that are located between two interconnected devices. NOTE 5 This is the minimum BER requirements for testing timing and voltage parameters listed in Input Clock Jitter, Rx CA Voltage Sensitivity, Rx Clock Jitter Sensitivity, Rx CA Stressed Eye, Tx Clock Jitter, Tx QCA Jitter, and Tx QCA Stressed EH/EW specifications. NOTE 6 The BER for DDR5 during normal operation is 10^{-16} . For validation purposes, the BER used is 10^{-9} .						

12.1.4 Unit Interval and Jitter Definitions

This clause describes the UI and NUI Jitter definitions associated with the Jitter parameters specified in Rx Stressed Eye, Tx QCK Jitter, and Input Clock Jitter specifications.

12.1.5 Unit Interval

The times at which the differential crossing points of the clock occur are defined at $t_1, t_2, \dots, t_{n-1}, t_n, \dots, t_K$.

The UI at index “n” is defined as shown in Figure 84 (with $n = 1, 2, \dots$) from an arbitrary time in steady state, where $n = 0$ is chosen as the starting crossing point.

Mathematical definition of UI is shown in Figure 84 and Figure 85.

$$UI_n = t_n - t_{n-1}$$

Figure 84 — UI Definition in Terms of Adjacent Edge Timings

For the Single-Ended data, the unit interval time starts when the signal crosses a pre-specified reference voltage. For the differential clock, the unit interval time starts when the DCK_t and DCK_c intersect (see Figure 85).

12.1.5 Unit Interval (cont'd)

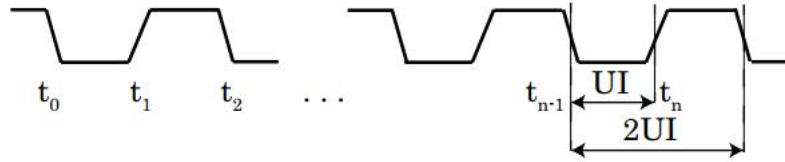


Figure 85 — UI Definition Using Clock Waveforms

12.1.6 UI Jitter Definition

If a number of UI edges are computed or measured at times $t_1, t_2, \dots, t_{n-1}, t_n, \dots, t_K$, where K is the maximum number of samples, then the UI jitter at any instance “ n ” is defined in Figure 86, where T = the ideal UI size.

$$UI_{(jit)n} = (t_n - t_{n-1}) - T; n = 1, 2, 3, \dots, K$$

Figure 86 — UI Jitter of the “ n th” UI Definition (in terms of ideal UI)

In a large sample with random Gaussian-like jitter (therefore very close to symmetric distribution), the average of all UI sizes usually turns out to be very close to the ideal UI size.

The equation described in Figure 86 assumes starting from an instant of steady state, where $n = 0$ is chosen as the starting point. 1 UI = one bit, which means 2 UI = one full cycle or time period of the forwarded clock.

Example: For 6.4 GT/s signaling, the forwarded clock frequency is 3.2 GHz, or 1 UI = 156.25 ps.

Deterministic jitter is analyzed in terms of the peak-to-peak value and in terms of specific frequency components present in the jitter, isolating the causes for each frequency. Random jitter is unbounded and analyzed in terms of statistical distribution to convert to a bit error rate (*BER*) for the link.

12.1.7 UI-UI Jitter Definition

UI-UI (read as “UI to UI”) jitter is defined to be the jitter between two consecutive UI as shown in Figure 87.

$$UI_n = UI_n - UI_{n-1}; n = 2, 3, \dots, K$$

Figure 87 — UI - UI Jitter Definitions

12.1.8 Accumulated Jitter (Over “N” UI)

Accumulated jitter is defined as the jitter accumulated over any consecutive “N” UI as shown in Figure 88.

$$T_{\text{acc}}^N = \sum_{p=m}^{m+N-1} (UI_p - \overline{UI}), \text{ where } m = 1, 2, \dots, K-N$$

Figure 88 — Definition of Accumulated Jitter (over “N” UI)

where UI is defined in the equation shown in Figure 89.

$$\overline{UI} = \frac{\sum_{p=1}^K UI_p}{K}, \text{ where } p = 1, 2, \dots, N, \dots, K$$

Figure 89 — Definition of UI

13 Electrical - Input AC and DC Specifications

13.1 Input Clock Architecture

The RCD is provided a clock by the entity that is controlling it, i.e., memory controller, or any test equipment. This clock is used by the RCD to generate all the IO specific timing, and is the same clock used by the controlling entity as the one and only deterministic source of all the response timings to and from the RCD. This allows the memory controller to have a deterministic control of every event in the RCD. Spread Spectrum Clocking (SSC) Capability

13.1.1 Input Clock SSC and PLL Loop Filter

The system platform uses a reference clock, which is used to synthesize the RCD clock. Spread Spectrum Clock (SSC) with up to 0.5% down-spread in frequency must be supported by the clocking system. The frequency of the reference clock, and therefore bit rate, can be modulated from 0% to -0.5% of the nominal data rate/frequency at a modulation rate in the range of 30 KHz to 33 KHz. The modulation profile of SSC must provide optimal or close to optimal EMI reduction. Typical profiles include a triangular profile. The RCD must ensure that it functions normally even in the presence of SSC and truthfully lets SSC related components pass through to its output signals.

Table 206 — SSC and PLL Loop Filter Characteristics

Symbol	Parameter	Conditions	DDR5-3200 to 5600		Unit
			Min	Max	
f_{SSC}	SSC modulation frequency ¹		30	33	KHz
a_{SSC}	SSC amplitude ¹		0	-0.5	%
f_{band}	PLL loop bandwidth ²	-3 dB bandwidth ³	$0.01 \times f_{clock}$	-	MHz
NOTE 1 The DDR5RCD02 must meet all parameters defined in this specification while supporting input clock SSC requirements described in this table					
NOTE 2 The DDR5RCD02 PLL must fulfill this loop filter requirement in order to track typical system clock synthesizer output clock signals.					
NOTE 3 Implies a jitter peaking of <3 dB					

13.1.2 Input Slew Rate for Differential Clock

Input slew rate for differential signals DCK_t/DCK_c are defined and measured as in Table 207 and Table 208.

Table 207 — Differential Input Slew Rate Definition for DCK_t/DCK_c

Symbol	Parameter	Measured		Units	Notes
		Min	Max		
V_{IHdiff_CK}	Differential Input High	$0.75 \times V_{DIFFpp}$	-	mV	1, 2, 3, 4
V_{ILdiff_CK}	Differential input Low	-	$0.25 \times V_{DIFFpp}$	mV	1, 2, 3, 4
NOTE 1 Clock V_{DIFFpp} , V_{ILdiff_CK} and V_{IHdiff_CK} are defined in Figure 90.					
NOTE 2 V_{DIFFpp} is the mean HIGH voltage minus the mean LOW voltage over 1e6 samples.					
NOTE 3 Differential signal rising edge from V_{ILdiff_CK} to V_{IHdiff_CK} must be monotonic slope.					
NOTE 4 Differential signal falling edge from V_{IHdiff_CK} to V_{ILdiff_CK} must be monotonic slope.					

13.1.2 Input Slew Rate for Differential Clock (cont'd)

Table 208 — Differential Input Slew Rate DCK_t/DCK_c

Symbol	Parameter	Measured		Units	Notes
		Min	Max		
SRIdiff	Differential Input Slew Rate	2	18	V/ns	
NOTE 1 All parameters are defined over the entire clock common mode range.					

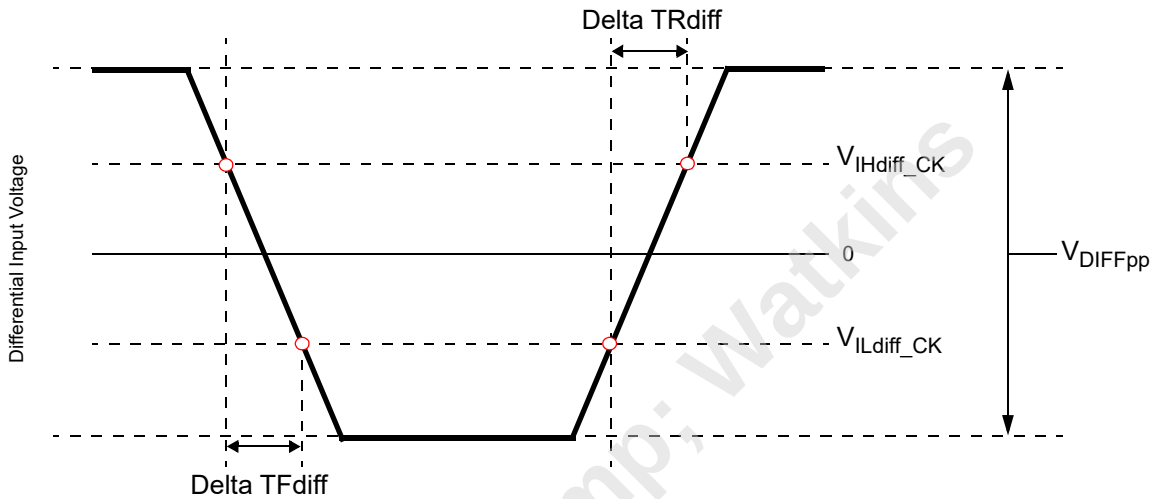


Figure 90 — Differential Input Slew Rate Definition for DCK_t/DCK_c

13.1.3 Differential Input Clock Cross Point Voltage

To achieve tight CA Rx Mask input requirements as well as output skew parameters with respect to clock, the cross point voltage of differential input clock signals (DCK_t, DCK_c) must meet the requirements in Figure 91, “VIX Definition (DCK)”. The differential input cross point voltage VIX_CK (VIX_CK_FR and VIX_CK_RF) is measured from the actual cross point of DCK_t, DCK_c relative to the Vswing/2 of the DCK_t and DCK_c signals.

13.1.3 Differential Input Clock Cross Point Voltage (cont'd)

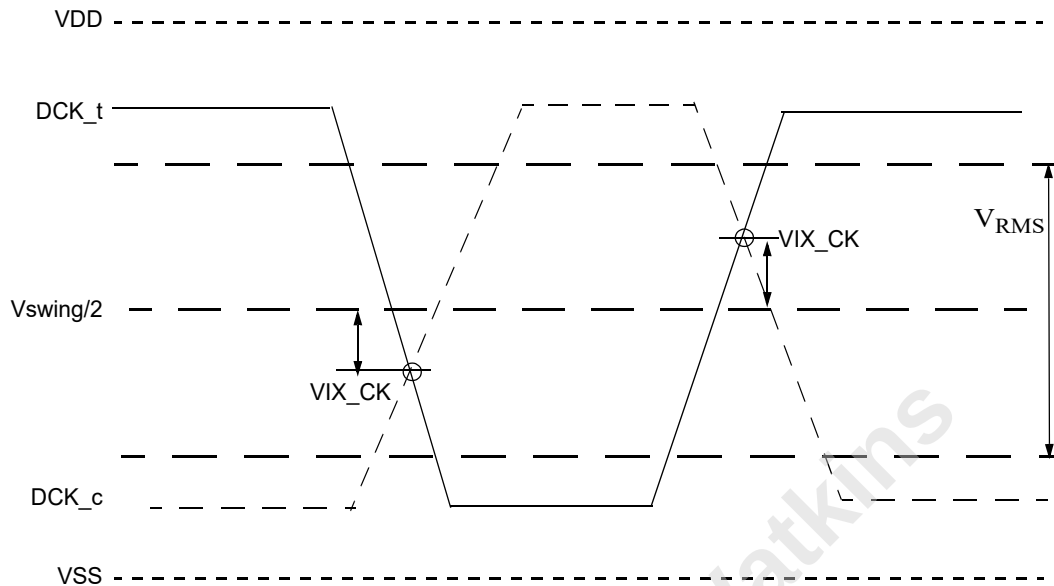


Figure 91 — VIX Definition (DCK)

Table 209 — Cross Point Voltage (Vix) for Differential Input Signals (DCK)

Symbol	Parameter	DDR5-3200~4800		DDR5-5200~5600		Unit	NOTE
		Min	Max	Min	Max		
VIX_CK_Ratio	Clock differential input cross-point voltage ratio	-	50	-	50	%	1, 2, 3
NOTE 1 The VIX_CK voltage is referenced to $V_{swing}/2(\text{mean}) = (\text{DCK_t voltage} + \text{DCK_c voltage}) / 2$, where the mean is over 8 UI.							
NOTE 2 $\text{VIX_CK_Ratio} = (\text{VIX_CK} / \text{V}_{\text{RMS}}) * 100\%$, where $\text{V}_{\text{RMS}} = \text{RMS}(\text{DCK_t voltage} - \text{DCK_c voltage})$.							
NOTE 3 Only applies when both DCK_t and DCK_c are transitioning.							

13.2 Input Voltage Levels for DERROR_IN_n Signals

Table 210 — Input Voltage Levels for DERROR_IN_n Signals

Symbol	Parameter	Min	Max	Unit	NOTE
V _{IH,ERROR}	HIGH-level input voltage	V _{ref} + 75	-	mV	1
V _{IL,ERROR}	LOW-level input voltage	-	V _{ref} - 75	mV	1
NOTE 1 V _{ref} refers to the target setting selected in Table 156, “RW4A: DERROR_IN_n Vref Control Word,” on page 165.					

13.3 Overshoot and Undershoot Specifications

TBD

13.4 Input Bus Termination (IBT) Characteristics for DCA, DPAR, DCS, DCK, and DERROR_IN_n

A functional representation of the input bus termination is shown in Figure 92, “Input Bus Termination: Definition of Voltages and Currents”.

$$RTT = \frac{V_{DD} - V_{IN}}{|I_{IN}|}$$

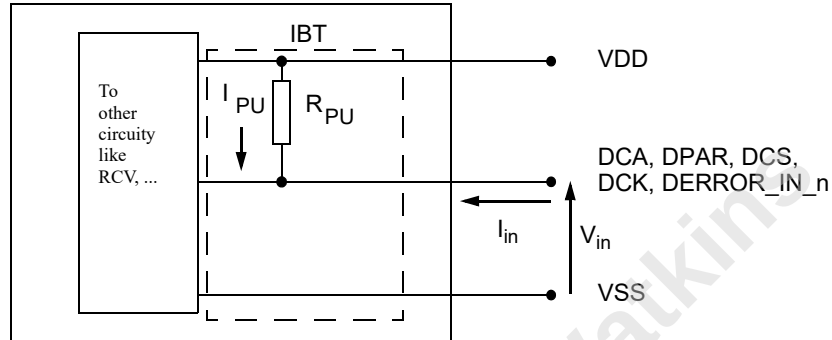


Figure 92 — Input Bus Termination: Definition of Voltages and Currents

The DDR5RCD02 device supports an effective Input Bus Termination R_{tt} value of 48 and 60 ohms (in addition to RTT_OFF).

Table 211 — IBT Electrical Characteristics for DCA, DPAR, DCS, DCK, and DERROR_IN_n

RTT	V _{in}	Min	Nom	Max	Unit	NOTE
60 Ω	V _{ILdc} = 0.5 * VDD	0.9	1	1.25	RZQ/4	1,2
	V _{IMdc} = 0.8 * VDD	0.9	1	1.1	RZQ/4	1,2
	V _{IHdc} = 0.95 * VDD	0.8	1	1.1	RZQ/4	1,2
DCA-DCA Mismatch within sub-channel	V _{IMdc} = 0.8 * VDD	0	-	8	%	1,3,4
NOTE 1 For the behavior of the tolerance limits if temperature or voltage changes, see Section 14.6, “Output Driver and Termination Resistor Temperature and Supply Voltage Sensitivity,” on page 239. NOTE 2 Applies to DCA[6:0]_[B:A], DPAR_[B:A], DCS[1:0]_[B:A]_n, DCK and DERROR_IN_[B:A]_n. NOTE 3 DCA to DCA mismatch within sub-channel variation for a given component including DCA[6:0], DPAR, and DCS[1:0]_n but excluding DCK_t, DCK_c, and DERROR_IN_n (characterized). NOTE 4 RTT variance range ratio to RTT Nominal value in a given component, including DCA[6:0], DPAR, and DCS[1:0]_n but excluding DCK_t, DCK_c and DERROR_IN_n. $\text{DCA-DCA Mismatch in a Sub-channel} = \frac{RTT_{MAX} - RTT_{MIN}}{RTT_{NOM}} \times 100$						

13.5 DLBD and DLBS Loopback Input Termination

The DDR5RCD02 device includes On-Die termination resistance for DLBD and DLBS Loopback inputs.

A functional representation of the on-die termination is shown in the figure below.

$$R_{TT} = \frac{V_{DD} - V_{IN}}{|I_{IN}|}$$

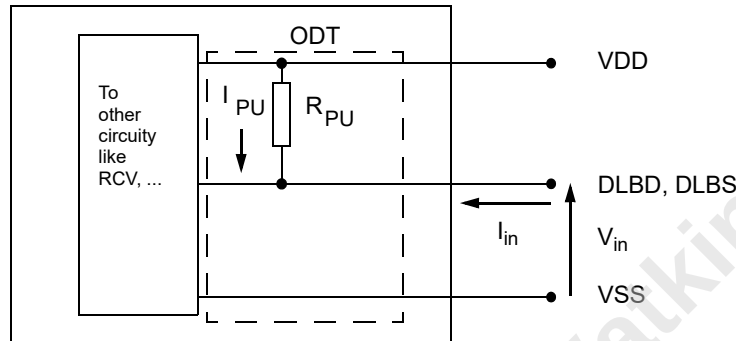


Figure 93 — DLBD and DLBS Termination: Definition of Voltages and Currents

The DDR5RCD02 device supports effective On-Die Termination R_{tt} values of 80, 60, and 40 ohms (in addition to R_{TT_OFF}).

Table 212 — Electrical Characteristics for DLBD and DLBS Input Termination

RTT	Vin	Min	Nom	Max	Unit	NOTE
80 Ω	VILdc= 0.5 * VDD	0.9	1	1.25	RZQ/3	1,2
	VIMdc= 0.8 * VDD	0.9	1	1.1	RZQ/3	1,2
	VIHdc= 0.95 * VDD	0.8	1	1.1	RZQ/3	1,2
60 Ω	VILdc= 0.5 * VDD	0.9	1	1.25	RZQ/4	1,2
	VIMdc= 0.8 * VDD	0.9	1	1.1	RZQ/4	1,2
	VIHdc= 0.95 * VDD	0.8	1	1.1	RZQ/4	1,2
40 Ω	VILdc= 0.5 * VDD	0.9	1	1.25	RZQ/6	1,2
	VIMdc= 0.8 * VDD	0.9	1	1.1	RZQ/6	1,2
	VIHdc= 0.95 * VDD	0.8	1	1.1	RZQ/6	1,2
DLBD-DLBS Mismatch within sub-channel	VIMdc= 0.8 * VDD	0	-	8	%	1,2,3

NOTE 1 For the behavior of the tolerance limits if temperature or voltage changes, see Section 14.6, “Output Driver and Termination Resistor Temperature and Supply Voltage Sensitivity,” on page 239.

NOTE 2 DLBD to DLBS mismatch within sub-channel for a given component (characterized).

NOTE 3 R_{TT} variance range ratio to R_{TT} Nominal value in a given component.

$$\text{DLBD-DLBS Mismatch in a Sub-channel} = \frac{R_{TT_MAX} - R_{TT_MIN}}{R_{TT_NOM}} \times 100$$

13.6 Input CMOS Rail-to-Rail Levels for DRST_n

Table 213 — CMOS Rail-to-Rail Input Levels for DRST_n

Symbol	Parameter	Min	Max	Unit	NOTE
VIH(AC)_RESET	AC Input High Voltage	$0.8 * V_{DD}$	V_{DD}	V	6
VIH(DC)_RESET	DC Input HIGH Voltage	$0.7 * V_{DD}$	V_{DD}	V	2
VIL(DC)_RESET	DC Input LOW Voltage	VSS	$0.3 * V_{DD}$	V	1
VIL(AC)_RESET	AC Input LOW Voltage	VSS	$0.2 * V_{DD}$	V	7
TR_RESET	Rise Time	-	1.0	μs	4
V_{SLPR_p2p}	Peak to peak voltage of slope reversal which must be suppressed	-	100	mV	4
t_{SLPRPW}	Peak to peak width of slope reversal which must be suppressed	-	10	ns	4
t_{PW_RESET}	RESET pulse width	1.0	-	μs	3, 5, 8

NOTE 1 After DRST_n is registered LOW, DRST_n level shall be maintained below VIL(DC)_RESET during t_{PW_RESET} , otherwise, the RCD may not be reset.

NOTE 2 Once DRST_n is registered HIGH, DRST_n level must be maintained above VIH(DC)_RESET, otherwise, RCD operation will not be guaranteed until it is reset asserting DRST_n signal LOW.

NOTE 3 DRST_n will clear all register settings to their default values with the exception of sticky registers listed in Section 8.5.

NOTE 4 Slope reversal (ringback) must remain below V_{SLPR_p2p} with the pulse width below t_{SLPRPW} .

NOTE 5 This definition is applied only “Reset Procedure with Power Stable.”

NOTE 6 Overshoot might occur. It should be limited by the Absolute Maximum DC Ratings.

NOTE 7 Undershoot might occur. It should be limited by Absolute Maximum DC Ratings.

NOTE 8 t_{PW_RESET} is equivalent to parameter t_{RINIT1} (see Table 199 on page 191).

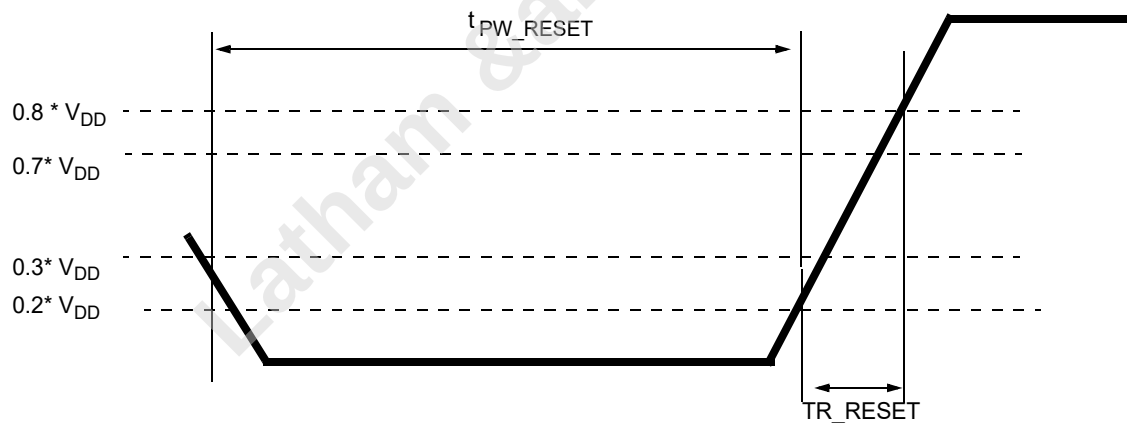


Figure 94 — DRST_n Input Slew Rate Definition

13.6 Input CMOS Rail-to-Rail Levels for DRST_n (cont'd)

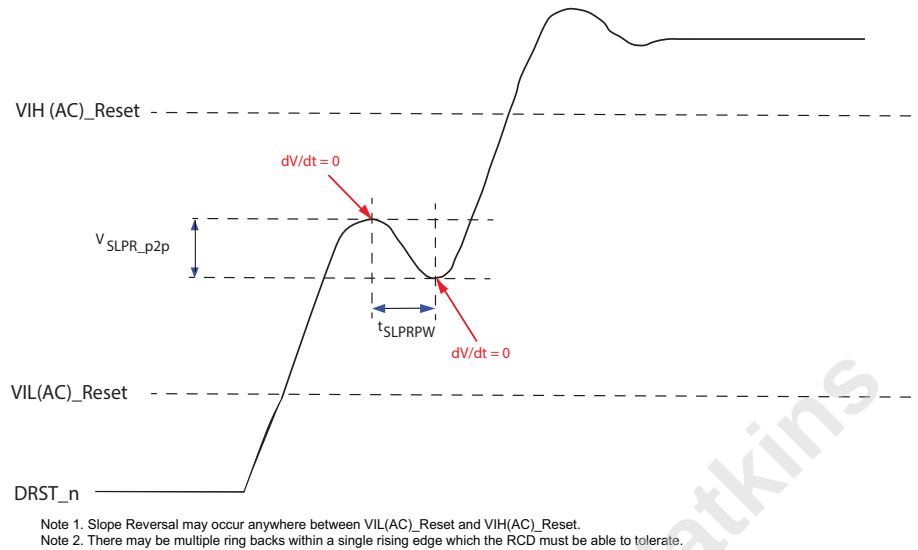


Figure 95 — DRST_n Slope Reversal Definition

13.7 Differential Input Clock and DCA Voltage Sensitivity

The differential input clock voltage sensitivity test provides the methodology for testing the receiver's sensitivity to clock by varying input voltage in the absence of Inter-Symbol Interference (ISI), jitter (Rj, Dj, DCD), and crosstalk noise.

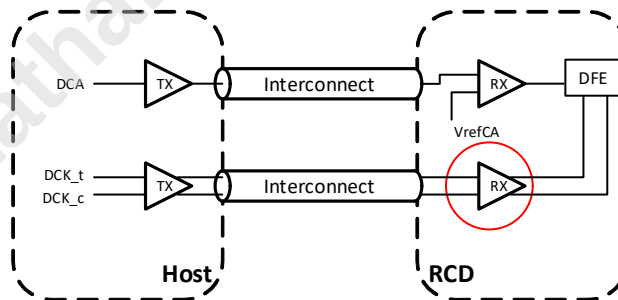


Figure 96 — Example of DDR5 Memory Interconnect

13.7.1 Differential Input Clock and DCA Voltage Sensitivity Parameter

Differential input clock (DCK_t, DCK_c) VR_x_CK is defined and measured as shown in Table 214. The clock receiver must pass the minimum BER requirements for DDR5.

Table 214 — Rx Input Voltage Sensitivity Parameters for DDR5-3200 to 5600

Symbol	Parameter	DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		DDR5-5200		DDR5-5600		Unit	Note
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
VR _x _CK_1	Differential DCK _t /DCK _c Rx input voltage sensitivity (differential p-p).	-	130	-	115	-	105	-	100	-	100	-	90	-	80	mV	1, 3, 4, 5, 7, 8
VR _x _CK_2	Differential DCK _t /DCK _c Rx input voltage sensitivity (differential p-p).	-	100	-	90	-	85	-	80	-	75	-	65	-	65	mV	1, 3, 4, 6, 7, 8
VR _x _DCA_V	Minimum Voltage Amplitude (p-p) that needs to be applied around VrefCA.	-	75	-	70	-	65	-	60	-	55	-	50	-	50	mV	1, 2, 3, 7, 8
VR _x _DLBS	Minimum Voltage Amplitude (p-p) that needs to be applied around DLBS Vref.	-	100	-	100	-	100	-	100	-	100	-	90	-	90	mV	1, 3, 7

NOTE 1 Refer to the minimum BER 1e-9 requirements for DDR5.
 NOTE 2 This Parameter applies to DCA[6:0] [B:A] and DPAR [B:A].
 NOTE 3 Test using clock-like pattern of repeating 3 “1s” and 3 “0s” without applying cross-talk or SSO conditions.
 NOTE 4 The DDR5RCD02 device is required to meet either VR_x_CK_1 or VR_x_CK_2.
 NOTE 5 VR_x_CK_1 only applies to devices that meet Option-1 Rx Clock Jitter Sensitivity specification defined in Table 216.
 NOTE 6 VR_x_CK_2 only applies to devices that meet Option-2 Rx Clock Jitter Sensitivity specification defined in Table 217.
 NOTE 7 This test should be done in typical temperature and voltage conditions (i.e., V_{DD} = 1.1 V, 25 °C).
 NOTE 8 tRx_DCA_tMargin condition does not need to be met.

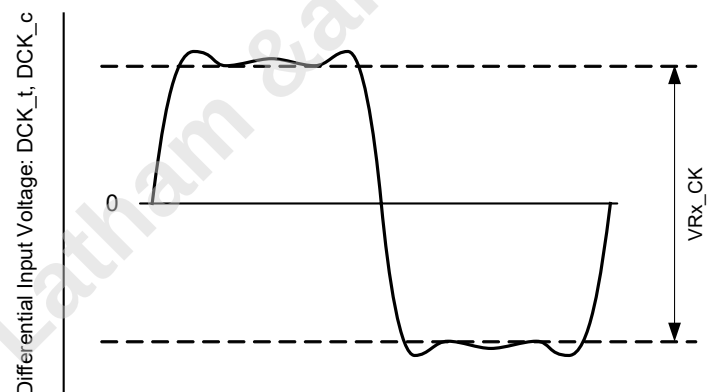


Figure 97 — VR_x_CK

13.7.1 Differential Input Clock and DCA Voltage Sensitivity Parameter (cont'd)

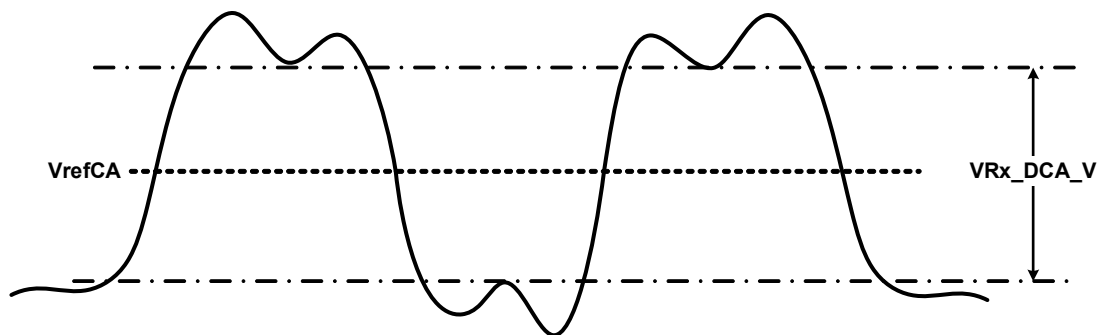


Figure 98 — VRx_DCA_V

13.8 Input Clock Jitter

13.8.1 Overview

The clock is being driven to the RCD for L/RDIMM modules, see Figure 99.

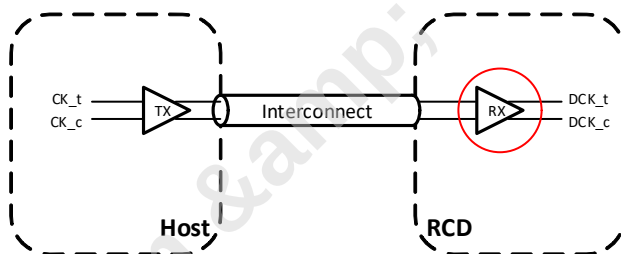


Figure 99 — Host Driving Clock Signals to the RCD

13.8.2 Specification for RCD Input Clock Jitter

The Random Jitter (Rj) specified is a random jitter meeting a Gaussian distribution. The Deterministic Jitter (Dj) specified is bounded. Input clock violating the min/max jitter values may result in malfunction of the DDR5RCD02 device.

Table 215 — Input Clock Differential Jitter¹

Symbol	Parameter	DDR5-3200 to 5600		Unit ²	NOTE
		Min	Max		
tCK	RCD Reference clock frequency	0.9999*f ₀	1.0001*f ₀	MHz	3,12
tCK_Duty_UI	Duty Cycle	47.5	52.5	%	6,12
tCK_Duty_UI_Error	Duty Cycle Error	-	0.05	UI	3,7,12
tCK_1UI_Rj_NoBUJ	Rj value of 1-UI Jitter	-	0.0037	UI (RMS)	5,8,12
tCK_1UI_Dj_NoBUJ	Dj pp value of 1-UI Jitter	-	0.030	UI	5,9,12
tCK_1UI_Tj_NoBUJ	Tj value of 1-UI Jitter	-	0.090	UI	5,9,12
tCK_NUI_Rj_NoBUJ	Rj value of N-UI Jitter	-	0.004	UI (RMS)	5,10,12,13
tCK_NUI_Dj_NoBUJ	Dj pp value of N-UI Jitter	-	0.074	UI	5,11,12,13
tCK_NUI_Tj_NoBUJ	Tj value of N-UI Jitter	-	0.140	UI	5,11,12,13
NOTE 1 [BUJ = Bounded Uncorrelated Jitter; DCD = Duty Cycle Distortion; Dj = Deterministic Jitter; Rj = Random Jitter; Tj = Total jitter; pp = Peak-to-Peak]					
NOTE 2 Unit UI = tCK(avg).min/2					
NOTE 3 f ₀ = Data Rate/2, example: if data rate is 3200MT/s, then f ₀ =1600					
NOTE 4 Rise and fall time slopes (V / nsec) are measured between +100 mV and -100 mV of the differential output of reference clock					
NOTE 5 When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or cannot be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining Tx lanes send patterns to the corresponding Rx receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility					
NOTE 6 Duty Cycle defined as the ratio between any even UI and tCK.					
NOTE 7 Duty Cycle Error defined as absolute difference between average value of all UI with that of average of odd UI, which in magnitude would equal absolute difference between average of all UI and average of all even UI.					
NOTE 8 Rj RMS value of 1-UI jitter without BUJ. This extraction is to be done after software correction of DCD					
NOTE 9 Dj pp value of 1-UI jitter (after software assisted DCC). Without BUJ. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD					
NOTE 10 Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 1 < N < 4. This extraction is to be done after software correction of DCD					
NOTE 11 Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 1 < N < 4. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD					
NOTE 12 The validation methodology for these parameters will be covered in future ballots					
NOTE 13 When frequency is less than or equal to 4800 MT/s 1 < N < 4, When frequency is greater than 4800 MT/s up to 5600 MT/s 1 < N < 6					

13.9 RX Clock Jitter Sensitivity

The receiver clock jitter sensitivity test provides the methodology for testing the receiver's clock sensitivity to an applied duty cycle distortion (DCD) and/or random jitter (Rj) at the forwarded clock input while keeping the data crosstalk free, input jitter free, noise free and ISI free. The receiver must pass the appropriate BER rate when the equivalent cross-talk pattern is applied through the combination of applied DCD and Rj.

13.9 RX Clock Jitter Sensitivity (cont'd)

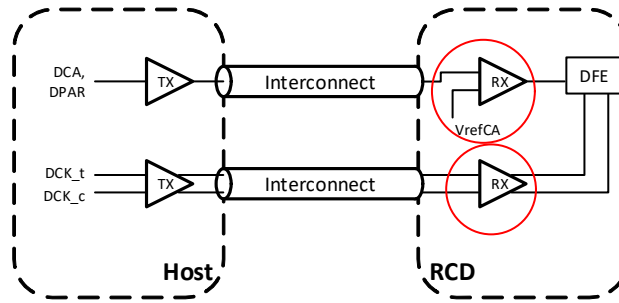


Figure 100 — RCD RX Forward Clock for Jitter Sensitivity

13.9.1 Rx Clock Jitter Sensitivity Specification

The following tables provide Rx DCK Jitter Sensitivity Specification for the DDR5RCD02 receivers when operating at various possible transfer rates. These parameters are tested with neither additive gain nor Rx Equalization set. The DDR5RCD02 device is required to meet only one of the following two options: the Option-1 requirements defined in Table 216 or the Option-2 requirements defined in Table 217 (as applicable for the implemented receiver architecture).

Table 216 — Rx Clock Jitter Sensitivity Option-1 Specification for DDR5-3200 to 5600

Symbol	Parameter	DDR5-3200 to 4800		DDR5-5200 to 5600		Unit ¹	NOTE
		Min	Max	Min	Max		
tRx_DCA_tMargin	DCA Timing Width	0.75	-	0.8	-	UI	2,3,4,9,10,11
ΔtRx_DCA_tMargin_DCD_DCK	Degradation of timing width compared to tRx_DCA_tMargin, with DCD injection in DCK	-	0.03	-	0.03	UI	2,5,9,10,11
ΔtRx_DCA_tMargin_Rj_DCK	Degradation of timing width compared to tRx_DCA_tMargin, with Rj injection in DCK	-	0.03	-	0.03	UI	2,6,9,10,11
ΔtRx_DCA_tMargin_DCD_Rj_DCK	Degradation of timing width compared to tRx_DCA_tMargin with both DCD and Rj injection in DCK	-	0.05	-	0.05	UI	2,3,7,9,10,11
tRx_DCK2DCA_Skew	Delay of any DCA or DCS lane relative to the DCK_t/DCK_c crossing within a sub-channel.	- 0.25	0.25	-0.25	0.25	UI	2,8,9,10,11
NOTE 1 Unit UI = tCK(avg).min/2 NOTE 2 Validation methodology will be defined in future ballots. NOTE 3 Each of ΔtRx_DCA_tMargin_DCD_DCK, ΔtRx_DCA_tMargin_DCD_Rj_DCK, and ΔtRx_DCA_tMargin_DCD_Rj_DCK can be relaxed by up to 5% if tRx_DCA_tMargin exceeds the spec by at least 5%. Conversely, tRx_DCA_tMargin can be relaxed by up to 5% if all of ΔtRx_DCA_tMargin_DCD_DCK, ΔtRx_DCA_tMargin_Rj_DCK, and ΔtRx_DCA_tMargin_DCD_Rj_DCK are better than spec by at least 5%. NOTE 4 DCA Timing Width - timing width for any data lane using repetitive patterns (check notes 5 to 7 for the pattern) measured at BER=E-9. NOTE 5 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD injection in forwarded clock compared to tRx_DCA_tMargin, measured at BER=E-9. The magnitude of DCD is specified under Test Conditions for Rx clock Jitter Sensitivity Testing. Test using clock-like pattern of repeating 3 “1s” and 3 “0s”. NOTE 6 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with only Rj injection in forwarded clock measured at BER=E-9, compared to tRx_tMargin. The magnitude of Rj is specified under Test Conditions for Rx Clock Jitter Sensitivity Testing. Test using clock-like pattern of repeating 3 “1s” and 3 “0s”. NOTE 7 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD and Rj injection in forwarded clock measured at BER=E-9, compared to tRx_tMargin. The magnitudes of DCD and Rj are specified under Test Conditions for Rx Clock Jitter Sensitivity Testing. NOTE 8 Delay of any data lane relative to the clock lane, as measured at the end of Tx+Channel. This parameter is a collective sum of effects of data clock mismatches in Tx and on the medium connecting Tx and Rx. Test using clock-like pattern of repeating 3 “1s” and 3 “0s”. NOTE 9 All measurements at BER=E-9. NOTE 10 This test should be done in typical temperature and voltage conditions (i.e., V _{DD} = 1.1 V, 25 °C) with default VrefCA settings. NOTE 11 The user has the freedom to set the voltage swing and slew rates for clock and DCA signals as long as they meet the specification.							

13.9.1 Rx Clock Jitter Sensitivity Specification (cont'd)

Table 217 — Rx Clock Jitter Sensitivity Option-2 Specification for DDR5-3200 to 5600

Symbol	Parameter	DDR5-3200 to 4800		DDR5-5200 to 5600		Unit ¹	NOTE
		Min	Max	Min	Max		
tRx_DCA_tMargin	DCA Timing Width	0.9	-	0.9	-	UI	2,3,4,9,10,11
$\Delta tRx_DCA_tMargin_DCD_DCK$	Degradation of timing width compared to tRx_DCA_tMargin, with DCD injection in DCK	-	0.05	-	0.05	UI	2,5,9,10,11
$\Delta tRx_DCA_tMargin_Rj_DCK$	Degradation of timing width compared to tRx_DCA_tMargin, with Rj injection in DCK	-	0.075	-	0.075	UI	2,6,9,10,11
$\Delta tRx_DCA_tMargin_DCD_Rj_DCK$	Degradation of timing width compared to tRx_DCA_tMargin with both DCD and Rj injection in DCK	-	0.1	-	0.125	UI	2,3,7,9,10,11
tRx_DCK2DCA_Skew	Delay of any DCA or DCS lane relative to the DCK_t/DCK_c crossing within a sub-channel.	0	1	0	1	UI	2,8,9,10,11
NOTE 1 Unit UI = tCK(avg).min/2 NOTE 2 Validation methodology will be defined in future ballots. NOTE 3 Each of $\Delta tRx_DCA_tMargin_DCD_DCK$, $\Delta tRx_DCA_tMargin_DCD_Rj_DCK$, and $\Delta tRx_DCA_tMargin_DCD_Rj_DCK$ can be relaxed by up to 5% if tRx_DCA_tMargin exceeds the spec by at least by 5%. Conversely, tRx_DCA_tMargin can be relaxed by up to 5% if all of $\Delta tRx_DCA_tMargin_DCD_DCK$, $\Delta tRx_DCA_tMargin_Rj_DCK$, and $\Delta tRx_DCA_tMargin_DCD_Rj_DCK$ are better than spec by at least 5%. NOTE 4 DCA Timing Width - timing width for any data lane using repetitive patterns (check notes 5 to 7 for the pattern) measured at BER=E-9. NOTE 5 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD injection in forwarded clock compared to tRx_DCA_tMargin, measured at BER=E-9. The magnitude of DCD is specified under Test Conditions for Rx clock Jitter Sensitivity Testing. Test using clock-like pattern of repeating 3 “1s” and 3 “0s”. NOTE 6 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with only Rj injection in forwarded clock measured at BER=E-9, compared to tRx_tMargin. The magnitude of Rj is specified under Test Conditions for Rx Clock Jitter Sensitivity Testing. Test using clock-like pattern of repeating 3 “1s” and 3 “0s”. NOTE 7 Magnitude of degradation of timing width for any data lane using repetitive no ISI patterns with DCD and Rj injection in forwarded clock measured at BER=E-9, compared to tRx_tMargin. The magnitudes of DCD and Rj are specified under Test Conditions for Rx Clock Jitter Sensitivity Testing. Test using clock-like pattern of repeating 3 “1s” and 3 “0s”. NOTE 8 Delay of any data lane relative to the clock lane, as measured at the end of Tx+Channel. This parameter is a collective sum of effects of data clock mismatches in Tx and on the medium connecting Tx and Rx. NOTE 9 All measurements at BER=E-9. NOTE 10 This test should be done in typical temperature and voltage conditions (i.e., V_{DD} = 1.1 V, 25 °C) with default VrefCA settings. NOTE 11 The user has the freedom to set the voltage swing and slew rates for clock and DCA signals as long as they meet the specification.							

13.9.2 Test Conditions for RX Clock Jitter Sensitivity Tests

Table 218 lists the amount of Duty Cycle Distortion (DCD) and/or Random Jitter (Rj) that must be applied to the forwarded clock when measuring the Rx Clock Jitter Sensitivity parameters specified in Table 217.

Table 218 — Rx Clock Jitter Sensitivity Specification for DDR5-3200 to 5600

Symbol	Parameter	DDR5-3200 to 5600		Unit ¹	NOTE
		Min	Max		
tRx_DCK_DCD	Applied DCD to the DCK	-	0.045	UI	2,3,4,7,8,11
tRx_DCK_Rj	Applied Rj RMS to the DCK	-	0.00625	UI (RMS)	2,3,5,7,9,11
tRx_DCK_DCD_Rj	Applied DCD and Rj RMS to the DCK	-	0.045 UI DCD + 0.00625 UI (RMS) Rj	UI	2,3,6,7,8,10,11
NOTE 1 Unit UI = tCK(avg).min/2					
NOTE 2 While imposing this spec, the clock lane is stressed, but the data input is kept large amplitude and no jitter or ISI injection. The specified voltages are at the Rx input pin. The DCK and DCA input voltage swing and/or slew rate can be adjusted, without exceeding the specifications, for this test.					
NOTE 3 The jitter response of the forwarded clock channel will depend on the input voltage, primarily due to bandwidth limitations of the clock receiver. For this revision, no separate specification of jitter as a function of input amplitude is specified, instead the response characterization done at the specified clock amplitude only. The specified voltages are at the Rx input pin.					
NOTE 4 Various DCD values should be tested, complying within the maximum limits.					
NOTE 5 Various Rj values should be tested, complying within the maximum limits.					
NOTE 6 Various combinations of DCD and Rj should be tested, complying within the maximum limits. The maximum timing margin degradation as a result of these injected jitter is specified in a separate table.					
NOTE 7 Although DDR5 has bursty traffic, in order to ensure that current available BERTs can be used for this test, a continuous clock and continuous DCA are used for this parameter. The clock like pattern repeating 3 “1s” and 3 “0s” is used for this test.					
NOTE 8 Duty Cycle Distortion (in UI DCD) as applied to the input forwarded clock from BERT (UI).					
NOTE 9 RMS value of Rj (specified as Edge jitter) applied to the input forwarded clock from BERT (values of the edge jitter RMS values specified as % of UI).					
NOTE 10 Duty cycle distortion (specified as UI DCD) and rms values of Rj (specified as edge jitter) applied to the input forwarded clock from BERT (values of the edge jitter RMS values specified as % of UI).					
NOTE 11 The user has the freedom to set the voltage swing and slew rates for clock and DCA signals as long as they meet the specification.					

13.10 Input Stressed Eye

13.10.1 Overview

The stressed eye tests provide the methodology for creating the appropriate stress for the RCD's receiver with the combination of ISI (both loss and reflective), jitter (Rj, Dj, DCD), and crosstalk noise. The receiver must pass the appropriate BER rate when the equivalent stressed eye is applied through the combination of ISI, jitter and crosstalk.

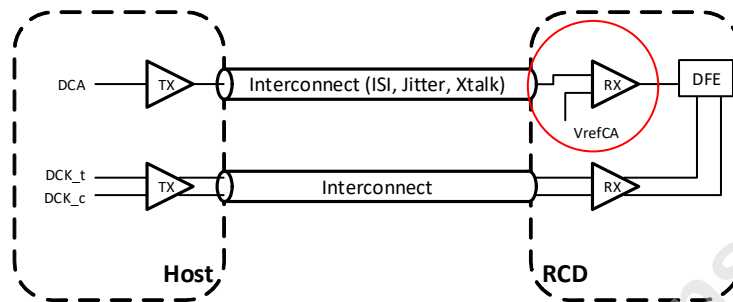


Figure 101 — Example of Rx Stressed Test Setup in Presence of ISI, Jitter, and Crosstalk

13.10.2 Input Stressed Eye Parameters

Table 219 — Test Conditions for Rx Stressed Eye Tests for DDR5-3200 to 5600¹

Symbol	Parameter	DDR5-3200 to 4000		DDR5-4400 to 4800		DDR5-5200 to 5600		Unit ²	NOTE
		Min	Max	Min	Max	Min	Max		
RxEH_Stressed_Eye	Eye height of stressed eye	-	65	-	55	-	50	mV	3,4,5, 6,7,8,9
RxEW_Stressed_Eye	Eye width of stressed eye	-	0.20	-	0.20	-	0.20	UI	3,4,5, 6,7,8,9
VSwing_Stressed_Eye	Vswing stress to meet above data eye	0	600	0	600	0	600	mV	3,4
Sj_Stressed_Eye	Injected sinusoidal jitter at 200 MHz to meet above data eye	0	0.45	0	0.45	0	0.45	UI (p-p)	3,4
Rj_Stressed_Eye	Injected Random wide band (10 MHz-1 GHz) Jitter to meet above data eye	0	0.04	0	0.04	0	0.04	UI RMS	3,4
V_Noise_Stressed_Eye	Injected voltage noise at 2.1 GHz to meet above eye	0	125	0	125	0	125	mV (p-p)	3,4
NOTE 1 [BER = Bit Error Rate; DCD = Duty Cycle Distortion; Rj = Random Jitter; Sj = Sinusoidal Jitter; p-p = peak to peak]									
NOTE 2 Unit UI = tCK(avg).min/2									
NOTE 3 Must meet minimum BER 1e-9 requirement with eye measured at the input of CA Slicer (i.e., node CA_Sum_p/n in Figure 49, “DFE Training Circuitry” on page 62).									
NOTE 4 These parameters are applied on the reference channel (TBD).									
NOTE 5 Evaluated with no DC supply voltage drift.									
NOTE 6 Evaluated with no temperature drift.									
NOTE 7 Supply voltage noise limited according to DC bandwidth spec, see Recommended DC Operating Conditions.									
NOTE 8 The stressed eye is to be assumed to have a diamond shape.									
NOTE 9 The VrefCA, DFE Gain Bias Step, and DFE Taps 1,2,3,4 Bias Step can be adjusted as needed, without exceeding the specifications, for this test.									

13.11 DCS Input Receiver Specification

The DCS input receiver mask for voltage and timing is shown in Figure 102, “DCS Receiver (RX) Mask”. The receiver mask (Rx Mask) defines area the input signal must not encroach in order for the DDR5RCD02 input receiver to successfully capture a valid input signal. The mask is a receiver property for each pin and it is not the valid data eye.

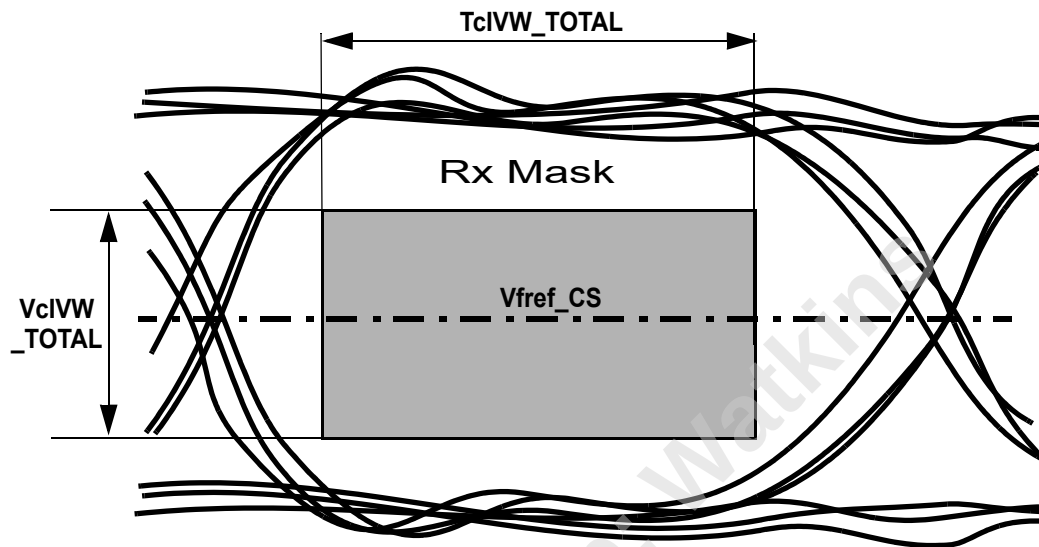


Figure 102 — DCS Receiver (RX) Mask

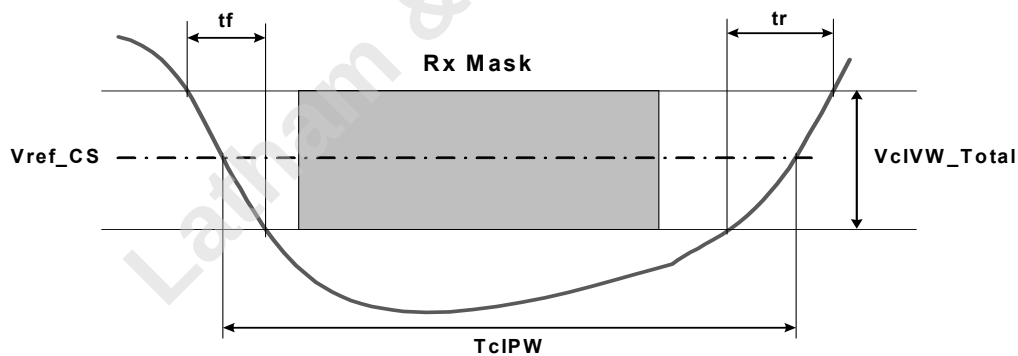


Figure 103 — DCS TcIPW and SRIN_cIVW Definition (for each Input Pulse)

13.11.1 DCS Input Receiver Voltage Margin and AC Timing by Speed Bin

Table 220 — DCS Input Receiver Voltage Margin and AC Timing by Speed Bin

Speed		DDR5- 3200		DDR5- 3600 to 4000		DDR5- 4400 to 4800		DDR5- 5200 to 5600		Unit ¹	NOTE
Symbol	Parameter	Min	Max	Min	Max	Min	Max	Min	Max		
VcIVW_TOTAL	Rx Mask p-p voltage total	-	100	-	95	-	90	-	85	mV	2,3,4,5
TcIVW_TOTAL	Rx timing window total	-	0.135	-	0.130	-	0.125	-	0.125	UI	2,5
VIHL_AC	DCS AC input pulse amplitude pk-pk	120	-	115	-	110	-	105	-	mV	6
TcIPW	DCS input pulse width	0.3	-	0.29	-	0.28	-	0.28	-	UI	7
SRIN_cIVW	Input Slew Rate over VcIVW_TOTAL	1	5	1	5	1	5	1	5	V/ns	8
NOTE 1 Unit UI = tCK(avg).min NOTE 2 DCS Rx mask voltage and timing total input valid window where VcIVW is centered around VrefCS). The data Rx mask is applied per bit and includes voltage and temperature drift terms. NOTE 3 Rx mask voltage AC swing peak-peak requirement over TcIVW_TOTAL with at least half of VcIVW_TOTAL(max) above VrefCS and at least half of VcIVW_TOTAL(max) below VrefCS. NOTE 4 The VcIVW voltage levels are centered around VrefCS. NOTE 5 Overshoot and Undershoot Specifications see Section 13.3 on page 215. NOTE 6 DCS input pulse signal swing into the receiver must meet or exceed VIHL_AC for at least one point over the duration of TcIPW for any UI during which there is a signal transition. No timing requirement above level. VIHL_AC is the peak to peak voltage centered around VrefCS, which is defined in Figure 102. NOTE 7 DCS minimum input pulse width defined at the VrefCS). NOTE 8 Input slew rate over VcIVW Mask centered at VrefCS. For a given measurement, under the same conditions, the applied slew rate for all transition edges (slowest to fastest) must be within 2V/ns of each other.											

13.12 DLBD Input Receiver Specification

The Loopback input receiver mask for voltage and timing is shown in Figure 104, “DLBD Receiver (RX) Mask”. The receiver mask (Rx Mask) defines area the input signal must not encroach in order for the DDR5RCD02 input receiver to successfully capture a valid input signal. The mask is a receiver property for each pin and it is not the valid data eye.

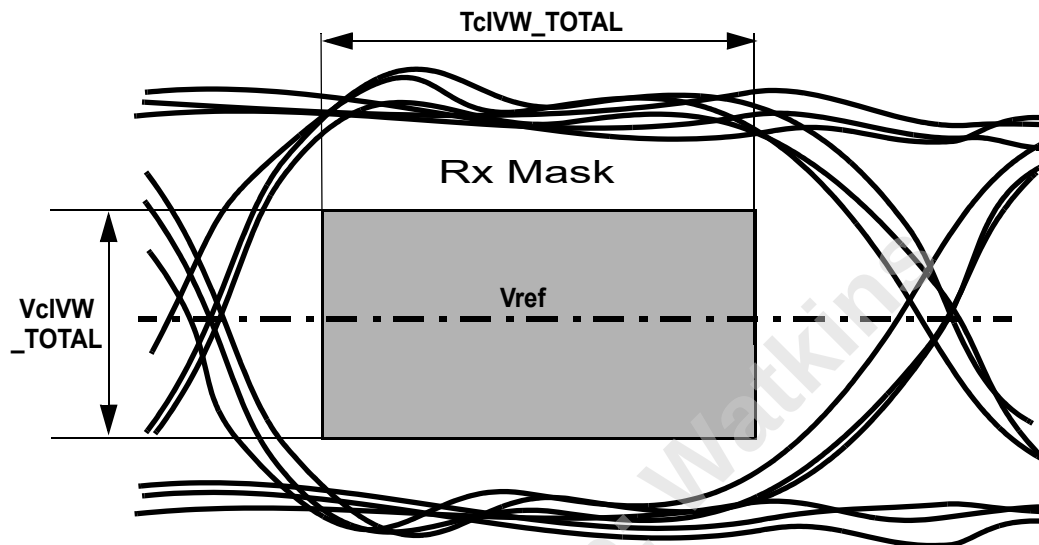
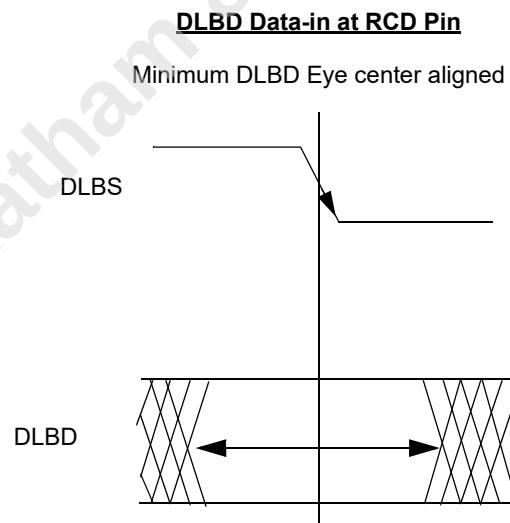


Figure 104 — DLBD Receiver (RX) Mask



NOTE T_{cIVW} for DLBD is defined as centered on the DLBS falling edge at the RCD pin

Figure 105 — DLBD Timings at the RCD

13.12 DLBD Input Receiver Specification (cont'd)

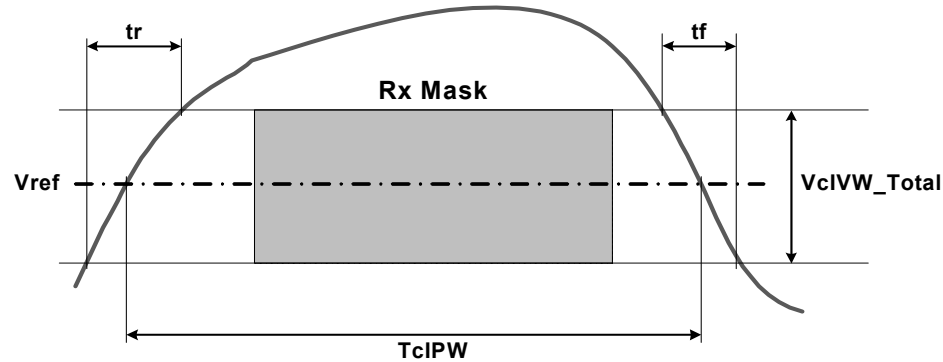


Figure 106 — DLBD TcIPW Definition (for each Input Pulse)

Table 221 — Loopback RX Receiver Voltage Margin and AC Timing by Speed Bin

Speed	Parameter	DDR5-3200-4800		DDR5-5200-5600		Unit	Note
Symbol		Min	Max	Min	Max		
Data Voltage and Timing							
VcIVW_TOTAL	DLBD Rx Mask p-p voltage total	-	90		100	mV	1,2,3,4,7
TcIVW_TOTAL	DLBD Rx timing window total	-	0.3	-	0.45	tCK(avg)min	1,4,7
tDLBS2DLBD	DLBS to DLBD offset	-	0.1	-	0.2	tCK(avg)min	5,7
TdlbsIPW_2PH	DLBS input pulse width in 2-phase loopback mode	0.25	-	N/A	N/A	tCK(avg)min	6,7
TdlbsIPW_4PH	DLBS input pulse width in 4-phase loopback mode	0.55	-	0.55	-	tCK(avg)min	6,7
NOTE 1	Loopback Rx mask voltage and timing total input valid window where VcIVW is centered around the effective Vref level. The Loopback Rx mask is applied per bit and includes voltage and temperature drift terms. The BER for DDR5 during normal operation is 1e-16. For validation purpose, the BER used is 1e-9.						
NOTE 2	Rx mask voltage AC swing peak-peak requirement over TcIVW_TOTAL with at least half of VcIVW_TOTAL(max) above effective Vref and at least half of VcIVW_TOTAL(max) below effective Vref.						
NOTE 3	The VcIVW voltage levels are centered around the effective Vref level.						
NOTE 4	DLBD minimum input pulse width defined at the effective Vref level.						
NOTE 5	DLBS to DLBD is defined as the input offset for each Loopback input in the DDR5RCD02 device measured at the package balls. Includes all DDR5RCD02 process, voltage and temperature variation. Measured from the falling edge of DLBS to the center (midpoint) of the DLBD TcIVW_TOTAL window. SeeFigure 105 on page 229.						
NOTE 6	DLBS minimum input pulse width defined at the effective Vref level.						
NOTE 7	At DDR5-5200 and DDR5-5600, only 1/4 rate loopback is required.						

14 Electrical - Output AC and DC Specifications

14.1 Single-Ended Output Slew Rate

With the reference load for timing measurements, output slew rate for falling and rising edges is defined and measured between 75% and 25% of the Total swing as shown in Table 222 and Figure 107.

Table 222 — Output Single-ended Slew Rate Definition

Description	Measured		Defined by
	From	To	
Single-ended output slew rate for rising edge	25%	75%	$[(25\% - 75\%)] / \Delta t_{Rse}$
Single-ended output slew rate for falling edge	75%	25%	$[(75\% - 25\%)] / \Delta t_{Fse}$
NOTE Output slew rate is verified by design and characterization, and may not be subject to production test.			

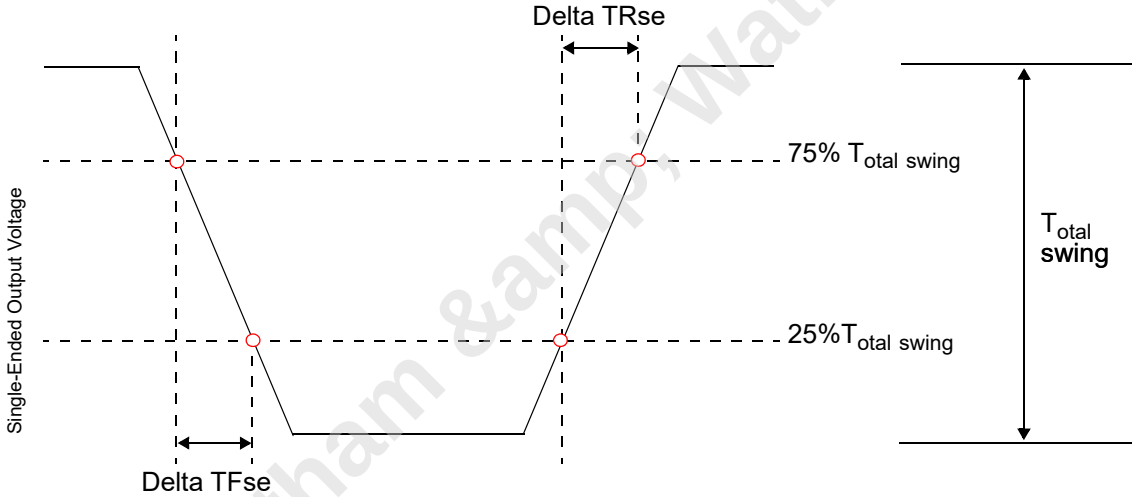


Figure 107 — Output Single-Ended Slew Rate Definition

14.1 Single-Ended Output Slew Rate (cont'd)

Table 223 — Output Single-Ended Edge Rates Over Specified Operating Temperature Range

Symbol	Parameter	Conditions	DDR5-3200 - 5600		Unit	Note
			Min	Max		
dV/dt_{rQCA}	QCA rising edge slew rate	1.1V operation	2.5	12.1	V/ns	1,2
dV/dt_{fQCA}	QCA falling edge slew rate	1.1V operation	2.5	12.1	V/ns	1,2
dV/dt_{rQCS}	QCS rising edge slew rate	1.1V operation	2.5	12.1	V/ns	2
dV/dt_{fQCS}	QCS falling edge slew rate	1.1V operation	2.5	12.1	V/ns	2
dV/dt_{fAlert_n}	Alert_n falling edge slew rate	1.1V operation	2.7	10	V/ns	2
dV/dt_{rQRST_n}	QRST_n rising edge slew rate	1.1V operation	0.1	3.0	V/ns	2
dV/dt_{fQRST_n}	QRST_n falling edge slew rate	1.1V operation	0.1	1.0	V/ns	2
dV/dt_{rBRST_n}	BRST_n rising edge slew rate	1.1V operation	0.1	3.0	V/ns	2
dV/dt_{fBRST_n}	BRST_n falling edge slew rate	1.1V operation	0.1	1.0	V/ns	2
dV/dt_D	Absolute difference between dV/dt_r and dV/dt_f	1.1V operation	-	2	V/ns	3
NOTE 1 These parameters are for the QCA _n , BCOM, and BCS _n outputs.						
NOTE 2 Measured into 50 Ω or 1 k Ω reference load terminated to V_{DD} , as shown in Figure 132.						
NOTE 3 Difference between dV/dt_r (rising edge rate) and dV/dt_f (falling edge rate)						

14.2 Output Clock Differential Slew Rate

Output slew rates for differential signals QnCK_{x_t} / QnCK_{x_c} and BnCK_t / BnCK_c are defined and measured as shown in Table 224 and Figure 108.

Table 224 — Output Clock Differential Slew Rate Definition for QnCK_t / QnCK_c and BnCK_t / BnCK_c

Description	Measured		Defined by
	From	To	
Differential output slew rate for rising edge (QnCK _t / QnCK _c and BnCK _t / BnCK _c).	25%	75%	$[25\% - 75\%] / \Delta TR_{diff}$
Differential output slew rate for falling edge (QnCK _t / QnCK _c and BnCK _t / BnCK _c).	75%	25%	$[75\% - 25\%] / \Delta TF_{diff}$
NOTE Differential output slew rate is verified by design and characterization, and may not be subject to production test.			

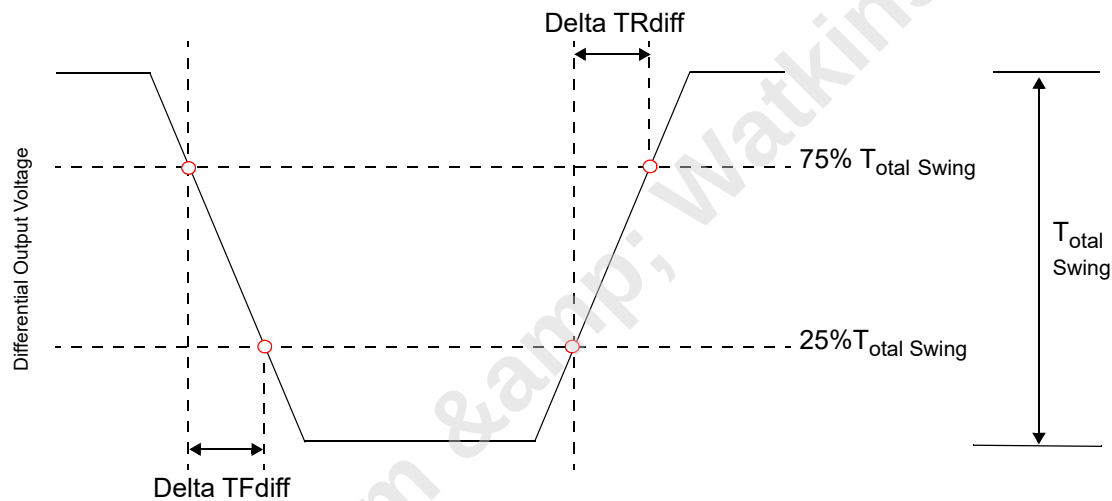


Figure 108 — Differential Output Slew Rate Definition for QnCK_t / QnCK_c and BnCK_t / BnCK_c

Table 225 — Output Differential Edge Rates¹

Symbol	Parameter	Conditions	DDR5-3200 - 5600		Unit
			Min	Max	
dV/dt _r	QCK rising edge differential Slew Rate	1.1V operation	12	37	V/ns
dV/dt _f	QCK falling edge differential Slew Rate	1.1V operation	12	37	V/ns
NOTE 1 Note: There are two selectable output slew rate ranges defined for QCK and BCK in RW0E and RW0F, respectively. See Table 114 on page 144 and Table 115 on page 145, for operating conditions.					

14.3 Differential Output Clock Cross Point Voltage

The differential output clock cross point voltage is defined as the cross point voltage measured on the differential signals QnCK_t / QnCK_c and BCK_t/BCK_c with respect to the output common mode voltage (V_{OCM}).

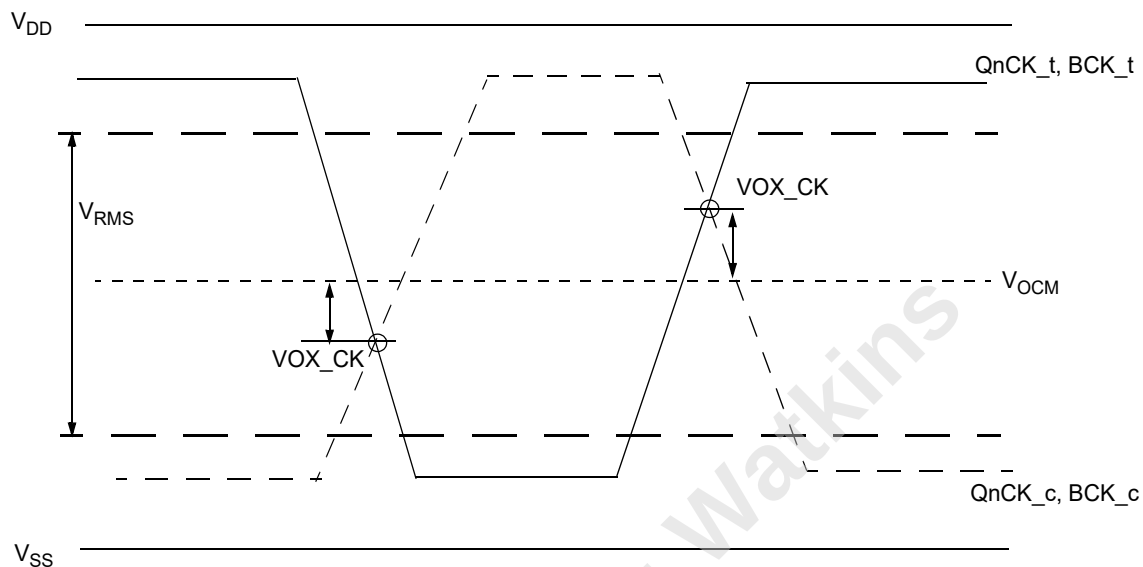


Figure 109 — VOX Definition (QnCKx and BCK)

Table 226 — Output Cross Point Voltage for QnCK_t/QnCK_c and BCK_t/BCK_c

Symbol	Parameter	DDR5-3200~4800		DDR5-5200~5600		Unit	NOTE
		min	max	min	max		
VOX_CK_Ratio	QnCK and BCK Differential Output Cross-Point Voltage Ratio	-	10	-	10	%	1, 2, 3
NOTE 1 Referenced to $V_{OCM} = \text{avg} (QnCK_t + QnCK_c)/2$ or $\text{avg} (BCK_t + BCK_c)/2$, where the average is over 8 UI.							
NOTE 2 $VOX_CK_Ratio = (VOX_CK / V_{RMS}) * 100\%$, where $V_{RMS} = \text{RMS}(QnCK_t \text{ or } BCK_t \text{ voltage} - QnCK_c \text{ or } BCK_c \text{ voltage})$.							
NOTE 3 Only applies when both (QnCK_t or BCK_t) and (QnCK_c or BCK_c) are transitioning. Measured into 50- Ω reference load terminated to V_{DD} , as shown in Figure 132							

14.4 QCK, QCA, QCS, BCK, BCOM, and BCS Output Slew Rate Specifications per Control Words Settings

Table 227 — QCK, QCA, QCS, BCK, BCOM, and BCS Output Slew Rate

Symbol	Parameter	Min	Max	Unit
QCK[D:A]_t / QCK[D:A]_c Differential Slew Rate				
SR _{QCK_M}	Moderate slew rate	16	30	V/ns
SR _{QCK_F}	Fast slew rate	20	37	V/ns
SR _{QCK_L}	Slow slew rate	12	22	V/ns
Q[B:A]CA[13:0] Single Ended Slew Rate				
SR _{QCA_M}	Moderate slew rate	4.0	8.6	V/ns
SR _{QCA_F}	Fast slew rate	6.0	12.1	V/ns
SR _{QCA_L}	Slow slew rate	2.5	6.7	V/ns
Q[B:A]CS[1:0]_n Single Ended Slew Rate				
SR _{QCS_M}	Moderate slew rate	4.0	8.6	V/ns
SR _{QCS_F}	Fast slew rate	6.0	12.1	V/ns
SR _{QCS_L}	Slow slew rate	2.5	6.7	V/ns
BCOM[2:0], BCS_n Single Ended Slew Rate				
SR _{BCOM_M}	Moderate slew rate	3.7	8.6	V/ns
SR _{BCOM_F}	Fast slew rate	5.2	12.1	V/ns
SR _{BCOM_L}	Slow slew rate	2.5	6.7	V/ns
BCK_t / BCK_c Differential Slew Rate				
SR _{BCK_M}	Moderate slew rate	16	30	V/ns
SR _{BCK_F}	Fast slew rate	20	37	V/ns
SR _{BCK_L}	Slow slew rate	12	22	V/ns

14.5 Output R-on Drive Specifications per Drive Strength Setting

Table 228 — Output Ron Drive

Symbol	Parameter	DDR5-3200 to 5600			Unit
		Min	Nom	Max	
ALERT_n					
R _{on} (ALERT) _{Pd}	ALERT_n Pull-down Impedance	see Table 229	R _{ZQ} /17	see Table 229	Ω
R _{on} (ALERT) _{Pu}	ALERT_n Pull-up Impedance		R _{ZQ}		Ω
QRST_n and BRST_n					
RON _{RSTPd}	QRST_n and BRST_n Pull-down Impedance	see Table 229	R _{ZQ} /8	see Table 229	Ω
RON _{RSTPu}	QRST_n and BRST_n Pull-up Impedance		R _{ZQ} /8		Ω
QLBS, QLBD					
R _{on_LBL}	Light Drive Impedance	see Table 229	R _{ZQ} /5	see Table 229	Ω
R _{on_LBM}	Moderate Drive Impedance		R _{ZQ} /7		Ω
BCOM[2:0]/BCS_n					
R _{on_BL}	Light Drive Impedance	see Table 229	R _{ZQ} /12	see Table 229	Ω
R _{on_BM}	Moderate Drive Impedance		R _{ZQ} /17		Ω
QxCA[13:0]/QxCS[1:0]_n/QnCK_t/QnCK_c/BCK_t/BCK_c					
R _{on_QL}	Light Drive Impedance	see Table 229	R _{ZQ} /12	see Table 229	Ω
R _{on_QM}	Moderate Drive Impedance		R _{ZQ} /17		Ω
R _{on_QS}	Strong Drive Impedance		R _{ZQ} /24		Ω
NOTE 1 A functional representation of the output buffer is shown in Figure 110. Output impedance RON is defined by the value of the external reference resistor R _{ZQ} as defined in Table 229.					
The individual pull-up and pull-down resistors (RON _{Pu} and RON _{Pd}) are defined as follows:					
$RON_{Pu} = \frac{V_{DD} - V_{Out}}{ I_{Out} }$ under the condition that RON _{Pd} is turned off. (1)					
$RON_{Pd} = \frac{V_{Out}}{ I_{Out} }$ under the condition that RON _{Pu} is turned off.					
NOTE 2 Assuming R _{ZQ} = 240 Ω +/- 1%					

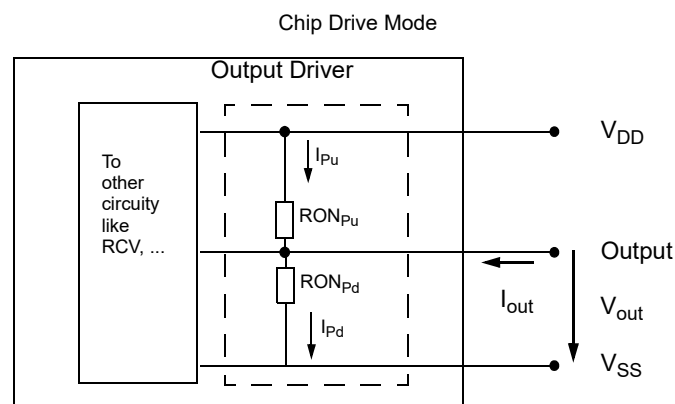


Figure 110 — Output Driver: Definition of Voltages and Currents

14.5 Output R-on Drive Specifications per Drive Strength Setting (cont'd)

Table 229 — Output Driver DC Electrical Characteristics, Entire Operating Temperature Range

RON _{Nom}	Resistor	V _{Out}	Min	Nom	Max	Unit	NOTE
48 Ω	RON _{48Pd}	V _{OLdc} = 0.5 x V _{DD}	0.8	1.0	1.1	R _{ZQ} /5	1, 2
		V _{OMdc} = 0.8 x V _{DD}	0.9	1.0	1.1	R _{ZQ} /5	1, 2
		V _{OHdc} = 0.95 x V _{DD}	0.9	1.0	1.25	R _{ZQ} /5	1, 2
	RON _{48Pu}	V _{OLdc} = 0.5 x V _{DD}	0.9	1.0	1.25	R _{ZQ} /5	1, 2
		V _{OMdc} = 0.8 x V _{DD}	0.9	1.0	1.1	R _{ZQ} /5	1, 2
		V _{OHdc} = 0.95 x V _{DD}	0.8	1.0	1.1	R _{ZQ} /5	1, 2
34 Ω	RON _{34Pd}	V _{OLdc} = 0.5 x V _{DD}	0.8	1.0	1.1	R _{ZQ} /7	1, 2
		V _{OMdc} = 0.8 x V _{DD}	0.9	1.0	1.1	R _{ZQ} /7	1, 2
		V _{OHdc} = 0.95 x V _{DD}	0.9	1.0	1.25	R _{ZQ} /7	1, 2
	RON _{34Pu}	V _{OLdc} = 0.5 x V _{DD}	0.9	1.0	1.25	R _{ZQ} /7	1, 2
		V _{OMdc} = 0.8 x V _{DD}	0.9	1.0	1.1	R _{ZQ} /7	1, 2
		V _{OHdc} = 0.95 x V _{DD}	0.8	1.0	1.1	R _{ZQ} /7	1, 2
20 Ω	RON _{20Pd}	V _{OLdc} = 0.5 x V _{DD}	0.8	1.0	1.1	R _{ZQ} /12	1, 2
		V _{OMdc} = 0.8 x V _{DD}	0.9	1.0	1.1	R _{ZQ} /12	1, 2
		V _{OHdc} = 0.95 x V _{DD}	0.9	1.0	1.25	R _{ZQ} /12	1, 2
	RON _{20Pu}	V _{OLdc} = 0.5 x V _{DD}	0.9	1.0	1.25	R _{ZQ} /12	1, 2
		V _{OMdc} = 0.8 x V _{DD}	0.9	1.0	1.1	R _{ZQ} /12	1, 2
		V _{OHdc} = 0.95 x V _{DD}	0.8	1.0	1.1	R _{ZQ} /12	1, 2
14 Ω	RON _{14Pd}	V _{OLdc} = 0.5 x V _{DD}	0.8	1.0	1.1	R _{ZQ} /17	1, 2
		V _{OMdc} = 0.8 x V _{DD}	0.9	1.0	1.1	R _{ZQ} /17	1, 2
		V _{OHdc} = 0.95 x V _{DD}	0.9	1.0	1.25	R _{ZQ} /17	1, 2
	RON _{14Pu}	V _{OLdc} = 0.5 x V _{DD}	0.9	1.0	1.25	R _{ZQ} /17	1, 2
		V _{OMdc} = 0.8 x V _{DD}	0.9	1.0	1.1	R _{ZQ} /17	1, 2
		V _{OHdc} = 0.95 x V _{DD}	0.8	1.0	1.1	R _{ZQ} /17	1, 2
10 Ω	RON _{10Pd}	V _{OLdc} = 0.5 x V _{DD}	0.8	1.0	1.1	R _{ZQ} /24	1, 2
		V _{OMdc} = 0.8 x V _{DD}	0.9	1.0	1.1	R _{ZQ} /24	1, 2
		V _{OHdc} = 0.95 x V _{DD}	0.9	1.0	1.25	R _{ZQ} /24	1, 2
	RON _{10Pu}	V _{OLdc} = 0.5 x V _{DD}	0.9	1.0	1.25	R _{ZQ} /24	1, 2
		V _{OMdc} = 0.8 x V _{DD}	0.9	1.0	1.1	R _{ZQ} /24	1, 2
		V _{OHdc} = 0.95 x V _{DD}	0.8	1.0	1.1	R _{ZQ} /24	1, 2

Table 229 — Output Driver DC Electrical Characteristics, Entire Operating Temperature Range (cont'd)

RON _{Nom}	Resistor	V _{Out}	Min	Nom	Max	Unit	NOTE
14 Ω (for ALERT_n pulldown only)	RON _{ALERTPd}	V _{OLdc} = 0.5 x V _{DD}	0.8	1.0	1.1	R _{ZQ} /17	1, 2
		V _{OMdc} = 0.8 x V _{DD}	0.9	1.0	1.1	R _{ZQ} /17	1, 2
		V _{OHdc} = 0.95 x V _{DD}	0.9	1.0	1.25	R _{ZQ} /17	1, 2
240 Ω (for ALERT_n pullup only)	RON _{ALERTPu}	V _{OLdc} = 0.5 x V _{DD}	0.85	1.0	1.25	R _{ZQ}	1, 2
		V _{OMdc} = 0.8 x V _{DD}	0.85	1.0	1.1	R _{ZQ}	1, 2
		V _{OHdc} = 0.95 x V _{DD}	0.6	1.0	1.1	R _{ZQ}	1, 2
30 Ω (for QRST_n and BRST_n only)	RON _{RSTPd}	V _{OLdc} = 0.2 x V _{DD}	0.6	1.0	1.25	R _{ZQ} /8	1, 2
		V _{OMdc} = 0.5 x V _{DD}	0.75	1.0	1.25	R _{ZQ} /8	1, 2
		V _{OHdc} = 0.8 x V _{DD}	0.75	1.0	1.4	R _{ZQ} /8	1, 2
	RON _{RSTPu}	V _{OLdc} = 0.2 x V _{DD}	0.75	1.0	1.4	R _{ZQ} /8	1, 2
		V _{OMdc} = 0.5 x V _{DD}	0.75	1.0	1.25	R _{ZQ} /8	1, 2
		V _{OHdc} = 0.8 x V _{DD}	0.6	1.0	1.25	R _{ZQ} /8	1, 2
Mismatch between pull-up and pull-down, MM _{PuPd}		V _{OMdc} = 0.8 x V _{DD}	-10		10	%	1, 2, 3
Mismatch within component variation pull-up, MM _{Pudd}		V _{OMdc} = 0.8 x V _{DD}	0		10	%	1, 2, 4
Mismatch within component variation pull-down, MM _{Pddd}		V _{OMdc} = 0.8 x V _{DD}	0		10	%	1, 2, 4
NOTE 1 The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see Section 14.6, “Output Driver and Termination Resistor Temperature and Supply Voltage Sensitivity,” on page 239.							
NOTE 2 Pull-up and pull-down output driver impedances are recommended to be calibrated at 0.5 * V _{DD} (for RON _{RSTPd} and RON _{RSTPu}), or 0.8 * V _{DD} (for the other impedances). Other calibration schemes may be used to achieve the linearity spec shown above, e.g., calibration at 0.5 * V _{DD} and 0.95 * V _{DD} . RON variance range ratio to RON nominal value in a given component.							
NOTE 3 Measurement definition for mismatch between pull-up and pull-down, MM _{PuPd} : Measure RON _{Pu} and RON _{Pd} , both at 0.5 * V _{DD} (for RON _{RSTPd} and RON _{RSTPu}), or 0.8 * V _{DD} (for the other impedances) separately. Ron-nom is the nominal Ron value:							
$MM_{PuPd} = \frac{RON_{Pu} - RON_{Pd}}{RON_{Nom}} \times 100$							
NOTE 4 RON variance range ratio to RON nominal value in a given component:							
$MM_{Pudd} = \frac{RON_{PuMax} - RON_{PuMin}}{RON_{Nom}} \times 100$							
$MM_{Pddd} = \frac{RON_{PdMax} - RON_{PdMin}}{RON_{Nom}} \times 100$							

14.6 Output Driver and Termination Resistor Temperature and Supply Voltage Sensitivity

If temperature and/or voltage change after calibration, the tolerance limits widen according to Table 230 and Table 231.

Table 230 — Output Driver and Termination Resistor Sensitivity Definition

Resistor	Definition Point	Min	Max	Unit	Note
R_{ONPD}	$0.8 \times VDD$	$90 - (dR_{ONPD}dT \times \Delta T) - (dR_{ONPD}dV \times \Delta V)$	$110 + (dR_{ONPD}dT \times \Delta T) + (dR_{ONPD}dV \times \Delta V)$	%	1,2
R_{ONPU}	$0.8 \times VDD$	$90 - (dR_{ONPU} \times \Delta T) - (dR_{ONPU} \times \Delta V)$	$110 + (dR_{ONPU} \times \Delta T) + (dR_{ONPU} \times \Delta V)$	%	1,2
R_{IBT}	$0.8 \times VDD$	$90 - (dR_{IBT}dT \times \Delta T) - (dR_{IBT}dV \times \Delta V)$	$110 + (dR_{IBT}dT \times \Delta T) + (dR_{IBT}dV \times \Delta V)$	%	1,2,3
NOTE 1 $\Delta T = T - T(@ \text{Calibration})$, $\Delta V = V - V(@ \text{Calibration})$					
NOTE 2 $dR_{ONPD}dT$, $dR_{ONPD}dV$, $dR_{ONPU}dT$, $dR_{ONPU}dV$, $dR_{IBT}dV$, and $dR_{IBT}dT$ are not subject to production test but are verified by design and characterization.					
NOTE 3 3. This parameter applies to Input pins such as DCA[6:0], DPAR, DCS[1:0]_n, DCK_t/DCK_c, DERROR_IN_n, DLBD, and DLBS.					

Table 231 — Output Driver and Termination Resistor Temperature and Voltage Sensitivity

Symbol	Parameter	Min	Max	Unit
$dR_{ONPD}dT$	R_{ONPD} Temperature Sensitivity	0.00	0.1	%/°C
$dR_{ONPD}dV$	R_{ONPD} Voltage Sensitivity	0.00	0.1	%/mV
$dR_{ONPU}dT$	R_{ONPU} Temperature Sensitivity	0.00	0.1	%/°C
$dR_{ONPU}dV$	R_{ONPU} Voltage Sensitivity	0.00	0.1	%/mV
$dR_{IBT}dT$	R_{IBT} Temperature Sensitivity	0.00	0.1	%/°C
$dR_{IBT}dV$	R_{IBT} Voltage Sensitivity	0.00	0.1	%/mV

14.7 Output Clock Driver Characteristics

Table 232 — Output Clock Driver Characteristics at Application Mode (Frequency Band 1)

Symbol	Parameter	Conditions	DDR5-3200		DDR5-3600		DDR5-4000/4400/4800		DDR5-5200/5600		Unit	Note
			Min	Max	Min	Max	Min	Max	Min	Max		
t_{STAB01}	RCD Stabilization time from start of DCK toggle to PLL lock	DCK_t/DCK_c stable	-	3.5	-	3.5	-	3.5	-	3.5	μs	1
t_{STAB02}	RCD Stabilization time from control word write to PLL lock	DCK_t/DCK_c stable	-	25.0	-	25.0	-	25.0	-	25.0	μs	2
t_{STAB_DB}	Data Buffer Stabilization time	BCK_t/BCK_c stable	See DDR5DB02 Specification								μs	
t_{STAB_DBCMD}	LRDIMM stabilization time for commands not requiring use of data path	DCK_t/DCK_c stable	-	3.5	-	3.5	-	3.5	-	3.5	μs	3,4

Table 232 — Output Clock Driver Characteristics at Application Mode (Frequency Band 1) (cont'd)

Symbol	Parameter	Conditions	DDR5-3200		DDR5-3600		DDR5-4000/ 4400/4800		DDR5-5200/ 5600		Unit	Note
			Min	Max	Min	Max	Min	Max	Min	Max		
tSTAB_DBDATA	LRDIMM stabilization time for commands requiring use of data path	DCK_t/DCK_c stable	-	4.14	-	4.14	-	4.14	-	4.14	μs	4,5
tCKsk	Fractional Clock Output Skew		-	20	-	20	-	20	-	20	ps	6
tstaoff	Clock delay through the register between the input clock and output clock		Min: (t _{PD} (min) + 1/2 tCK); Max: (t _{PD} (max) + 1/2 tCK)								ps	7,8
tdynoff	Maximum re-driven dynamic clock offset		-	40	-	35	-	35	-	35	ps	9,10
NOTE 1 At Down-bin Data Rate (1980 MT/s ≤ f ≤ 2100 MT/s), the max value of tSTAB01 is 10μs.												
NOTE 2 This parameter applies to control word writes to RW05/RW06 and Bit OP0 in RW00. Since application of RW05/RW06 is postponed in the frequency change sequence defined in Section 5.4.2, a tSTAB01 waiting time is always sufficient after exit from Self Refresh with Clock Stop (i.e., tSTAB02 is not required). See timing diagrams in Figure 35 on page 48 and Figure 36 on page 49.												
NOTE 3 Measured from start of DCK after Exit from Self Refresh with Clock Stop to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface but does not require the use of data path (e.g., SRX, NOP, etc.).												
NOTE 4 Applies only to Self Refresh Exit without frequency change.												
NOTE 5 Measured from start of DCK after Exit from Self Refresh with Clock Stop to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface and requires the use of data path (e.g., WR, RD).												
NOTE 6 This skew represents the absolute output clock skew and contains the pad skew and package skew (See Figure 113). This parameter is specified for the clock pairs on each side of the register independently. The skew is applicable to left side clock pairs between QACK_A_t/QACK_A_c and QCCK_A_t/QCCK_A_c, as well as right side of the clock pairs between QACK_B_t/QACK_B_c and QCCK_B_t/QCCK_B_c. Similarly, to left side clock pairs between QBCK_A_t/QBCK_A_c and QDCK_A_t/QDCK_A_c, as well as right side of the clock pairs between QBCK_B_t/QBCK_B_c and QDCK_B_t/QDCK_B_c. This is not a tested parameter and has to be considered as a design goal only.												
NOTE 7 RW05/RW06 must be set to correct speed and the device must run at one of the speed nodes defined in Table 199 on page 191. See Figure 111 and Figure 112.												
NOTE 8 This measures the delay from the rising differential input clock which samples incoming DCA to the rising differential output clock that will be used to sample the same QCA data. The parameter describes the maximum and minimum QxCK clock t _{PD} .												
NOTE 9 V _{DD} measurement DC bandwidth is limited to 20 MHz.												
NOTE 10 See Figure 111 and Figure 112.												

14.7 Output Clock Driver Characteristics (cont'd)

Table 233 — Output Clock Driver Characteristics at Test Mode (Frequency Band 2)

Symbol	Parameter	Conditions	Test Frequency (140 to 990 MHz)		Unit	Note
			Min	Max		
t _{STAB01}	RCD Stabilization time from start of DCK toggle to PLL lock	DCK_t/DCK_c stable	-	50	μs	
t _{STAB02}	RCD Stabilization time from control word write to PLL lock	DCK_t/DCK_c stable	-	50	μs	1
t _{CKsk}	Fractional Clock Output Skew		-	100	ps	2
t _{stao}	Clock delay through the register between the input clock and output clock		Min: (t _{PD} (min) + 1/2 t _{CK}); Max: (t _{PD} (max) + 1/2 t _{CK})		ps	3,4
t _{dyno}	Maximum re-driven dynamic clock offset		-	500	ps	5,6

NOTE 1 This parameter applies to control word writes to **RW05** and Bit OP0 in **RW00**.

NOTE 2 This skew represents the absolute output clock skew and contains the pad skew and package skew (See Figure 113). This parameter is specified for the clock pairs on each side of the register independently. The skew is applicable to left side clock pairs between QACK_A_t/QACK_A_c and QCCK_A_t/QCCK_A_c, as well as right side of the clock pairs between QACK_B_t/QACK_B_c and QCCK_B_t/QCCK_B_c. Similarly, to left side clock pairs between QBCK_A_t/QBCK_A_c and QDCK_A_t/QDCK_A_c, as well as right side of the clock pairs between QBCK_B_t/QBCK_B_c and QDCK_B_t/QDCK_B_c. This is not a tested parameter and has to be considered as a design goal only.

NOTE 3 **RW05** must not be set to enable PLL bypass mode and the device must run within the Test Frequency range defined in Table 199 on page 191. See Figure 111 and Figure 112.

NOTE 4 This measures the delay from the rising differential input clock which samples incoming DCA to the rising differential output clock that will be used to sample the same QCA data. The parameter describes the maximum and minimum QxCK clock t_{pp}.

NOTE 5 V_{DD} measurement DC bandwidth is limited to 20 MHz.

NOTE 6 See Figure 111 and Figure 112.

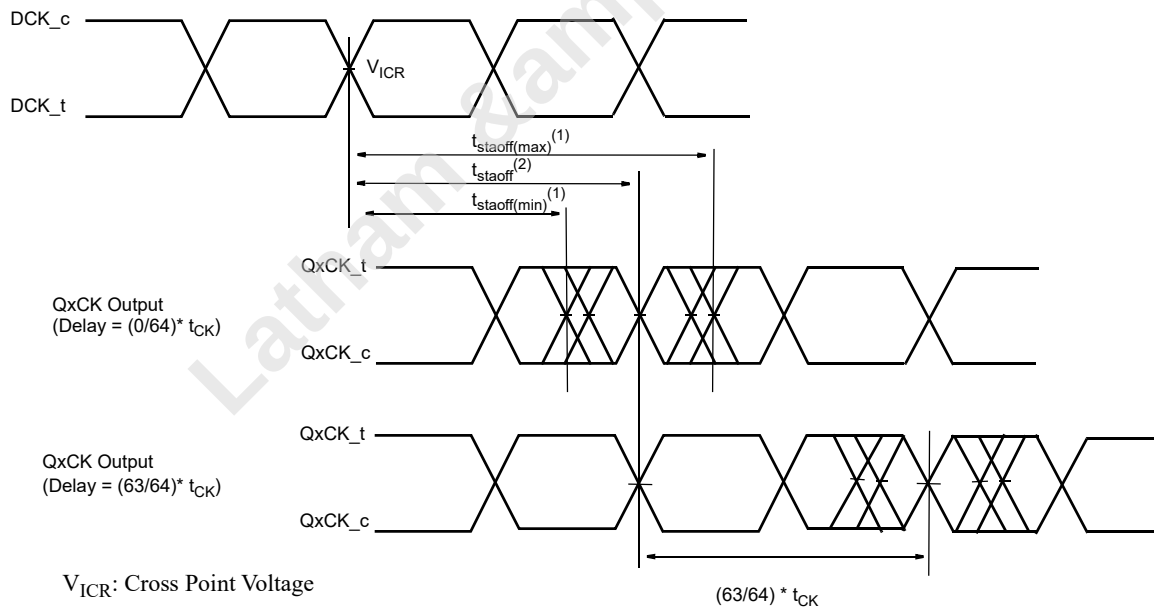


Figure 111 — Definition for t_{stao} and t_{dyno} in DDR Mode

- NOTE 1 t_{stao(max/min)} = propagation delay specification limits for clock signal (falling DCK input clock edge to rising QxCK output clock edge for corresponding data cycles) over process, voltage and temperature.
- NOTE 2 t_{stao} = measured propagation delay for clock signal (falling DCK input clock edge to rising QxCK output clock edge for corresponding data cycles).
- NOTE 3 t_{dyno} = maximum propagation delay variation over voltage and temperature within t_{stao} window. This includes all sources of jitter and drift (e.g. thermal noise, supply noise, voltage/temperature drift, SSC tracking, SSO, etc.) except reference clock noise.

14.7 Output Clock Driver Characteristics (cont'd)

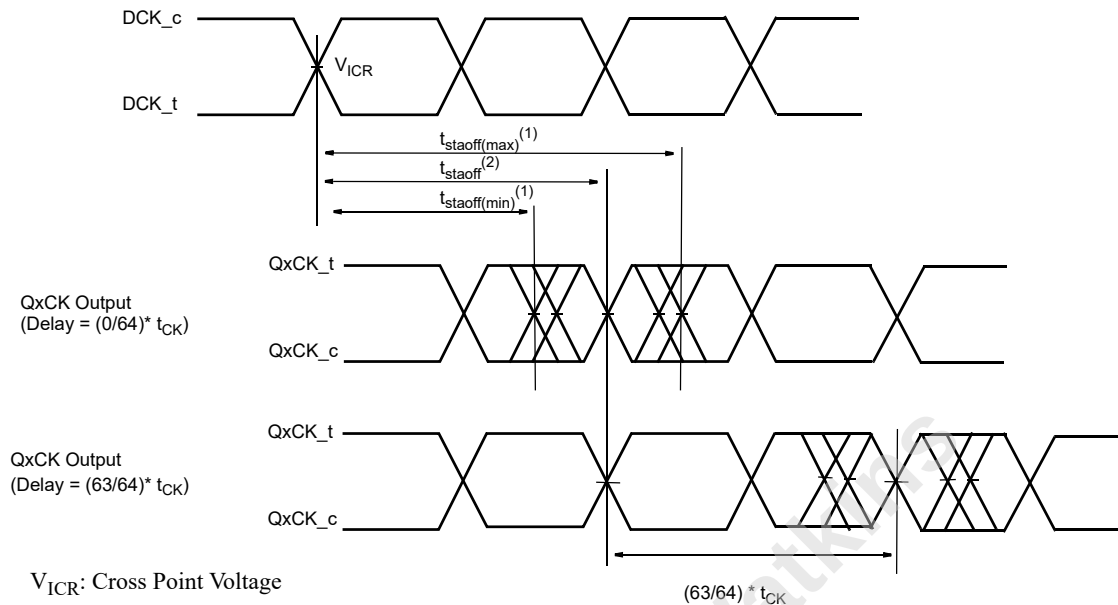


Figure 112 — Definition for t_{staoff} and t_{dynoff} in SDR Mode

- NOTE 1 $t_{staoff(max/min)}$ = propagation delay specification limits for clock signal (rising DCK input clock edge to rising QxCK output clock edge for corresponding data cycles) over process, voltage and temperature.
- NOTE 2 t_{staoff} = measured propagation delay for clock signal (rising DCK input clock edge to rising QxCK output clock edge for corresponding data cycles).
- NOTE 3 t_{dynoff} = maximum propagation delay variation over voltage and temperature within t_{staoff} window. This includes all sources of jitter and drift (e.g. thermal noise, supply noise, voltage/temperature drift, SSC tracking, SSO, etc.) except reference clock noise.

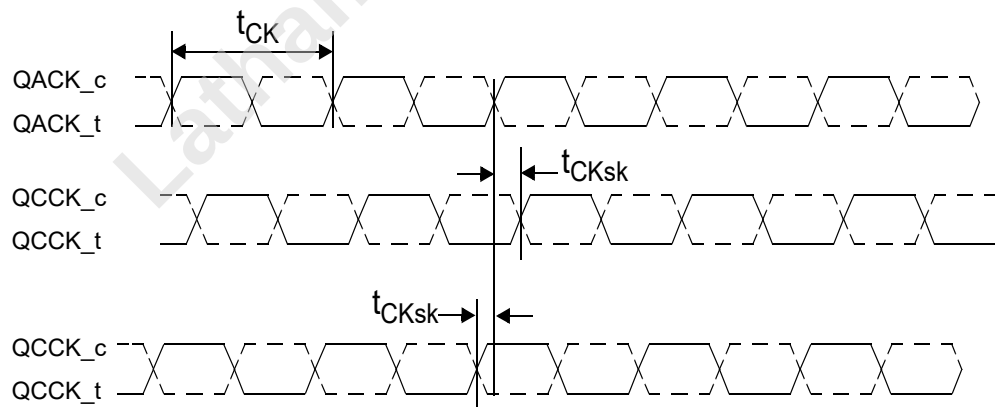


Figure 113 — Clock Output (QxCK) Skew

14.8 Differential Output Clock Driver Jitter Characteristics

14.8.1 Overview

The Random Jitter (Rj) specified is a random jitter meeting Gaussian distribution. The Deterministic Jitter (Dj) specified is bounded. The DDR5 device output jitter must not exceed maximum values specified in Table 215.

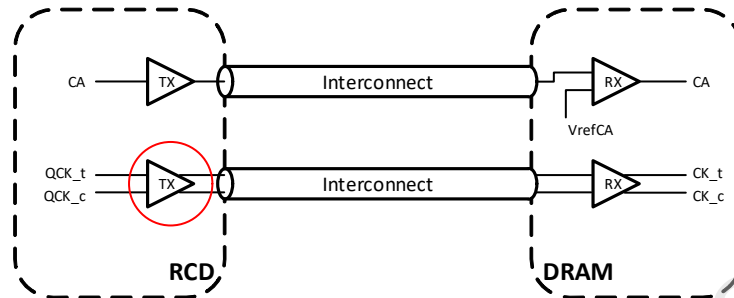


Figure 114 — Example of QCK TX Interconnect

14.8.2 Output Clock Jitter Parameters

Table 234 — Tx QCK Jitter Specifications for DDR5-3200 to 5600¹

Symbol	Parameter	DDR5-3200 to 4800		DDR5-5200 to 5600		Unit ²	NOTE
		Min	Max	Min	Max		
tTx_QCK_1UI_Rj_NoBUJ	Rj value of 1-UI Jitter without BUJ	-	0.0035	-	0.003	UI (RMS)	3,4,5,10,11
tTx_QCK_1UI_Dj_NoBUJ	Dj pp value of 1-UI Jitter without BUJ	-	0.025	-	0.025	UI	4,5,6,10,11
tTx_QCK_NUI_Rj_NoBUJ	Rj value of N-UI Jitter without BUJ	-	0.0035	-	0.003	UI (RMS)	3,5,7,9,10,11
tTx_QCK_NUI_Dj_NoBUJ	Dj pp value of N-UI Jitter without BUJ	-	0.07	-	0.07	UI	4,5,8,9,10,11
NOTE 1 [BUJ = Bounded Uncorrelated Jitter; DCD = Duty Cycle Distortion; Dj = Deterministic Jitter; Rj = Random Jitter; pp = Peak-to-Peak]							
NOTE 2 Unit UI = tCK(avg).min/2							
NOTE 3 The test should be run under quiet conditions with single output switching.							
NOTE 4 When there is no socket in transmitter measurement setup, in many cases, the contribution of the cross-talk is not significant or cannot be estimated within tolerable error even with all the transmitter lanes sending patterns. When a socket is present, such as DUT being DRAM component, the contribution of the cross-talk could be significant. To minimize the impact of crosstalk on the measurement results, a small group of selected lanes in the vicinity of the lane under test may be turned off (sending DC), while the remaining Tx lanes send patterns to the corresponding Rx receivers so as to excite realistic on-die noise profile from device switching. Note that there may be cases when one of Dj and Rj specs is met and another violated in which case the signaling analysis should be run to determine link feasibility. Rj RMS value of 1-UI jitter without BUJ. This extraction is to be done after software correction of DCD.							
NOTE 5 This test should be done in typical temperature and voltage conditions (i.e., V _{DD} = 1.1 V, 25 °C).							
NOTE 6 Dj pp value of 1-UI jitter (after software assisted DCC). Without BUJ. Dj indicates Djdd of dual-Dirac fitting.							
NOTE 7 Rj RMS value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 1 < N < 4. This extraction is to be done after software correction of DCD.							
NOTE 8 Dj pp value of N-UI jitter without BUJ, but on-die system like noise present. Evaluated for 1 < N < 4. Dj indicates Djdd of dual-Dirac fitting, after software correction of DCD.							
NOTE 9 When frequency is less than or equal to 4800 MT/s 1 < N < 4. When frequency is greater than 4800 MT/s up to 5600 MT/s 1 < N < 6.							
NOTE 10 Measurements are done with default R-on and fast slew rate settings for all parameters in Table 1.							
NOTE 11 All parameters in Table 234 may not be subject to production test.							

14.9 ALERT_n Output Driver DC Electrical Characteristic

Table 235 — ALERT_n Output Driver DC Electrical Characteristics

Symbol	Parameter	Applicable Signals	Condition	Min	Nom	Max	Unit
I _{OL}	LOW-level output current ALERT _n	Measured at V _{OL} of 0.4 V	-	25	-	-	mA
V _{OL}	Output LOW voltage ALERT _n	Measured at I _{OL} = 25 mA	-	-	-	0.4	V

15 IDD Specification Parameters

15.1 IDD Specification Parameters and Test Conditions

In this chapter, IDD measurement conditions such as test load and patterns are defined. Figure 115 shows the setup and test load for IDD measurements.

- IDD currents (such as IDD3NA, IDD3NB, IDD3P1, IDD3P2, IDD4A, IDD4B, IDD6R and IDD6S) are measured as time-averaged currents with all V_{DD} balls of the DDR5RCD02 under test tied together.
- IDD currents can be measured for each speed bin. Each measurement shall use the minimum tCK (avg) for each speed bin.
- **ATTENTION:** IDD values cannot be directly used to calculate IO power of the DDR5RCD02. They can be used to support correlation of simulated IO power to actual IO power as outlined in Figure 116.

For IDD measurements, the following definitions apply:

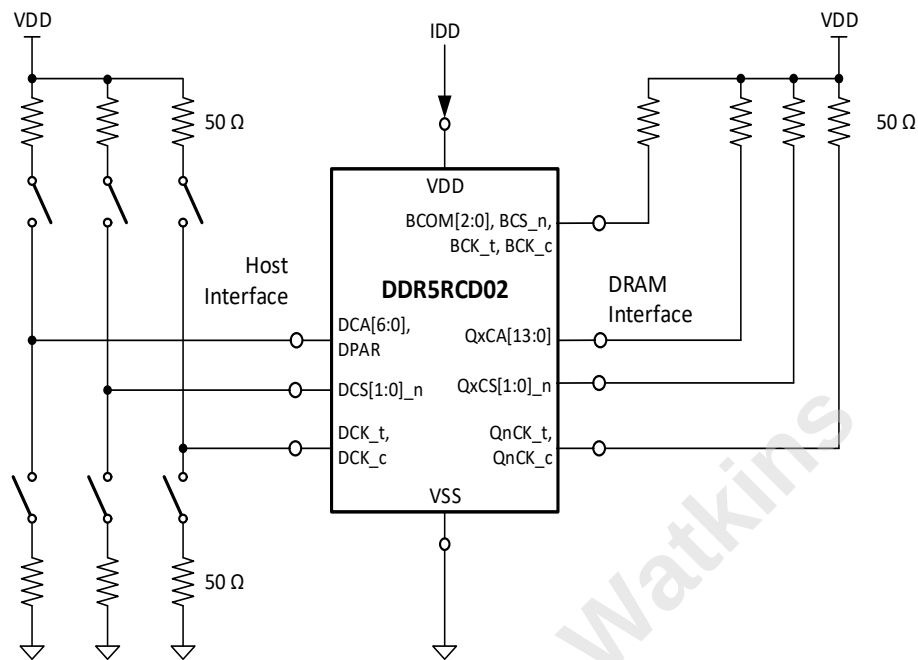
- “0” and “LOW” is defined as $V_{IN} \leq V_{IL(AC).max}$.
- “1” and “HIGH” is defined as $V_{IN} \geq V_{IH(AC).min}$.
- Basic IDD Measurement Conditions are described in Table 236.
- Detailed IDD Measurement-Loop Patterns are described in Table 237 and Table 238.

IDD Measurements are done after properly initializing the DDR5RCD02. This includes but is not limited to setting

- All four clock outputs enabled per channel in RW08;
- IBT enabled in RW10
- All output driver strength set to RZQ/17;
- Parity checking disabled in RW01[0];
- SidebandBus interface enabled;
- QCA13 enabled;
- 60 Ω IBT enabled for DCA[6:0], DPAR, DCS[1:0].

ATTENTION: The IDD Measurement-Loop Patterns need to be executed at least one time before actual IDD measurement is started.

15.1 IDD Specification Parameters and Test Conditions (cont'd)



Note: Host drives HIGH and drives LOW

Figure 115 — Measurement Setup and Test Load for IDD Measurements

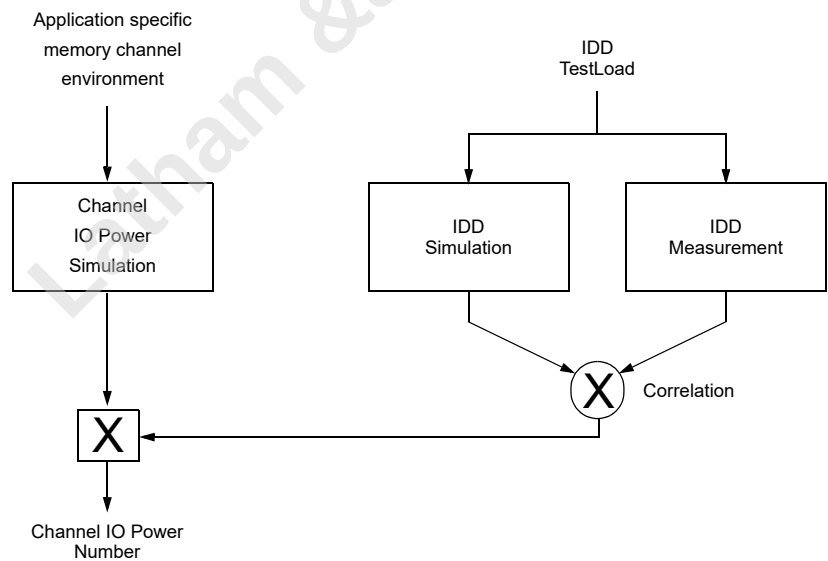


Figure 116 — Correlation from Simulated Channel IO Power to Actual Channel IO Power Supported by IDD Measurement

15.1 IDD Specification Parameters and Test Conditions (cont'd)

Table 236 — Basic IDD Measurement Conditions¹

Symbol	Description
IDD3NA	Active Idle Current DB interface: Disabled; Q[D:A]CK: Enabled; Data IO & Pattern Details: see Table
IDD3NB	Active Idle Current DB interface: Enabled; Q[D:A]CK: Enabled; Data IO & Pattern Details: see Table on page 247
IDD3P1	Power Down with ODT Control DB interface: Enabled; Q[D:A]CK: Enabled; Data IO & Pattern Details: see Table on page 247
IDD3P2	Power Down without ODT Control DB interface: Enabled; Q[D:A]CK: Enabled; Data IO & Pattern Details: see Table on page 247
IDD4A	Active Current DB interface: Disabled; Q[D:A]CK: Enabled; Data IO & Pattern Details: see Table 238 on page 247
IDD4B	Active Current DB interface: Enabled; Q[D:A]CK: Enabled; Data IO & Pattern Details: see Table 238 on page 247
IDD6R	Static Reset Current DRST_n: LOW; DCK_t and DCK_c: LOW; DCS: LOW; Outputs Hi-Z (except QRST, BRST, QCS)
IDD6S	Clock Stopped Power Down Current DRST_n: HIGH after being LOW for t_{RINT1} with described input conditions (i.e., DCK_t = DCK_c = DCS_n = LOW) applied; DCK_t and DCK_c: LOW; DCS: LOW; Outputs Hi-Z (except QCS: LOW, BCK: HIGH or HI-Z ² , BCS & BCOM: HIGH)
NOTE 1 All tests use termination setup depicted in Figure 115	
NOTE 2 BCK output is HIGH or HI-Z depending on vendor implementation.	

15.1 IDD Specification Parameters and Test Conditions (cont'd)

Table 237 — IDD3NA, IDD3NB, IDD3P1, and IDD3P2 Measurement-Loop Pattern

DCCK_t, DCK_c	Cycle Number	UI Number	Command	DCS[0]_n	DCS[1]_n	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	DPAR
Toggling	0	0	None	1	1	1	1	1	1	1	1	1	1
		1		1	1	1	1	1	1	1	1	1	1

Table 238 — IDD4A, IDD4B Measurement-Loop Pattern

DCCK_t, DCK_c	Sub-Loop	Cycle Number	UI Number	Command	DCS[0]_n	DCS[1]_n	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	DPAR
Toggling	0	0	0	RD	0	1	1	0	1	1	1	1	0	1
			1		1	1	1	0	1	0	0	0	0	0
		1	0		1	1	1	0	1	0	1	0	1	0
			1		1	1	1	0	1	1	1	1	0	1
		2	0	ACT	1	1	1	1	1	1	1	1	1	1
			1		1	1	1	1	1	1	1	1	1	1
		3	0		0	1	0	0	0	1	0	1	0	0
			1		1	1	1	0	1	0	0	0	0	0
		4	0		1	1	1	0	1	0	1	0	1	0
			1		1	1	0	1	0	1	0	1	0	1
		5	0		1	1	1	1	1	1	1	1	1	1
			1		1	1	1	1	1	1	1	1	1	1
		6	0		1	1	1	1	1	1	1	1	1	1
			1		1	1	1	1	1	1	1	1	1	1
		7	0		1	1	1	1	1	1	1	1	1	1
			1		1	1	1	1	1	1	1	1	1	1
	1	8	0	RD	1	0	1	0	1	1	1	1	0	1
			1		1	1	1	0	1	0	0	0	0	0
		9	0		1	1	1	0	1	0	1	0	1	0
			1		1	1	1	0	1	1	1	1	0	1
		10	0		1	1	1	1	1	1	1	1	1	1
			1		1	1	1	1	1	1	1	1	1	1
		11	0		1	0	0	0	0	1	0	1	0	0
			1		1	1	1	0	1	0	0	0	0	0
		12	0		1	1	1	0	1	0	1	0	1	0
			1		1	1	0	1	0	1	0	1	0	1
		13	0		1	1	1	1	1	1	1	1	1	1
			1		1	1	1	1	1	1	1	1	1	1
		14	0		1	1	1	1	1	1	1	1	1	1
			1		1	1	1	1	1	1	1	1	1	1
		15	0		1	1	1	1	1	1	1	1	1	1
			1		1	1	1	1	1	1	1	1	1	1

16 Input/Output Capacitance

Table 239 — Silicon Pad I/O Capacitance Values

Symbol	Parameter	DDR5 - 3200-5600		Unit	Notes
		Min	Max		
C_I	Input capacitance, Data inputs	0.2	0.95	pF	1, 2
C_O	Output capacitance	0.45	2.5	pF	1,3
C_{CK}	Input capacitance, DCK_t, DCK_c	0.2	0.95	pF	1
C_{DCK}	Input capacitance delta DCK_t and DCK_c	-	0.1	pF	1, 4
C_{IDCA}	Delta capacitance over all DCA inputs, per sub-channel	-	0.2	pF	1
C_{IDCS}	Delta capacitance over all DCS inputs, per sub-channel	-	0.2	pF	1
C_{ILB}	Input capacitance DLB	0.2	1.0	pF	1
C_{OLB}	Output capacitance QLB	0.45	2.0	pF	1
C_{IR}	Input capacitance, DRST_n	0.5	2	pF	1, 5
C_{OR}	Output Capacitance of QRST_n & BRST_n	0.4	2.0	pF	1
C_{ALERT}	Output capacitance of ALERT_n	0.4	2.5	pF	1
C_{DERROR_IN}	Input capacitance of DERROR_IN_n	0.4	1.5	pF	1
NOTE 1 This parameter does not include package capacitance.					
NOTE 2 Data inputs are DCAy_N, DPAR_N, DCSy_N_n.					
NOTE 3 Applies to QxCAY_N, QxCSy_N_n,, BCS_N_n, BCOMy_N, BCK_N_t, BCK_N_c, QxCKy_N_t, and QxCKy_N_c.					
NOTE 4 Absolute value of DCK_t - DCK_c					
NOTE 5 Parameter specified at VI = VDD or VSS; VDD = 1.1 V					

16 Input/Output Capacitance (cont'd)

Table 240 — Package Electrical Parameters

Symbol	Parameter	DDR5 - 3200 to 5600		Unit	Notes
		Min	Max		
ZI _{DCS}	Input DCS pins Zpkg	40	70	Ω	1, 2, 4
TdI _{DCS}	Input DCS pins Pkg Delay	14	55	ps	1, 3, 4
DZI _{DCS}	Delta DCS pins Zpkg	-	15	Ω	1, 2, 4, 5
DTdI _{DCS}	Delta DCS pins Pkg Delay	-	20	ps	1, 3, 4, 6
ZI _{DCA}	Input DCA pins Zpkg	40	70	Ω	1, 2, 7
TdI _{DCA}	Input DCA pins Pkg Delay	14	55	ps	1, 3, 7
DZI _{DCA}	Delta DCA pins Zpkg	-	15	Ω	1, 2, 7, 8
DTdI _{DCA}	Delta DCA input pins Pkg Delay	-	20	ps	1, 3, 7, 9
ZO _{QCS}	Output QCS pins Zpkg	40	70	Ω	1, 2, 10
TdO _{QCS}	Output QCS pins Pkg Delay	14	65	ps	1, 3, 10
DZO _{QCS}	Delta QCS pins Zpkg	-	15	Ω	1, 2, 10, 11
DTdO _{QCS}	Delta QCS input pins Pkg Delay	-	20	ps	1, 3, 10, 12
ZO _{QCA}	Output QCA pins Zpkg	40	70	Ω	1, 2, 13
TdO _{QCA}	Output QCA pins Pkg Delay	14	65	ps	1, 3, 13
DZO _{QCA}	Delta QCA pins Zpkg	-	15	Ω	1, 3, 13, 14
DTdO _{QCA}	Delta QCA output pins Pkg Delay	-	20	ps	1, 3, 13, 15
ZO _{BC}	Output QBC pins Zpkg	40	70	Ω	1, 3, 16
TdO _{BC}	Output BC pins Pkg Delay	14	65	ps	1, 3, 16
DZO _{BC}	Delta BC pins Zpkg	-	15	Ω	1, 3, 16, 17
DTdO _{BC}	Delta BC output pins Pkg Delay	-	20	ps	1, 3, 16, 18
ZI _{DCK}	Input DCK pins Zpkg	40	70	Ω	1, 2, 19
TdI _{DCK}	Input DCK pins Pkg Delay	14	55	ps	1, 3
ZO _{QCK}	Output QnCK pins Zpkg	40	70	Ω	1, 2, 19
TdO _{QCK}	Output QnCK pins Pkg Delay	14	65	ps	1, 3
DTdO _{QCK}	Delta Delay between all QxCKy_N and BCK_N for a sub-channel	-	20	ps	12, 14
DZI _{DCK}	Delta Zpkg DCK_t and DCK_c	-	10	Ω	1, 2, 19, 20
DTdI _{DCK}	Delta Delay DCK_t and DCK_c	-	5	ps	1, 3, 21
DZO _{QCK}	Delta Zpkg QxCK_t and QxCK_c	-	10	Ω	1, 2, 10, 19, 22
DTdO _{QCK}	Delta Delay QxCK_t and QxCKn_c	-	5	ps	1, 3, 11, 23
ZO _{BCK}	Output BCK pins Zpkg	40	70	Ω	1, 2
TdO _{BCK}	Output BCK pins Pkg Delay	14	65	ps	1, 3
DZO _{BCK}	Delta Zpkg BCK_t and BCK_c	-	10	Ω	1, 2
DTdO _{BCK}	Delta Delay BCK_t and BCK_c	-	5	ps	1, 3
ZO _{ZQ}	Output ZQCAL Zpkg	30	70	Ω	1, 2
ZI _{DLB}	Input DLB pins Zpkg	40	70	Ω	1, 2
TdI _{DLB}	Input DLB pins Pkg Delay	14	75	ps	1, 3
ZO _{QLB}	Output QLB pins Zpkg	40	70	Ω	1, 2
TdO _{QLB}	Output QLB pins Pkg Delay	14	50	ps	1, 3
ZO _{ALERT}	Output ALERT_n Zpkg	40	70	Ω	1, 2
TdO _{ALERT}	Output ALERT_n Pkg Delay	14	75	ps	1, 3
ZI _{ERROR_IN}	Input Error in Zpkg	40	70	Ω	1, 2

Table 240 — Package Electrical Parameters (cont'd)

Symbol	Parameter	DDR5 - 3200 to 5600		Unit	Notes
		Min	Max		
TdI _{DERROR_IN}	Input Error in Pkg Delay	14	55	ps	1,3
ZO _{QRST}	Output QRST Zpkg	40	70	Ω	1,2
TdO _{QRST}	Output QRST Pkg Delay	14	55	ps	1,3
ZI _{DRST}	Input DRST Zpkg	40	70	Ω	1,2
TdI _{DRST}	Input DRST Pkg Delay	14	55	ps	1,3
ZI _{SDA_SCL}	Input SDA/SCL Zpkg	40	70	Ω	1,2
TdI _{SDA_SCL}	Input SDA/SCL Pkg Delay	14	55	ps	1,3
NOTE 1 This parameter is not subject to production test. It is verified by design and characterization. The package parasitics (L and C) are determined using package only samples. Package capacitance and inductance are computed from S-parameters. The capacitance is derived with V _{DD} , V _{SS} shorted and all other signals floating. The inductance is derived with V _{DD} , V _{SS} shorted and all other signals shorted at the die side (not pin).					
NOTE 2 Package only impedance (Zpkg) is calculated based on the computed Lpkg and Cpkg total for a given pin where: Zpkg (total per pin) = SQRT(Lpkg/Cpkg) (Lpkg and Cpkg for trace only, i.e. without including the ball).					
NOTE 3 Package only delay (Tdpkg) is calculated based on computed Lpkg and Zpkg total for a given pin where: Tdpkg (total per pin) = SQRT(Lpkg*Cpkg) (Lpkg and Cpkg for trace only, i.e. without including the ball).					
NOTE 4 This value applies to DCS.					
NOTE 5 Absolute value of MAX(ZI _{DCS})-MIN(ZI _{DCS}) per sub-channel.					
NOTE 6 Absolute value of MAX(TdI _{DCS})-MIN(TdI _{DCS}) per sub-channel.					
NOTE 7 This value applies to DCA and DPAR.					
NOTE 8 Absolute value of MAX(ZI _{DCA})-MIN(ZI _{DCA}) per sub-channel.					
NOTE 9 Absolute value of MAX(TdI _{DCA})-MIN(TdI _{DCA}) per sub-channel.					
NOTE 10 This value applies to QxCS.					
NOTE 11 Absolute value of MAX(ZI _{QCS})-MIN(ZI _{QCS}) per sub-channel.					
NOTE 12 Absolute value of MAX(TdI _{QCS})-MIN(TdI _{QCS}) within A or B output group in sub-channel.					
NOTE 13 This value applies to QxCA.					
NOTE 14 Absolute value of MAX(ZI _{QCA})-MIN(ZI _{QCA}) per sub-channel.					
NOTE 15 Absolute value of MAX(TdI _{QCA})-MIN(TdI _{QCA}) within A or B output group in sub-channel.					
NOTE 16 This value applies to BCOM, BCS, and BRST.					
NOTE 17 Absolute value of MAX(ZI _{BC})-MIN(ZI _{BC}) per sub-channel.					
NOTE 18 Absolute value of MAX(TdI _{BC})-MIN(TdI _{BC}) per sub-channel.					
NOTE 19 Single-ended impedance.					
NOTE 20 Absolute value of ZIDCK _t - ZIDCK _c .					
NOTE 21 Absolute value of TdIDCK _t - TdIDCK _c .					
NOTE 22 Absolute value of ZOQxCK _{N_t} - ZOQxCK _{N_c} .					
NOTE 23 Absolute value of TdIQxCK _{N_t} - TdIQxCK _{N_c} .					

16.1 Electrostatic Discharge Sensitivity Characteristics

Table 241 — Electrostatic Discharge Sensitivity Characteristics

Parameter	Symbol	Min	Max	Unit	Notes
Human body model (HBM)	ESD _{HBM}	1000	-	V	1, 2
Charged-device model (CDM)	ESD _{CDM}	250	-	V	1, 3
NOTE 1 State-of-the-art basic ESD control measures have to be in place when handling devices.					
NOTE 2 Refer to JEDEC / ESDA Joint Standard JS-001 for measurement procedures.					
NOTE 3 Refer to ANSI / ESDA / JEDEC Joint Standard JS-002 f for measurement procedures.					

17 SidebandBus Interface Electrical and Timing Specifications

This clause summarizes the operating and measurement conditions, and the DC and AC characteristics of the DDR5RCD02 SidebandBus interface. The parameter in the DC and AC Characteristics tables that follow are derived from tests performed under the Measurement Conditions summarized in the relevant tables. Designers should check that the operating conditions in their circuits match the measurement conditions when relying on the quoted parameters.

17.1 Operating Conditions

Table 242 — Operating Conditions for the SidebandBus Interface

Symbol	Parameter	Min	Typ	Max	Unit
V_{DDIO}	DDR5RCD02 Sideband Interface I/O Supply Voltage	0.95	1.0	1.05	V

17.2 Measurement Test Conditions

Table 243 — AC Measurement Conditions for SidebandBus interface

Symbol	Parameter	Min	Max	Unit
C_L	Load capacitance	40		pF
	Input rise and fall times (Open Drain)	--	120	ns
	Input rise and fall times (Push-Pull)	--	5	ns
	Input Signal (Swing) Levels	0.2 to 0.8		V
	Input Levels for Timing Reference	0.3 to 0.7		V

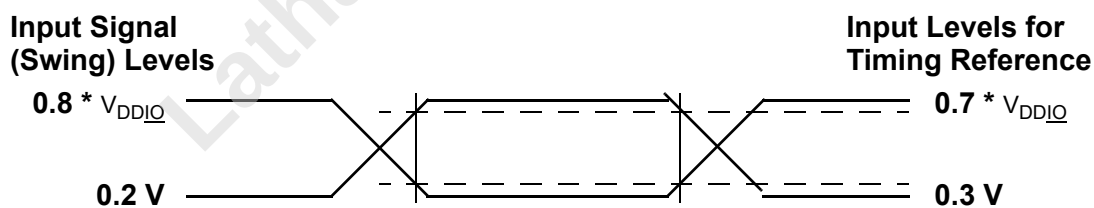


Figure 117 — AC Measurement I/O Waveform

17.3 DC Electrical Characteristics

Table 244 — Output Ron Spec

Symbol	Parameter	Min	Max	Unit	Notes
R_{ON}	SDA Output Pullup and Pulldown Driver Impedance	20	100	Ω	1
NOTE 1 Pull-down $R_{on} = V_{out}/I_{out}$; Pull-up $R_{on} = (V_{DDIO} - V_{out})/I_{out}$					

Table 245 — Input Parameters for SidebandBus interface

Symbol	Parameter ^{1,2}	Test Condition	Min	Max	Unit
C_{IN}	Input capacitance (SDA)	--	--	5	pF
C_{IN}	Input capacitance (SCL)	--	--	5	pF
NOTE 1 $T_A = 25^\circ\text{C}$, $f = 1000\text{ kHz}$					
NOTE 2 Verified by design and characterization, not necessarily tested on all devices					

Table 246 — DC Characteristics for SidebandBus Interface

Symbol	Parameter ^{1,2}	Test Condition (in addition to those in Table 242 on page 251)	Min	Max	Unit
I_{LI}	Input leakage current (SCL, SDA)	$V_{IN} = V_{SS}$ or V_{DDIO}	--	± 5	μA
I_{LO}	Output leakage current	$V_{OUT} = V_{SS}$ or V_{DDIO} ; SDA in Hi-Z	--	± 5	μA
I_{DDIO}	Supply current	$V_{DDIO} = 1.0\text{ V}$, $f_C = 12.5\text{ MHz}$ (rise/fall time $< 5\text{ ns}$)	--	5	mA
V_{IL}	Input LOW voltage (SCL, SDA)	--	-0.35	0.3	V
V_{IH}	Input HIGH voltage (SCL, SDA)	--	0.7	1.3	V
V_{OL}	Output LOW voltage (SDA)	3-mA sink current	--	0.3	V
V_{OH}	Output HIGH voltage (SDA)	3-mA source current	$V_{DDIO} - 0.3$	--	V
I_{OL}	LOW-level output current	$V_{OL} = 0.3\text{ V}$	3	--	mA
I_{OH}	HIGH-level output current	$V_{OH} = V_{DDIO} - 0.3$	--	-3	mA
Slew_Rate	Output Slew Rate (SDA)	See NOTE 3	0.1	1.0	V/ns
NOTE 1 Undershoot might occur. It should be limited by the Absolute Maximum DC Ratings.					
NOTE 2 Overshoot might occur. It should be limited by Absolute Maximum DC Ratings					
NOTE 3 Output slew rate is guaranteed by design and/or characterization. The output slew rate reference load is shown in Figure 123 and Figure 122 shows the timing measurement points.					

17.4 AC Electrical Characteristics

Table 247 — AC Characteristics for SidebandBus Interface

Symbol	Parameter	I ² C Mode (Open Drain) ¹		I3C Basic Push-Pull ² (12.5 MHz)		Unit
		Min	Max	Min	Max	
f _{SCL}	Clock frequency	0.01	1	0.01	12.5	MHz
t _{HIGH}	Clock pulse width HIGH time	260	--	35	--	ns
t _{LOW} ³	Clock pulse width LOW time	500	--	35	--	ns
t _{TIMEOUT} ^{4,5}	Detect clock LOW timeout	10	50	10	50	ms
t _R ^{6,7,8}	SDA rise time	--	120	--	5	ns
t _F ^{6,7,8}	SDA fall time	--	120	--	5	ns
t _{SU:DAT} ^{6,9}	Data in setup time	50	--	8	--	ns
t _{HD:DI} ^{6,9}	Data in hold time	0	--	3	--	ns
t _{SU:STA} ^{6,10}	START condition setup time	260	--	12	--	ns
t _{HD:STA} ⁶	START condition hold time	260	--	30	--	ns
t _{SU:STO} ⁶	STOP condition setup time	260	--	12	--	ns
t _{BUF} ⁶	Time between STOP Condition and next START Condition	500	--	500	--	ns
t _{HD:DAT} ¹¹	Data out hold time	0.5	350	N/A	N/A	ns
t _{DOUT} ^{12,13}	SCL Falling Clock In to Valid SDA Data Out Time	N/A	N/A	0.5	12	ns
t _{DOFFT} ^{12,14}	SCL Rising Clock In to Target SDA Output Off	N/A	N/A	0.5	12	ns
t _{DOFFC} ^{12,15}	SCL Rising Clock In to Controller SDA Output Off	N/A	N/A	0.5	12	ns
t _{CL_r_DAT_f} ¹⁶	SCL Rising Clock In to Controller Driving Data Signal LOW	N/A	N/A	40	--	ns
t _{AVAIL}	Bus Available Time (no edges seen on SDA and SCL)	N/A	N/A	1	--	μs
t _{IBI_Issue}	Time to issue IBI after an event is detected when Bus is available	-	-	-	15	μs
t _{CLR_I3C_CMD_Delay}	Time from Clear Register Status to any I3C operation with Start condition to avoid IBI generation (PEC Disabled)	-	-	4	-	μs
	Time from Clear Register Status to any I3C operation with Start condition to avoid IBI generation (PEC Enabled)	-	-	15	-	μs
t _{DEVCTRLCCC_DELAY_PEC_DIS}	DEVCTRL CCC Followed by DEVCTRL CCC or Register Read/Write Command Delay ^{17,18,19}	3	--	3	--	μs
t _{WR_RD_DELAY_PEC_EN}	Register Write Command Followed by Register Read Command Delay in PEC Enabled Mode ^{18,19,20}	N/A	N/A	8	--	μs
t _{I2C_CCC_Update_Delay}	Minimum delay from SETHID CCC or SETAASA CCC to any other CCC or Read/Write Command	2.5	-	N/A	N/A	μs
t _{I3C_CCC_Update_Delay}	Minimum delay from ENEC/DISEC CCC or RSTDAA CCC to any other CCC or Read/Write Command	N/A	N/A	2.5	-	μs
t _{CCC_Delay}	Delay from any CCC to RSTDAA CCC	N/A	N/A	2.5	-	μs

17.4 AC Electrical Characteristics

Table 247 — AC Characteristics for SidebandBus Interface (cont'd)

Symbol	Parameter	I ² C Mode (Open Drain) ¹		I ³ C Basic Push-Pull ² (12.5 MHz)		Unit
		Min	Max	Min	Max	
NOTE 1	The timing parameters listed under this column correspond to Fast-mode Plus from the I2C specification. The DDR5RCD02 device must also support the values and conditions corresponding to the other (lower) speed modes supported by I2C specification, namely, Standard-mode and Fast-mode.					
NOTE 2	I3C Basic mode with Open Drain operation follows timing values as shown in I2C Mode - Open Drain column.					
NOTE 3	The DDR5RCD02 I2C/I3C Basic interface logic shall not initiate clock stretching.					
NOTE 4	The DDR5RCD02 must support bus timeout on I2C/I3C Basic access.					
NOTE 5	Devices participating in a transfer can abort the transfer in progress and release the bus by forcing SCL LOW. A timeout condition can only be ensured if the device forcing the timeout holds SCL LOW for tTIMEOUT,MAX or longer. After the controller in a transaction detects this condition, it must generate a STOP condition within or after the current data byte in the transfer process. Devices that have detected this condition must reset their communication and be able to receive a new START condition no later than 10ms after the controller generates the STOP condition.					
NOTE 6	See Figure 118 for input timing definitions.					
NOTE 7	See Figure 121 for voltage threshold definitions for rise and fall times.					
NOTE 8	Guaranteed by design and characterization, not necessarily tested.					
NOTE 9	The input setup time is referenced from SDA VIL or VIH threshold to SCL VIL threshold as shown in Figure 118. The input hold time is referenced from SCL VIL threshold to SDA VIL or VIH threshold as shown in Figure 118.					
NOTE 10	For a reSTART condition, or following a write cycle.					
NOTE 11	The DDR5RCD02 device guarantees tHD:DAT value in I2C mode of operation. See Figure 119 for I2C output timing definitions as well as SCL clock input threshold level and SDA data output threshold levels.					
NOTE 12	This timing parameter is guaranteed into output timing reference load as shown in Figure 123.					
NOTE 13	The DDR5RCD02 must be configured in I3C Basic mode to guarantee tDOUT value. See Figure 120 for I3C Basic output timing definition as well as SCL clock input threshold level and SDA data output threshold levels. See Figure 123 for output timing parameter measurement reference load.					
NOTE 14	The DDR5RCD02 must be configured in I3C Basic mode to guarantee tDOFFT value. See Figure 124. See Figure 123 for output timing parameter measurement reference load.					
NOTE 15	The DDR5RCD02 must be configured in I3C Basic mode to guarantee tDOFFC value. See Figure 125. See Figure 123 for output timing parameter measurement reference load.					
NOTE 16	The DDR5RCD02 must be configured in I3C Basic mode to guarantee tCL_r_DAT_f value. See Figure 127.					
NOTE 17	From STOP condition of DEVCTRL CCC to START condition for Register Read or Register Write Command Data Packet delay.					
NOTE 18	The device may send NACK if Host does not satisfy this timing parameter.					
NOTE 19	This timing constraint is only applicable when PEC is disabled. If PEC is enabled, this timing parameter does not apply.					
NOTE 20	From STOP condition for Register Write Command Data Packet to START condition for Register Read Command Data Packet delay.					

17.4 AC Electrical Characteristics (cont'd)

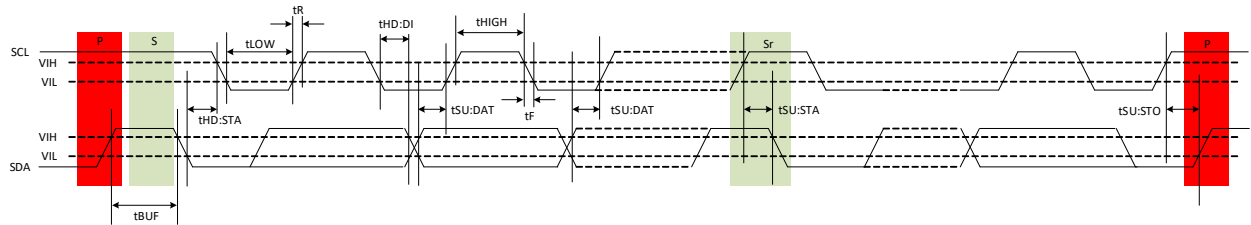


Figure 118 — I²C and I3C Basic AC Input Timing Parameter Definitions

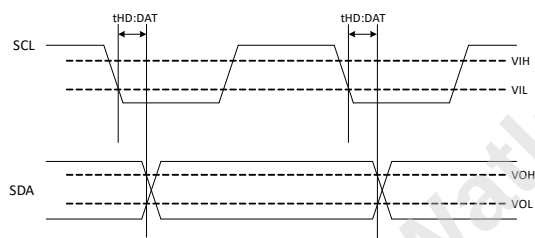


Figure 119 — I²C Bus AC Output Timing Parameter Definitions

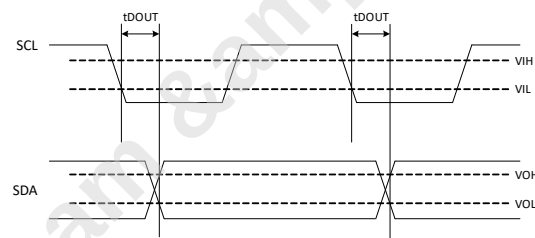


Figure 120 — I3C Basic AC Output Timing Parameter Definitions

17.4 AC Electrical Characteristics (cont'd)

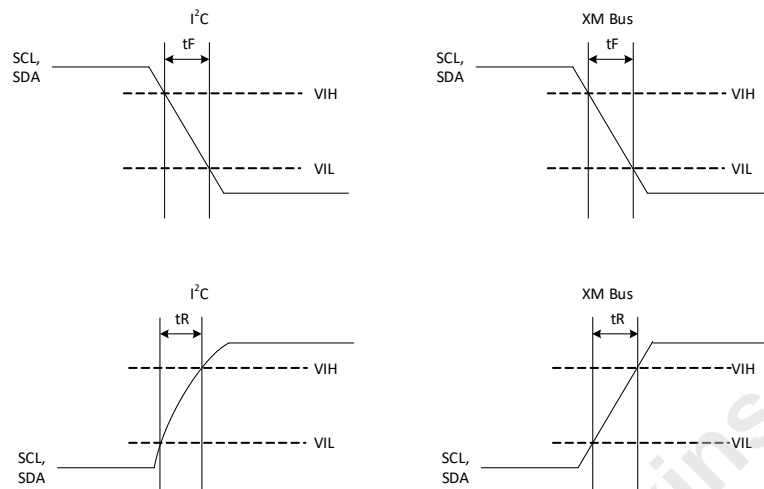


Figure 121 — Rise and Fall Timing Parameter Definitions

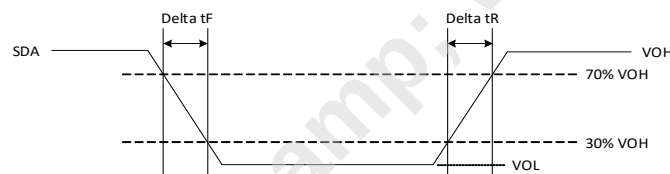


Figure 122 — Output Slew Rate Measurement Points

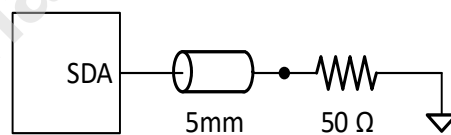


Figure 123 — Test Load for Output Slew Rate and Output Timing Measurement

17.4 AC Electrical Characteristics (cont'd)

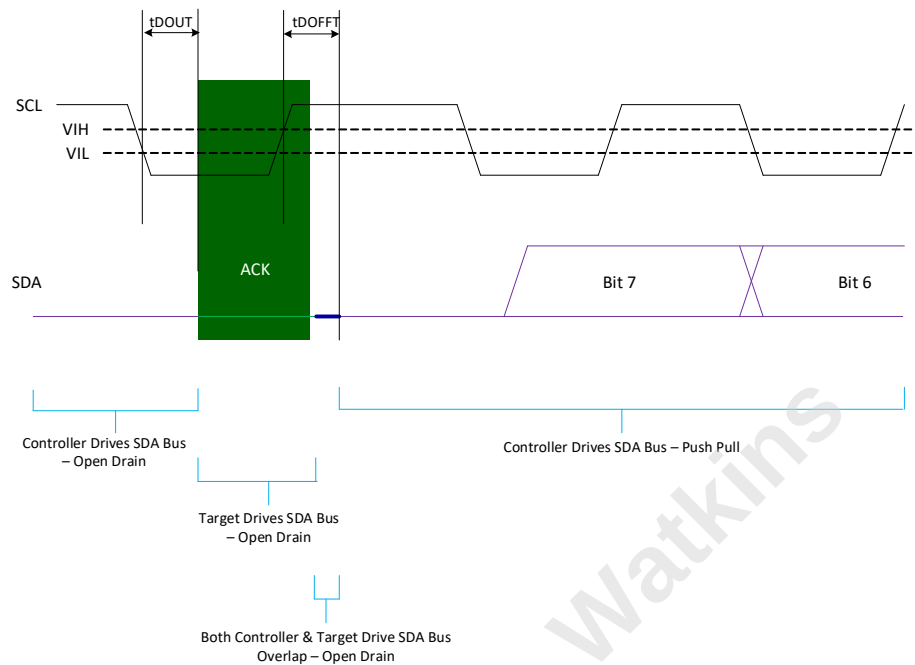


Figure 124 — I3C Basic Hand-off Operation - RCD Open Drain (ACK) to Controller Push-Pull

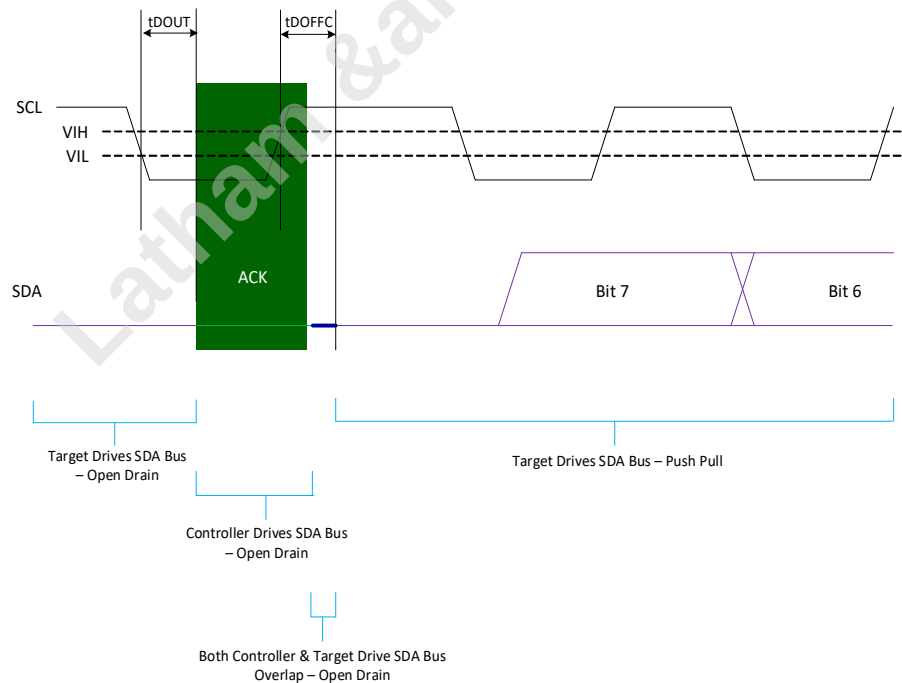


Figure 125 — I3C Basic Hand-off Operation - Controller Open Drain (ACK) to RCD Push-Pull

17.4 AC Electrical Characteristics (cont'd)

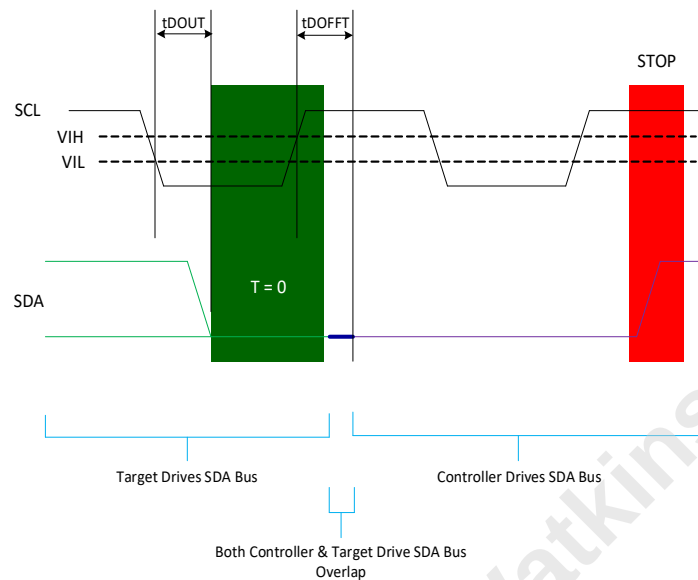


Figure 126 — I3C Basic T=0 Transition from RCD Read Operation to Controller STOP Bit

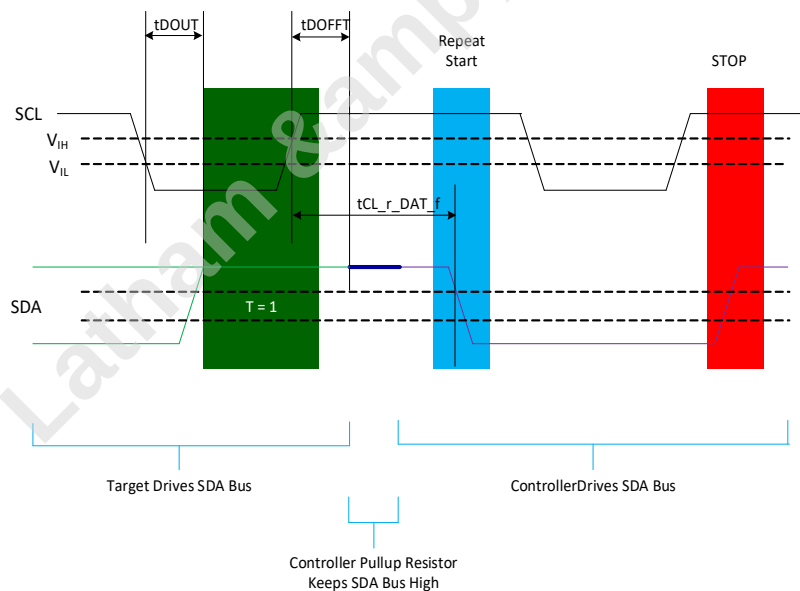


Figure 127 — I3C Basic T = 1 Transition from RCD Read Operation Interrupted by Controller Repeated START and STOP Bit

17.4 AC Electrical Characteristics (cont'd)

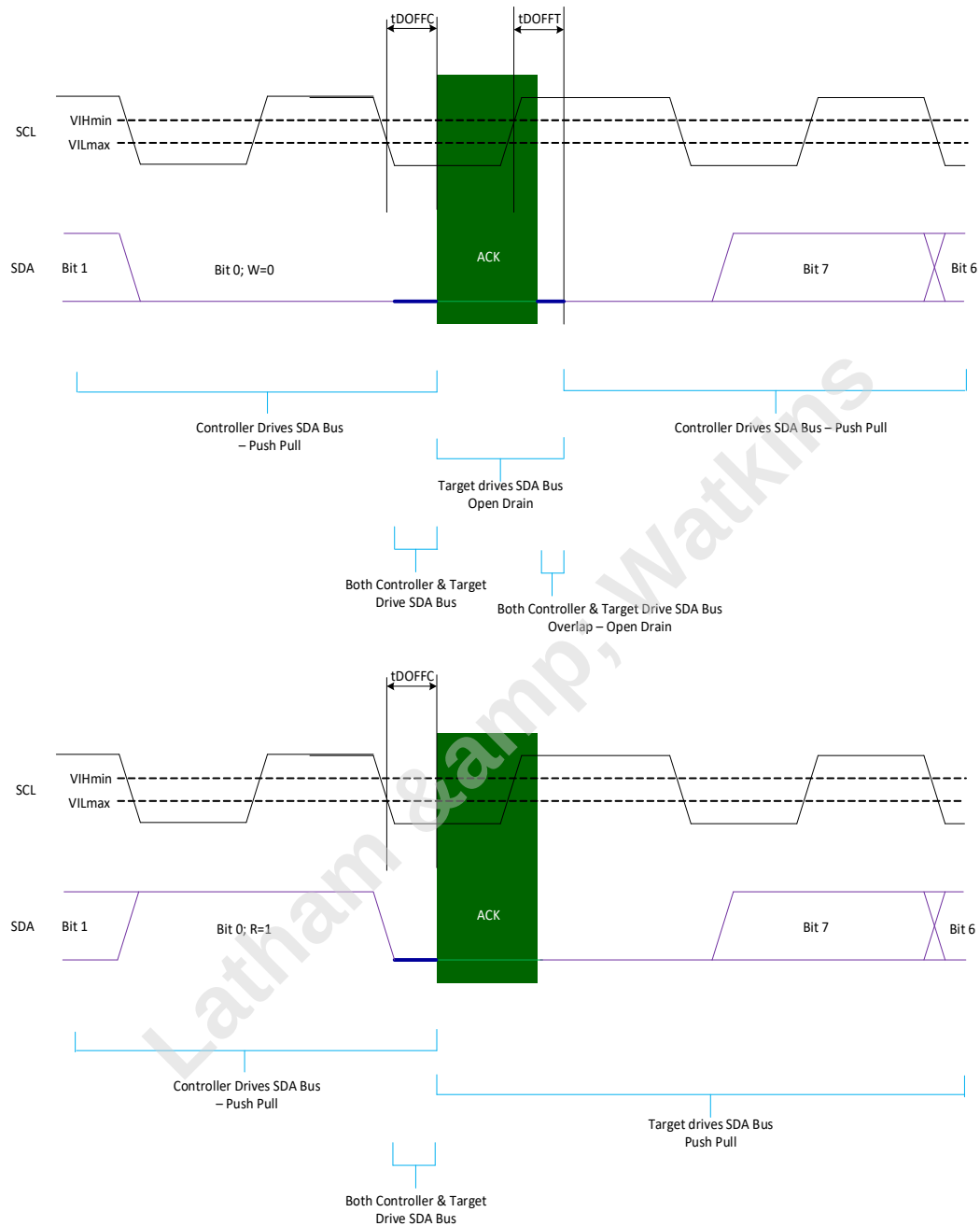
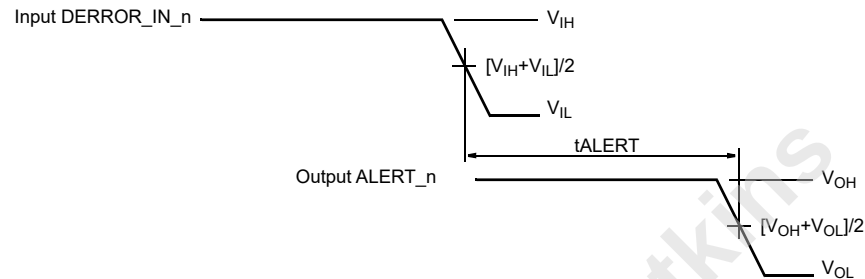


Figure 128 — Controller Push Pull to Target Open Drain Hand Off Operation

18 Test Circuits and Switching Waveforms

18.1 Alert Output Voltage Measurement Information

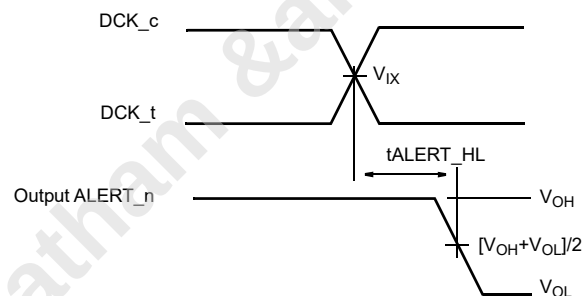
All input pulses are supplied by generators having the following characteristics: $1000 \text{ MHz} \leq \text{PRR}$ (Pulse Repetition Rate) $\leq 2436 \text{ MHz}$; $Z_o = 50 \Omega$; input slew rate = $1 \text{ V/ns} \pm 20\%$, unless otherwise specified.



NOTE 1 V_{IH} and V_{IL} represent the applied Logic HIGH and LOW input DC voltage levels resulting from $50\text{-}\Omega$ driver into the default on-die termination (60Ω).

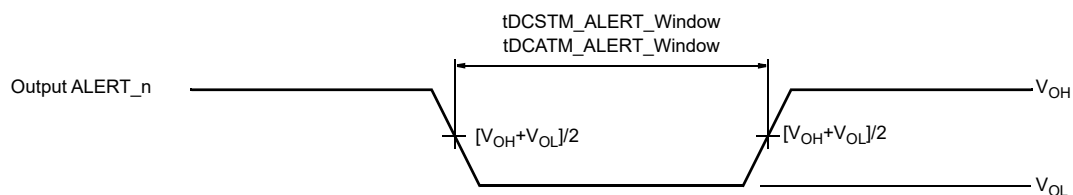
NOTE 2 V_{OH} and V_{OL} represent the measured Logic HIGH and LOW output DC voltage levels into the test load (50Ω).

Figure 129 — Voltage Waveforms, tALERT Measurement



NOTE 1 V_{OH} and V_{OL} represent the measured Logic HIGH and LOW output DC voltage levels into the test load (50Ω).

Figure 130 — Voltage Waveforms, tALERT_HL Measurement



NOTE 1 V_{OH} and V_{OL} represent the measured Logic HIGH and LOW output DC voltage levels into the test load (50Ω).

Figure 131 — Voltage Waveforms, tDCSTM_ALERT_Window, and tDCATM_ALERT_Window Measurements

18.2 DDR5RCD02 Reference Load

The circuit in Figure 132 represents the effective reference load of 50 ohms used in defining the relevant AC timing parameters of the device, as well as output slew rate measurements.

It is not intended as a precise representation of any particular system environment or a depiction of the actual load presented by a production tester. System designers should use IBIS or other simulation tools to correlate the timing reference load to a system environment. Manufacturers correlate to their production test conditions, generally one or more coaxial transmission lines terminated at the tester electronics.

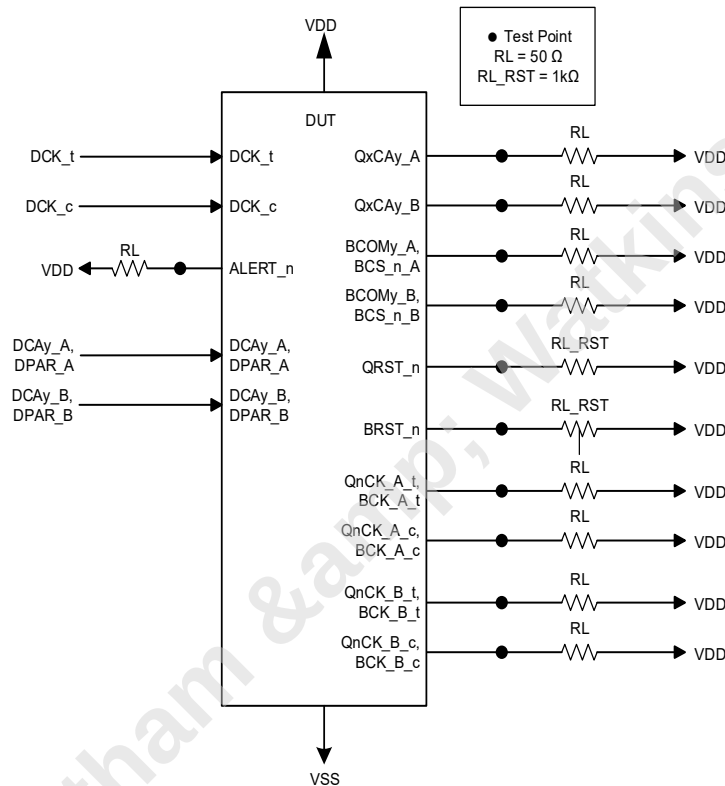


Figure 132 — Test Load

19 Reference to Other Applicable JEDEC Standards and Publications

- JEP95, JEDEC Registered and Standard Outlines for Solid State and Related Products.
- JEP104, Reference Guide to Letter Symbols for Semiconductor Devices.
- JESD21-C, Configuration for Solid State Memories.
- JESD8-11A, Definition of wide range non-terminated logic.
- JESD79-5, DDR5 SDRAM Specification.
- JESD82-522, DDR5DB02 Specification.
- MO-330A, Package Mechanical Outline.
- I3C Basic, Specification for I3C Basic Version 1.0 – 19 July 2018.
- JESD403-1, Specification for JEDEC Module SidebandBus.
- JS-001-2017, Joint JEDEC/ESDA Standard for Electrostatic Discharge Sensitivity Test – Human Body Model (HBM) – Component Level.
- JS-002-2018, ANSI/ESDA/JEDEC Joint Standard for Electrostatic Discharge Sensitivity Testing – Charged Device Model (CDM) – Device Level.



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1. I recommend changes to the following:

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☐ Test method number _____ Clause number _____

The referenced clause number has proven to be:

☐ Unclear ☐ Too Rigid ☐ In Error

☐ Other _____

2. Recommendations for correction:

3. Other suggestions for document improvement:

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